

ISC Class XII Mathematics

Chapter 11 Three Dimensional Geometry

189 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

- I. 11.1 – Direction Cosines and Direction Ratios of a Line
- II. 11.2 – Equation of a Line in Space
- III. 11.3 – Angle Between Two Lines
- IV. 11.4 – Shortest Distance Between Two Lines
- V. 11.5 – Equation of a Plane
- VI. 11.6 – Intercept and Three-Point Forms of a Plane
- VII. 11.7 – Angle Between Two Planes; Angle Between a Line and a Plane
- VIII. 11.8 – Distance of a Point from a Plane

Compiled Answer Key – Step-by-step ISC board-style solutions

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Index of Problems

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
Exercise 11.1 – Direction Cosines and Direction Ratios of a Line				
1				
(i)	Find the distance of the point $P(2, -3, 4)$ from the x -axis.	1	Numerical	11.1
(ii)	Find the distance of the point $Q(5, -4, 12)$ from the y -axis.	1	Numerical	11.1
(iii)	Find the distance of the point $P(5, -6, 3)$ from the XOY plane.	1	Numerical	11.1
(iv)	Find the distance of the point $Q(2, -4, 6)$ from the ZOX plane.	1	Numerical	11.1
2				
(i)	If a line makes angles 90° , 60° and 30° with the positive directions of x ...	2	Numerical	11.1
(ii)	Write the direction cosines of z -axis.	2	Numerical	11.1
3				

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
(i)	Can the numbers $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ be the directio...	3	Numerical	11.1
(ii)	Can a line have direction angles $45^\circ, 60^\circ, 120^\circ$?	3	Numerical	11.1
(iii)	Prove that 1, 1, 1 cannot be direction cosines of a straight line.	3	Numerical	11.1
4	If the direction cosines of a line are $\left\langle \frac{1}{c}, \frac{1}{c}, \frac{1}{c} \right\rangle \dots$	3a	Numerical	11.1
5				
(i)	If a line makes angles $90^\circ, 60^\circ$ and θ with x, y and z -axes respectively...	4	Numerical	11.1
(ii)	If a line makes angles of 90° and 60° respectively with the positive directions o...	4	Numerical	11.1
(iii)	A line makes an angle of $\frac{\pi}{4}$ with each of X-axis and Y-axis. What angle does it ma...	4	Numerical	11.1
(iv)	If a straight line makes angles of 45° and 120° with the positive directions of x ...	4	Numerical	11.1
(v)	If the direction cosines of a line are $\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}$, then find...	4	Numerical	11.1
6				
(i)	If a line has direction ratios $2, -1, -2$, determine its direction cosines.	2	Numerical	11.1
(ii)	If direction numbers of a line are $\langle 2, -6, 3 \rangle$, find its direction cosines.	2	Numerical	11.1
7				
(i)	Write the direction cosines of the line joining the points $(1, 0, 0)$ and $(0, 1, 1)$.	2a	Numerical	11.1
(ii)	Find the direction cosines of the line passing through the points $P(2, 3, 5)$ and $Q(-1, 2, 4)$...	2a	Numerical	11.1
8				
(i)	Show that the line joining the origin to the point $(2, 1, 1)$ is perpendicular to the line det...	5	Proof	11.1
(ii)	Show that the line through the points $(1, -1, 2)$ and $(3, 4, -2)$ is perpendicular to the lin...	5	Proof	11.1

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
9				
(i)	For what value of p will the line through $(4, 1, 2)$ and $(5, p, 0)$ be perpendicular to the...	6	Numerical	11.1
(ii)	Find the values of p and q so that the line joining the points $(7, p, 2)$ and $(q, -2, 5)$...	6	Numerical	11.1
10	Show that the points $(2, 3, 4)$, $(-1, -2, 1)$ and $(5, 8, 7)$ are collinear.	7	Proof	11.1
11	Find the direction ratios and the direction cosines of the line joining the points $(5, 4, -3)$...	8	Numerical	11.1
12				
(i)	Find the angle between the lines whose direction cosines are...	9	Numerical	11.1
(ii)	Find the angle between the lines whose direction numbers are $\langle 1, 2, 1 \rangle$, $\langle 2, -3, 4 \rangle$...	9	Numerical	11.1
(iii)	Find the angle between the lines whose direction ratios are $\langle 1, 1, 2 \rangle$ and $\langle \sqrt{3} - 1, -\sqrt{3} - 1, 4 \rangle$...	9	Numerical	11.1
13	Show that the lines with direction cosines...	5a	Proof	11.1
14	If A, B are the points $(2, 1, -2)$, $(3, -4, 5)$, then find the angle that OA makes with OB ...	9	Numerical	11.1
15	If the coordinates of the points A, B, C and D are $(1, 2, 3)$, $(4, 5, 7)$, $(-4, 3, -6)$ and...	9	Numerical	11.1
16	Show that the points $A(4, 7, 8)$, $B(2, 3, 4)$, $C(-1, -2, 1)$ and $D(1, 2, 5)$, taken in order, a...	10	Proof	11.1
17	Find the direction cosines of a line which is perpendicular to the two lines formed by joining...	11	Numerical	11.1
18	Find the angle between the lines whose direction cosines satisfy the relations $l + m + n = 0$...	12	Numerical	11.1
19 (i)	Show that the straight lines whose direction cosines are given by $2l + 2m - n = 0$ and $mn + nl + lm = 0$...	12a	Proof	11.1

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
19 (ii)	Find the angle between the lines whose direction cosines are given by the equations $3l + m + 5n = 0...$	12b	Numerical	11.1
20	Find the point in which the join of $A(-9, 4, 5)$ and $B(11, 0, -1)$ is met by the perpendicular...	13	Numerical	11.1
21	Find the coordinates of the foot of perpendicular drawn from the point $A(1, 2, 1)$ on the line...	13	Numerical	11.1
22	The adjoining figure shows a rectangular box of size a , b and c units along x , y and...	14	Numerical	11.1

Exercise 11.2 – Equation of a Line in Space

1 (i)	Find the vector equation of the line passing through the point $(3, 4, 5)$ and parallel to the...	15	Numerical	11.2
1 (ii)	Find the vector equation of the line which is parallel to the vector $3\hat{i} - 2\hat{j} + 6\hat{k}...$	15	Numerical	11.2
2	Find the cartesian equation of the line which passes through the point with position vector $2\hat{i} - \hat{j} + 4\hat{k}...$	15	Numerical	11.2
3 (i)	Write the vector equation of the line $\frac{x+3}{2} = \frac{y-5}{4} = \frac{z+6}{2}$.	16	Numerical	11.2
3 (ii)	Write the vector equation of the line $\frac{x-5}{3} = \frac{y+4}{7} = \frac{6-z}{2}$.	16	Numerical	11.2
4 (i)	Write the cartesian equation of the following line given in vector form...	16	Numerical	11.2
4 (ii)	Find the cartesian equation of the line which passes through the point $(-2, 4, -5)$ and parallel to the vector $2\hat{i} - \hat{j} + 4\hat{k}...$	15	Numerical	11.2
5 (i)	Find the vector equation of the line passing through the points $(-1, 0, 2)$ and $(3, 4, 6)$.	17	Numerical	11.2
5 (ii)	Find the vector and the cartesian equations of the line that passes through origin and the point $(2, -1, 3)$.	17	Numerical	11.2
6 (i)	If the equation of a line AB is $\frac{x-3}{1} = \frac{y+2}{-2} = \frac{z-5}{4}$, find the direction ratios of the line.	16	Numerical	11.2

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
6 (ii)	The cartesian equations of a line AB are $\frac{2x-1}{2} = \frac{4-y}{7} = \frac{z+1}{2}$. W...	16	Numerical	11.2
6 (iii)	Write the direction cosines of a vector parallel to the line $\frac{4-x}{2} = \frac{y+3}{3} = \frac{z+2}{6}$...	16	Numerical	11.2
6 (iv)	The cartesian equation of a line AB is $\frac{2x-1}{\sqrt{3}} = \frac{y+2}{2} = \frac{z-3}{3}$...	16	Numerical	11.2
7 (i)	The cartesian equations of a line are $3x-3=2y+1=5-6z$. Write the direction ratios of...	16	Numerical	11.2
7 (ii)	Find the vector equation of the X-axis.	15	Numerical	11.2
7 (iii)	The equations of a line are $5x-3=15y+7=3-10z$. Write the direction cosines of the li...	16	Numerical	11.2
8 (i)	Find the vector and the cartesian equations of the line through the point $(5, 2, -4)$ and whic...	15	Numerical	11.2
8 (ii)	A line passes through the point with position vector $2\hat{i} - \hat{j} + 4\hat{k}$ and is in...	15	Numerical	11.2
9 (i)	Find the coordinates of the point where the line $\frac{x-1}{3} = \frac{y+4}{7} = \frac{z+4}{2}$...	18	Numerical	11.2
9 (ii)	Find the coordinates of the point where the line through $(-1, 1, -8)$ and $(5, -2, 10)$ crosses...	18	Numerical	11.2
9 (iii)	Find the coordinates of the point where the line through $(5, 1, 6)$ and $(3, 4, 1)$ crosses th...	18	Numerical	11.2
10	Find the coordinates of the point where the line through the points $(-1, 4, 4)$ and $(2, 1, -5)$...	18	Numerical	11.2
11	The cartesian equations of a line are $x = ay + b, z = cy + d$. Find direction numbers of the l...	16	Numerical	11.2
12 (i)	Find the vector equation of the line passing through the point $A(2, -1, 1)$ and parallel to th...	15	Numerical	11.2
12 (ii)	Find the vector equation of a line passing through the point with position vector $\hat{i} - 2\hat{j} - 3\hat{k}$...	15	Numerical	11.2

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
12 (iii)	Find the vector equation of the line passing through the point $(2, 3, -5)$ and making equal an...	19	Numerical	11.2
13	Find the vector equation of the line joining the points whose position vectors are $2\hat{i} - \hat{j} + \hat{k}$...	17	Numerical	11.2
14	Find the vector equation of the line passing through the point $A(1, 2, -1)$ and parallel to th...	15	Numerical	11.2
15	The points $A(4, 5, 10)$, $B(2, 3, 4)$ and $C(1, 2, -1)$ are three vertices of a parallelogram...	20	Numerical	11.2
16 (i)	Find the equation of the line passing through the points $A(-2, 4, 7)$, $B(3, -6, -8)$. Hence...	7	Proof	11.2
16 (ii)	If the points $(-1, 3, 2)$, $(-4, 2, -2)$ and $(5, 5, \lambda)$ are collinear, then find the va...	7a	Numerical	11.2
17 (i)	Find the coordinates of the points on the line $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{6}$...	21	Numerical	11.2
17 (ii)	Find the coordinates of points on line $\frac{x}{1} = \frac{y-1}{2} = \frac{z+1}{2}$ which a...	21	Numerical	11.2
18	Find the points on the line through the points $A(3, 4, -1)$ and $B(5, 1, 5)$ at a distance of...	21	Numerical	11.2
19 (i)	Show that the lines $\frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7}$ and $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$...	22	Both	11.2
19 (ii)	Show that the lines $\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$...	22	Both	11.2
20 (i)	Prove that the line through $A(0, -1, -1)$ and $B(4, 5, 1)$ intersects the line through $C(3, 9, 4)$...	22	Both	11.2
20 (ii)	Show that the lines $\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-1}{5}$ and $\frac{x-2}{4} = \frac{y-1}{3} = \frac{z+1}{-2}$...	22a	Proof	11.2
21	A line with direction numbers $\langle 2, 7, -5 \rangle$ is drawn to intersect the lines...	23	Numerical	11.2

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
22	The vertices of $\triangle ABC$ are $A(1, 1, 0)$, $B(1, 2, 1)$ and $C(-2, 2, -1)$. Find the equati...	24	Numerical	11.2
Exercise 11.3 – Angle Between Two Lines				
1 (i)	Find the angle between the following pair of lines...	25	Numerical	11.3
1 (ii)	Find the angle between the following pair of lines...	25	Numerical	11.3
1 (iii)	Find the angle between the following pair of lines...	25	Numerical	11.3
2 (i)	Find the angle between the following pair of lines...	26	Numerical	11.3
2 (ii)	Find the angle between the following pair of lines...	27	Numerical	11.3
3 (i)	Show that the lines $\frac{x-5}{7} = \frac{2-y}{5} = \frac{z}{3}$ and $\frac{3-x}{-1} = \frac{y}{2} = \frac{z+1}{3}$...	28	Proof	11.3
3 (ii)	Check whether the lines given by the equations $x = 2\lambda + 2$, $y = 7\lambda + 1$, $z = -3\lambda - 3$...	28	Numerical	11.3
4 (i)	If the lines $\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-3}{2}$ and $\frac{x-1}{3k} = \frac{y-1}{1} = \frac{z-6}{-5}$...	29	Numerical	11.3
4 (ii)	Find the value of k so that the lines $x = -y = kz$ and $x - 2 = 2y + 1 = -z + 1$ are perpend...	29	Numerical	11.3
4 (iii)	Find the value of p so that the lines $\frac{x-1}{-2} = \frac{y-4}{3p} = \frac{z-3}{4}$ and...	29	Numerical	11.3
5 (i)	Find the angle between the following pair of lines...	26	Numerical	11.3
5 (ii)	Find the angle between the following pair of lines...	26	Numerical	11.3
6 (i)	If the lines $\frac{x-1}{-3} = \frac{y-2}{2\lambda} = \frac{z-3}{2}$ and $\frac{x-1}{3\lambda} = \frac{y-1}{2} = \frac{z-6}{-5}$...	28	Numerical	11.3
6 (ii)	Find the value of λ so that the lines $\frac{1-x}{3} = \frac{7y-14}{\lambda} = \frac{z-3}{2}$...	28	Numerical	11.3
7	Find the equations of the perpendicular drawn from the point $A(2, 4, -1)$ to the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$...	30	Numerical	11.3
8 (i)	Find the foot of perpendicular drawn from the point $(0, 2, 3)$ on the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$...	31	Numerical	11.3
8 (ii)	Find the coordinates of the foot of the perpendicular drawn from the point $(5, 7, 3)$ to the l...	31	Numerical	11.3

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
8 (iii)	Vertices B and C of $\triangle ABC$ lie on the line $\frac{x+2}{2} = \frac{y-1}{1} = \frac{z}{4} \dots$	32	Numerical	11.3
8 (iv)	Find the distance of the point $(2, 4, -1)$ from the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9} \dots$	33	Numerical	11.3
9	Find the image of the point $P(1, 2, 3)$ in the line...	31a	Numerical	11.3
10 (i)	Find the equations of the line passing through the point $(2, 1, 3)$ and perpendicular to the l...	34	Numerical	11.3
10 (ii)	Determine the vector and cartesian equations of the line passing through the point $(1, 2, -4) \dots$	34	Numerical	11.3
10 (iii)	Find the vector and cartesian equations of the line which is perpendicular to the lines with eq...	34a	Numerical	11.3
10 (iv)	Find the equation of the line which bisects the line segment joining the points $A(2, 3, 4)$ an...	34	Numerical	11.3
10 (v)	If the lines $\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-3}{2}$ and $\frac{x-1}{3k} = \frac{y-1}{1} = \frac{z-6}{-7} \dots$	34b	Numerical	11.3
11	Find the equation of a line passing through the point $(1, 2, -4)$ and perpendicular to two lin...	34	Numerical	11.3
12	Find the equations of all the sides of the parallelogram $ABCD$ whose vertices are $A(4, 7, 8) \dots$	35	Numerical	11.3
13	Find the value of p for which the lines $\vec{r} = \lambda \hat{i} + (2\lambda + 1)\hat{j} + (3\lambda + 2)\hat{k} \dots$	36	Numerical	11.3

Exercise 11.4 – Shortest Distance Between Two Lines

1				
(i)	$\vec{r} = \hat{i} + 2\hat{j} + \hat{k} + \lambda(\hat{i} - \hat{j} + \hat{k})$ and...	37	Numerical	11.4
(ii)	$\vec{r} = 6\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k})$ and...	37	Numerical	11.4
(iii)	$\vec{r} = 4\hat{i} - \hat{j} + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$ and...	37	Numerical	11.4
(iv)	$\vec{r} = (8 + 3\lambda)\hat{i} - (9 + 16\lambda)\hat{j} + (10 + 7\lambda)\hat{k}$ and...	37	Numerical	11.4
(v)	$\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + \lambda(3\hat{i} - 2\hat{j} + 5\hat{k})$ and...	37	Numerical	11.4

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
2	Find the shortest distance between the lines...	38	Numerical	11.4
3	Find the shortest distance between the lines...	38a	Numerical	11.4
4				
(i)	$\vec{r} = \hat{i} - \hat{j} + \lambda(2\hat{i} + \hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} + \mu(\hat{i} + \hat{j} - \hat{k})...$	39	Numerical	11.4
(ii)	$\vec{r} = \hat{i} + \hat{j} - \hat{k} + \lambda(3\hat{i} - \hat{j})$ and $\vec{r} = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k})...$	39	Numerical	11.4
(iii)	$\frac{x-1}{2} = \frac{y+1}{3} = z$ and $\frac{x+1}{5} = \frac{y-2}{1}, z = 2$	39	Numerical	11.4
5	Prove that the lines $2x = y = -z$ and $x - 1 = -y - 2 = -2z + 1$ are skew.	40	Proof	11.4
6	Find the shortest distance between the lines...	41	Numerical	11.4
7	Find the vector equation of the line passing through the point $(2, 1, -1)$ and parallel to the...	41a	Numerical	11.4
8	Find the vector equation of the line parallel to the line $\frac{x-1}{5} = \frac{3-y}{2} = \frac{z+1}{4}...$	41a	Numerical	11.4
9	Find the distance between the lines...	41	Numerical	11.4
10	Find the vector and cartesian equations of a line passing through the point $(1, 2, -4)$ and pa...	41a	Numerical	11.4
11	Determine the vector equation of the line passing through the point $(-1, 3, -2)$ and perpendic...	34	Numerical	11.4
12	Find the shortest distance between the lines L_1 and L_2 given below: (system given in the...	37	Numerical	11.4

Exercise 11.5 – Equation of a Plane

1				
(i)	Find the direction cosines of the normal to the plane $2x + 3y - z - 7 = 0$.	42	Numerical	11.5
(ii)	Find a unit normal vector to the plane $2x - 2y - z - 5 = 0$.	42	Numerical	11.5
2				
(i)	$\vec{r} \cdot (\hat{i} + \hat{j} - \hat{k}) = 2$	16	Numerical	11.5
(ii)	$\vec{r} \cdot (5\hat{i} + 7\hat{j} - 11\hat{k}) = 14$	16	Numerical	11.5

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
3				
(i)	$3x + 7y - 6z = 5$	16	Numerical	11.5
(ii)	$9x - 2y - 5z + 3 = 0$	16	Numerical	11.5
4				
(i)	Find the equation of the plane passing through the point $(2, -3, 1)$ and perpendicular to the...	43	Numerical	11.5
(ii)	Find the equation of a plane which bisects the line segment joining the points $A(2, 3, 5)$ and...	43	Numerical	11.5
(iii)	Find the equation of the plane passing through the point $(4, -1, 2)$ and perpendicular to the...	43	Numerical	11.5
5				
(i)	Show that the line $\frac{x-2}{1} = \frac{y+2}{-1} = \frac{z-3}{4}$ is parallel to the plane...	45	Numerical	11.5
(ii)	If the line $\frac{x+1}{3} = \frac{y-2}{4} = \frac{z+6}{5}$ is parallel to the plane $2x - 3y + kz = 0$...	45	Numerical	11.5
6				
(i)	Write the image of the point $(-2, 3, 5)$ in the plane XOY.	46	Numerical	11.5
(ii)	Write the image of the point $(-4, -3, 6)$ in the plane YOZ.	46	Numerical	11.5
(iii)	Write the coordinates of the point which is the reflection of the point (α, β, γ) ...	46	Numerical	11.5
(iv)	Write the image of the point $(2, -3, -1)$ in the plane $\vec{r} \cdot \hat{j} = 0$.	46	Numerical	11.5
7	If the vector equation of a plane is $\vec{r} \cdot (2\hat{i} + 2\hat{j} - \hat{k}) = 21$, then...	47	Numerical	11.5
8				
(i)	Find the vector equation of the plane which is at a distance of $\frac{6}{\sqrt{29}}$ units fr...	44	Numerical	11.5
(ii)	Find the vector equation of a plane which is at a distance of 5 units from the origin and has...	44	Numerical	11.5
9				
(i)	$2x - 3y + 4z - 6 = 0$	42	Numerical	11.5
(ii)	$3x + 5 = 0$	42	Numerical	11.5

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
10				
(i)	Find the vector and the cartesian equations of the plane that passes through the point $(1, 0, -2)$...	43	Numerical	11.5
(ii)	Find the vector and the cartesian equations of a plane through the point with position vector $2\hat{i} - \hat{j} + \hat{k}$...	43	Numerical	11.5
11				
(i)	If the foot of perpendicular drawn from the origin to a plane is $(12, -4, -3)$, find the equat...	48	Numerical	11.5
(ii)	Find the equation of the plane passing through the point $(2, -3, 1)$ and perpendicular to the...	48	Numerical	11.5
12	What is the length of perpendicular drawn from origin to the plane $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) + 35 = 0$...	48a	Numerical	11.5
13				
(i)	Find the vector equation of the line passing through the point $(1, -1, 2)$ and perpendicular t...	49	Numerical	11.5
(ii)	Find the equation of the line passing through the point with position vector $\hat{i} - \hat{j} + 2\hat{k}$...	49	Numerical	11.5
14	Show that the plane $x - 5y - 2z = 1$ contains the line $\frac{x-5}{3} = y = 2-z$.	45a	Proof	11.5
15				
(i)	Show that the line $\vec{r} = \hat{i} + \hat{j} + \lambda(2\hat{i} + \hat{j} + 4\hat{k})$ lies...	45a	Numerical	11.5
(ii)	If the line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lies in the plane $x + 3y - \alpha z + \beta = 0$...	45a	Numerical	11.5
(iii)	Show that the line $\vec{r} = (2\hat{i} - 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + 4\hat{k})$...	45a	Numerical	11.5
16				
(i)	Find the point where the line $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+3}{4}$ meets the pla...	50	Numerical	11.5

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
(ii)	Find the coordinates of the point where the line through $(3, -4, -5)$ and $(2, -3, 1)$ crosses...	50	Numerical	11.5
(iii)	Find the point of intersection of line $\vec{r} = 2\hat{i} - \hat{j} + 2\hat{k} + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})...$	50	Numerical	11.5
17				
(i)	Find the distance between the point $(-1, -5, -10)$ and the point of intersection of the line...	50	Numerical	11.5
(ii)	Find the distance of the point $(2, 12, 5)$ from the point of intersection of the line...	50	Numerical	11.5
(iii)	Find the distance of the point $P(-2, -4, 7)$ from the point of intersection Q of the line...	50	Numerical	11.5
18	Find the equation of the line passing through the point $P(4, 6, 2)$ and the point of intersect...	50	Numerical	11.5
19	Find the point where the line joining the points $(1, 3, 4)$ and $(-3, 5, 2)$ intersects the pl...	50	Numerical	11.5
20				
(i)	Find the distance of the point $(3, 4, 5)$ from the plane $x + y + z = 2$ measured parallel to...	51	Numerical	11.5
(ii)	Find the distance of the point $(2, 3, 4)$ measured along the line $\frac{x-4}{3} = \frac{y+5}{6} = \frac{z+1}{2}...$	51	Numerical	11.5
21	A line with positive direction cosines passes through the point $P(-2, 3, 2)$ and makes equal a...	52	Numerical	11.5
22				
(i)	$x + y + z = 1$	48	Numerical	11.5
(ii)	$2x + 3y + 4z - 12 = 0$	48	Numerical	11.5
(iii)	$5y + 8 = 0$	48	Numerical	11.5
23				
(i)	Find the length and the foot of perpendicular drawn from the point $(2, 3, 7)$ to the plane $3x - y - z = 7...$	53	Numerical	11.5
(ii)	Find the length and the foot of perpendicular drawn from the point $(1, 1, 2)$ to the plane...	53	Numerical	11.5

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
24	From the point $P(1, 2, 4)$, a perpendicular is drawn on the plane $2x + y - 2z + 3 = 0$. Find...	53	Numerical	11.5
25				
(i)	Find the image of the point $(1, 2, 3)$ in the plane $x + 2y + 4z = 38$.	46	Numerical	11.5
(ii)	Find the image of the point having position vector $3\hat{i} - 2\hat{j} + \hat{k}$ in the plane...	46	Numerical	11.5
26	Prove that the normals to the planes $3x - 2y + 5z = 7$ and $4x + 11y + 2z + 1 = 0$ are perpend...	54	Proof	11.5

Exercise 11.6 – Intercept and Three-Point Forms of a Plane

1				
(i)	Write the intercept cut off by the plane $2x + y - z = 5$ on x -axis.	55	Numerical	11.6
(ii)	Write the intercept cut off by the plane $3x - 4y + z = 6$ on y -axis.	55	Numerical	11.6
(iii)	Write the intercept cut off by the plane $5x - 2y + 3z + 7 = 0$ on z -axis.	55	Numerical	11.6
2				
(i)	Find the intercepts made by the plane $2x + y - z = 5$ on the coordinate axes.	55	Numerical	11.6
(ii)	Write the sum of intercepts cut off by the plane $\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) - 5 = 0$...	55	Numerical	11.6
3				
(i)	Write the equation of the plane which has intercepts 2, 3, and 4 on the x -axis, y -axis, and z -axis.	56	Numerical	11.6
(ii)	Write the equation of the plane with intercepts 2, -3, and 5 on the x -axis, y -axis, and z -axis.	56	Numerical	11.6
(iii)	Find the equation of the plane passing through the intersection of the planes $x + 2y + 3z + 4 = 0$ and $2x + y - z = 5$.	56	Numerical	11.6
4				
(i)	Find the equation of the plane passing through the points $(2, 3, -4)$ and $(1, -1, 3)$ and perpendicular to the plane $x + 2y + 3z + 4 = 0$.	57	Numerical	11.6

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
(ii)	Find the equation of the plane through the points $(2, 2, -1)$ and $(3, 4, 2)$ and parallel to...	57	Numerical	11.6
5	Find the equation of the plane which passes through the point $(3, 2, 0)$ and contains the line...	58	Numerical	11.6
6	Find the foot of perpendicular drawn from the point $P(1, 2, 3)$ on the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$...	60	Numerical	11.6
7				
(i)	Find the equation of the plane passing through the points $(-1, 2, 0)$, $(2, 2, -1)$ and parall...	57	Numerical	11.6
(ii)	Find the vector equation of the plane passing through the points $(0, -4, 1)$ and $(2, 3, -2)$...	57	Numerical	11.6
8	Find the equation of the plane passing through the point $(1, 2, -4)$ and parallel to the lines...	59	Numerical	11.6
9	Find the vector as well as the cartesian equation of the plane passing through the point $(1, 2, -4)$...	59	Numerical	11.6
10	Find the equation of the plane containing the line $\frac{x-4}{2} = \frac{y-2}{3} = \frac{z-3}{4}$...	61	Numerical	11.6
11	Find the vector equation of the plane which contains the two parallel lines $\frac{x-4}{1} = \frac{y-3}{-4} = \frac{z-2}{5}$...	62	Numerical	11.6
12	If the lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$...	39	Numerical	11.6
13	Show that the lines $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$ and $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$...	63	Proof	11.6
14	Show that the lines $\frac{x-2}{1} = \frac{y-2}{3} = \frac{z-3}{1}$ and $\frac{x-2}{1} = \frac{y-3}{4} = \frac{z-4}{2}$...	63	Proof	11.6

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
15	If the lines $\frac{x-1}{-3} = \frac{y-2}{-2k} = \frac{z-3}{1}$ and $\frac{x-1}{k} = \frac{y-2}{1} = \frac{z-3}{5}$...	64	Numerical	11.6
16				
(i)	$(-2, 6, 6)$, $(1, -1, 0)$, and $(1, 2, -1)$	65	Numerical	11.6
(ii)	$(2, 1, 0)$, $(3, -2, -2)$, and $(3, 1, 7)$	65	Numerical	11.6
17	Find the vector equation of the plane passing through points with position vectors $2\hat{i} + 5\hat{j} - 3\hat{k}$...	65	Numerical	11.6
18				
(i)	Show that the points $(1, -1, 1)$, $(2, 3, 1)$, $(1, 2, 3)$, and $(0, -2, 3)$ are coplanar. Als...	66	Proof	11.6
(ii)	Find k such that the four points $A(4, 4, 4)$, $B(5, k, 8)$, $C(5, 4, 1)$, and $D(7, 7, 2)$ a...	66	Proof	11.6
19				
(i)	Find the equations of the line passing through $(2, -1, 2)$ and $(5, 3, 4)$ and of the plane pa...	50	Numerical	11.6
(ii)	Find the vector and the cartesian equations of the plane passing through the points $(2, 5, -3)$...	50	Numerical	11.6
20				
(i)	Find the equation of the plane passing through the point $(2, 4, 6)$ and making equal intercept...	55	Numerical	11.6
(ii)	Find the equation of the plane which passes through the point $(1, -2, -1)$ and makes intercept...	55	Numerical	11.6
21	A plane meets the coordinates axes at A , B , and C respectively such that centroid of triang...	67	Numerical	11.6
22	Find the equation of the plane containing the line $\frac{x}{-2} = \frac{y-1}{3} = \frac{1-z}{1}$...	58	Numerical	11.6
Exercise 11.7 – Angle Between Two Planes; Angle Between a Line and a Plane				
1				
(i)	$\vec{r} \cdot (\hat{i} + \hat{j}) = 1$ and $\vec{r} \cdot (\hat{i} + \hat{k}) = 3$	68	Numerical	11.7
(ii)	$\vec{r} \cdot (\hat{i} - 2\hat{j} - 2\hat{k}) = 1$ and $\vec{r} \cdot (3\hat{i} - 6\hat{j} + 2\hat{k}) = 0$...	68	Numerical	11.7

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
(iii)	$\vec{r} \cdot (3\hat{i} - 4\hat{j} + 5\hat{k}) = 0$ and $\vec{r} \cdot (2\hat{i} - \hat{j} - 2\hat{k}) = 7...$	68	Numerical	11.7
(iv)	$2x - y + z = 6$ and $x + y + 2z = 7$	68	Numerical	11.7
2				
(i)	$3x - 5y + 3z - 1 = 0$ and $2x + 3y + 3z = 7$	69	Proof	11.7
(ii)	$\vec{r} \cdot (2\hat{i} + 6\hat{j} + 6\hat{k}) = 13$ and $\vec{r} \cdot (3\hat{i} + 4\hat{j} - 5\hat{k}) + 7 = 0...$	69	Proof	11.7
3				
(i)	$\vec{r} \cdot (p\hat{i} + 2\hat{j} + 3\hat{k}) = 5$ and $\vec{r} \cdot (\hat{i} + 2\hat{j} - 7\hat{k}) + 11 = 0...$	70	Numerical	11.7
(ii)	$x - 4y + pz + 3 = 0$ and $2x + 2y + 3z = 5$	70	Numerical	11.7
(iii)	$x - 2y + 4z = 10$ and $18x + 17y + pz = 50$	70	Numerical	11.7
4				
(i)	Write the equation of the plane passing through origin and parallel to the plane $5x - 3y + 7z + 13 = 0...$	72	Numerical	11.7
(ii)	Find the equation of the plane through the point $(1, 4, -2)$ and parallel to the plane $-2x + y - 3z = 7...$	72	Numerical	11.7
(iii)	Find the equation of the plane with intercept 3 on the y-axis and parallel to xz-plane.	72	Numerical	11.7
5				
(i)	Find the equation of the plane through the point $3\hat{i} + 4\hat{j} - \hat{k}$ and parallel to...	72	Numerical	11.7
(ii)	Find the equation of the plane passing through the point (a, b, c) and parallel to the plane...	72	Numerical	11.7
6				
(i)	Find the angle between the line $\vec{r} = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda(3\hat{i} - \hat{j} + 2\hat{k})...$	73	Numerical	11.7
(ii)	Find the angle between the line $\vec{r} = 2\hat{i} - 3\hat{j} + \lambda(\hat{k})$ and the plan...	73	Numerical	11.7
7				
(i)	Find the angle between the line $\frac{x-2}{3} = \frac{y+1}{-1} = \frac{z-3}{2}$ and the plane...	73	Numerical	11.7
(ii)	Find the angle between the line $\frac{x+1}{2} = \frac{3y+5}{9} = \frac{3-z}{-6}$ and the plane...	73	Numerical	11.7
8				

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
(i)	If the plane $2x - 3y - 6z = 13$ makes an angle $\sin^{-1}(\lambda)$ with the x-axis, then find...	71	Numerical	11.7
(ii)	If the angle between the line $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and the plane $2x - y + \sqrt{\lambda}z = -4$...	71	Numerical	11.7
9				
(i)	$2x + y + 3z - 2 = 0$ and $x - 2y + 5 = 0$	68	Numerical	11.7
(ii)	$4x + 8y + z - 8 = 0$ and $y + z - 4 = 0$	68	Numerical	11.7
(iii)	$7x + 5y + 6z + 30 = 0$ and $3x - y - 10z + 4 = 0$	68	Numerical	11.7
(iv)	$2x - 2y + 4z + 5 = 0$ and $3x - 3y + 6z - 1 = 0$	68	Numerical	11.7
10	Prove that the plane $2x + 3y - 4z = 11$ is perpendicular to each of the planes $x + 2y + 2z - 7 = 0$...	69	Proof	11.7
11	Find the equation of the plane passing through the points $(1, -1, 2)$, $(2, -2, 2)$ and perpendicular to...	74	Numerical	11.7
12				
(i)	Find the equation of the plane through the point $(1, 2, 3)$ and perpendicular to each of the p...	76	Numerical	11.7
(ii)	Find the equation of the plane passing through the points $2\hat{i} - 3\hat{j} + \hat{k}$, $5\hat{i} + 2\hat{j} - \hat{k}$...	76	Numerical	11.7
13	Find the equation of the plane passing through the point $(1, 1, -1)$ and perpendicular to the...	75	Numerical	11.7
14	Find the equation of the plane through the point $\hat{i} - 3\hat{j} - 2\hat{k}$ and perpendicular to...	75	Numerical	11.7
15	Find the vector equation of the plane passing through the point $5\hat{i} + 2\hat{j} - 3\hat{k}$...	76	Numerical	11.7
16	Find the equation of the plane passing through the points $(2, 3, 1)$ and $(4, -5, 3)$ and para...	77	Numerical	11.7
17	Show that the equation $ax + by + d = 0$ represents a plane parallel to the z-axis. Hence find...	77	Both	11.7

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
18	Show that the equation $ax + cz + d = 0$ represents a plane parallel to y-axis. Hence find the...	77	Both	11.7
19	Find the equation of a plane passing through the point $(2, -1, 5)$, perpendicular to the plane...	78	Numerical	11.7

Exercise 11.8 – Distance of a Point from a Plane

1				
(i)	from the origin to the plane $3x - 4y + 12z = 3$	79	Numerical	11.8
(ii)	from the point $(-6, 0, 0)$ to the plane $2x - 3y + 6z - 2 = 0$	79	Numerical	11.8
(iii)	from the point $(3, -2, 1)$ to the plane $2x - y + 2z + 3 = 0$	79	Numerical	11.8
2				
(i)	Find the distance of the point whose position vector is $2\hat{i} + \hat{j} - \hat{k}$ from the...	79	Numerical	11.8
(ii)	Find the length of the perpendicular from origin to the plane $\vec{r} \cdot (3\hat{i} - 4\hat{j} - 12\hat{k}) + 39 = 0$...	79	Numerical	11.8
(iii)	Find the distance of the point $(3, 4, 5)$ from the plane $\vec{r} \cdot (2\hat{i} - 5\hat{j} + 3\hat{k}) = 13$...	79	Numerical	11.8
(iv)	Write the distance from the point $(2, 3, -5)$ to the plane XY-plane.	79	Numerical	11.8
3	Show that the points with position vectors $\hat{i} - \hat{j} + 3\hat{k}$ and $3(\hat{i} + \hat{j} + \hat{k})$...	80	Proof	11.8
4				
(i)	If the distance from the point $(k, 3, -5)$ to the plane $x + 2y - 2z = 9$ is 3 units, then find...	81	Numerical	11.8
(ii)	If the distance from the point $(-6, 0, 0)$ to the plane $2x - 3y + z - k = 0$ is $\sqrt{14}$ u...	81	Numerical	11.8
5				
(i)	Find the distance between the planes $2x + 3y + 4z = 4$ and $4x + 6y + 8z = 12$.	82	Numerical	11.8

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
(ii)	Find the distance between the planes $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) - 4 = 0$ a...	82	Numerical	11.8
6	If the distance between the planes $2x - 2y + z = 3$ and $2x - 2y + z = k$ is 2 units, find the...	82	Numerical	11.8
7	If the points $(1, 1, p)$ and $(-3, 0, 1)$ are equidistant from the plane $\vec{r} \cdot (3\hat{i} + 4\hat{j} - 12\hat{k}) + 13 = 0$...	80	Numerical	11.8
8	Given below are equations of two planes....	83	Numerical	11.8
9	Find the distance from the point $(7, 2, 4)$ to the plane determined by the points $A(2, 5, -3)$...	79	Numerical	11.8
10	Find the vector equation of the plane determined by the points $A(3, -1, 2)$, $B(5, 2, 4)$ and...	84	Numerical	11.8
11	Find the equation of the plane passing through the points $(1, 0, -2)$, $(3, -1, 0)$ and perpen...	85	Numerical	11.8
12	Find the equation of the plane passing through the intersection of the planes $4x - y + z = 10$...	86	Numerical	11.8
13	Find the equations of the planes parallel to the plane $x - 2y + 2z = 3$ and at a unit distance...	87	Numerical	11.8
14	Find the equation of the plane through the point $(3, 4, -1)$ which is parallel to the plane...	87	Numerical	11.8
15	Find the equations of the planes parallel to the plane $3x - 6y + 2z = 12$ and 6 units away fro...	87	Numerical	11.8
16	Find the shortest distance between the planes $6x + 2y - 3z = 12$ and $12x + 4y - 6z = 17$.	82	Numerical	11.8
17	Find the equation of the planes mid-parallel to the planes $2x - 2y + z + 3 = 0$ and $2x - 2y + z + 9 = 0$...	88	Numerical	11.8

Q. No.	Topic / Statement Summary	Type	Kind	Sec.
18	Find the equation of the planes passing through the intersection of the planes $\vec{r} \cdot (2\hat{i} + 6\hat{j}) + 12 = 0...$	86	Numerical	11.8
19	State when the line $\vec{r} = \vec{a} + \lambda\vec{b}$ is parallel to the plane $\vec{r} \cdot \vec{n} = d...$	45	Numerical	11.8

Formula & Standard Results Sheet

11.1 – Direction Cosines and Direction Ratios of a Line

Direction Cosines: Definition

$$l = \cos \alpha, m = \cos \beta, n = \cos \gamma$$

α, β, γ are the angles the line makes with the positive x, y, z -axes.

Fundamental Identity

$$l^2 + m^2 + n^2 = 1$$

Always holds for direction cosines; used to test whether a given triple can be direction cosines.

Direction Ratios to Direction Cosines

$$l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

Divide direction ratios $\langle a, b, c \rangle$ by their magnitude to normalise into direction cosines.

Direction Ratios of a Line Joining Two Points

$$a = x_2 - x_1, b = y_2 - y_1, c = z_2 - z_1$$

For points (x_1, y_1, z_1) and (x_2, y_2, z_2) .

Distance from Coordinate Axes/Planes

$$\text{From } x\text{-axis: } \sqrt{y^2 + z^2} \quad \text{From } XOY\text{-plane: } |z|$$

Analogous formulas hold for the y, z -axes and the other two coordinate planes.

Perpendicularity of Two Lines

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

Using direction ratios $\langle a_1, b_1, c_1 \rangle$ and $\langle a_2, b_2, c_2 \rangle$.

Parallelism of Two Lines

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Direction ratios of parallel lines are proportional.

Angle Between Two Lines

$$\cos \theta = |l_1 l_2 + m_1 m_2 + n_1 n_2| = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Acute angle between two lines, in terms of direction cosines or direction ratios.

11.2 – Equation of a Line in Space**Vector Equation – Point & Direction**

$$\vec{r} = \vec{a} + \lambda \vec{b}$$

Line through point with position vector $\vec{a}$, parallel to $\vec{b}$; $\lambda \in \mathbb{R}$.

Cartesian Equation – Point & Direction

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Line through (x_1, y_1, z_1) with direction ratios $\langle a, b, c \rangle$.

Vector Equation – Two Points

$$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$$

Line through two points with position vectors $\vec{a}$ and $\vec{b}$.

Cartesian Equation – Two Points

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

Line through (x_1, y_1, z_1) and (x_2, y_2, z_2) .

General Point on a Line

$$P(x_1 + \lambda a, y_1 + \lambda b, z_1 + \lambda c)$$

Any point on the line, parametrised by λ – used to find intersections with axes/planes or points at a given distance.

Collinearity Test

Direction ratios of $AB \parallel$ direction ratios of BC

Three points are collinear iff the line through any two of them also passes through the third.

11.3 – Angle Between Two Lines

Perpendicularity (Vector Form)

$$\vec{b}_1 \cdot \vec{b}_2 = 0$$

Direction vectors of two perpendicular lines.

Foot of Perpendicular from a Point to a Line

$$N = (x_1 + \lambda a, y_1 + \lambda b, z_1 + \lambda c), \quad \vec{AN} \cdot \vec{b} = 0$$

Solve for λ using the perpendicularity condition, then substitute back.

Distance of a Point from a Line

$$d = \frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}$$

P any point on the line, $\vec{b}$ its direction vector, A the external point.

Image (Reflection) of a Point in a Line

If N is the foot of perpendicular, N is the midpoint of AA'

Used to find the mirror image A' of point A in a line.

11.4 – Shortest Distance Between Two Lines

Shortest Distance – Skew Lines (Vector Form)

$$d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

For lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$.

Lines Intersect iff

$$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$$

Shortest distance is zero – equivalent to the scalar triple product vanishing.

Shortest Distance – Parallel Lines

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$$

For lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}$ (same direction $\vec{b}$).

11.5 – Equation of a Plane

Normal Form (Vector)

$$\vec{r} \cdot \hat{n} = d$$

$\hat{n}$ unit normal, $d \geq 0$ distance of plane from origin.

Normal Form (Cartesian)

$$lx + my + nz = d$$

l, m, n direction cosines of the normal.

Point & Normal Form

$$\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n} \quad \text{or} \quad a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

Plane through point $\vec{a}$ (or (x_1, y_1, z_1)) with normal $\vec{n} = \langle a, b, c \rangle$.

Plane Through Three Points

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

Determinant form; expand along the first row to get the cartesian equation.

Coplanarity of a Point and a Plane

Substitute point into plane equation – holds iff on the plane

Used to test if given points are coplanar (plane through 3 of them, then check the 4th).

Plane Containing a Line and Parallel to Another

$$\vec{r} \cdot (\vec{b}_1 \times \vec{b}_2) = \vec{a}_1 \cdot (\vec{b}_1 \times \vec{b}_2)$$

Normal to the required plane is $\vec{b}_1 \times \vec{b}_2$.

11.6 – Intercept and Three-Point Forms of a Plane

Intercept Form

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

a, b, c are the intercepts made by the plane on the x -, y -, z -axes respectively.

Plane Through Line of Intersection of Two Planes

$$P_1 + \lambda P_2 = 0$$

Family of planes through the intersection of $P_1 = 0$ and $P_2 = 0$; λ fixed using an extra condition (e.g. passes through a given point).

Foot of Perpendicular from a Point to a Plane

$N = P + t\vec{n}$, substitute into plane equation to find t
 $\vec{n}$ is the plane's normal; solve for parameter t , then back-substitute.

Coplanarity of Two Lines

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

Two lines are coplanar iff this determinant vanishes; the same determinant (as a linear equation in x, y, z) gives the containing plane.

11.7 – Angle Between Two Planes; Angle Between a Line and a Plane**Angle Between Two Planes**

$$\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|} = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Angle between two planes equals the angle between their normals.

Perpendicularity of Two Planes

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

Normals are perpendicular.

Parallelism of Two Planes

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Normals are parallel.

Angle Between a Line and a Plane

$$\sin \phi = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|}$$

Complementary to the angle between the line and the normal; note sin, not cos.

Line Parallel to a Plane

$$\vec{b} \cdot \vec{n} = 0 \text{ and point of line not on plane}$$

Direction vector of the line is perpendicular to the plane's normal.

Line Lies in a Plane

$\vec{b} \cdot \vec{n} = 0$ and a point of the line satisfies the plane's equation
Both conditions must hold simultaneously.

11.8 – Distance of a Point from a Plane**Distance of a Point from a Plane (Vector)**

$$d = \frac{|\vec{r}_1 \cdot \vec{n} - d_0|}{|\vec{n}|}$$

Plane $\vec{r} \cdot \vec{n} = d_0$; point with position vector $\vec{r}_1$.

Distance of a Point from a Plane (Cartesian)

$$d = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

Plane $ax + by + cz + d = 0$; point (x_1, y_1, z_1) .

Distance Between Two Parallel Planes

$$d = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$$

Planes $ax + by + cz = d_1$ and $ax + by + cz = d_2$ (same normal direction).

Distance Measured Parallel to a Given Line

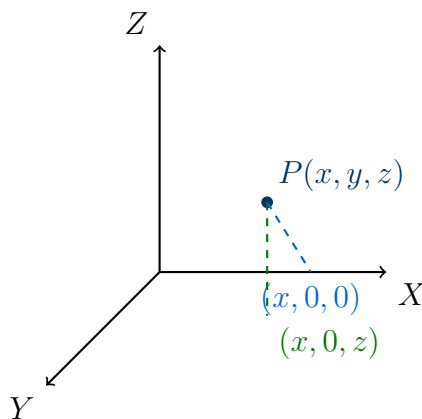
Find t where the line through the point meets the plane, then compute length.
Used when the distance is required along a specified direction rather than perpendicular to the plane.

11.1 – Direction Cosines and Direction Ratios of a Line

Question 1

- Find the distance of the point $P(2, -3, 4)$ from the x -axis.
- Find the distance of the point $Q(5, -4, 12)$ from the y -axis.
- Find the distance of the point $P(5, -6, 3)$ from the XOY plane.
- Find the distance of the point $Q(2, -4, 6)$ from the ZOX plane.

TYPE 1 Distance of a Point from Coordinate Axes/Planes



Point $P(x, y, z)$ in 3D coordinate frame showing perpendicular projection to axis and plane

Given: Points $P_1(2, -3, 4)$, $Q_1(5, -4, 12)$, $P_2(5, -6, 3)$, $Q_2(2, -4, 6)$

Concept / Formula Used: (x, y, z) x $\sqrt{y^2 + z^2}$ y $\sqrt{x^2 + z^2}$ XOY $|z|$ ZOY

Solution:

- For point $P(2, -3, 4)$, we have $x = 2$, $y = -3$, $z = 4$.

Distance from x -axis:

$$\begin{aligned}
 d &= \sqrt{y^2 + z^2} \\
 &= \sqrt{(-3)^2 + 4^2} \\
 &= \sqrt{9 + 16} \\
 &= \sqrt{25} = 5 \text{ units}
 \end{aligned}$$

- For point $Q(5, -4, 12)$, we have $x = 5$, $y = -4$, $z = 12$.

Distance from y -axis:

$$\begin{aligned} d &= \sqrt{x^2 + z^2} \\ &= \sqrt{5^2 + 12^2} \\ &= \sqrt{25 + 144} \\ &= \sqrt{169} = 13 \text{ units} \end{aligned}$$

(iii) For point $P(5, -6, 3)$, we have $x = 5$, $y = -6$, $z = 3$.

Distance from XOY plane ($z = 0$):

$$\begin{aligned} d &= |z| \\ &= |3| = 3 \text{ units} \end{aligned}$$

(iv) For point $Q(2, -4, 6)$, we have $x = 2$, $y = -4$, $z = 6$.

Distance from ZOX plane ($y = 0$):

$$\begin{aligned} d &= |y| \\ &= |-4| = 4 \text{ units} \end{aligned}$$

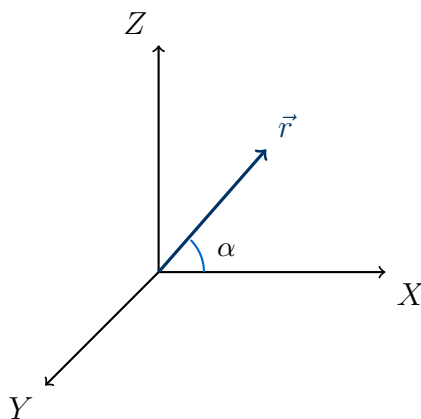
Final Answer

(i) 5 units (ii) 13 units (iii) 3 units (iv) 4 units

Question 2

- If a line makes angles 90° , 60° and 30° with the positive directions of x , y and z -axes respectively, find its direction cosines.
- Write the direction cosines of z -axis.

TYPE 2 Direction Cosines from Given Direction Ratios/Angles



Direction angles α, β, γ made by a vector with coordinate axes

Given: Direction angles $\alpha = 90^\circ$, $\beta = 60^\circ$, $\gamma = 30^\circ$

Concept / Formula Used: $l = \cos \alpha$ $m = \cos \beta$ $n = \cos \gamma$

Solution:

(i) The direction cosines l, m, n are given by:

$$\begin{aligned} l &= \cos 90^\circ = 0 \\ m &= \cos 60^\circ = \frac{1}{2} \\ n &= \cos 30^\circ = \frac{\sqrt{3}}{2} \end{aligned}$$

Thus, the direction cosines are $\left\langle 0, \frac{1}{2}, \frac{\sqrt{3}}{2} \right\rangle$.

(ii) The z -axis makes angles $\alpha = 90^\circ$, $\beta = 90^\circ$, and $\gamma = 0^\circ$ with the positive x , y , and z -axes respectively.

Therefore, its direction cosines are:

$$\begin{aligned} l &= \cos 90^\circ = 0 \\ m &= \cos 90^\circ = 0 \\ n &= \cos 0^\circ = 1 \end{aligned}$$

Final Answer

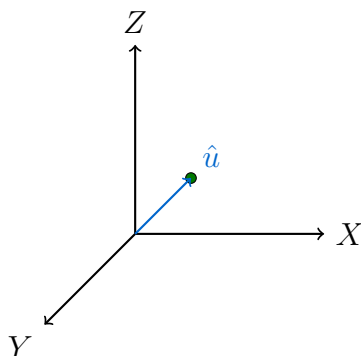
(i) $\left\langle 0, \frac{1}{2}, \frac{\sqrt{3}}{2} \right\rangle$

(ii) $\langle 0, 0, 1 \rangle$

Question 3

- (i) Can the numbers $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ be the direction cosines of a line?
- (ii) Can a line have direction angles $45^\circ, 60^\circ, 120^\circ$?
- (iii) Prove that $1, 1, 1$ cannot be direction cosines of a straight line.

TYPE 3 Fundamental Identity $l^2 + m^2 + n^2 = 1$ – Verification



Verification of identity $l^2 + m^2 + n^2 = 1$ in 3D space

Given: Proposed direction cosines/angles for a line

Concept / Formula Used: l, m, n $l^2 + m^2 + n^2 = 1$ $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

Solution:

(i) Let $l = \frac{1}{\sqrt{2}}, m = \frac{1}{\sqrt{2}}, n = \frac{1}{\sqrt{2}}$.

Calculate $l^2 + m^2 + n^2$:

$$\begin{aligned} l^2 + m^2 + n^2 &= \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 \\ &= \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \\ &= \frac{3}{2} \neq 1 \end{aligned}$$

Since $l^2 + m^2 + n^2 \neq 1$, these numbers cannot be direction cosines.

(ii) Let $\alpha = 45^\circ, \beta = 60^\circ, \gamma = 120^\circ$.

Calculate $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$:

$$\begin{aligned} \cos^2 45^\circ + \cos^2 60^\circ + \cos^2 120^\circ &= \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 \\ &= \frac{1}{2} + \frac{1}{4} + \frac{1}{4} \\ &= \frac{2}{4} + \frac{1}{4} + \frac{1}{4} \\ &= \frac{4}{4} = 1 \end{aligned}$$

Since $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$, a line can have these direction angles.

(iii) Let $l = 1, m = 1, n = 1$.

Calculate $l^2 + m^2 + n^2$:

$$\begin{aligned} l^2 + m^2 + n^2 &= 1^2 + 1^2 + 1^2 \\ &= 1 + 1 + 1 = 3 \neq 1 \end{aligned}$$

Since $l^2 + m^2 + n^2 = 3 \neq 1$, 1, 1, 1 cannot be the direction cosines of any straight line.

Final Answer

(i) No, because $l^2 + m^2 + n^2 = \frac{3}{2} \neq 1$.

(ii) Yes, because $\cos^2 45^\circ + \cos^2 60^\circ + \cos^2 120^\circ = 1$.

(iii) Proved ($1^2 + 1^2 + 1^2 = 3 \neq 1$).

Question 4

If the direction cosines of a line are $\left\langle \frac{1}{c}, \frac{1}{c}, \frac{1}{c} \right\rangle$, then find value(s) of c .

TYPE 3a Fundamental Identity – Solve for Unknown Parameter

Given: Direction cosines $l = \frac{1}{c}, m = \frac{1}{c}, n = \frac{1}{c}$

Concept / Formula Used: $l^2 + m^2 + n^2 = 1$

Solution:

Substitute $l = \frac{1}{c}, m = \frac{1}{c}, n = \frac{1}{c}$ into the identity:

$$\begin{aligned} l^2 + m^2 + n^2 &= 1 \\ \left(\frac{1}{c}\right)^2 + \left(\frac{1}{c}\right)^2 + \left(\frac{1}{c}\right)^2 &= 1 \\ \frac{1}{c^2} + \frac{1}{c^2} + \frac{1}{c^2} &= 1 \\ \frac{3}{c^2} &= 1 \\ c^2 &= 3 \\ c &= \pm\sqrt{3} \end{aligned}$$

Final Answer

$$c = \pm\sqrt{3}$$

Question 5

- If a line makes angles $90^\circ, 60^\circ$ and θ with x, y and z -axes respectively, where θ is acute, then find θ .
- If a line makes angles of 90° and 60° respectively with the positive directions of x and y axes, find the angle which it makes with the positive direction of z -axis.
- A line makes an angle of $\frac{\pi}{4}$ with each of X -axis and Y -axis. What angle does it make with Z -axis?
- If a straight line makes angles of 45° and 120° with the positive directions of x and z -axes respectively, then find the acute angle which it makes with the positive direction of y -axis.
- If the direction cosines of a line are $\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}$, then find its direction ratios.

TYPE 4 Angle Relations Between the Coordinate Axes

Given: Given angles α, β, γ or d.c.s for each sub-part

Concept / Formula Used: $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

Solution:

- (i) Given $\alpha = 90^\circ$, $\beta = 60^\circ$, $\gamma = \theta$ (acute).

Applying the identity:

$$\begin{aligned}\cos^2 90^\circ + \cos^2 60^\circ + \cos^2 \theta &= 1 \\ 0^2 + \left(\frac{1}{2}\right)^2 + \cos^2 \theta &= 1 \\ \frac{1}{4} + \cos^2 \theta &= 1 \\ \cos^2 \theta &= 1 - \frac{1}{4} = \frac{3}{4} \\ \cos \theta &= \frac{\sqrt{3}}{2} \quad (\text{since } \theta \text{ is acute, } \cos \theta > 0) \\ \theta &= 30^\circ \text{ or } \frac{\pi}{6}\end{aligned}$$

- (ii) Given $\alpha = 90^\circ$, $\beta = 60^\circ$, let angle with z -axis be γ .

Applying the identity:

$$\begin{aligned}\cos^2 90^\circ + \cos^2 60^\circ + \cos^2 \gamma &= 1 \\ 0 + \frac{1}{4} + \cos^2 \gamma &= 1 \\ \cos^2 \gamma &= \frac{3}{4} \\ \cos \gamma &= \pm \frac{\sqrt{3}}{2}\end{aligned}$$

Thus, $\gamma = 30^\circ$ or 150° ($\frac{\pi}{6}$ or $\frac{5\pi}{6}$).

- (iii) Given $\alpha = \frac{\pi}{4}$, $\beta = \frac{\pi}{4}$, let angle with Z -axis be γ .

Applying the identity:

$$\begin{aligned}\cos^2 \left(\frac{\pi}{4}\right) + \cos^2 \left(\frac{\pi}{4}\right) + \cos^2 \gamma &= 1 \\ \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \cos^2 \gamma &= 1 \\ \frac{1}{2} + \frac{1}{2} + \cos^2 \gamma &= 1 \\ 1 + \cos^2 \gamma &= 1 \\ \cos^2 \gamma &= 0 \implies \cos \gamma = 0 \implies \gamma = \frac{\pi}{2} \text{ or } 90^\circ\end{aligned}$$

- (iv) Given $\alpha = 45^\circ$, $\gamma = 120^\circ$, let acute angle with y -axis be β .

Applying the identity:

$$\begin{aligned}
 \cos^2 45^\circ + \cos^2 \beta + \cos^2 120^\circ &= 1 \\
 \left(\frac{1}{\sqrt{2}}\right)^2 + \cos^2 \beta + \left(-\frac{1}{2}\right)^2 &= 1 \\
 \frac{1}{2} + \cos^2 \beta + \frac{1}{4} &= 1 \\
 \frac{3}{4} + \cos^2 \beta &= 1 \\
 \cos^2 \beta &= \frac{1}{4} \\
 \cos \beta &= \frac{1}{2} \quad (\text{since } \beta \text{ is acute}) \\
 \beta &= 60^\circ \text{ or } \frac{\pi}{3}
 \end{aligned}$$

(v) Given d.c.s $l = \frac{2}{3}$, $m = -\frac{1}{3}$, $n = -\frac{2}{3}$.

Direction ratios a, b, c are proportional to l, m, n :

$$\langle a, b, c \rangle = k \cdot \left\langle \frac{2}{3}, -\frac{1}{3}, -\frac{2}{3} \right\rangle$$

Taking $k = 3$, we get direction ratios $\langle 2, -1, -2 \rangle$.

Final Answer

(i) $\theta = 30^\circ$ (ii) 30° or 150° (iii) $\frac{\pi}{2}$ (iv) 60° (v) $\langle 2, -1, -2 \rangle$

Question 6

- (i) If a line has direction ratios $2, -1, -2$, determine its direction cosines.
(ii) If direction numbers of a line are $\langle 2, -6, 3 \rangle$, find its direction cosines.

TYPE 2 Direction Cosines from Given Direction Ratios/Angles

Given: Direction ratios $\langle a, b, c \rangle$

Concept	/	Formula	Used:	a, b, c	l	$=$	$\frac{a}{\sqrt{a^2 + b^2 + c^2}}$	m	$=$
$\frac{b}{\sqrt{a^2 + b^2 + c^2}}$		n	$=$	$\frac{c}{\sqrt{a^2 + b^2 + c^2}}$					

Solution:

(i) Given $a = 2, b = -1, c = -2$.

Calculate magnitude:

$$\begin{aligned}\sqrt{a^2 + b^2 + c^2} &= \sqrt{2^2 + (-1)^2 + (-2)^2} \\ &= \sqrt{4 + 1 + 4} \\ &= \sqrt{9} = 3\end{aligned}$$

Therefore, direction cosines are:

$$l = \frac{2}{3}, \quad m = -\frac{1}{3}, \quad n = -\frac{2}{3}$$

(ii) Given $a = 2, b = -6, c = 3$.

Calculate magnitude:

$$\begin{aligned}\sqrt{a^2 + b^2 + c^2} &= \sqrt{2^2 + (-6)^2 + 3^2} \\ &= \sqrt{4 + 36 + 9} \\ &= \sqrt{49} = 7\end{aligned}$$

Therefore, direction cosines are:

$$l = \frac{2}{7}, \quad m = -\frac{6}{7}, \quad n = \frac{3}{7}$$

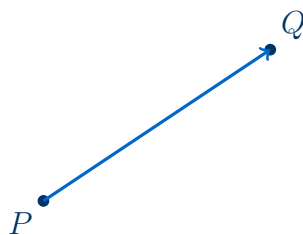
Final Answer

- (i) $\left\langle \frac{2}{3}, -\frac{1}{3}, -\frac{2}{3} \right\rangle$
 (ii) $\left\langle \frac{2}{7}, -\frac{6}{7}, \frac{3}{7} \right\rangle$

Question 7

- (i) Write the direction cosines of the line joining the points $(1, 0, 0)$ and $(0, 1, 1)$.
 (ii) Find the direction cosines of the line passing through the points $P(2, 3, 5)$ and $Q(-1, 2, 4)$.

TYPE 2a Direction Cosines of a Line Joining Two Points



Line passing through two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$

Given: Points $A(1, 0, 0), B(0, 1, 1)$ and $P(2, 3, 5), Q(-1, 2, 4)$

Concept / Formula Used: (x_1, y_1, z_1) (x_2, y_2, z_2) $a = x_2 - x_1, b = y_2 - y_1, c = z_2 - z_1$ $d = \sqrt{a^2 + b^2 + c^2}$

Solution:

(i) For points $A(1, 0, 0)$ and $B(0, 1, 1)$:

$$a = 0 - 1 = -1$$

$$b = 1 - 0 = 1$$

$$c = 1 - 0 = 1$$

Magnitude:

$$d = \sqrt{(-1)^2 + 1^2 + 1^2} = \sqrt{1 + 1 + 1} = \sqrt{3}$$

Direction cosines are:

$$\left\langle -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$$

(ii) For points $P(2, 3, 5)$ and $Q(-1, 2, 4)$:

$$a = -1 - 2 = -3$$

$$b = 2 - 3 = -1$$

$$c = 4 - 5 = -1$$

Magnitude:

$$d = \sqrt{(-3)^2 + (-1)^2 + (-1)^2} = \sqrt{9 + 1 + 1} = \sqrt{11}$$

Direction cosines are:

$$\left\langle -\frac{3}{\sqrt{11}}, -\frac{1}{\sqrt{11}}, -\frac{1}{\sqrt{11}} \right\rangle$$

Final Answer

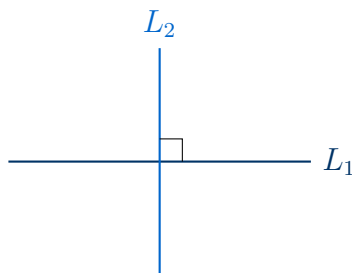
(i) $\left\langle -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$

(ii) $\left\langle -\frac{3}{\sqrt{11}}, -\frac{1}{\sqrt{11}}, -\frac{1}{\sqrt{11}} \right\rangle$

Question 8

- (i) Show that the line joining the origin to the point $(2, 1, 1)$ is perpendicular to the line determined by the points $(3, 5, -1)$ and $(4, 3, -1)$.
- (ii) Show that the line through the points $(1, -1, 2)$ and $(3, 4, -2)$ is perpendicular to the line through the points $(0, 3, 2)$ and $(3, 5, 6)$.

TYPE 5 Perpendicularity Proof via Direction Ratios



Two perpendicular lines intersecting at right angle

To Prove: Lines L_1 and L_2 are perpendicular, i.e., $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

Concept / Formula Used: $\langle a_1, b_1, c_1 \rangle$ $\langle a_2, b_2, c_2 \rangle$ $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Solution:

(i) Line 1 joins $O(0, 0, 0)$ and $P(2, 1, 1)$.

Direction ratios of line 1:

$$a_1 = 2 - 0 = 2$$

$$b_1 = 1 - 0 = 1$$

$$c_1 = 1 - 0 = 1$$

Line 2 joins $A(3, 5, -1)$ and $B(4, 3, -1)$.

Direction ratios of line 2:

$$a_2 = 4 - 3 = 1$$

$$b_2 = 3 - 5 = -2$$

$$c_2 = -1 - (-1) = 0$$

Evaluating the scalar product of direction ratios:

$$\begin{aligned} a_1a_2 + b_1b_2 + c_1c_2 &= (2)(1) + (1)(-2) + (1)(0) \\ &= 2 - 2 + 0 = 0 \end{aligned}$$

Since the sum of products of direction ratios is zero, the lines are perpendicular.

(ii) Line 1 joins $P(1, -1, 2)$ and $Q(3, 4, -2)$.

Direction ratios of line 1:

$$a_1 = 3 - 1 = 2$$

$$b_1 = 4 - (-1) = 5$$

$$c_1 = -2 - 2 = -4$$

Line 2 joins $R(0, 3, 2)$ and $S(3, 5, 6)$.

Direction ratios of line 2:

$$a_2 = 3 - 0 = 3$$

$$b_2 = 5 - 3 = 2$$

$$c_2 = 6 - 2 = 4$$

Evaluating the scalar product of direction ratios:

$$\begin{aligned} a_1a_2 + b_1b_2 + c_1c_2 &= (2)(3) + (5)(2) + (-4)(4) \\ &= 6 + 10 - 16 \\ &= 16 - 16 = 0 \end{aligned}$$

Hence, the two lines are perpendicular.

Hence Proved

Both pairs of lines satisfy $a_1a_2 + b_1b_2 + c_1c_2 = 0$, hence proved to be perpendicular.

Question 9

- (i) For what value of p will the line through $(4, 1, 2)$ and $(5, p, 0)$ be perpendicular to the line through $(2, 1, 1)$ and $(3, 3, -1)$?
- (ii) Find the values of p and q so that the line joining the points $(7, p, 2)$ and $(q, -2, 5)$ may be parallel to the line joining the points $(2, -3, 5)$ and $(-6, -15, 11)$.

TYPE 6 Solve for Unknown Parameter via Perpendicularity/Parallelism

Given: Coordinates of points defining two lines

Concept / Formula Used: $a_1a_2 + b_1b_2 + c_1c_2 = 0$ $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Solution:

- (i) Line 1 joins $A(4, 1, 2)$ and $B(5, p, 0)$.

Direction ratios of line 1:

$$a_1 = 5 - 4 = 1$$

$$b_1 = p - 1$$

$$c_1 = 0 - 2 = -2$$

Line 2 joins $C(2, 1, 1)$ and $D(3, 3, -1)$.

Direction ratios of line 2:

$$a_2 = 3 - 2 = 1$$

$$b_2 = 3 - 1 = 2$$

$$c_2 = -1 - 1 = -2$$

Since the lines are perpendicular:

$$\begin{aligned}
 a_1a_2 + b_1b_2 + c_1c_2 &= 0 \\
 (1)(1) + (p-1)(2) + (-2)(-2) &= 0 \\
 1 + 2p - 2 + 4 &= 0 \\
 2p + 3 &= 0 \\
 2p = -3 &\implies p = -\frac{3}{2}
 \end{aligned}$$

(ii) Line 1 joins $A(7, p, 2)$ and $B(q, -2, 5)$.

Direction ratios of line 1:

$$\begin{aligned}
 a_1 &= q - 7 \\
 b_1 &= -2 - p \\
 c_1 &= 5 - 2 = 3
 \end{aligned}$$

Line 2 joins $C(2, -3, 5)$ and $D(-6, -15, 11)$.

Direction ratios of line 2:

$$\begin{aligned}
 a_2 &= -6 - 2 = -8 \\
 b_2 &= -15 - (-3) = -12 \\
 c_2 &= 11 - 5 = 6
 \end{aligned}$$

Since lines are parallel:

$$\begin{aligned}
 \frac{a_1}{a_2} &= \frac{b_1}{b_2} = \frac{c_1}{c_2} \\
 \frac{q-7}{-8} &= \frac{-2-p}{-12} = \frac{3}{6}
 \end{aligned}$$

Simplifying $\frac{3}{6} = \frac{1}{2}$, we set up simultaneous equations:

$$\begin{aligned}
 \frac{q-7}{-8} &= \frac{1}{2} \implies q-7 = -4 \implies q = 3 \\
 \frac{-2-p}{-12} &= \frac{1}{2} \implies -2-p = -6 \implies -p = -4 \implies p = 4
 \end{aligned}$$

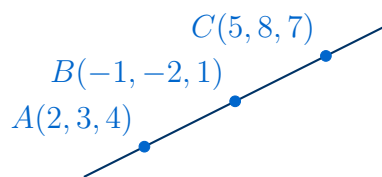
Final Answer

- (i) $p = -\frac{3}{2}$
 (ii) $p = 4, q = 3$

Question 10

Show that the points $(2, 3, 4)$, $(-1, -2, 1)$ and $(5, 8, 7)$ are collinear.

TYPE 7 Collinearity of Three Points



Three points A, B, C lying on a single straight line

To Prove: Points $A(2, 3, 4)$, $B(-1, -2, 1)$, and $C(5, 8, 7)$ are collinear.

Concept / Formula Used: A, B, C AB BC

Solution:

Let $A = (2, 3, 4)$, $B = (-1, -2, 1)$, $C = (5, 8, 7)$.

Direction ratios of AB :

$$a_1 = -1 - 2 = -3$$

$$b_1 = -2 - 3 = -5$$

$$c_1 = 1 - 4 = -3$$

Direction ratios of BC :

$$a_2 = 5 - (-1) = 6$$

$$b_2 = 8 - (-2) = 10$$

$$c_2 = 7 - 1 = 6$$

Checking the ratio of corresponding direction ratios:

$$\frac{a_1}{a_2} = \frac{-3}{6} = -\frac{1}{2}$$

$$\frac{b_1}{b_2} = \frac{-5}{10} = -\frac{1}{2}$$

$$\frac{c_1}{c_2} = \frac{-3}{6} = -\frac{1}{2}$$

Since $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = -\frac{1}{2}$, the direction ratios of AB and BC are proportional. Furthermore, B is a common point to both segments.

Hence Proved

The direction ratios of AB and BC are proportional and share point B ; hence A, B, C are collinear.

Question 11

Find the direction ratios and the direction cosines of the line joining the points $(5, 4, -3)$ and $(3, -2, 6)$, the angle between the line and X-axis taken acute.

TYPE 8 Point at a Given Distance with Given Direction Ratios

Given: Points $P(5, 4, -3)$ and $Q(3, -2, 6)$

Concept / Formula Used: $l \quad a$

Solution:

Let $P = (5, 4, -3)$ and $Q = (3, -2, 6)$.

The vector from P to Q is:

$$\begin{aligned}\vec{PQ} &= (3 - 5)\hat{i} + (-2 - 4)\hat{j} + (6 - (-3))\hat{k} \\ &= -2\hat{i} - 6\hat{j} + 9\hat{k}\end{aligned}$$

The x -component of $\vec{PQ}$ is $-2 < 0$, which implies an obtuse angle with the positive X -axis.

To make the angle with the positive X -axis acute, we take the reverse vector $\vec{QP}$:

$$\vec{QP} = 2\hat{i} + 6\hat{j} - 9\hat{k}$$

Thus, direction ratios are $\langle 2, 6, -9 \rangle$.

Now calculate the magnitude:

$$\begin{aligned}d &= \sqrt{2^2 + 6^2 + (-9)^2} \\ &= \sqrt{4 + 36 + 81} \\ &= \sqrt{121} = 11\end{aligned}$$

The direction cosines are:

$$l = \frac{2}{11}, \quad m = \frac{6}{11}, \quad n = -\frac{9}{11}$$

Final Answer

Direction ratios: $\langle 2, 6, -9 \rangle$

Direction cosines: $\left\langle \frac{2}{11}, \frac{6}{11}, -\frac{9}{11} \right\rangle$

Question 12

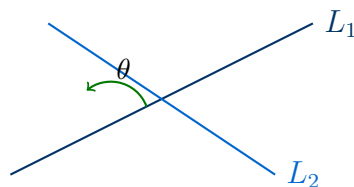
(i) Find the angle between the lines whose direction cosines are $\left\langle \frac{\sqrt{3}}{2}, \frac{1}{4}, \frac{\sqrt{3}}{4} \right\rangle$ and

$$\left\langle -\frac{\sqrt{3}}{2}, \frac{1}{4}, \frac{\sqrt{3}}{4} \right\rangle.$$

(ii) Find the angle between the lines whose direction numbers are $\langle 1, 2, 1 \rangle, \langle 2, -3, 4 \rangle$.

(iii) Find the angle between the lines whose direction ratios are $\langle 1, 1, 2 \rangle$ and $\langle \sqrt{3} - 1, -\sqrt{3} - 1, 4 \rangle$.

TYPE 9 Angle Between Two Lines – Direction Cosines/Ratios/Vectors

Angle θ between two lines in space

Given: Direction cosines or direction ratios for pairs of lines

Concept / Formula Used: $\cos \theta = \frac{|l_1 l_2 + m_1 m_2 + n_1 n_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$ $\cos \theta =$

Solution:

(i) Given $l_1 = \frac{\sqrt{3}}{2}, m_1 = \frac{1}{4}, n_1 = \frac{\sqrt{3}}{4}$ and $l_2 = -\frac{\sqrt{3}}{2}, m_2 = \frac{1}{4}, n_2 = \frac{\sqrt{3}}{4}$.

Calculate $\cos \theta$:

$$\begin{aligned} \cos \theta &= |l_1 l_2 + m_1 m_2 + n_1 n_2| \\ &= \left| \left(\frac{\sqrt{3}}{2} \right) \left(-\frac{\sqrt{3}}{2} \right) + \left(\frac{1}{4} \right) \left(\frac{1}{4} \right) + \left(\frac{\sqrt{3}}{4} \right) \left(\frac{\sqrt{3}}{4} \right) \right| \\ &= \left| -\frac{3}{4} + \frac{1}{16} + \frac{3}{16} \right| \\ &= \left| -\frac{12}{16} + \frac{4}{16} \right| \\ &= \left| -\frac{8}{16} \right| = \left| -\frac{1}{2} \right| = \frac{1}{2} \\ \theta &= \cos^{-1} \left(\frac{1}{2} \right) = 60^\circ \text{ or } \frac{\pi}{3} \end{aligned}$$

(ii) Given $a_1 = 1, b_1 = 2, c_1 = 1$ and $a_2 = 2, b_2 = -3, c_2 = 4$.

Calculate dot product of direction ratios:

$$\begin{aligned} a_1 a_2 + b_1 b_2 + c_1 c_2 &= (1)(2) + (2)(-3) + (1)(4) \\ &= 2 - 6 + 4 = 0 \end{aligned}$$

Since the scalar product is 0, $\cos \theta = 0$, so $\theta = 90^\circ$ or $\frac{\pi}{2}$.

(iii) Given $a_1 = 1, b_1 = 1, c_1 = 2$ and $a_2 = \sqrt{3} - 1, b_2 = -\sqrt{3} - 1, c_2 = 4$.

Magnitudes:

$$\begin{aligned}\sqrt{a_1^2 + b_1^2 + c_1^2} &= \sqrt{1^2 + 1^2 + 2^2} = \sqrt{1 + 1 + 4} = \sqrt{6} \\ \sqrt{a_2^2 + b_2^2 + c_2^2} &= \sqrt{(\sqrt{3} - 1)^2 + (-\sqrt{3} - 1)^2 + 4^2} \\ &= \sqrt{(3 - 2\sqrt{3} + 1) + (3 + 2\sqrt{3} + 1) + 16} \\ &= \sqrt{4 - 2\sqrt{3} + 4 + 2\sqrt{3} + 16} \\ &= \sqrt{24} = 2\sqrt{6}\end{aligned}$$

Numerator:

$$\begin{aligned}a_1a_2 + b_1b_2 + c_1c_2 &= 1(\sqrt{3} - 1) + 1(-\sqrt{3} - 1) + 2(4) \\ &= \sqrt{3} - 1 - \sqrt{3} - 1 + 8 = 6\end{aligned}$$

Cos theta:

$$\begin{aligned}\cos \theta &= \frac{6}{\sqrt{6} \cdot 2\sqrt{6}} = \frac{6}{2 \times 6} = \frac{1}{2} \\ \theta &= \cos^{-1} \left(\frac{1}{2} \right) = 60^\circ \text{ or } \frac{\pi}{3}\end{aligned}$$

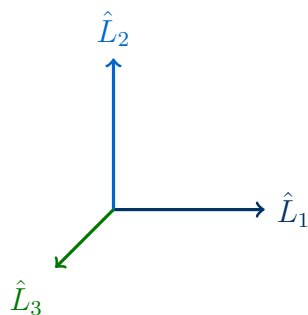
Final Answer

(i) 60° (ii) 90° (iii) 60°

Question 13

Show that the lines with direction cosines $\left\langle \frac{12}{13}, -\frac{3}{13}, -\frac{4}{13} \right\rangle$, $\left\langle \frac{4}{13}, \frac{12}{13}, \frac{3}{13} \right\rangle$ and $\left\langle \frac{3}{13}, -\frac{4}{13}, \frac{12}{13} \right\rangle$ are mutually perpendicular.

TYPE 5a Mutually Perpendicular Lines – Pairwise Check



Three mutually perpendicular direction vectors forming an orthogonal frame

To Prove: The three given direction vectors are mutually perpendicular.

Concept / Formula Used: $l_1l_2 + m_1m_2 + n_1n_2 = 0$

Solution:

Let the direction cosines of the three lines be L_1, L_2, L_3 respectively:

$$L_1 = \left\langle \frac{12}{13}, -\frac{3}{13}, -\frac{4}{13} \right\rangle, \quad L_2 = \left\langle \frac{4}{13}, \frac{12}{13}, \frac{3}{13} \right\rangle, \quad L_3 = \left\langle \frac{3}{13}, -\frac{4}{13}, \frac{12}{13} \right\rangle.$$

1. For lines 1 and 2:

$$\begin{aligned} l_1l_2 + m_1m_2 + n_1n_2 &= \left(\frac{12}{13}\right)\left(\frac{4}{13}\right) + \left(-\frac{3}{13}\right)\left(\frac{12}{13}\right) + \left(-\frac{4}{13}\right)\left(\frac{3}{13}\right) \\ &= \frac{48 - 36 - 12}{169} \\ &= \frac{0}{169} = 0 \end{aligned}$$

Thus, line 1 is perpendicular to line 2.

2. For lines 2 and 3:

$$\begin{aligned} l_2l_3 + m_2m_3 + n_2n_3 &= \left(\frac{4}{13}\right)\left(\frac{3}{13}\right) + \left(\frac{12}{13}\right)\left(-\frac{4}{13}\right) + \left(\frac{3}{13}\right)\left(\frac{12}{13}\right) \\ &= \frac{12 - 48 + 36}{169} \\ &= \frac{0}{169} = 0 \end{aligned}$$

Thus, line 2 is perpendicular to line 3.

3. For lines 3 and 1:

$$\begin{aligned} l_3l_1 + m_3m_1 + n_3n_1 &= \left(\frac{3}{13}\right)\left(\frac{12}{13}\right) + \left(-\frac{4}{13}\right)\left(-\frac{3}{13}\right) + \left(\frac{12}{13}\right)\left(-\frac{4}{13}\right) \\ &= \frac{36 + 12 - 48}{169} \\ &= \frac{0}{169} = 0 \end{aligned}$$

Thus, line 3 is perpendicular to line 1.

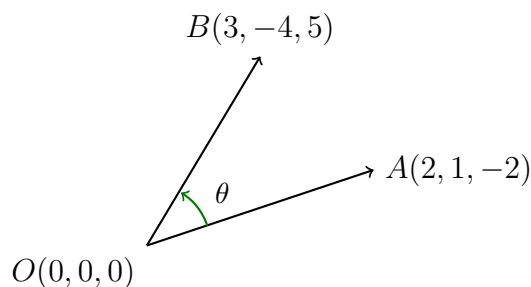
Hence Proved

All pair-wise scalar products equal zero; hence the three lines are mutually perpendicular.

Question 14

If A, B are the points $(2, 1, -2), (3, -4, 5)$, then find the angle that OA makes with OB , where O is the origin.

TYPE 9 Angle Between Two Lines – Direction Cosines/Ratios/Vectors



Angle between vectors $\vec{OA}$ and $\vec{OB}$

Given: Points $O(0, 0, 0)$, $A(2, 1, -2)$, $B(3, -4, 5)$

Concept / Formula Used: $\cos \theta = \frac{\vec{OA} \cdot \vec{OB}}{|\vec{OA}||\vec{OB}|}$

Solution:

Position vector of A : $\vec{OA} = 2\hat{i} + \hat{j} - 2\hat{k}$.

Position vector of B : $\vec{OB} = 3\hat{i} - 4\hat{j} + 5\hat{k}$.

Magnitudes:

$$|\vec{OA}| = \sqrt{2^2 + 1^2 + (-2)^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

$$|\vec{OB}| = \sqrt{3^2 + (-4)^2 + 5^2} = \sqrt{9 + 16 + 25} = \sqrt{50} = 5\sqrt{2}$$

Dot product:

$$\begin{aligned}\vec{OA} \cdot \vec{OB} &= (2)(3) + (1)(-4) + (-2)(5) \\ &= 6 - 4 - 10 = -8\end{aligned}$$

Calculating angle θ :

$$\begin{aligned}\cos \theta &= \frac{\vec{OA} \cdot \vec{OB}}{|\vec{OA}||\vec{OB}|} \\ &= \frac{-8}{3 \times 5\sqrt{2}} \\ &= -\frac{8}{15\sqrt{2}} = -\frac{4\sqrt{2}}{15} \\ \theta &= \cos^{-1} \left(-\frac{4\sqrt{2}}{15} \right)\end{aligned}$$

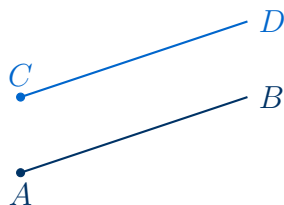
Final Answer

$$\theta = \cos^{-1} \left(-\frac{4\sqrt{2}}{15} \right)$$

Question 15

If the coordinates of the points A, B, C and D are $(1, 2, 3), (4, 5, 7), (-4, 3, -6)$ and $(2, 9, 2)$ respectively, then find the angle between the lines AB and CD .

TYPE 9 Angle Between Two Lines – Direction Cosines/Ratios/Vectors



Parallel lines AB and CD in 3D space

Given: Points $A(1, 2, 3), B(4, 5, 7), C(-4, 3, -6), D(2, 9, 2)$

Concept / Formula Used: $\cos \theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$

Solution:

Direction ratios of line AB :

$$a_1 = 4 - 1 = 3$$

$$b_1 = 5 - 2 = 3$$

$$c_1 = 7 - 3 = 4$$

Direction ratios of line CD :

$$a_2 = 2 - (-4) = 6$$

$$b_2 = 9 - 3 = 6$$

$$c_2 = 2 - (-6) = 8$$

Notice that $\langle a_2, b_2, c_2 \rangle = \langle 6, 6, 8 \rangle = 2\langle 3, 3, 4 \rangle = 2\langle a_1, b_1, c_1 \rangle$.

Since the direction ratios are proportional:

$$\begin{aligned} \frac{a_1}{a_2} &= \frac{3}{6} = \frac{1}{2} \\ \frac{b_1}{b_2} &= \frac{3}{6} = \frac{1}{2} \\ \frac{c_1}{c_2} &= \frac{4}{8} = \frac{1}{2} \end{aligned}$$

Since the direction ratios of line AB and line CD are proportional, line AB is parallel to line CD .

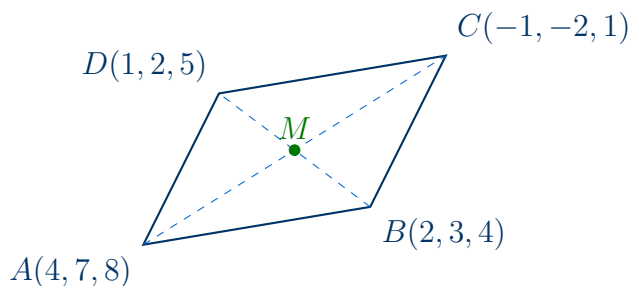
Thus, the angle between the lines is 0° .

Final Answer

0° (or lines are parallel)

Question 16

Show that the points $A(4, 7, 8)$, $B(2, 3, 4)$, $C(-1, -2, 1)$ and $D(1, 2, 5)$, taken in order, are the vertices of a parallelogram.

TYPE 10 Verifying Vertices of a Parallelogram

Parallelogram ABCD with intersecting diagonals AC and BD

To Prove: Quadrilateral $ABCD$ is a parallelogram.

Concept / Formula Used: AC BD

Solution:

Let $A = (4, 7, 8)$, $B = (2, 3, 4)$, $C = (-1, -2, 1)$, $D = (1, 2, 5)$.

Mid-point of diagonal AC :

$$\begin{aligned} M_{AC} &= \left(\frac{4 + (-1)}{2}, \frac{7 + (-2)}{2}, \frac{8 + 1}{2} \right) \\ &= \left(\frac{3}{2}, \frac{5}{2}, \frac{9}{2} \right) \end{aligned}$$

Mid-point of diagonal BD :

$$\begin{aligned} M_{BD} &= \left(\frac{2 + 1}{2}, \frac{3 + 2}{2}, \frac{4 + 5}{2} \right) \\ &= \left(\frac{3}{2}, \frac{5}{2}, \frac{9}{2} \right) \end{aligned}$$

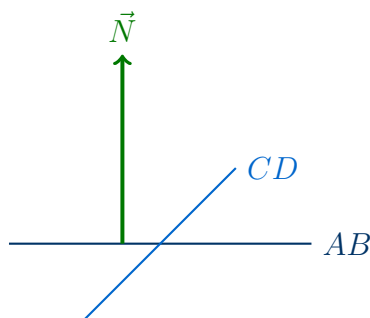
Since $M_{AC} = M_{BD}$, the diagonals AC and BD bisect each other.

Hence Proved

The mid-points of diagonals AC and BD coincide at $\left(\frac{3}{2}, \frac{5}{2}, \frac{9}{2} \right)$; hence $ABCD$ is a parallelogram.

Question 17

Find the direction cosines of a line which is perpendicular to the two lines formed by joining $A(2, 3, -4)$ to $B(-3, 3, -2)$ and $C(-1, 4, 2)$ to $D(3, 5, 1)$.

TYPE 11 Direction Cosines of a Line Perpendicular to Two Given Lines

$$\text{Vector } \vec{N} = \vec{AB} \times \vec{CD} \text{ perpendicular to both lines}$$

Given: Points $A(2, 3, -4)$, $B(-3, 3, -2)$, $C(-1, 4, 2)$, $D(3, 5, 1)$

Concept / Formula Used: $\vec{AB} \quad \vec{CD} \quad \vec{N} = \vec{AB} \times \vec{CD} \quad \pm \frac{\vec{N}}{|\vec{N}|}$

Solution:

Vector $\vec{AB}$:

$$\begin{aligned} \vec{AB} &= (-3 - 2)\hat{i} + (3 - 3)\hat{j} + (-2 - (-4))\hat{k} \\ &= -5\hat{i} + 0\hat{j} + 2\hat{k} \end{aligned}$$

Vector $\vec{CD}$:

$$\begin{aligned} \vec{CD} &= (3 - (-1))\hat{i} + (5 - 4)\hat{j} + (1 - 2)\hat{k} \\ &= 4\hat{i} + 1\hat{j} - 1\hat{k} \end{aligned}$$

Cross product $\vec{N} = \vec{AB} \times \vec{CD}$:

$$\begin{aligned} \vec{N} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -5 & 0 & 2 \\ 4 & 1 & -1 \end{vmatrix} \\ &= \hat{i}((0)(-1) - (2)(1)) - \hat{j}((-5)(-1) - (2)(4)) + \hat{k}((-5)(1) - (0)(4)) \\ &= \hat{i}(0 - 2) - \hat{j}(5 - 8) + \hat{k}(-5 - 0) \\ &= -2\hat{i} + 3\hat{j} - 5\hat{k} \end{aligned}$$

Magnitude of $\vec{N}$:

$$\begin{aligned} |\vec{N}| &= \sqrt{(-2)^2 + 3^2 + (-5)^2} \\ &= \sqrt{4 + 9 + 25} = \sqrt{38} \end{aligned}$$

Therefore, the direction cosines of the common perpendicular line are:

$$\pm \left\langle -\frac{2}{\sqrt{38}}, \frac{3}{\sqrt{38}}, -\frac{5}{\sqrt{38}} \right\rangle$$

Final Answer

$$\left\langle -\frac{2}{\sqrt{38}}, \frac{3}{\sqrt{38}}, -\frac{5}{\sqrt{38}} \right\rangle \text{ or } \left\langle \frac{2}{\sqrt{38}}, -\frac{3}{\sqrt{38}}, \frac{5}{\sqrt{38}} \right\rangle$$

Question 18

Find the angle between the lines whose direction cosines satisfy the relations $l+m+n = 0$ and $l^2 = m^2 + n^2$.

TYPE 12 System of Direction-Cosine Relations – Elimination

Given: Relations $l + m + n = 0$ and $l^2 = m^2 + n^2$

Concept / Formula Used: l_1, m_1, n_1 l_2, m_2, n_2 $\cos \theta = |l_1 l_2 + m_1 m_2 + n_1 n_2|$

Solution:

From $l + m + n = 0$, express l in terms of m and n :

$$l = -(m + n)$$

Substitute $l = -(m + n)$ into $l^2 = m^2 + n^2$:

$$\begin{aligned} [-(m + n)]^2 &= m^2 + n^2 \\ (m + n)^2 &= m^2 + n^2 \\ m^2 + 2mn + n^2 &= m^2 + n^2 \\ 2mn &= 0 \implies m = 0 \text{ or } n = 0 \end{aligned}$$

Case 1: $m = 0$

Substitute $m = 0$ into $l = -(m + n)$:

$$l = -n \implies \frac{l}{-1} = \frac{m}{0} = \frac{n}{1}$$

Direction ratios of first line: $\langle -1, 0, 1 \rangle$.

Normalized direction cosines L_1 :

$$l_1 = -\frac{1}{\sqrt{2}}, \quad m_1 = 0, \quad n_1 = \frac{1}{\sqrt{2}}$$

Case 2: $n = 0$

Substitute $n = 0$ into $l = -(m + n)$:

$$l = -m \implies \frac{l}{-1} = \frac{m}{1} = \frac{n}{0}$$

Direction ratios of second line: $\langle -1, 1, 0 \rangle$.

Normalized direction cosines L_2 :

$$l_2 = -\frac{1}{\sqrt{2}}, \quad m_2 = \frac{1}{\sqrt{2}}, \quad n_2 = 0$$

Now calculate the angle θ between the two lines:

$$\begin{aligned} \cos \theta &= |l_1 l_2 + m_1 m_2 + n_1 n_2| \\ &= \left| \left(-\frac{1}{\sqrt{2}} \right) \left(-\frac{1}{\sqrt{2}} \right) + (0) \left(\frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{2}} \right) (0) \right| \\ &= \left| \frac{1}{2} + 0 + 0 \right| = \frac{1}{2} \\ \theta &= \cos^{-1} \left(\frac{1}{2} \right) = 60^\circ \text{ or } \frac{\pi}{3} \end{aligned}$$

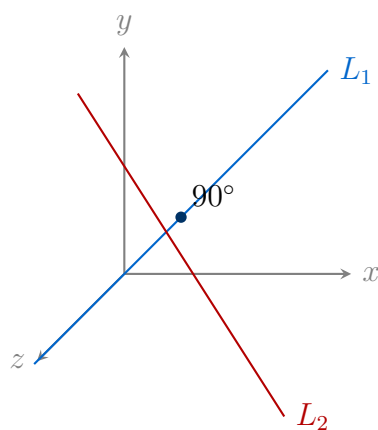
Final Answer

$$\theta = 60^\circ \text{ (or } \frac{\pi}{3} \text{)}$$

Question 19 (i)

Show that the straight lines whose direction cosines are given by $2l + 2m - n = 0$ and $mn + nl + lm = 0$ are at right angles.

TYPE 12a DC System – Perpendicularity Proof



Perpendicular lines in 3D space

Given: System of equations in direction cosines: $2l + 2m - n = 0$ and $mn + nl + lm = 0$.

To Prove: The angle between the two lines is 90° , i.e., $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$.

Concept / Formula Used: (a_1, b_1, c_1) (a_2, b_2, c_2) $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

Solution:

From the linear equation $2l + 2m - n = 0$, express n in terms of l and m :

$$n = 2l + 2m = 2(l + m)$$

Substitute this expression for n into the second equation $mn + nl + lm = 0$:

$$\begin{aligned} m(2l + 2m) + (2l + 2m)l + lm &= 0 \\ 2lm + 2m^2 + 2l^2 + 2lm + lm &= 0 \\ 2l^2 + 5lm + 2m^2 &= 0 \end{aligned}$$

Factorize the quadratic equation in l and m :

$$\begin{aligned} 2l^2 + 4lm + lm + 2m^2 &= 0 \\ 2l(l + 2m) + m(l + 2m) &= 0 \\ (2l + m)(l + 2m) &= 0 \end{aligned}$$

This yields two cases:

Case 1: $2l + m = 0 \implies m = -2l$. Substitute $m = -2l$ back into the expression for n :

$$\begin{aligned} n &= 2l + 2(-2l) \\ n &= 2l - 4l = -2l \end{aligned}$$

Thus, the direction ratios of the first line (a_1, b_1, c_1) are proportional to:

$$\frac{l}{1} = \frac{m}{-2} = \frac{n}{-2} \implies (a_1, b_1, c_1) = (1, -2, -2)$$

Case 2: $l + 2m = 0 \implies l = -2m$. Substitute $l = -2m$ back into the expression for n :

$$\begin{aligned} n &= 2(-2m) + 2m \\ n &= -4m + 2m = -2m \end{aligned}$$

Thus, the direction ratios of the second line (a_2, b_2, c_2) are proportional to:

$$\frac{l}{-2} = \frac{m}{1} = \frac{n}{-2} \implies (a_2, b_2, c_2) = (-2, 1, -2)$$

Now, evaluate the dot product $a_1a_2 + b_1b_2 + c_1c_2$:

$$\begin{aligned} a_1a_2 + b_1b_2 + c_1c_2 &= (1)(-2) + (-2)(1) + (-2)(-2) \\ &= -2 - 2 + 4 \\ &= 0 \end{aligned}$$

Since $a_1a_2 + b_1b_2 + c_1c_2 = 0$, the two lines are perpendicular to each other.

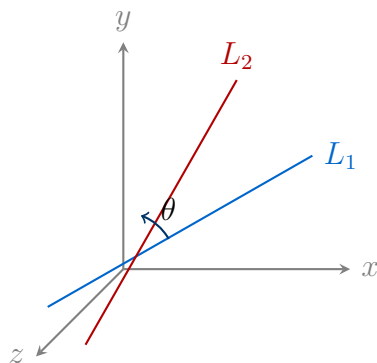
Hence Proved

Since $a_1a_2 + b_1b_2 + c_1c_2 = 0$, the straight lines are at right angles.

Question 19 (ii)

Find the angle between the lines whose direction cosines are given by the equations $3l + m + 5n = 0$ and $6mn - 2nl + 5lm = 0$.

TYPE 12b DC System – Angle Between Lines

★ **Likely Exam Question**

Two lines intersecting at angle θ

Given: System of equations: $3l + m + 5n = 0$ and $6mn - 2nl + 5lm = 0$.

To Find: The angle θ between the two lines.

Concept / Formula Used: $\cos \theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$

Solution:

From the linear equation $3l + m + 5n = 0$, express m in terms of l and n :

$$m = -(3l + 5n)$$

Substitute $m = -(3l + 5n)$ into the second equation $6mn - 2nl + 5lm = 0$:

$$\begin{aligned} 6[-(3l + 5n)]n - 2nl + 5l[-(3l + 5n)] &= 0 \\ -6n(3l + 5n) - 2nl - 5l(3l + 5n) &= 0 \\ -18ln - 30n^2 - 2nl - 15l^2 - 25ln &= 0 \\ -15l^2 - 45ln - 30n^2 &= 0 \end{aligned}$$

Divide the entire equation by -15 :

$$l^2 + 3ln + 2n^2 = 0$$

Factorize the quadratic equation:

$$(l + n)(l + 2n) = 0$$

This gives two cases:

Case 1: $l + n = 0 \implies l = -n$. Substitute $l = -n$ back into $m = -(3l + 5n)$:

$$\begin{aligned} m &= -[3(-n) + 5n] \\ m &= -[-3n + 5n] = -2n \end{aligned}$$

So the direction ratios (a_1, b_1, c_1) of the first line are:

$$\frac{l}{-1} = \frac{m}{-2} = \frac{n}{1} \implies (a_1, b_1, c_1) = (1, 2, -1)$$

Case 2: $l + 2n = 0 \implies l = -2n$. Substitute $l = -2n$ back into $m = -(3l + 5n)$:

$$\begin{aligned} m &= -[3(-2n) + 5n] \\ m &= -[-6n + 5n] = n \end{aligned}$$

So the direction ratios (a_2, b_2, c_2) of the second line are:

$$\frac{l}{-2} = \frac{m}{1} = \frac{n}{1} \implies (a_2, b_2, c_2) = (-2, 1, 1)$$

Using the formula for the angle between two lines:

$$\begin{aligned} a_1a_2 + b_1b_2 + c_1c_2 &= (1)(-2) + (2)(1) + (-1)(1) \\ &= -2 + 2 - 1 = -1 \end{aligned}$$

$$\sqrt{a_1^2 + b_1^2 + c_1^2} = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{1 + 4 + 1} = \sqrt{6}$$

$$\sqrt{a_2^2 + b_2^2 + c_2^2} = \sqrt{(-2)^2 + 1^2 + 1^2} = \sqrt{4 + 1 + 1} = \sqrt{6}$$

Substitute these into the cosine formula:

$$\begin{aligned} \cos \theta &= \frac{|-1|}{\sqrt{6} \cdot \sqrt{6}} \\ \cos \theta &= \frac{1}{6} \end{aligned}$$

Taking the inverse cosine:

$$\theta = \cos^{-1}\left(\frac{1}{6}\right)$$

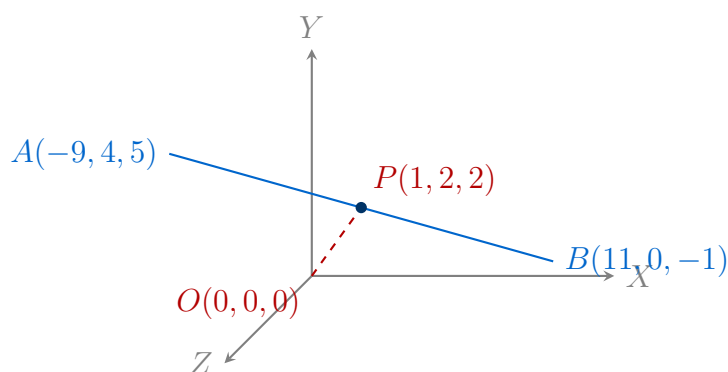
Final Answer

$$\theta = \cos^{-1}\left(\frac{1}{6}\right)$$

Question 20

Find the point in which the join of $A(-9, 4, 5)$ and $B(11, 0, -1)$ is met by the perpendicular from the origin.

TYPE 13 Foot of Perpendicular from a Point to a Line (via Two Points)



Perpendicular from origin O to line AB at point P

Given: Points $A(-9, 4, 5)$, $B(11, 0, -1)$, and the origin $O(0, 0, 0)$.

To Find: The coordinates of the foot of the perpendicular P from O to line AB .

Concept / Formula Used: $P = \left(\frac{kx_2 + x_1}{k + 1}, \frac{ky_2 + y_1}{k + 1}, \frac{kz_2 + z_1}{k + 1} \right)$ $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Solution:

Let P be the foot of the perpendicular from the origin $O(0, 0, 0)$ on the line joining $A(-9, 4, 5)$ and $B(11, 0, -1)$. Let P divide AB in the ratio $k : 1$.

Using the section formula:

$$\begin{aligned} P &= \left(\frac{11k - 9}{k + 1}, \frac{0 \cdot k + 4}{k + 1}, \frac{-1 \cdot k + 5}{k + 1} \right) \\ &= \left(\frac{11k - 9}{k + 1}, \frac{4}{k + 1}, \frac{5 - k}{k + 1} \right) \end{aligned}$$

The direction ratios of OP (joining $O(0, 0, 0)$ to P) are:

$$(a_1, b_1, c_1) = \left(\frac{11k - 9}{k + 1}, \frac{4}{k + 1}, \frac{5 - k}{k + 1} \right)$$

The direction ratios of line AB joining $A(-9, 4, 5)$ and $B(11, 0, -1)$ are:

$$\begin{aligned} (a_2, b_2, c_2) &= (11 - (-9), 0 - 4, -1 - 5) \\ &= (20, -4, -6) \end{aligned}$$

Since $OP \perp AB$, the dot product of their direction ratios is zero:

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

$$20 \left(\frac{11k-9}{k+1} \right) - 4 \left(\frac{4}{k+1} \right) - 6 \left(\frac{5-k}{k+1} \right) = 0$$

Multiplying the entire equation by $(k+1)$:

$$20(11k-9) - 16 - 6(5-k) = 0$$

$$220k - 180 - 16 - 30 + 6k = 0$$

$$226k - 226 = 0$$

$$226k = 226$$

$$k = 1$$

Substitute $k = 1$ back into the coordinates of P :

$$x = \frac{11(1)-9}{1+1} = \frac{2}{2} = 1$$

$$y = \frac{4}{1+1} = \frac{4}{2} = 2$$

$$z = \frac{5-1}{1+1} = \frac{4}{2} = 2$$

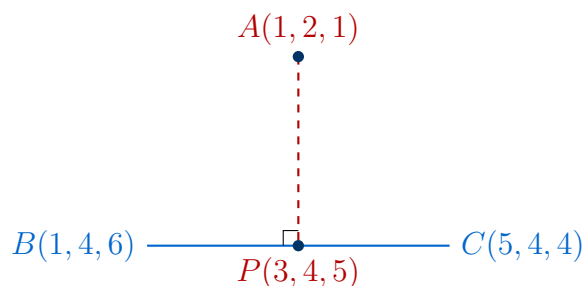
Final Answer

The required point is $(1, 2, 2)$.

Question 21

Find the coordinates of the foot of perpendicular drawn from the point $A(1, 2, 1)$ on the line joining the points $B(1, 4, 6)$ and $C(5, 4, 4)$.

TYPE 13 Foot of Perpendicular from a Point to a Line (via Two Points)



Foot of perpendicular P from A to line BC

Given: Points $A(1, 2, 1)$, $B(1, 4, 6)$, and $C(5, 4, 4)$.

To Find: The coordinates of the foot of perpendicular P from A on line BC .

Concept / Formula Used: $P = \left(\frac{kx_2 + x_1}{k+1}, \frac{ky_2 + y_1}{k+1}, \frac{kz_2 + z_1}{k+1} \right)$ $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Solution:

Let P be the foot of the perpendicular drawn from $A(1, 2, 1)$ to line BC . Let P divide line segment BC in the ratio $k : 1$.

Using the section formula with $B(1, 4, 6)$ and $C(5, 4, 4)$:

$$\begin{aligned} P &= \left(\frac{5k+1}{k+1}, \frac{4k+4}{k+1}, \frac{4k+6}{k+1} \right) \\ &= \left(\frac{5k+1}{k+1}, 4, \frac{4k+6}{k+1} \right) \end{aligned}$$

The direction ratios of line AP (where $A = (1, 2, 1)$) are:

$$\begin{aligned} a_1 &= \frac{5k+1}{k+1} - 1 = \frac{5k+1-(k+1)}{k+1} = \frac{4k}{k+1} \\ b_1 &= 4 - 2 = 2 \\ c_1 &= \frac{4k+6}{k+1} - 1 = \frac{4k+6-(k+1)}{k+1} = \frac{3k+5}{k+1} \end{aligned}$$

The direction ratios of line BC joining $B(1, 4, 6)$ and $C(5, 4, 4)$ are:

$$\begin{aligned} a_2 &= 5 - 1 = 4 \\ b_2 &= 4 - 4 = 0 \\ c_2 &= 4 - 6 = -2 \end{aligned}$$

Since $AP \perp BC$, the dot product of their direction ratios is zero:

$$\begin{aligned} a_1a_2 + b_1b_2 + c_1c_2 &= 0 \\ 4 \left(\frac{4k}{k+1} \right) + 0(2) - 2 \left(\frac{3k+5}{k+1} \right) &= 0 \end{aligned}$$

Multiplying the entire equation by $(k+1)$:

$$\begin{aligned} 16k - 2(3k+5) &= 0 \\ 16k - 6k - 10 &= 0 \\ 10k - 10 &= 0 \\ 10k &= 10 \\ k &= 1 \end{aligned}$$

Substitute $k = 1$ back into the coordinates of P :

$$\begin{aligned} x &= \frac{5(1)+1}{1+1} = \frac{6}{2} = 3 \\ y &= 4 \\ z &= \frac{4(1)+6}{1+1} = \frac{10}{2} = 5 \end{aligned}$$

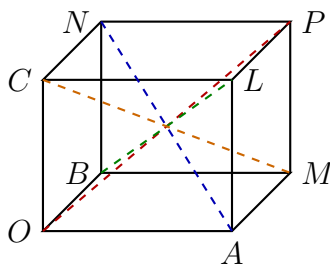
Final Answer

The coordinates of the foot of perpendicular are $(3, 4, 5)$.

Question 22

The adjoining figure shows a rectangular box of size a , b and c units along x , y and z -axes respectively. Prove that the angles between the four diagonals are given by $\cos^{-1} \left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2} \right)$ where all signs cannot be +ve or -ve.

TYPE 14 Angles Between Diagonals of a Rectangular Box



Rectangular box with vertices and diagonals

Given: A rectangular box with edges of lengths a , b , and c along the x , y , and z axes respectively, with origin at $O(0, 0, 0)$.

To Prove: The angles between the four body diagonals are given by $\theta = \cos^{-1} \left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2} \right)$, where all signs cannot be all positive or all negative.

Concept / Formula Used: $\cos \theta = \frac{\vec{d}_1 \cdot \vec{d}_2}{|\vec{d}_1||\vec{d}_2|} = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$

Solution:

Let one vertex of the rectangular box be at the origin $O(0, 0, 0)$. The coordinates of the 8 vertices are:

$$\begin{aligned} O(0, 0, 0), \quad A(a, 0, 0), \quad B(0, b, 0), \quad C(0, 0, c), \\ M(a, b, 0), \quad L(a, 0, c), \quad N(0, b, c), \quad P(a, b, c) \end{aligned}$$

The four main body diagonals of the rectangular box are OP , AN , BL , and CM .

Direction ratios of the four diagonals are obtained by taking the coordinate differences:

$$\begin{aligned} \text{Diagonal } OP : (a_1, b_1, c_1) &= (a - 0, b - 0, c - 0) = (a, b, c) \\ \text{Diagonal } AN : (a_2, b_2, c_2) &= (0 - a, b - 0, c - 0) = (-a, b, c) \\ \text{Diagonal } BL : (a_3, b_3, c_3) &= (a - 0, 0 - b, c - 0) = (a, -b, c) \\ \text{Diagonal } CM : (a_4, b_4, c_4) &= (a - 0, b - 0, 0 - c) = (a, b, -c) \end{aligned}$$

The magnitude of each diagonal is equal:

$$|\vec{d}_1| = |\vec{d}_2| = |\vec{d}_3| = |\vec{d}_4| = \sqrt{a^2 + b^2 + c^2}$$

Let θ be the angle between any pair of diagonals.

For instance, taking diagonals $OP(a, b, c)$ and $AN(-a, b, c)$:

$$\begin{aligned}\cos \theta &= \frac{(a)(-a) + (b)(b) + (c)(c)}{\sqrt{a^2 + b^2 + c^2}\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{-a^2 + b^2 + c^2}{a^2 + b^2 + c^2}\end{aligned}$$

Similarly, taking diagonals $AN(-a, b, c)$ and $BL(a, -b, c)$:

$$\begin{aligned}\cos \theta &= \frac{(-a)(a) + (b)(-b) + (c)(c)}{\sqrt{a^2 + b^2 + c^2}\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{-a^2 - b^2 + c^2}{a^2 + b^2 + c^2}\end{aligned}$$

Evaluating the angle between any of the $\binom{4}{2} = 6$ possible pairs of diagonals, the numerator of $\cos \theta$ is formed by taking products of corresponding components $(\pm a)(\pm a) + (\pm b)(\pm b) + (\pm c)(\pm c)$, which gives:

$$\cos \theta = \frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2}$$

Note that if all signs were positive, $\cos \theta = \frac{a^2+b^2+c^2}{a^2+b^2+c^2} = 1 \implies \theta = 0^\circ$ (the angle of a diagonal with itself). If all signs were negative, $\cos \theta = -1 \implies \theta = 180^\circ$. Hence, for distinct intersecting diagonals, all signs cannot be simultaneously positive or simultaneously negative.

Hence Proved

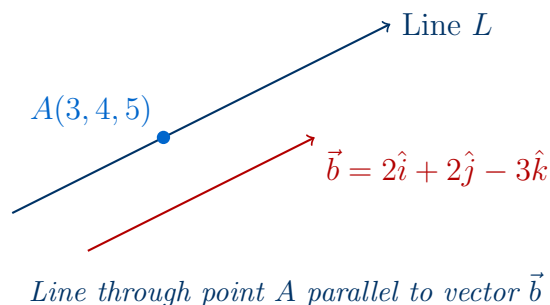
The angles between the four diagonals are given by $\theta = \cos^{-1}\left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2}\right)$, where all signs cannot be all positive or all negative.

11.2 – Equation of a Line in Space

Question 1 (i)

Find the vector equation of the line passing through the point $(3, 4, 5)$ and parallel to the vector $2\hat{i} + 2\hat{j} - 3\hat{k}$.

TYPE 15 Vector/Cartesian Equation of a Line – Point & Direction



Given: Passing point $(3, 4, 5)$ and parallel vector $\vec{b} = 2\hat{i} + 2\hat{j} - 3\hat{k}$.

Concept / Formula Used: $\vec{a}$ $\vec{b}$ $\vec{r} = \vec{a} + \lambda\vec{b}$

Solution:

Let the position vector of the given point $(3, 4, 5)$ be $\vec{a}$:

$$\vec{a} = 3\hat{i} + 4\hat{j} + 5\hat{k}$$

The vector parallel to the required line is given as:

$$\vec{b} = 2\hat{i} + 2\hat{j} - 3\hat{k}$$

Using the standard vector equation formula $\vec{r} = \vec{a} + \lambda\vec{b}$:

$$\vec{r} = (3\hat{i} + 4\hat{j} + 5\hat{k}) + \lambda(2\hat{i} + 2\hat{j} - 3\hat{k})$$

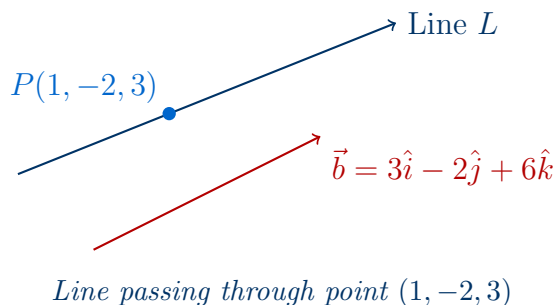
Final Answer

$$\vec{r} = (3\hat{i} + 4\hat{j} + 5\hat{k}) + \lambda(2\hat{i} + 2\hat{j} - 3\hat{k}), \text{ where } \lambda \in \mathbb{R}$$

Question 1 (ii)

Find the vector equation of the line which is parallel to the vector $3\hat{i} - 2\hat{j} + 6\hat{k}$ and passes through the point $(1, -2, 3)$.

TYPE 15 Vector/Cartesian Equation of a Line – Point & Direction



Given: Passing point $(1, -2, 3)$ and parallel vector $\vec{b} = 3\hat{i} - 2\hat{j} + 6\hat{k}$.

Concept / Formula Used: $\vec{r} = \vec{a} + \lambda \vec{b}$

Solution:

Position vector of the point $(1, -2, 3)$ is:

$$\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$$

The direction vector parallel to the line is:

$$\vec{b} = 3\hat{i} - 2\hat{j} + 6\hat{k}$$

Substituting $\vec{a}$ and $\vec{b}$ into the standard formula $\vec{r} = \vec{a} + \lambda \vec{b}$:

$$\vec{r} = (\hat{i} - 2\hat{j} + 3\hat{k}) + \lambda(3\hat{i} - 2\hat{j} + 6\hat{k})$$

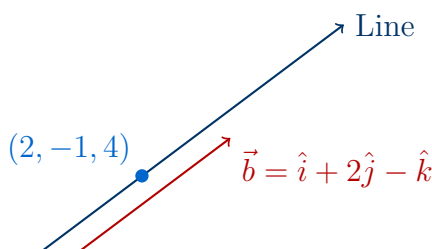
Final Answer

$$\vec{r} = (\hat{i} - 2\hat{j} + 3\hat{k}) + \lambda(3\hat{i} - 2\hat{j} + 6\hat{k}), \text{ where } \lambda \in \mathbb{R}$$

Question 2

Find the cartesian equation of the line which passes through the point with position vector $2\hat{i} - \hat{j} + 4\hat{k}$ and is in the direction of the vector $\hat{i} + 2\hat{j} - \hat{k}$.

TYPE 15 Vector/Cartesian Equation of a Line – Point & Direction



Line passing through position vector $\vec{a}$ in direction $\vec{b}$

Given: Point $(x_1, y_1, z_1) = (2, -1, 4)$ and direction ratios $(a, b, c) = (1, 2, -1)$.

Concept / Formula Used: $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$

Solution:

The line passes through the point with position vector $2\hat{i} - \hat{j} + 4\hat{k}$, so:

$$(x_1, y_1, z_1) = (2, -1, 4)$$

The direction vector of the line is $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$, giving direction ratios:

$$a = 1, \quad b = 2, \quad c = -1$$

Using the standard cartesian form $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$:

$$\frac{x - 2}{1} = \frac{y - (-1)}{2} = \frac{z - 4}{-1}$$

$$\frac{x - 2}{1} = \frac{y + 1}{2} = \frac{z - 4}{-1}$$

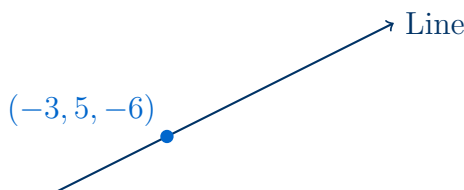
Final Answer

$$\frac{x - 2}{1} = \frac{y + 1}{2} = \frac{z - 4}{-1}$$

Question 3 (i)

Write the vector equation of the line $\frac{x + 3}{2} = \frac{y - 5}{4} = \frac{z + 6}{2}$.

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction



Conversion of cartesian equation to vector form

Given: Cartesian equation $\frac{x + 3}{2} = \frac{y - 5}{4} = \frac{z + 6}{2}$.

Concept / Formula Used: $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$ $\vec{r} = (x_1\hat{i} + y_1\hat{j} + z_1\hat{k}) + \lambda(a\hat{i} + b\hat{j} + c\hat{k})$

Solution:

Rewrite the given cartesian equation in standard form $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$:

$$\frac{x - (-3)}{2} = \frac{y - 5}{4} = \frac{z - (-6)}{2}$$

Comparing with standard form:

$$(x_1, y_1, z_1) = (-3, 5, -6)$$

Position vector of the point on the line:

$$\vec{a} = -3\hat{i} + 5\hat{j} - 6\hat{k}$$

Direction ratios are $a = 2, b = 4, c = 2$, so direction vector:

$$\vec{b} = 2\hat{i} + 4\hat{j} + 2\hat{k}$$

Vector equation $\vec{r} = \vec{a} + \lambda\vec{b}$:

$$\vec{r} = (-3\hat{i} + 5\hat{j} - 6\hat{k}) + \lambda(2\hat{i} + 4\hat{j} + 2\hat{k})$$

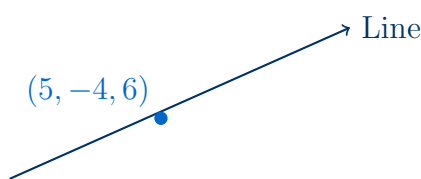
Final Answer

$$\vec{r} = (-3\hat{i} + 5\hat{j} - 6\hat{k}) + \lambda(2\hat{i} + 4\hat{j} + 2\hat{k})$$

Question 3 (ii)

Write the vector equation of the line $\frac{x-5}{3} = \frac{y+4}{7} = \frac{6-z}{2}$.

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction



Standardizing cartesian terms to find vector equation

Given: Cartesian equation $\frac{x-5}{3} = \frac{y+4}{7} = \frac{6-z}{2}$.

Concept / Formula Used: x, y, z + 1

Solution:

First, convert $\frac{6-z}{2}$ to standard form with a positive coefficient of z :

$$\frac{6-z}{2} = \frac{-(z-6)}{2} = \frac{z-6}{-2}$$

The standard cartesian equation is:

$$\frac{x-5}{3} = \frac{y-(-4)}{7} = \frac{z-6}{-2}$$

Point on the line:

$$(x_1, y_1, z_1) = (5, -4, 6)$$

Position vector of the point:

$$\vec{a} = 5\hat{i} - 4\hat{j} + 6\hat{k}$$

Direction ratios are $a = 3, b = 7, c = -2$, so direction vector:

$$\vec{b} = 3\hat{i} + 7\hat{j} - 2\hat{k}$$

Vector equation $\vec{r} = \vec{a} + \lambda\vec{b}$:

$$\vec{r} = (5\hat{i} - 4\hat{j} + 6\hat{k}) + \lambda(3\hat{i} + 7\hat{j} - 2\hat{k})$$

Final Answer

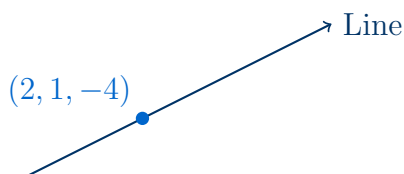
$$\vec{r} = (5\hat{i} - 4\hat{j} + 6\hat{k}) + \lambda(3\hat{i} + 7\hat{j} - 2\hat{k})$$

Common Mistake: Leaving $\frac{6-z}{2}$ unchanged leads to incorrect sign in direction vector; always rewrite it as $\frac{z-6}{-2}$.

Question 4 (i)

Write the cartesian equation of the following line given in vector form: $\vec{r} = 2\hat{i} + \hat{j} - 4\hat{k} + \lambda(\hat{i} - \hat{j} - \hat{k})$.

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction



Vector equation to cartesian symmetric form

Given: Vector equation $\vec{r} = (2\hat{i} + \hat{j} - 4\hat{k}) + \lambda(\hat{i} - \hat{j} - \hat{k})$.

Concept / Formula Used: $\vec{r} = (x_1\hat{i} + y_1\hat{j} + z_1\hat{k}) + \lambda(a\hat{i} + b\hat{j} + c\hat{k})$ $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$

Solution:

Comparing given vector equation with $\vec{r} = \vec{a} + \lambda\vec{b}$: Position vector $\vec{a} = 2\hat{i} + \hat{j} - 4\hat{k} \Rightarrow (x_1, y_1, z_1) = (2, 1, -4)$ Direction vector $\vec{b} = \hat{i} - \hat{j} - \hat{k} \Rightarrow (a, b, c) = (1, -1, -1)$
Substituting into cartesian formula:

$$\frac{x-2}{1} = \frac{y-1}{-1} = \frac{z-(-4)}{-1}$$

$$\frac{x-2}{1} = \frac{y-1}{-1} = \frac{z+4}{-1}$$

Final Answer

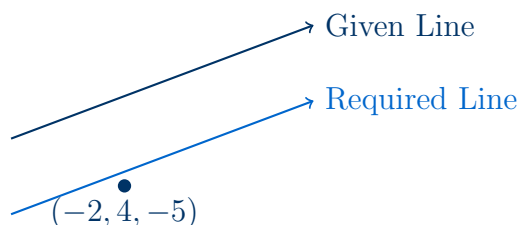
$$\frac{x-2}{1} = \frac{y-1}{-1} = \frac{z+4}{-1}$$

Question 4 (ii)

Find the cartesian equation of the line which passes through the point $(-2, 4, -5)$ and

parallel to the line $\frac{x+3}{3} = \frac{4-y}{5} = \frac{z+8}{6}$.

TYPE 15 Vector/Cartesian Equation of a Line – Point & Direction



Parallel lines in 3D space

Given: Passing point $(-2, 4, -5)$ and parallel line $\frac{x+3}{3} = \frac{4-y}{5} = \frac{z+8}{6}$.

Solution:

Convert the given parallel line into standard cartesian form:

$$\frac{4-y}{5} = \frac{-(y-4)}{5} = \frac{y-4}{-5}$$

So the given line is:

$$\frac{x+3}{3} = \frac{y-4}{-5} = \frac{z+8}{6}$$

Direction ratios of given line are $(3, -5, 6)$. Since the required line is parallel, its direction ratios are also $(a, b, c) = (3, -5, 6)$. The line passes through $(x_1, y_1, z_1) = (-2, 4, -5)$. Using cartesian equation formula:

$$\frac{x - (-2)}{3} = \frac{y - 4}{-5} = \frac{z - (-5)}{6}$$

$$\frac{x + 2}{3} = \frac{y - 4}{-5} = \frac{z + 5}{6}$$

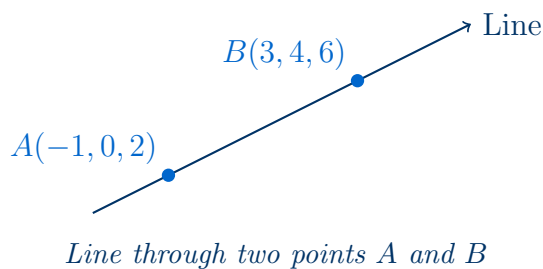
Final Answer

$$\frac{x+2}{3} = \frac{y-4}{-5} = \frac{z+5}{6}$$

Question 5 (i)

Find the vector equation of the line passing through the points $(-1, 0, 2)$ and $(3, 4, 6)$.

TYPE 17 Line Equation Through Two Given Points



Given: Points $A(-1, 0, 2)$ and $B(3, 4, 6)$.

Concept / Formula Used: $\vec{a}$ $\vec{b}$ $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$

Solution:

Position vector of point A:

$$\vec{a} = -\hat{i} + 0\hat{j} + 2\hat{k} = -\hat{i} + 2\hat{k}$$

Position vector of point B:

$$\vec{b} = 3\hat{i} + 4\hat{j} + 6\hat{k}$$

Direction vector $\vec{b} - \vec{a}$:

$$\begin{aligned}\vec{b} - \vec{a} &= (3 - (-1))\hat{i} + (4 - 0)\hat{j} + (6 - 2)\hat{k} \\ &= 4\hat{i} + 4\hat{j} + 4\hat{k}\end{aligned}$$

Vector equation $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$:

$$\vec{r} = (-\hat{i} + 2\hat{k}) + \lambda(4\hat{i} + 4\hat{j} + 4\hat{k})$$

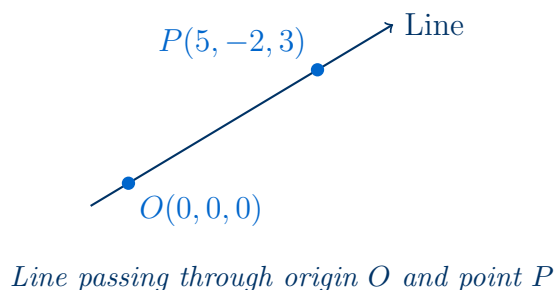
Final Answer

$$\vec{r} = (-\hat{i} + 2\hat{k}) + \lambda(4\hat{i} + 4\hat{j} + 4\hat{k})$$

Question 5 (ii)

Find the vector and the cartesian equations of the line that passes through origin and the point $(5, -2, 3)$.

TYPE 17 Line Equation Through Two Given Points



Given: Points $O(0, 0, 0)$ and $P(5, -2, 3)$.

Concept / Formula Used: $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$ $\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$

Solution:

Position vector of origin O :

$$\vec{a} = 0\hat{i} + 0\hat{j} + 0\hat{k} = \vec{0}$$

Position vector of P :

$$\vec{b} = 5\hat{i} - 2\hat{j} + 3\hat{k}$$

Direction vector:

$$\vec{b} - \vec{a} = 5\hat{i} - 2\hat{j} + 3\hat{k}$$

Vector equation $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$:

$$\vec{r} = \vec{0} + \lambda(5\hat{i} - 2\hat{j} + 3\hat{k})$$

$$\vec{r} = \lambda(5\hat{i} - 2\hat{j} + 3\hat{k})$$

Cartesian equation using $(x_1, y_1, z_1) = (0, 0, 0)$ and direction ratios $(5, -2, 3)$:

$$\frac{x - 0}{5} = \frac{y - 0}{-2} = \frac{z - 0}{3}$$

$$\frac{x}{5} = \frac{y}{-2} = \frac{z}{3}$$

Final Answer

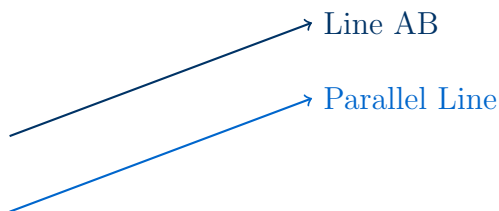
Vector equation: $\vec{r} = \lambda(5\hat{i} - 2\hat{j} + 3\hat{k})$

Cartesian equation: $\frac{x}{5} = \frac{y}{-2} = \frac{z}{3}$

Question 6 (i)

If the equation of a line AB is $\frac{x - 3}{1} = \frac{y + 2}{-2} = \frac{z - 5}{4}$, find the direction ratios of a line parallel to AB.

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction



Parallel lines having proportional direction ratios

Given: Line equation $\frac{x-3}{1} = \frac{y+2}{-2} = \frac{z-5}{4}$.

Concept / Formula Used: $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$ (a, b, c)

Solution:

Comparing line equation $\frac{x-3}{1} = \frac{y-(-2)}{-2} = \frac{z-5}{4}$ with standard form: Direction ratios of line AB are $a = 1, b = -2, c = 4$. Parallel lines have direction ratios proportional to those of line AB.

Final Answer

Direction ratios are 1, -2, 4

Question 6 (ii)

The cartesian equations of a line AB are $\frac{2x-1}{2} = \frac{4-y}{7} = \frac{z+1}{2}$. Write the direction ratios of a line parallel to AB.

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction



Standardizing equations to extract direction ratios

Given: Cartesian equations $\frac{2x-1}{2} = \frac{4-y}{7} = \frac{z+1}{2}$.

Concept / Formula Used: x, y, z $+ 1$

Solution:

Convert each expression to standard form:

$$\frac{2x-1}{2} = \frac{2\left(x-\frac{1}{2}\right)}{2} = \frac{x-\frac{1}{2}}{1}$$

$$\frac{4-y}{7} = \frac{-(y-4)}{7} = \frac{y-4}{-7}$$

$$\frac{z+1}{2} = \frac{z-(-1)}{2}$$

Rewriting in standard cartesian form:

$$\frac{x-1/2}{1} = \frac{y-4}{-7} = \frac{z+1}{2}$$

Direction ratios of line AB are $1, -7, 2$. Any line parallel to AB has direction ratios proportional to $1, -7, 2$.

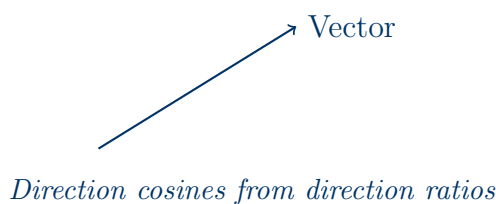
Final Answer

Direction ratios are $1, -7, 2$

Question 6 (iii)

Write the direction cosines of a vector parallel to the line $\frac{4-x}{2} = \frac{y+3}{3} = \frac{z+2}{6}$.

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction



Given: Line equation $\frac{4-x}{2} = \frac{y+3}{3} = \frac{z+2}{6}$.

Concept / Formula Used: a, b, c $l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$

Solution:

Standardize the first term:

$$\frac{4-x}{2} = \frac{-(x-4)}{2} = \frac{x-4}{-2}$$

Standard cartesian equation:

$$\frac{x-4}{-2} = \frac{y+3}{3} = \frac{z+2}{6}$$

Direction ratios are $a = -2, b = 3, c = 6$. Calculate magnitude $\sqrt{a^2 + b^2 + c^2}$:

$$\begin{aligned}\sqrt{a^2 + b^2 + c^2} &= \sqrt{(-2)^2 + 3^2 + 6^2} \\ &= \sqrt{4 + 9 + 36} \\ &= \sqrt{49} = 7\end{aligned}$$

Direction cosines:

$$\begin{aligned}l &= \frac{-2}{7} \\ m &= \frac{3}{7} \\ n &= \frac{6}{7}\end{aligned}$$

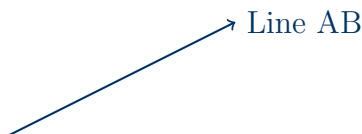
Final Answer

Direction cosines are $-\frac{2}{7}, \frac{3}{7}, \frac{6}{7}$

Question 6 (iv)

The cartesian equation of a line AB is $\frac{2x-1}{\sqrt{3}} = \frac{y+2}{2} = \frac{z-3}{3}$. Find the direction cosines of a line parallel to AB.

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction



Extracting direction cosines for fractional direction ratios

Given: Line equation $\frac{2x-1}{\sqrt{3}} = \frac{y+2}{2} = \frac{z-3}{3}$.

Concept / Formula Used: $l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$

Solution:

Standardize the first term:

$$\frac{2x-1}{\sqrt{3}} = \frac{2(x-\frac{1}{2})}{\sqrt{3}} = \frac{x-\frac{1}{2}}{\frac{\sqrt{3}}{2}}$$

Standard cartesian equation:

$$\frac{x-1/2}{\sqrt{3}/2} = \frac{y+2}{2} = \frac{z-3}{3}$$

Direction ratios are $a = \frac{\sqrt{3}}{2}, b = 2, c = 3$. Multiplying by 2 to clear fraction gives integer direction ratios:

$$a' = \sqrt{3}, \quad b' = 4, \quad c' = 6$$

Calculate magnitude:

$$\begin{aligned} \sqrt{(a')^2 + (b')^2 + (c')^2} &= \sqrt{(\sqrt{3})^2 + 4^2 + 6^2} \\ &= \sqrt{3 + 16 + 36} \\ &= \sqrt{55} \end{aligned}$$

Direction cosines:

$$l = \frac{\sqrt{3}}{\sqrt{55}}, \quad m = \frac{4}{\sqrt{55}}, \quad n = \frac{6}{\sqrt{55}}$$

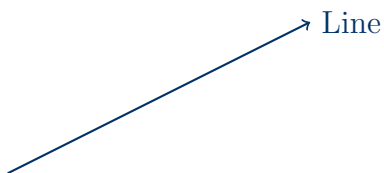
Final Answer

Direction cosines are $\frac{\sqrt{3}}{\sqrt{55}}, \frac{4}{\sqrt{55}}, \frac{6}{\sqrt{55}}$

Question 7 (i)

The cartesian equations of a line are $3x - 3 = 2y + 1 = 5 - 6z$. Write the direction ratios of the line.

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction



Reducing general equality to standard symmetric form

Given: Equations $3x - 3 = 2y + 1 = 5 - 6z$.

Concept / Formula Used: $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$

Solution:

Factor variable coefficients:

$$3(x - 1) = 2\left(y + \frac{1}{2}\right) = -6\left(z - \frac{5}{6}\right)$$

Divide each term by 6 (lcm of 3, 2, 6):

$$\frac{3(x - 1)}{6} = \frac{2(y + 1/2)}{6} = \frac{-6(z - 5/6)}{6}$$

$$\frac{x - 1}{2} = \frac{y + 1/2}{3} = \frac{z - 5/6}{-1}$$

Comparing with standard form, direction ratios are:

$$a = 2, \quad b = 3, \quad c = -1$$

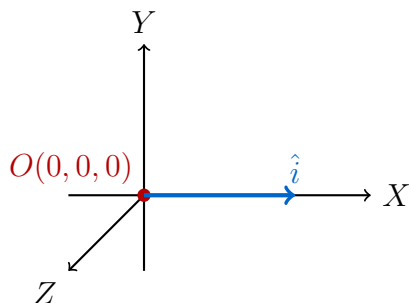
Final Answer

Direction ratios are 2, 3, -1

Question 7 (ii)

Find the vector equation of the X-axis.

TYPE 15 Vector/Cartesian Equation of a Line – Point & Direction



X-axis in 3D coordinate system

Given: X-axis passing through origin $(0,0,0)$ along unit vector $\hat{i}$.

Concept / Formula Used: $\vec{r} = \vec{a} + \lambda\vec{b}$

Solution:

The X-axis passes through the origin $(0,0,0)$, so its position vector is:

$$\vec{a} = 0\hat{i} + 0\hat{j} + 0\hat{k} = \vec{0}$$

The unit vector parallel to the X-axis is:

$$\vec{b} = \hat{i}$$

Substituting $\vec{a}$ and $\vec{b}$ into vector equation formula $\vec{r} = \vec{a} + \lambda\vec{b}$:

$$\vec{r} = \vec{0} + \lambda\hat{i}$$

$$\vec{r} = \lambda\hat{i}$$

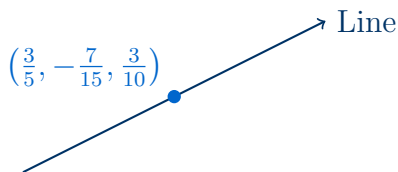
Final Answer

$$\vec{r} = \lambda\hat{i}, \text{ where } \lambda \in \mathbb{R}$$

Question 7 (iii)

The equations of a line are $5x - 3 = 15y + 7 = 3 - 10z$. Write the direction cosines of the line and find the coordinates of a point through which it passes.

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction



Extracting point and direction cosines

Given: Equations $5x - 3 = 15y + 7 = 3 - 10z$.

Concept / Formula Used: $l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$

Solution:

Factor variable coefficients:

$$5\left(x - \frac{3}{5}\right) = 15\left(y + \frac{7}{15}\right) = -10\left(z - \frac{3}{10}\right)$$

Divide each expression by 30 (lcm of 5, 15, 10):

$$\frac{5(x - 3/5)}{30} = \frac{15(y + 7/15)}{30} = \frac{-10(z - 3/10)}{30}$$

$$\frac{x - 3/5}{6} = \frac{y + 7/15}{2} = \frac{z - 3/10}{-3}$$

Coordinates of a point on the line:

$$(x_1, y_1, z_1) = \left(\frac{3}{5}, -\frac{7}{15}, \frac{3}{10}\right)$$

Direction ratios are $a = 6, b = 2, c = -3$. Calculate magnitude:

$$\sqrt{6^2 + 2^2 + (-3)^2} = \sqrt{36 + 4 + 9} = \sqrt{49} = 7$$

Direction cosines:

$$l = \frac{6}{7}, \quad m = \frac{2}{7}, \quad n = -\frac{3}{7}$$

Final Answer

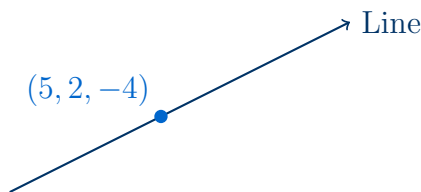
Point coordinates: $\left(\frac{3}{5}, -\frac{7}{15}, \frac{3}{10}\right)$

Direction cosines: $\left(\frac{6}{7}, \frac{2}{7}, -\frac{3}{7}\right)$

Question 8 (i)

Find the vector and the cartesian equations of the line through the point $(5, 2, -4)$ and which is parallel to the vector $3\hat{i} + 2\hat{j} - 8\hat{k}$.

TYPE 15 Vector/Cartesian Equation of a Line – Point & Direction



Line from given point parallel to direction vector

Given: Point $(5, 2, -4)$ and parallel vector $\vec{b} = 3\hat{i} + 2\hat{j} - 8\hat{k}$.

Concept / Formula Used: $\vec{r} = \vec{a} + \lambda\vec{b}$ $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$

Solution:

Position vector of point $(5, 2, -4)$:

$$\vec{a} = 5\hat{i} + 2\hat{j} - 4\hat{k}$$

Direction vector:

$$\vec{b} = 3\hat{i} + 2\hat{j} - 8\hat{k}$$

Vector equation:

$$\vec{r} = (5\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(3\hat{i} + 2\hat{j} - 8\hat{k})$$

Cartesian equation using $(x_1, y_1, z_1) = (5, 2, -4)$ and direction ratios $(3, 2, -8)$:

$$\frac{x - 5}{3} = \frac{y - 2}{2} = \frac{z - (-4)}{-8}$$

$$\frac{x - 5}{3} = \frac{y - 2}{2} = \frac{z + 4}{-8}$$

Final Answer

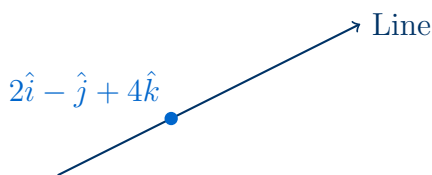
Vector equation: $\vec{r} = (5\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(3\hat{i} + 2\hat{j} - 8\hat{k})$

Cartesian equation: $\frac{x - 5}{3} = \frac{y - 2}{2} = \frac{z + 4}{-8}$

Question 8 (ii)

A line passes through the point with position vector $2\hat{i} - \hat{j} + 4\hat{k}$ and is in the direction of $\hat{i} + \hat{j} - 2\hat{k}$. Find the equations of the line in vector and in cartesian form.

TYPE 15 Vector/Cartesian Equation of a Line – Point & Direction



Line from position vector and direction

Given: Position vector $\vec{a} = 2\hat{i} - \hat{j} + 4\hat{k}$ and direction vector $\vec{b} = \hat{i} + \hat{j} - 2\hat{k}$.

Concept / Formula Used: $\vec{r} = \vec{a} + \lambda\vec{b}$ $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$

Solution:

Position vector of point:

$$\vec{a} = 2\hat{i} - \hat{j} + 4\hat{k} \implies (x_1, y_1, z_1) = (2, -1, 4)$$

Direction vector:

$$\vec{b} = \hat{i} + \hat{j} - 2\hat{k} \implies (a, b, c) = (1, 1, -2)$$

Vector equation:

$$\vec{r} = (2\hat{i} - \hat{j} + 4\hat{k}) + \lambda(\hat{i} + \hat{j} - 2\hat{k})$$

Cartesian equation:

$$\frac{x - 2}{1} = \frac{y - (-1)}{1} = \frac{z - 4}{-2}$$

$$\frac{x - 2}{1} = \frac{y + 1}{1} = \frac{z - 4}{-2}$$

Final Answer

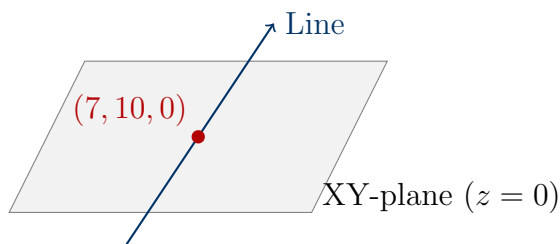
Vector equation: $\vec{r} = (2\hat{i} - \hat{j} + 4\hat{k}) + \lambda(\hat{i} + \hat{j} - 2\hat{k})$

Cartesian equation: $\frac{x - 2}{1} = \frac{y + 1}{1} = \frac{z - 4}{-2}$

Question 9 (i)

Find the coordinates of the point where the line $\frac{x - 1}{3} = \frac{y + 4}{7} = \frac{z + 4}{2}$ cuts the xy-plane.

TYPE 18 Point Where a Line Meets a Coordinate Plane



Intersection of line with XY-plane ($z = 0$)

Given: Line $\frac{x-1}{3} = \frac{y+4}{7} = \frac{z+4}{2}$ and plane $z = 0$.

Concept / Formula Used: $z = 0$

Solution:

Set the cartesian equation equal to parameter λ :

$$\frac{x-1}{3} = \frac{y+4}{7} = \frac{z+4}{2} = \lambda$$

Express general coordinates in terms of λ :

$$x = 3\lambda + 1$$

$$y = 7\lambda - 4$$

$$z = 2\lambda - 4$$

Since the point lies on the xy-plane, $z = 0$:

$$2\lambda - 4 = 0$$

$$2\lambda = 4 \implies \lambda = 2$$

Substitute $\lambda = 2$ back into the coordinate equations:

$$x = 3(2) + 1 = 6 + 1 = 7$$

$$y = 7(2) - 4 = 14 - 4 = 10$$

$$z = 2(2) - 4 = 4 - 4 = 0$$

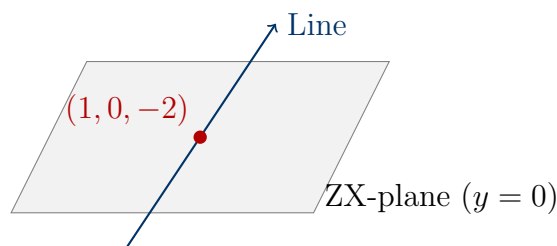
Final Answer

Point of intersection is $(7, 10, 0)$

Question 9 (ii)

Find the coordinates of the point where the line through $(-1, 1, -8)$ and $(5, -2, 10)$ crosses the zx-plane.

TYPE 18 Point Where a Line Meets a Coordinate Plane



Intersection of line with ZX-plane ($y = 0$)

Given: Points $A(-1, 1, -8)$ and $B(5, -2, 10)$, plane $y = 0$.

Concept / Formula Used: $y = 0$

Solution:

Find direction ratios of line AB :

$$a = 5 - (-1) = 6$$

$$b = -2 - 1 = -3$$

$$c = 10 - (-8) = 18$$

Equation of line AB :

$$\frac{x - (-1)}{6} = \frac{y - 1}{-3} = \frac{z - (-8)}{18} = \lambda$$

$$\frac{x + 1}{6} = \frac{y - 1}{-3} = \frac{z + 8}{18} = \lambda$$

General point on line:

$$x = 6\lambda - 1$$

$$y = -3\lambda + 1$$

$$z = 18\lambda - 8$$

Since the line crosses the zx -plane, $y = 0$:

$$-3\lambda + 1 = 0 \implies 3\lambda = 1 \implies \lambda = \frac{1}{3}$$

Substitute $\lambda = \frac{1}{3}$ into coordinates:

$$x = 6\left(\frac{1}{3}\right) - 1 = 2 - 1 = 1$$

$$y = -3\left(\frac{1}{3}\right) + 1 = -1 + 1 = 0$$

$$z = 18\left(\frac{1}{3}\right) - 8 = 6 - 8 = -2$$

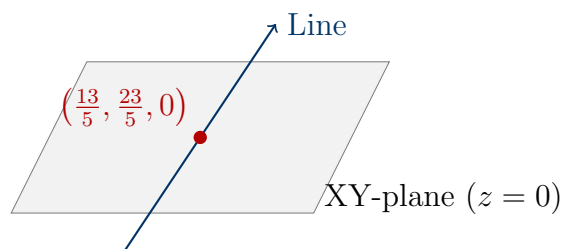
Final Answer

Point of intersection is $(1, 0, -2)$

Question 9 (iii)

Find the coordinates of the point where the line through $(5, 1, 6)$ and $(3, 4, 1)$ crosses the xy -plane.

TYPE 18 Point Where a Line Meets a Coordinate Plane



Intersection of line with XY-plane ($z = 0$)

Given: Points $(5, 1, 6)$ and $(3, 4, 1)$, plane $z = 0$.

Concept / Formula Used: $z = 0$

Solution:

Direction ratios of the line:

$$a = 3 - 5 = -2$$

$$b = 4 - 1 = 3$$

$$c = 1 - 6 = -5$$

Cartesian equation of line:

$$\frac{x - 5}{-2} = \frac{y - 1}{3} = \frac{z - 6}{-5} = \lambda$$

General point on line:

$$x = -2\lambda + 5$$

$$y = 3\lambda + 1$$

$$z = -5\lambda + 6$$

Crosses the xy-plane where $z = 0$:

$$-5\lambda + 6 = 0 \implies 5\lambda = 6 \implies \lambda = \frac{6}{5}$$

Substitute $\lambda = \frac{6}{5}$:

$$x = -2\left(\frac{6}{5}\right) + 5 = -\frac{12}{5} + 5 = \frac{13}{5}$$

$$y = 3\left(\frac{6}{5}\right) + 1 = \frac{18}{5} + 1 = \frac{23}{5}$$

$$z = -5\left(\frac{6}{5}\right) + 6 = -6 + 6 = 0$$

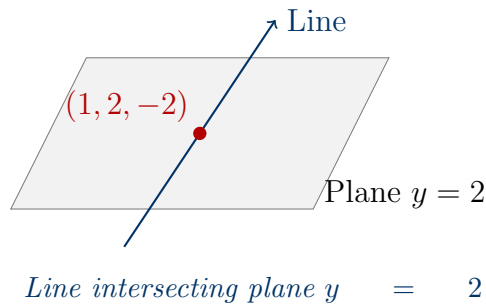
Final Answer

Point of intersection is $\left(\frac{13}{5}, \frac{23}{5}, 0\right)$

Question 10

Find the coordinates of the point where the line through the points $(-1, 4, 4)$ and $(2, 1, -5)$ intersects the plane $y = 2$.

TYPE 18 Point Where a Line Meets a Coordinate Plane



Given: Points $A(-1, 4, 4)$ and $B(2, 1, -5)$, plane $y = 2$.

Solution:

Direction ratios of line AB :

$$a = 2 - (-1) = 3$$

$$b = 1 - 4 = -3$$

$$c = -5 - 4 = -9$$

Simplifying direction ratios by dividing by 3 gives $(1, -1, -3)$. Cartesian equation of line:

$$\frac{x - (-1)}{1} = \frac{y - 4}{-1} = \frac{z - 4}{-3} = \lambda$$

$$\frac{x + 1}{1} = \frac{y - 4}{-1} = \frac{z - 4}{-3} = \lambda$$

General point on line:

$$x = \lambda - 1$$

$$y = -\lambda + 4$$

$$z = -3\lambda + 4$$

Substitute $y = 2$ into the y -coordinate equation:

$$-\lambda + 4 = 2 \implies \lambda = 2$$

Substitute $\lambda = 2$ into coordinates:

$$x = 2 - 1 = 1$$

$$y = 2$$

$$z = -3(2) + 4 = -6 + 4 = -2$$

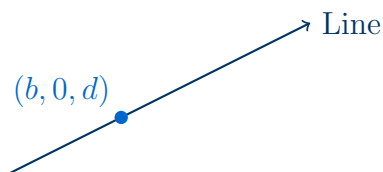
Final Answer

Point of intersection is $(1, 2, -2)$

Question 11

The cartesian equations of a line are $x = ay + b, z = cy + d$. Find direction numbers of the line, also find its equation in vector form.

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction



Representation of line given in asymmetric form

Given: Equations $x = ay + b$ and $z = cy + d$.

Concept / Formula Used: $\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$

Solution:

Express x and z in terms of y :

$$\begin{aligned} x = ay + b &\implies x - b = ay \implies \frac{x - b}{a} = y \\ z = cy + d &\implies z - d = cy \implies \frac{z - d}{c} = y \end{aligned}$$

Equating the expressions:

$$\frac{x - b}{a} = \frac{y - 0}{1} = \frac{z - d}{c}$$

Direction numbers (ratios) are $a, 1, c$. Position vector of point on line $(b, 0, d)$:

$$\vec{a} = b\hat{i} + 0\hat{j} + d\hat{k} = b\hat{i} + d\hat{k}$$

Direction vector:

$$\vec{v} = a\hat{i} + \hat{j} + c\hat{k}$$

Vector equation $\vec{r} = \vec{a} + \lambda\vec{v}$:

$$\vec{r} = (b\hat{i} + d\hat{k}) + \lambda(a\hat{i} + \hat{j} + c\hat{k})$$

Final Answer

Direction numbers: $a, 1, c$

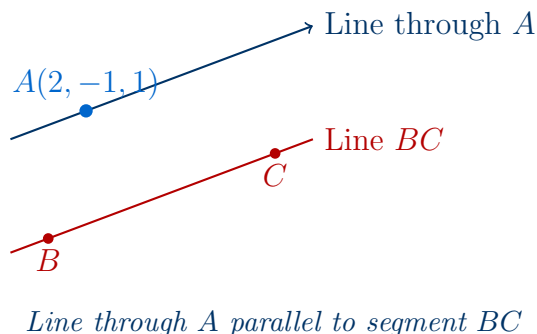
Vector equation: $\vec{r} = (b\hat{i} + d\hat{k}) + \lambda(a\hat{i} + \hat{j} + c\hat{k})$

Question 12 (i)

Find the vector equation of the line passing through the point $A(2, -1, 1)$ and parallel to the line joining the points $B(-1, 4, 1)$ and $C(1, 2, 2)$. Also find the cartesian

equation of the line.

TYPE 15 Vector/Cartesian Equation of a Line – Point & Direction



Given: Point $A(2, -1, 1)$ and points $B(-1, 4, 1)$, $C(1, 2, 2)$.

Concept / Formula Used: $BC \quad \vec{BC} = \vec{r}_C - \vec{r}_B$

Solution:

Calculate direction vector $\vec{BC}$:

$$\begin{aligned}\vec{BC} &= (1 - (-1))\hat{i} + (2 - 4)\hat{j} + (2 - 1)\hat{k} \\ &= 2\hat{i} - 2\hat{j} + \hat{k}\end{aligned}$$

Position vector of point A:

$$\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$$

Vector equation $\vec{r} = \vec{a} + \lambda\vec{BC}$:

$$\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + \lambda(2\hat{i} - 2\hat{j} + \hat{k})$$

Cartesian equation using point $(2, -1, 1)$ and direction ratios $(2, -2, 1)$:

$$\begin{aligned}\frac{x - 2}{2} &= \frac{y - (-1)}{-2} = \frac{z - 1}{1} \\ \frac{x - 2}{2} &= \frac{y + 1}{-2} = \frac{z - 1}{1}\end{aligned}$$

Final Answer

Vector equation: $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + \lambda(2\hat{i} - 2\hat{j} + \hat{k})$

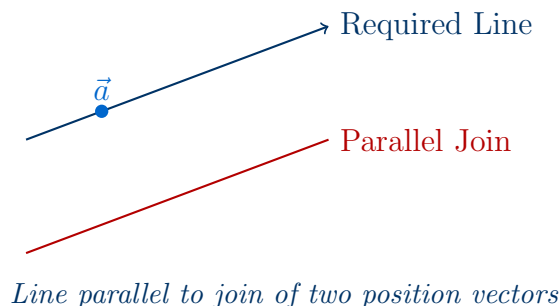
Cartesian equation: $\frac{x - 2}{2} = \frac{y + 1}{-2} = \frac{z - 1}{1}$

Question 12 (ii)

Find the vector equation of a line passing through the point with position vector $\hat{i} - 2\hat{j} - 3\hat{k}$ and parallel to the line joining the points with position vectors $\hat{i} - \hat{j} + 4\hat{k}$

and $2\hat{i} + \hat{j} + 2\hat{k}$. Also find the cartesian equations of the line.

TYPE 15 Vector/Cartesian Equation of a Line – Point & Direction



Given: Passing position vector $\vec{a} = \hat{i} - 2\hat{j} - 3\hat{k}$, parallel join vectors $\vec{v}_1 = \hat{i} - \hat{j} + 4\hat{k}$ and $\vec{v}_2 = 2\hat{i} + \hat{j} + 2\hat{k}$.

Concept / Formula Used: $\vec{b} = \vec{v}_2 - \vec{v}_1$

Solution:

Calculate direction vector $\vec{b} = \vec{v}_2 - \vec{v}_1$:

$$\begin{aligned}\vec{b} &= (2\hat{i} + \hat{j} + 2\hat{k}) - (\hat{i} - \hat{j} + 4\hat{k}) \\ &= (2 - 1)\hat{i} + (1 - (-1))\hat{j} + (2 - 4)\hat{k} \\ &= \hat{i} + 2\hat{j} - 2\hat{k}\end{aligned}$$

Vector equation $\vec{r} = \vec{a} + \lambda\vec{b}$:

$$\vec{r} = (\hat{i} - 2\hat{j} - 3\hat{k}) + \lambda(\hat{i} + 2\hat{j} - 2\hat{k})$$

Cartesian equation using point $(1, -2, -3)$ and direction ratios $(1, 2, -2)$:

$$\begin{aligned}\frac{x-1}{1} &= \frac{y-(-2)}{2} = \frac{z-(-3)}{-2} \\ \frac{x-1}{1} &= \frac{y+2}{2} = \frac{z+3}{-2}\end{aligned}$$

Final Answer

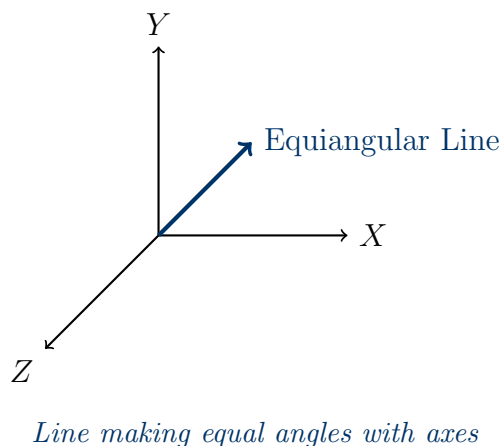
Vector equation: $\vec{r} = (\hat{i} - 2\hat{j} - 3\hat{k}) + \lambda(\hat{i} + 2\hat{j} - 2\hat{k})$

Cartesian equation: $\frac{x-1}{1} = \frac{y+2}{2} = \frac{z+3}{-2}$

Question 12 (iii)

Find the vector equation of the line passing through the point $(2, 3, -5)$ and making equal angles with coordinate axes.

TYPE 19 Equiangular Line – Equal Angles with All Axes



Given: Passing point $(2, 3, -5)$ and equal angles $\alpha = \beta = \gamma$.

Concept / Formula Used: $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1 \implies 3 \cos^2 \alpha = 1 \implies \cos \alpha = \pm \frac{1}{\sqrt{3}}$

Solution:

Let the line make equal angles α, β, γ with the coordinate axes:

$$\alpha = \beta = \gamma$$

Using direction cosines relation:

$$\begin{aligned} \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma &= 1 \\ 3 \cos^2 \alpha &= 1 \\ \cos \alpha &= \pm \frac{1}{\sqrt{3}} \end{aligned}$$

Direction cosines are $l = m = n = \frac{1}{\sqrt{3}}$, which implies direction ratios are proportional to:

$$a = 1, \quad b = 1, \quad c = 1$$

Direction vector:

$$\vec{b} = \hat{i} + \hat{j} + \hat{k}$$

Position vector of passing point $(2, 3, -5)$:

$$\vec{a} = 2\hat{i} + 3\hat{j} - 5\hat{k}$$

Vector equation $\vec{r} = \vec{a} + \lambda \vec{b}$:

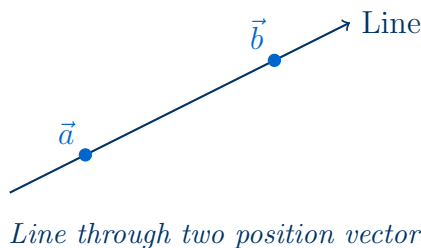
$$\vec{r} = (2\hat{i} + 3\hat{j} - 5\hat{k}) + \lambda(\hat{i} + \hat{j} + \hat{k})$$

Final Answer

$$\vec{r} = (2\hat{i} + 3\hat{j} - 5\hat{k}) + \lambda(\hat{i} + \hat{j} + \hat{k})$$

Question 13

Find the vector equation of the line joining the points whose position vectors are $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} - 3\hat{k}$. Also find its cartesian equations.

TYPE 17 Line Equation Through Two Given Points

Given: Position vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 3\hat{k}$.

Concept / Formula Used: $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$ $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$

Solution:

Calculate direction vector $\vec{d} = \vec{b} - \vec{a}$:

$$\begin{aligned}\vec{d} &= (\hat{i} + 2\hat{j} - 3\hat{k}) - (2\hat{i} - \hat{j} + \hat{k}) \\ &= (1 - 2)\hat{i} + (2 - (-1))\hat{j} + (-3 - 1)\hat{k} \\ &= -\hat{i} + 3\hat{j} - 4\hat{k}\end{aligned}$$

Vector equation $\vec{r} = \vec{a} + \lambda\vec{d}$:

$$\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + \lambda(-\hat{i} + 3\hat{j} - 4\hat{k})$$

Point coordinates from $\vec{a}$: $(x_1, y_1, z_1) = (2, -1, 1)$. Direction ratios: $(a, b, c) = (-1, 3, -4)$.

Cartesian equation:

$$\begin{aligned}\frac{x - 2}{-1} &= \frac{y - (-1)}{3} = \frac{z - 1}{-4} \\ \frac{x - 2}{-1} &= \frac{y + 1}{3} = \frac{z - 1}{-4}\end{aligned}$$

Final Answer

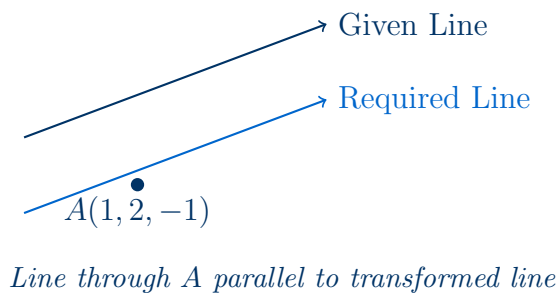
Vector equation: $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + \lambda(-\hat{i} + 3\hat{j} - 4\hat{k})$

Cartesian equation: $\frac{x - 2}{-1} = \frac{y + 1}{3} = \frac{z - 1}{-4}$

Question 14

Find the vector equation of the line passing through the point $A(1, 2, -1)$ and parallel to the line $5x - 25 = 14 - 7y = 35z$.

TYPE 15 Vector/Cartesian Equation of a Line – Point & Direction



Given: Passing point $A(1, 2, -1)$ and line equation $5x - 25 = 14 - 7y = 35z$.

Solution:

Factor variable coefficients:

$$5(x - 5) = -7(y - 2) = 35z$$

Divide each expression by 35 (lcm of 5, 7, 35):

$$\frac{5(x - 5)}{35} = \frac{-7(y - 2)}{35} = \frac{35z}{35}$$

$$\frac{x - 5}{7} = \frac{y - 2}{-5} = \frac{z}{1}$$

Direction ratios of given line are $(7, -5, 1)$. Direction vector of required line:

$$\vec{b} = 7\hat{i} - 5\hat{j} + \hat{k}$$

Position vector of point $A(1, 2, -1)$:

$$\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$$

Vector equation $\vec{r} = \vec{a} + \lambda\vec{b}$:

$$\vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + \lambda(7\hat{i} - 5\hat{j} + \hat{k})$$

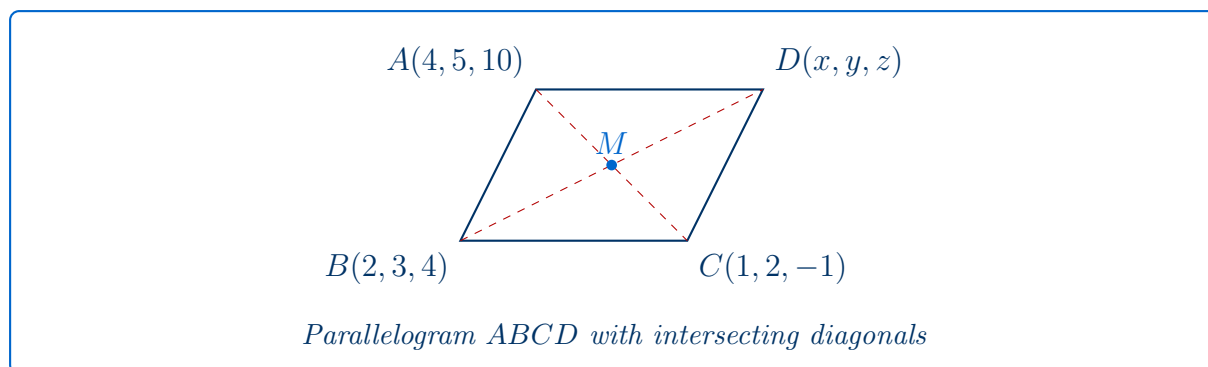
Final Answer

$$\vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + \lambda(7\hat{i} - 5\hat{j} + \hat{k})$$

Question 15

The points $A(4, 5, 10)$, $B(2, 3, 4)$ and $C(1, 2, -1)$ are three vertices of a parallelogram $ABCD$. Find the vector equations of the sides AB and BC . Also find the coordinates of D .

TYPE 20 Vector Equations of Parallelogram Sides from Vertices



Given: Vertices $A(4, 5, 10)$, $B(2, 3, 4)$, $C(1, 2, -1)$.

Concept / Formula Used: AC BD

Solution:

Let the coordinates of vertex D be (x, y, z) . Midpoint M of diagonal AC :

$$M = \left(\frac{4+1}{2}, \frac{5+2}{2}, \frac{10+(-1)}{2} \right) = \left(\frac{5}{2}, \frac{7}{2}, \frac{9}{2} \right)$$

Midpoint M of diagonal BD :

$$M = \left(\frac{2+x}{2}, \frac{3+y}{2}, \frac{4+z}{2} \right)$$

Equating corresponding coordinates:

$$\begin{aligned} \frac{2+x}{2} &= \frac{5}{2} \implies 2+x=5 \implies x=3 \\ \frac{3+y}{2} &= \frac{7}{2} \implies 3+y=7 \implies y=4 \\ \frac{4+z}{2} &= \frac{9}{2} \implies 4+z=9 \implies z=5 \end{aligned}$$

Coordinates of D are $(3, 4, 5)$.

Vector equation of side AB : Position vector of A : $\vec{a} = 4\hat{i} + 5\hat{j} + 10\hat{k}$. Direction vector $\vec{d}_{AB} = \vec{B} - \vec{A}$:

$$\begin{aligned} \vec{d}_{AB} &= (2-4)\hat{i} + (3-5)\hat{j} + (4-10)\hat{k} \\ &= -2\hat{i} - 2\hat{j} - 6\hat{k} \end{aligned}$$

Using parallel vector $\hat{i} + \hat{j} + 3\hat{k}$:

$$\vec{r}_{AB} = (4\hat{i} + 5\hat{j} + 10\hat{k}) + \lambda(\hat{i} + \hat{j} + 3\hat{k})$$

Vector equation of side BC : Position vector of B : $\vec{b} = 2\hat{i} + 3\hat{j} + 4\hat{k}$. Direction vector $\vec{d}_{BC} = \vec{C} - \vec{B}$:

$$\begin{aligned} \vec{d}_{BC} &= (1-2)\hat{i} + (2-3)\hat{j} + (-1-4)\hat{k} \\ &= -\hat{i} - \hat{j} - 5\hat{k} \end{aligned}$$

Using parallel vector $\hat{i} + \hat{j} + 5\hat{k}$:

$$\vec{r}_{BC} = (2\hat{i} + 3\hat{j} + 4\hat{k}) + \mu(\hat{i} + \hat{j} + 5\hat{k})$$

Final Answer

Coordinates of D : $(3, 4, 5)$

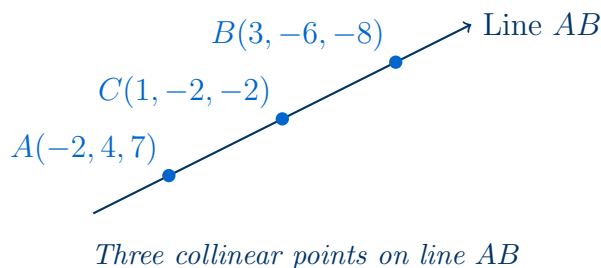
Vector eq of AB : $\vec{r} = (4\hat{i} + 5\hat{j} + 10\hat{k}) + \lambda(\hat{i} + \hat{j} + 3\hat{k})$

Vector eq of BC : $\vec{r} = (2\hat{i} + 3\hat{j} + 4\hat{k}) + \mu(\hat{i} + \hat{j} + 5\hat{k})$

Question 16 (i)

Find the equation of the line passing through the points $A(-2, 4, 7)$, $B(3, -6, -8)$. Hence, show that the points A , B and $C(1, -2, -2)$ are collinear.

TYPE 7 Collinearity of Three Points



To Prove: Equation of line AB and collinearity of $A, B, C(1, -2, -2)$.

Concept / Formula Used: $C(x_3, y_3, z_3)$ $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$

Solution:

Direction ratios of line passing through $A(-2, 4, 7)$ and $B(3, -6, -8)$:

$$a = 3 - (-2) = 5$$

$$b = -6 - 4 = -10$$

$$c = -8 - 7 = -15$$

Dividing by 5 gives simplified direction ratios $(1, -2, -3)$. Cartesian equation of line AB :

$$\frac{x - (-2)}{1} = \frac{y - 4}{-2} = \frac{z - 7}{-3}$$

$$\frac{x + 2}{1} = \frac{y - 4}{-2} = \frac{z - 7}{-3} = \lambda$$

Substitute coordinates of $C(1, -2, -2)$ into the line ratios:

$$\text{For } x = 1 : \frac{1 + 2}{1} = 3$$

$$\text{For } y = -2 : \frac{-2 - 4}{-2} = \frac{-6}{-2} = 3$$

$$\text{For } z = -2 : \frac{-2 - 7}{-3} = \frac{-9}{-3} = 3$$

Since all three ratios are equal to 3, point $C(1, -2, -2)$ lies on line AB .

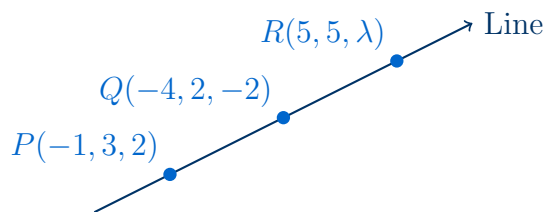
Hence Proved

Line equation is $\frac{x+2}{1} = \frac{y-4}{-2} = \frac{z-7}{-3}$, and points A, B, C are collinear.

Question 16 (ii)

If the points $(-1, 3, 2)$, $(-4, 2, -2)$ and $(5, 5, \lambda)$ are collinear, then find the value of λ .

TYPE 7a Collinearity – Solve for Unknown Parameter



Collinear points P, Q, R with unknown parameter

Given: Collinear points $P(-1, 3, 2)$, $Q(-4, 2, -2)$, and $R(5, 5, \lambda)$.

Concept / Formula Used: P, Q, R PQ QR $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Solution:

Calculate direction ratios of segment PQ :

$$a_1 = -4 - (-1) = -3$$

$$b_1 = 2 - 3 = -1$$

$$c_1 = -2 - 2 = -4$$

Calculate direction ratios of segment QR :

$$a_2 = 5 - (-4) = 9$$

$$b_2 = 5 - 2 = 3$$

$$c_2 = \lambda - (-2) = \lambda + 2$$

Equating direction ratio proportions:

$$\frac{a_2}{a_1} = \frac{b_2}{b_1} = \frac{c_2}{c_1}$$

$$\frac{9}{-3} = \frac{3}{-1} = \frac{\lambda + 2}{-4}$$

$$-3 = \frac{\lambda + 2}{-4}$$

Multiply both sides by -4 :

$$12 = \lambda + 2 \implies \lambda = 10$$

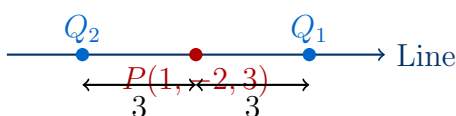
Final Answer

$$\lambda = 10$$

Question 17 (i)

Find the coordinates of the points on the line $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{6}$ which are at a distance of 3 units from the point $(1, -2, 3)$.

TYPE 21 Points on a Line at a Given Distance from a Point



Points Q_1 and Q_2 on line at distance 3 from point P

Given: Line $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{6}$, point $P(1, -2, 3)$, distance $d = 3$.

Concept / Formula Used: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

Solution:

Set line equation equal to parameter λ :

$$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{6} = \lambda$$

General point Q on line:

$$Q(2\lambda + 1, 3\lambda - 2, 6\lambda + 3)$$

Note that for $\lambda = 0$, Q coincides with $P(1, -2, 3)$. Distance PQ :

$$PQ = \sqrt{(2\lambda + 1 - 1)^2 + (3\lambda - 2 - (-2))^2 + (6\lambda + 3 - 3)^2} = 3$$

$$\sqrt{(2\lambda)^2 + (3\lambda)^2 + (6\lambda)^2} = 3$$

$$\sqrt{4\lambda^2 + 9\lambda^2 + 36\lambda^2} = 3$$

$$\sqrt{49\lambda^2} = 3$$

$$7|\lambda| = 3 \implies \lambda = \pm \frac{3}{7}$$

Case 1: When $\lambda = \frac{3}{7}$:

$$\begin{aligned}x &= 2\left(\frac{3}{7}\right) + 1 = \frac{6}{7} + 1 = \frac{13}{7} \\y &= 3\left(\frac{3}{7}\right) - 2 = \frac{9}{7} - 2 = -\frac{5}{7} \\z &= 6\left(\frac{3}{7}\right) + 3 = \frac{18}{7} + 3 = \frac{39}{7}\end{aligned}$$

Case 2: When $\lambda = -\frac{3}{7}$:

$$\begin{aligned}x &= 2\left(-\frac{3}{7}\right) + 1 = -\frac{6}{7} + 1 = \frac{1}{7} \\y &= 3\left(-\frac{3}{7}\right) - 2 = -\frac{9}{7} - 2 = -\frac{23}{7} \\z &= 6\left(-\frac{3}{7}\right) + 3 = -\frac{18}{7} + 3 = \frac{3}{7}\end{aligned}$$

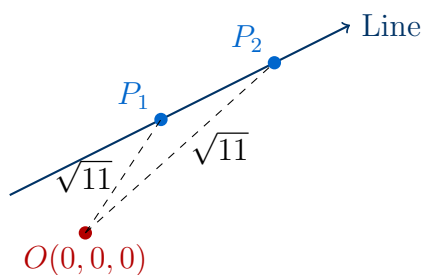
Final Answer

$$\left(\frac{13}{7}, -\frac{5}{7}, \frac{39}{7}\right) \text{ and } \left(\frac{1}{7}, -\frac{23}{7}, \frac{3}{7}\right)$$

Question 17 (ii)

Find the coordinates of points on line $\frac{x}{1} = \frac{y-1}{2} = \frac{z+1}{2}$ which are at a distance of $\sqrt{11}$ units from origin.

TYPE 21 Points on a Line at a Given Distance from a Point



Points on line at distance $\sqrt{11}$ from origin O

Given: Line $\frac{x}{1} = \frac{y-1}{2} = \frac{z+1}{2}$, origin $O(0, 0, 0)$, distance $OP = \sqrt{11}$.

Concept / Formula Used: $OP^2 = x^2 + y^2 + z^2$

Solution:

Set line equation equal to parameter λ :

$$\frac{x}{1} = \frac{y-1}{2} = \frac{z+1}{2} = \lambda$$

General point P :

$$P(\lambda, 2\lambda + 1, 2\lambda - 1)$$

Given $OP^2 = (\sqrt{11})^2 = 11$:

$$\lambda^2 + (2\lambda + 1)^2 + (2\lambda - 1)^2 = 11$$

Expand the squared terms:

$$\begin{aligned}\lambda^2 + (4\lambda^2 + 4\lambda + 1) + (4\lambda^2 - 4\lambda + 1) &= 11 \\ 9\lambda^2 + 2 &= 11 \\ 9\lambda^2 &= 9 \\ \lambda^2 = 1 &\implies \lambda = \pm 1\end{aligned}$$

Case 1: When $\lambda = 1$:

$$x = 1, \quad y = 2(1) + 1 = 3, \quad z = 2(1) - 1 = 1$$

Point is $(1, 3, 1)$.

Case 2: When $\lambda = -1$:

$$x = -1, \quad y = 2(-1) + 1 = -1, \quad z = 2(-1) - 1 = -3$$

Point is $(-1, -1, -3)$.

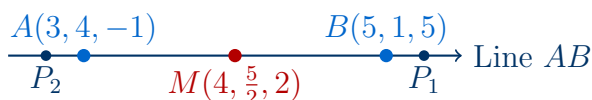
Final Answer

$(1, 3, 1)$ and $(-1, -1, -3)$

Question 18

Find the points on the line through the points $A(3, 4, -1)$ and $B(5, 1, 5)$ at a distance of 7 units from the mid-point of line segment AB .

TYPE 21 Points on a Line at a Given Distance from a Point



Points on line AB at distance 7 from midpoint M

Given: Points $A(3, 4, -1)$ and $B(5, 1, 5)$, distance from midpoint M is $d = 7$.

Concept / Formula Used: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$ $\vec{r} = \vec{r}_M \pm d\hat{u}$

Solution:

Calculate coordinates of midpoint M of AB :

$$M = \left(\frac{3+5}{2}, \frac{4+1}{2}, \frac{-1+5}{2} \right) = \left(4, \frac{5}{2}, 2 \right)$$

Position vector of M :

$$\vec{r}_M = 4\hat{i} + \frac{5}{2}\hat{j} + 2\hat{k}$$

Direction vector of line AB :

$$\begin{aligned} \vec{d} &= (5-3)\hat{i} + (1-4)\hat{j} + (5-(-1))\hat{k} \\ &= 2\hat{i} - 3\hat{j} + 6\hat{k} \end{aligned}$$

Magnitude of $\vec{d}$:

$$|\vec{d}| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$$

Unit direction vector along line AB :

$$\hat{u} = \frac{2}{7}\hat{i} - \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k}$$

Points at distance 7 units from midpoint M :

$$\begin{aligned} \vec{r} &= \vec{r}_M \pm 7\hat{u} \\ &= \left(4\hat{i} + \frac{5}{2}\hat{j} + 2\hat{k} \right) \pm 7 \left(\frac{2}{7}\hat{i} - \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k} \right) \\ &= \left(4\hat{i} + \frac{5}{2}\hat{j} + 2\hat{k} \right) \pm (2\hat{i} - 3\hat{j} + 6\hat{k}) \end{aligned}$$

First point (using + sign):

$$\begin{aligned} x &= 4 + 2 = 6 \\ y &= \frac{5}{2} - 3 = -\frac{1}{2} \\ z &= 2 + 6 = 8 \end{aligned}$$

Second point (using - sign):

$$\begin{aligned} x &= 4 - 2 = 2 \\ y &= \frac{5}{2} - (-3) = \frac{5}{2} + 3 = \frac{11}{2} \\ z &= 2 - 6 = -4 \end{aligned}$$

Final Answer

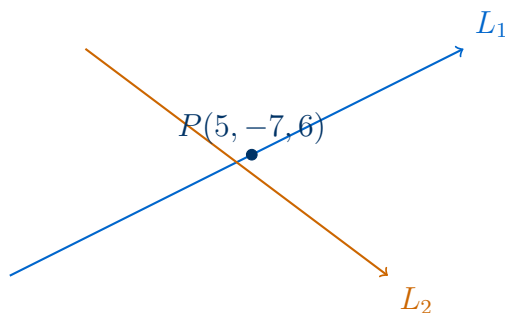
$$\left(6, -\frac{1}{2}, 8 \right) \text{ and } \left(2, \frac{11}{2}, -4 \right)$$

Question 19 (i)

Show that the lines $\frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7}$ and $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ intersect. Also find the coordinates of their point of intersection.

TYPE 22 Two Lines – Test Intersection & Find Point

★ Likely Exam Question



Intersection of two 3-D lines at point $P(5, -7, 6)$

To Prove: The lines $L_1 : \frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7}$ and $L_2 : \frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ intersect.

To Find: The coordinates of the point of intersection of L_1 and L_2 .

Concept / Formula Used: $\lambda \quad \mu \quad x \quad y \quad z$

Solution:

Let the given lines be:

$$L_1 : \frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7} = \lambda$$

$$L_2 : \frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8} = \mu$$

Expressing any general point P_1 on L_1 in terms of λ :

$$x = \lambda + 4$$

$$y = -4\lambda - 3$$

$$z = 7\lambda - 1$$

Expressing any general point P_2 on L_2 in terms of μ :

$$x = 2\mu + 1$$

$$y = -3\mu - 1$$

$$z = 8\mu - 10$$

If the lines intersect, P_1 and P_2 must coincide for some parameters λ and μ . Equating the x , y , and z coordinates:

$$\lambda + 4 = 2\mu + 1 \implies \lambda - 2\mu = -3 \quad \text{--- (1)}$$

$$-4\lambda - 3 = -3\mu - 1 \implies 4\lambda - 3\mu = -2 \quad \text{--- (2)}$$

$$7\lambda - 1 = 8\mu - 10 \quad \text{--- (3)}$$

From equation (1), expressing λ in terms of μ :

$$\lambda = 2\mu - 3$$

Substituting $\lambda = 2\mu - 3$ into equation (2):

$$4(2\mu - 3) - 3\mu = -2$$

$$8\mu - 12 - 3\mu = -2$$

$$5\mu - 12 = -2$$

$$5\mu = 10$$

$$\mu = 2$$

Substituting $\mu = 2$ back to find λ :

$$\lambda = 2(2) - 3$$

$$\lambda = 1$$

Now, substituting $\lambda = 1$ and $\mu = 2$ into the left-hand side and right-hand side of equation (3):

$$\text{LHS} = 7(1) - 1 = 6$$

$$\text{RHS} = 8(2) - 10 = 16 - 10 = 6$$

Since $\text{LHS} = \text{RHS} = 6$, equation (3) is satisfied by $\lambda = 1$ and $\mu = 2$. Therefore, the lines L_1 and L_2 intersect.

Substituting $\lambda = 1$ into the coordinates of P_1 :

$$x = 1 + 4 = 5$$

$$y = -4(1) - 3 = -7$$

$$z = 7(1) - 1 = 6$$

Thus, the coordinates of the point of intersection are $(5, -7, 6)$.

Hence Proved

LHS = RHS for z -coordinates; hence the given lines intersect.

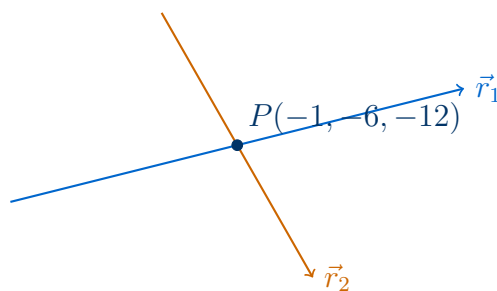
Final Answer

The coordinates of the point of intersection are $(5, -7, 6)$.

Common Mistake: Forgetting to verify the third coordinate equation after finding λ and μ will result in a deduction; proving intersection requires satisfying all three equations.

Question 19 (ii)

Show that the lines $\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$ and $\vec{r} = 5\hat{i} - 2\hat{j} + \mu(3\hat{i} + 2\hat{j} + 6\hat{k})$ are intersecting. Hence, find their point of intersection.

TYPE 22 Two Lines – Test Intersection & Find Point

Point of intersection of lines given in vector form

To Prove: The lines $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$ and $\vec{r}_2 = 5\hat{i} - 2\hat{j} + \mu(3\hat{i} + 2\hat{j} + 6\hat{k})$ intersect.

To Find: The point of intersection of the two lines.

Concept / Formula Used: $\vec{r}_1 = \vec{a}_1 + \lambda\vec{b}_1$ $\vec{r}_2 = \vec{a}_2 + \mu\vec{b}_2$ λ μ $\vec{r}_1 = \vec{r}_2$

Solution:

Writing position vectors of general points on the two lines:

$$\vec{r}_1 = (3 + \lambda)\hat{i} + (2 + 2\lambda)\hat{j} + (-4 + 2\lambda)\hat{k}$$

$$\vec{r}_2 = (5 + 3\mu)\hat{i} + (-2 + 2\mu)\hat{j} + (6\mu)\hat{k}$$

Equating the coefficients of $\hat{i}$, $\hat{j}$, and $\hat{k}$:

$$3 + \lambda = 5 + 3\mu \implies \lambda - 3\mu = 2 \quad \text{--- (1)}$$

$$2 + 2\lambda = -2 + 2\mu \implies 2\lambda - 2\mu = -4 \implies \lambda - \mu = -2 \quad \text{--- (2)}$$

$$-4 + 2\lambda = 6\mu \implies 2\lambda - 6\mu = 4 \implies \lambda - 3\mu = 2 \quad \text{--- (3)}$$

Subtracting equation (1) from equation (2):

$$(\lambda - \mu) - (\lambda - 3\mu) = -2 - 2$$

$$2\mu = -4$$

$$\mu = -2$$

Substituting $\mu = -2$ into equation (2):

$$\lambda - (-2) = -2$$

$$\lambda + 2 = -2$$

$$\lambda = -4$$

Now checking the $\hat{k}$ -component equation (equation 3) with $\lambda = -4$ and $\mu = -2$:

$$\text{LHS} = -4 + 2(-4) = -4 - 8 = -12$$

$$\text{RHS} = 6(-2) = -12$$

Since $\text{LHS} = \text{RHS} = -12$, the values $\lambda = -4$ and $\mu = -2$ satisfy all three component equations. Thus, the lines intersect.

Substituting $\lambda = -4$ into the expression for $\vec{r}_1$:

$$\begin{aligned}\vec{r} &= (3 - 4)\hat{i} + (2 + 2(-4))\hat{j} + (-4 + 2(-4))\hat{k} \\ &= -\hat{i} + (2 - 8)\hat{j} + (-4 - 8)\hat{k} \\ &= -\hat{i} - 6\hat{j} - 12\hat{k}\end{aligned}$$

Therefore, the point of intersection in coordinate form is $(-1, -6, -12)$.

Hence Proved

LHS = RHS for the $\hat{k}$ component; hence the lines intersect.

Final Answer

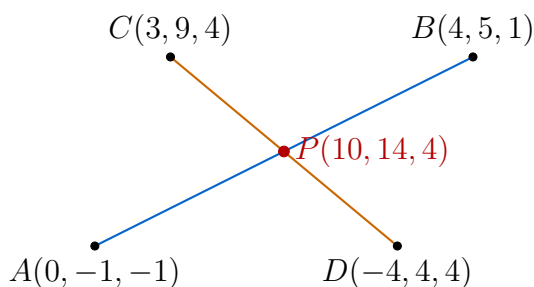
The point of intersection is $(-1, -6, -12)$ or vector position $-\hat{i} - 6\hat{j} - 12\hat{k}$.

Question 20 (i)

Prove that the line through $A(0, -1, -1)$ and $B(4, 5, 1)$ intersects the line through $C(3, 9, 4)$ and $D(-4, 4, 4)$. Also find their point of intersection.

TYPE 22 Two Lines – Test Intersection & Find Point

★ Likely Exam Question



Intersection of line AB and line CD

Given: Points $A(0, -1, -1)$, $B(4, 5, 1)$, $C(3, 9, 4)$, and $D(-4, 4, 4)$.

To Prove: The line joining A and B intersects the line joining C and D .

To Find: The coordinates of the point of intersection.

Concept / Formula Used: (x_1, y_1, z_1) (x_2, y_2, z_2) $\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$

Solution:

Direction ratios of line AB :

$$\langle 4 - 0, 5 - (-1), 1 - (-1) \rangle = \langle 4, 6, 2 \rangle \equiv \langle 2, 3, 1 \rangle$$

Equation of line AB passing through $A(0, -1, -1)$:

$$\frac{x - 0}{2} = \frac{y + 1}{3} = \frac{z + 1}{1} = \lambda$$

Any general point P on line AB is:

$$(2\lambda, 3\lambda - 1, \lambda - 1)$$

Direction ratios of line CD :

$$\langle -4 - 3, 4 - 9, 4 - 4 \rangle = \langle -7, -5, 0 \rangle$$

Equation of line CD passing through $C(3, 9, 4)$:

$$\frac{x - 3}{-7} = \frac{y - 9}{-5} = \frac{z - 4}{0} = \mu$$

Any general point Q on line CD is:

$$(-7\mu + 3, -5\mu + 9, 4)$$

Equating the coordinates of P and Q :

$$2\lambda = -7\mu + 3 \quad \text{--- (1)}$$

$$3\lambda - 1 = -5\mu + 9 \implies 3\lambda + 5\mu = 10 \quad \text{--- (2)}$$

$$\lambda - 1 = 4 \quad \text{--- (3)}$$

From equation (3):

$$\lambda - 1 = 4$$

$$\lambda = 5$$

Substituting $\lambda = 5$ into equation (1):

$$2(5) = -7\mu + 3$$

$$10 = -7\mu + 3$$

$$7\mu = 3 - 10$$

$$7\mu = -7$$

$$\mu = -1$$

Now checking equation (2) with $\lambda = 5$ and $\mu = -1$:

$$\text{LHS} = 3(5) + 5(-1) = 15 - 5 = 10$$

$$\text{RHS} = 10$$

Since $\text{LHS} = \text{RHS} = 10$, the values $\lambda = 5$ and $\mu = -1$ satisfy all three coordinate equations. Therefore, line AB and line CD intersect.

Substituting $\lambda = 5$ into the coordinates of point P :

$$x = 2(5) = 10$$

$$y = 3(5) - 1 = 14$$

$$z = 5 - 1 = 4$$

Thus, the point of intersection is $(10, 14, 4)$.

Hence Proved

LHS = RHS for the y -coordinate equation; hence the lines intersect.

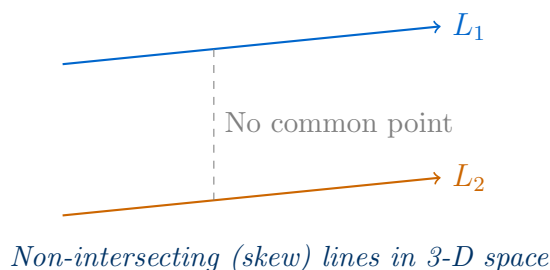
Final Answer

The coordinates of the point of intersection are $(10, 14, 4)$.

Question 20 (ii)

Show that the lines $\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-1}{5}$ and $\frac{x-2}{4} = \frac{y-1}{3} = \frac{z+1}{-2}$ do not intersect.

TYPE 22a Two Lines – Prove Non-Intersection (Skew)



To Prove: The lines $L_1 : \frac{x-1}{3} = \frac{y+1}{2} = \frac{z-1}{5}$ and $L_2 : \frac{x-2}{4} = \frac{y-1}{3} = \frac{z+1}{-2}$ do not intersect.

Concept / Formula Used: (λ, μ)

Solution:

Let the given lines be:

$$L_1 : \frac{x-1}{3} = \frac{y+1}{2} = \frac{z-1}{5} = \lambda$$

$$L_2 : \frac{x-2}{4} = \frac{y-1}{3} = \frac{z+1}{-2} = \mu$$

Any general point P_1 on L_1 is $(3\lambda + 1, 2\lambda - 1, 5\lambda + 1)$. Any general point P_2 on L_2 is $(4\mu + 2, 3\mu + 1, -2\mu - 1)$.

Equating the x , y , and z coordinates for intersection:

$$3\lambda + 1 = 4\mu + 2 \implies 3\lambda - 4\mu = 1 \quad \text{--- (1)}$$

$$2\lambda - 1 = 3\mu + 1 \implies 2\lambda - 3\mu = 2 \quad \text{--- (2)}$$

$$5\lambda + 1 = -2\mu - 1 \implies 5\lambda + 2\mu = -2 \quad \text{--- (3)}$$

Multiplying equation (1) by 2 and equation (2) by 3 to eliminate λ :

$$6\lambda - 8\mu = 2 \quad \text{--- (4)}$$

$$6\lambda - 9\mu = 6 \quad \text{--- (5)}$$

Subtracting equation (5) from equation (4):

$$(6\lambda - 8\mu) - (6\lambda - 9\mu) = 2 - 6$$

$$\mu = -4$$

Substituting $\mu = -4$ into equation (2):

$$2\lambda - 3(-4) = 2$$

$$2\lambda + 12 = 2$$

$$2\lambda = -10$$

$$\lambda = -5$$

Now checking equation (3) with $\lambda = -5$ and $\mu = -4$:

$$\text{LHS} = 5(-5) + 2(-4) = -25 - 8 = -33$$

$$\text{RHS} = -2$$

Since $\text{LHS} \neq \text{RHS}$ ($-33 \neq -2$), equation (3) is not satisfied by $\lambda = -5$ and $\mu = -4$. Therefore, there exists no set of values for λ and μ satisfying all three coordinate equations simultaneously.

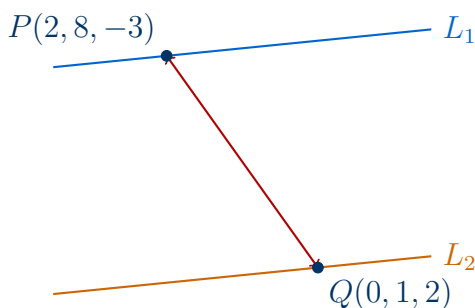
Hence Proved

Since the system of equations is inconsistent ($\text{LHS} \neq \text{RHS}$), the given lines do not intersect.

Question 21

A line with direction numbers $\langle 2, 7, -5 \rangle$ is drawn to intersect the lines $\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1}$ and $\frac{x+3}{-3} = \frac{y-3}{2} = \frac{z-6}{4}$. Find the coordinates of the points of intersection and the length intercepted on it.

TYPE 23 Construct a Line Intersecting Two Given Lines

★ Likely Exam Question

Line segment PQ with given direction numbers intersecting L_1 and L_2

Given: Intersecting line has direction numbers $\langle 2, 7, -5 \rangle$. Line $L_1 : \frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1}$ and Line $L_2 : \frac{x+3}{-3} = \frac{y-3}{2} = \frac{z-6}{4}$.

To Find: Coordinates of points of intersection P and Q , and the distance PQ .

Concept / Formula Used: $P(x_1, y_1, z_1)$ $Q(x_2, y_2, z_2)$ $\langle a, b, c \rangle$ $\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{z_2 - z_1}{c}$ $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

Solution:

Let P be a general point on line L_1 :

$$\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1} = \lambda \implies P = (3\lambda + 5, -\lambda + 7, \lambda + 2)$$

Let Q be a general point on line L_2 :

$$\frac{x+3}{-3} = \frac{y-3}{2} = \frac{z-6}{4} = \mu \implies Q = (-3\mu - 3, 2\mu + 3, 4\mu + 6)$$

Direction ratios of line PQ :

$$\begin{aligned}\text{DRs of } PQ &= \langle (-3\mu - 3) - (3\lambda + 5), (2\mu + 3) - (-\lambda + 7), (4\mu + 6) - (\lambda - 2) \rangle \\ &= \langle -3\mu - 3\lambda - 8, 2\mu + \lambda - 4, 4\mu - \lambda + 8 \rangle\end{aligned}$$

Since the line PQ has direction numbers proportional to $\langle 2, 7, -5 \rangle$:

$$\frac{-3\mu - 3\lambda - 8}{2} = \frac{2\mu + \lambda - 4}{7} = \frac{4\mu - \lambda + 8}{-5}$$

Equating the first two ratios:

$$\begin{aligned}7(-3\mu - 3\lambda - 8) &= 2(2\mu + \lambda - 4) \\ -21\mu - 21\lambda - 56 &= 4\mu + 2\lambda - 8 \\ -23\lambda - 25\mu &= 48 \\ 23\lambda + 25\mu &= -48 \quad \text{--- (1)}\end{aligned}$$

Equating the second and third ratios:

$$\begin{aligned}-5(2\mu + \lambda - 4) &= 7(4\mu - \lambda + 8) \\ -10\mu - 5\lambda + 20 &= 28\mu - 7\lambda + 56 \\ 2\lambda - 38\mu &= 36 \\ \lambda - 19\mu &= 18 \implies \lambda = 19\mu + 18 \quad \text{--- (2)}\end{aligned}$$

Substituting $\lambda = 19\mu + 18$ into equation (1):

$$\begin{aligned}23(19\mu + 18) + 25\mu &= -48 \\ 437\mu + 414 + 25\mu &= -48 \\ 462\mu &= -48 - 414 \\ 462\mu &= -462 \\ \mu &= -1\end{aligned}$$

Substituting $\mu = -1$ into equation (2):

$$\begin{aligned}\lambda &= 19(-1) + 18 \\ \lambda &= -1\end{aligned}$$

Finding the coordinates of point P by substituting $\lambda = -1$:

$$\begin{aligned}x_P &= 3(-1) + 5 = 2 \\ y_P &= -(-1) + 7 = 8 \\ z_P &= -1 - 2 = -3\end{aligned}$$

Thus, $P = (2, 8, -3)$.

Finding the coordinates of point Q by substituting $\mu = -1$:

$$x_Q = -3(-1) - 3 = 0$$

$$y_Q = 2(-1) + 3 = 1$$

$$z_Q = 4(-1) + 6 = 2$$

Thus, $Q = (0, 1, 2)$.

Calculating the length of intercept PQ :

$$\begin{aligned} PQ &= \sqrt{(0 - 2)^2 + (1 - 8)^2 + (2 - (-3))^2} \\ &= \sqrt{(-2)^2 + (-7)^2 + (5)^2} \\ &= \sqrt{4 + 49 + 25} \\ &= \sqrt{78} \text{ units} \end{aligned}$$

Final Answer

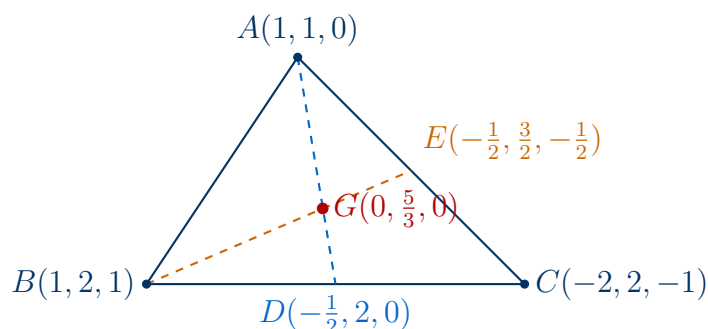
The points of intersection are $(2, 8, -3)$ and $(0, 1, 2)$.

The length intercepted on the line is $\sqrt{78}$ units.

Question 22

The vertices of $\triangle ABC$ are $A(1, 1, 0)$, $B(1, 2, 1)$ and $C(-2, 2, -1)$. Find the equation of the medians through A and B . Use the equations so obtained to find the coordinates of the centroid.

TYPE 24 Equations of Medians & Centroid of a Triangle



Triangle ABC with medians AD and BE intersecting at centroid G

Given: Vertices of $\triangle ABC$: $A(1, 1, 0)$, $B(1, 2, 1)$, and $C(-2, 2, -1)$.

To Find: Equations of medians AD and BE , and the coordinates of centroid G .

Concept / Formula Used: $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$ (x_1, y_1, z_1) $\langle a, b, c \rangle$

$$\frac{y - y_1}{b} = \frac{z - z_1}{c}$$

$$\frac{x - x_1}{a} =$$

Solution:

Let D be the midpoint of side BC and E be the midpoint of side AC .

Coordinates of midpoint D of BC :

$$\begin{aligned} D &= \left(\frac{1 + (-2)}{2}, \frac{2 + 2}{2}, \frac{1 + (-1)}{2} \right) \\ &= \left(-\frac{1}{2}, 2, 0 \right) \end{aligned}$$

Coordinates of midpoint E of AC :

$$\begin{aligned} E &= \left(\frac{1 + (-2)}{2}, \frac{1 + 2}{2}, \frac{0 + (-1)}{2} \right) \\ &= \left(-\frac{1}{2}, \frac{3}{2}, -\frac{1}{2} \right) \end{aligned}$$

1. Equation of median AD (through $A(1, 1, 0)$ and $D(-\frac{1}{2}, 2, 0)$):

$$\begin{aligned} \text{Direction ratios of } AD &= \left\langle -\frac{1}{2} - 1, 2 - 1, 0 - 0 \right\rangle \\ &= \left\langle -\frac{3}{2}, 1, 0 \right\rangle \equiv \langle -3, 2, 0 \rangle \end{aligned}$$

Equation of line AD :

$$\frac{x - 1}{-3} = \frac{y - 1}{2} = \frac{z - 0}{0} = \lambda$$

2. Equation of median BE (through $B(1, 2, 1)$ and $E(-\frac{1}{2}, \frac{3}{2}, -\frac{1}{2})$):

$$\begin{aligned} \text{Direction ratios of } BE &= \left\langle -\frac{1}{2} - 1, \frac{3}{2} - 2, -\frac{1}{2} - 1 \right\rangle \\ &= \left\langle -\frac{3}{2}, -\frac{1}{2}, -\frac{3}{2} \right\rangle \equiv \langle 3, 1, 3 \rangle \end{aligned}$$

Equation of line BE :

$$\frac{x - 1}{3} = \frac{y - 2}{1} = \frac{z - 1}{3} = \mu$$

3. Finding coordinates of centroid G from intersection of AD and BE : General point on AD : $(-3\lambda + 1, 2\lambda + 1, 0)$. General point on BE : $(3\mu + 1, \mu + 2, 3\mu + 1)$.

Equating the z -coordinates of both points:

$$\begin{aligned} 0 &= 3\mu + 1 \\ 3\mu &= -1 \\ \mu &= -\frac{1}{3} \end{aligned}$$

Substituting $\mu = -\frac{1}{3}$ into the general point on BE :

$$\begin{aligned}x &= 3 \left(-\frac{1}{3} \right) + 1 = -1 + 1 = 0 \\y &= -\frac{1}{3} + 2 = \frac{5}{3} \\z &= 3 \left(-\frac{1}{3} \right) + 1 = -1 + 1 = 0\end{aligned}$$

Verifying using line AD : Equating the x -coordinate to find λ :

$$\begin{aligned}-3\lambda + 1 &= 0 \implies \lambda = \frac{1}{3} \\y &= 2 \left(\frac{1}{3} \right) + 1 = \frac{5}{3} \\z &= 0\end{aligned}$$

Thus, the centroid G has coordinates $\left(0, \frac{5}{3}, 0 \right)$.

Final Answer

Equation of median AD : $\frac{x-1}{-3} = \frac{y-1}{2} = \frac{z}{0}$

Equation of median BE : $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-1}{3}$

Coordinates of centroid: $\left(0, \frac{5}{3}, 0 \right)$

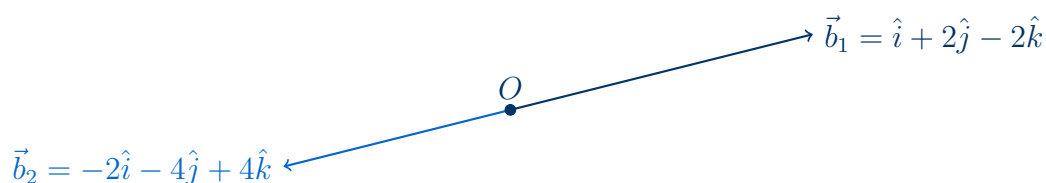
11.3 – Angle Between Two Lines

Question 1 (i)

Find the angle between the following pair of lines:

$$\vec{r} = 4\hat{i} - \hat{j} + \lambda(\hat{i} + 2\hat{j} - 2\hat{k}) \quad \text{and} \quad \vec{r} = \hat{i} - \hat{j} + 2\hat{k} - \mu(2\hat{i} + 4\hat{j} - 4\hat{k})$$

TYPE 25 Angle Between Two Lines – Vector Form



Angle between vector lines

Given: Line 1 direction vector $\vec{b}_1 = \hat{i} + 2\hat{j} - 2\hat{k}$ and Line 2 direction vector $\vec{b}_2 = -2\hat{i} - 4\hat{j} + 4\hat{k}$.

Concept / Formula Used: $\theta \quad \cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|}$

Solution:

Extract the direction vectors $\vec{b}_1$ and $\vec{b}_2$ from the given vector equations of the lines:

$$\vec{b}_1 = \hat{i} + 2\hat{j} - 2\hat{k}$$

$$\vec{b}_2 = -(2\hat{i} + 4\hat{j} - 4\hat{k}) = -2\hat{i} - 4\hat{j} + 4\hat{k}$$

Calculate the dot product $\vec{b}_1 \cdot \vec{b}_2$:

$$\begin{aligned}\vec{b}_1 \cdot \vec{b}_2 &= (1)(-2) + (2)(-4) + (-2)(4) \\ &= -2 - 8 - 8 \\ &= -18\end{aligned}$$

Calculate the magnitude of $\vec{b}_1$:

$$\begin{aligned}|\vec{b}_1| &= \sqrt{1^2 + 2^2 + (-2)^2} \\ &= \sqrt{1 + 4 + 4} \\ &= \sqrt{9} = 3\end{aligned}$$

Calculate the magnitude of $\vec{b}_2$:

$$\begin{aligned}|\vec{b}_2| &= \sqrt{(-2)^2 + (-4)^2 + 4^2} \\ &= \sqrt{4 + 16 + 16} \\ &= \sqrt{36} = 6\end{aligned}$$

Apply the angle formula:

$$\begin{aligned}\cos \theta &= \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|} \\ &= \frac{|-18|}{3 \times 6} \\ &= \frac{18}{18} = 1\end{aligned}$$

Since $\cos \theta = 1$, we get:

$$\theta = 0^\circ$$

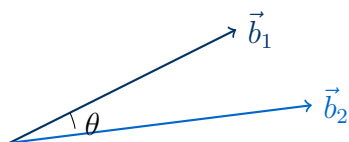
Final Answer

The angle between the pair of lines is $\theta = 0^\circ$ (the lines are parallel).

Question 1 (ii)

Find the angle between the following pair of lines:

$$\vec{r} = 3\hat{i} - 2\hat{j} + 6\hat{k} + \lambda(2\hat{i} + \hat{j} + 2\hat{k}) \quad \text{and} \quad \vec{r} = 2\hat{j} - 5\hat{k} + \mu(6\hat{i} + 3\hat{j} + 2\hat{k})$$

TYPE 25 Angle Between Two Lines – Vector Form

Angle between two 3D vector lines

Given: Line 1 direction vector $\vec{b}_1 = 2\hat{i} + \hat{j} + 2\hat{k}$ and Line 2 direction vector $\vec{b}_2 = 6\hat{i} + 3\hat{j} + 2\hat{k}$.

Concept / Formula Used: $\theta \quad \vec{b}_1 \quad \vec{b}_2 \quad \cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|}$

Solution:

Identify the direction vectors from the equations of the lines:

$$\vec{b}_1 = 2\hat{i} + \hat{j} + 2\hat{k}$$

$$\vec{b}_2 = 6\hat{i} + 3\hat{j} + 2\hat{k}$$

Compute the dot product $\vec{b}_1 \cdot \vec{b}_2$:

$$\begin{aligned} \vec{b}_1 \cdot \vec{b}_2 &= (2)(6) + (1)(3) + (2)(2) \\ &= 12 + 3 + 4 \\ &= 19 \end{aligned}$$

Compute the magnitudes $|\vec{b}_1|$ and $|\vec{b}_2|$:

$$|\vec{b}_1| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

$$|\vec{b}_2| = \sqrt{6^2 + 3^2 + 2^2} = \sqrt{36 + 9 + 4} = \sqrt{49} = 7$$

Substitute these values into the angle formula:

$$\cos \theta = \frac{19}{3 \times 7}$$

$$\cos \theta = \frac{19}{21}$$

Taking the inverse cosine:

$$\theta = \cos^{-1} \left(\frac{19}{21} \right)$$

Final Answer

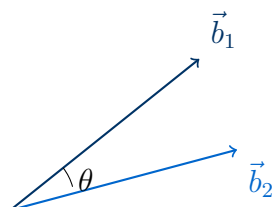
$$\theta = \cos^{-1} \left(\frac{19}{21} \right)$$

Question 1 (iii)

Find the angle between the following pair of lines:

$$\vec{r} = 2\hat{i} - 5\hat{j} + \hat{k} + \lambda(3\hat{i} + 2\hat{j} + 6\hat{k}) \quad \text{and} \quad \vec{r} = 7\hat{i} - 6\hat{k} + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$$

TYPE 25 Angle Between Two Lines – Vector Form



Angle between two lines in space

Given: Direction vector $\vec{b}_1 = 3\hat{i} + 2\hat{j} + 6\hat{k}$ and direction vector $\vec{b}_2 = \hat{i} + 2\hat{j} + 2\hat{k}$.

Concept / Formula Used: $\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|}$

Solution:

Extract direction vectors:

$$\vec{b}_1 = 3\hat{i} + 2\hat{j} + 6\hat{k}$$

$$\vec{b}_2 = \hat{i} + 2\hat{j} + 2\hat{k}$$

Calculate dot product:

$$\begin{aligned} \vec{b}_1 \cdot \vec{b}_2 &= (3)(1) + (2)(2) + (6)(2) \\ &= 3 + 4 + 12 \\ &= 19 \end{aligned}$$

Calculate magnitudes:

$$\begin{aligned} |\vec{b}_1| &= \sqrt{3^2 + 2^2 + 6^2} = \sqrt{9 + 4 + 36} = \sqrt{49} = 7 \\ |\vec{b}_2| &= \sqrt{1^2 + 2^2 + 2^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3 \end{aligned}$$

Calculate $\cos \theta$:

$$\begin{aligned} \cos \theta &= \frac{19}{7 \times 3} \\ \cos \theta &= \frac{19}{21} \end{aligned}$$

Thus, we obtain:

$$\theta = \cos^{-1} \left(\frac{19}{21} \right)$$

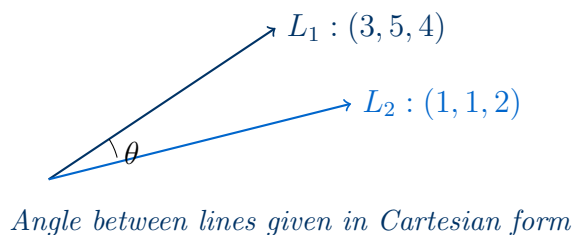
Final Answer

$$\theta = \cos^{-1} \left(\frac{19}{21} \right)$$

Question 2 (i)

Find the angle between the following pair of lines:

$$\frac{x+3}{3} = \frac{y-1}{5} = \frac{z+3}{4} \quad \text{and} \quad \frac{x+1}{1} = \frac{y-4}{1} = \frac{z-5}{2}$$

TYPE 26 Angle Between Two Lines – Cartesian Form

Given: Direction ratios of Line 1: $(a_1, b_1, c_1) = (3, 5, 4)$ and Line 2: $(a_2, b_2, c_2) = (1, 1, 2)$.

Concept / Formula Used: $\cos \theta = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$

Solution:

Identify the direction ratios of the two given lines:

$$\begin{aligned} a_1 &= 3, & b_1 &= 5, & c_1 &= 4 \\ a_2 &= 1, & b_2 &= 1, & c_2 &= 2 \end{aligned}$$

Calculate the numerator $a_1 a_2 + b_1 b_2 + c_1 c_2$:

$$\begin{aligned} a_1 a_2 + b_1 b_2 + c_1 c_2 &= (3)(1) + (5)(1) + (4)(2) \\ &= 3 + 5 + 8 \\ &= 16 \end{aligned}$$

Calculate the terms in the denominator:

$$\begin{aligned} \sqrt{a_1^2 + b_1^2 + c_1^2} &= \sqrt{3^2 + 5^2 + 4^2} = \sqrt{9 + 25 + 16} = \sqrt{50} = 5\sqrt{2} \\ \sqrt{a_2^2 + b_2^2 + c_2^2} &= \sqrt{1^2 + 1^2 + 2^2} = \sqrt{1 + 1 + 4} = \sqrt{6} \end{aligned}$$

Substitute into the formula:

$$\begin{aligned}\cos \theta &= \frac{16}{5\sqrt{2} \times \sqrt{6}} \\ &= \frac{16}{5\sqrt{12}} \\ &= \frac{16}{5 \times 2\sqrt{3}} \\ &= \frac{8}{5\sqrt{3}} \\ &= \frac{8\sqrt{3}}{15}\end{aligned}$$

Thus:

$$\theta = \cos^{-1} \left(\frac{8\sqrt{3}}{15} \right)$$

Final Answer

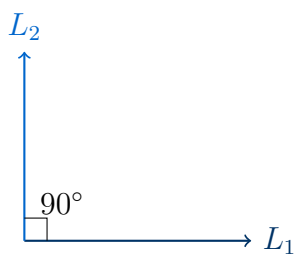
$$\theta = \cos^{-1} \left(\frac{8\sqrt{3}}{15} \right)$$

Question 2 (ii)

Find the angle between the following pair of lines:

$$\frac{x+1}{2} = \frac{y-2}{5} = \frac{z+3}{4} \quad \text{and} \quad \frac{x-1}{5} = \frac{y+2}{2} = \frac{z-1}{-5}$$

TYPE 27 Perpendicularity – Compute/Verify Angle is 90°



Perpendicular lines in 3D

Given: Direction ratios of Line 1: $(2, 5, 4)$ and Line 2: $(5, 2, -5)$.

Concept / Formula Used: $\cos \theta = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$

Solution:

Extract direction ratios of both lines:

$$\begin{aligned}(a_1, b_1, c_1) &= (2, 5, 4) \\ (a_2, b_2, c_2) &= (5, 2, -5)\end{aligned}$$

Calculate $a_1a_2 + b_1b_2 + c_1c_2$:

$$\begin{aligned}a_1a_2 + b_1b_2 + c_1c_2 &= (2)(5) + (5)(2) + (4)(-5) \\ &= 10 + 10 - 20 \\ &= 0\end{aligned}$$

Since the dot product of the direction vectors is zero:

$$\begin{aligned}\cos \theta &= 0 \\ \theta &= 90^\circ\end{aligned}$$

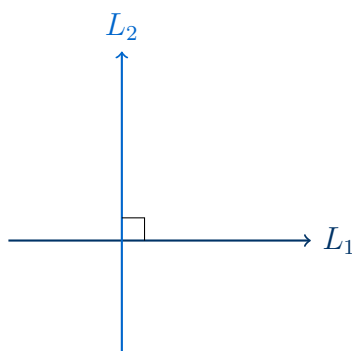
Final Answer

$$\theta = 90^\circ$$

Question 3 (i)

Show that the lines $\frac{x-5}{7} = \frac{2-y}{5} = \frac{z}{1}$ and $\frac{3-x}{-1} = \frac{y}{2} = \frac{z+1}{3}$ are at right angles.

TYPE 28 Perpendicularity Proof Between Two Lines



Right-angled lines verification

To Prove: The given lines are at right angles, i.e., $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

Concept / Formula Used: $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$ $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Solution:

First rewrite both lines in standard Cartesian form $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$:

For Line 1:

$$\begin{aligned}\frac{x-5}{7} = \frac{2-y}{5} = \frac{z}{1} &\implies \frac{x-5}{7} = \frac{-(y-2)}{5} = \frac{z-0}{1} \\ &\implies \frac{x-5}{7} = \frac{y-2}{-5} = \frac{z-0}{1}\end{aligned}$$

Direction ratios of Line 1 are $(a_1, b_1, c_1) = (7, -5, 1)$.

For Line 2:

$$\begin{aligned}\frac{3-x}{-1} = \frac{y}{2} = \frac{z+1}{3} &\implies \frac{-(x-3)}{-1} = \frac{y-0}{2} = \frac{z+1}{3} \\ &\implies \frac{x-3}{1} = \frac{y-0}{2} = \frac{z+1}{3}\end{aligned}$$

Direction ratios of Line 2 are $(a_2, b_2, c_2) = (1, 2, 3)$.

Now calculate the sum $a_1a_2 + b_1b_2 + c_1c_2$:

$$\begin{aligned}a_1a_2 + b_1b_2 + c_1c_2 &= (7)(1) + (-5)(2) + (1)(3) \\ &= 7 - 10 + 3 \\ &= 0\end{aligned}$$

Since $a_1a_2 + b_1b_2 + c_1c_2 = 0$, the direction vectors of the two lines are orthogonal.

Hence Proved

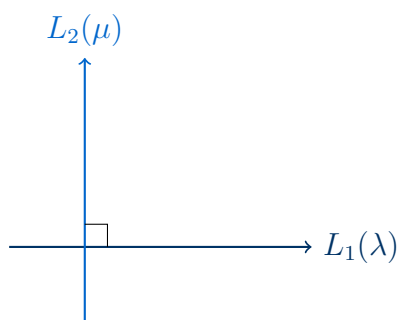
LHS = 0, hence the given lines are at right angles.

Common Mistake: Failing to write $\frac{2-y}{5}$ as $\frac{y-2}{-5}$ or $\frac{3-x}{-1}$ as $\frac{x-3}{1}$ flips the sign of direction ratios, leading to an incorrect dot product.

Question 3 (ii)

Check whether the lines given by the equations $x = 2\lambda + 2, y = 7\lambda + 1, z = -3\lambda - 3$ and $x = -\mu - 2, y = 2\mu + 8, z = 4\mu + 5$ are perpendicular to each other or not.

TYPE 28 Perpendicularity Proof Between Two Lines



Perpendicularity of parametric lines

Given: Line 1 in parametric form: $x = 2\lambda + 2, y = 7\lambda + 1, z = -3\lambda - 3$ and Line 2: $x = -\mu - 2, y = 2\mu + 8, z = 4\mu + 5$.

Concept / Formula Used: $x = a\lambda + x_0, y = b\lambda + y_0, z = c\lambda + z_0$ (a, b, c) $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Solution:

Extract direction ratios of Line 1 from coefficients of parameter λ :

$$a_1 = 2, \quad b_1 = 7, \quad c_1 = -3$$

Extract direction ratios of Line 2 from coefficients of parameter μ :

$$a_2 = -1, \quad b_2 = 2, \quad c_2 = 4$$

Calculate the dot product of direction ratios:

$$\begin{aligned} a_1 a_2 + b_1 b_2 + c_1 c_2 &= (2)(-1) + (7)(2) + (-3)(4) \\ &= -2 + 14 - 12 \\ &= 12 - 12 \\ &= 0 \end{aligned}$$

Since the condition $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$ is satisfied, the two lines are perpendicular to each other.

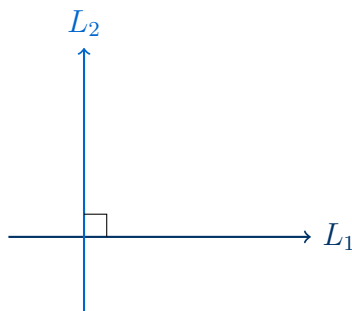
Final Answer

The given lines are perpendicular to each other.

Question 4 (i)

If the lines $\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-3}{2}$ and $\frac{x-1}{3k} = \frac{y-1}{1} = \frac{z-6}{-5}$ are perpendicular, then find the value of k .

TYPE 29 Solve for Unknown Parameter Using Perpendicularity



Perpendicular lines with parameter k

Given: Line 1 direction ratios $(-3, 2k, 2)$ and Line 2 direction ratios $(3k, 1, -5)$. Lines are perpendicular.

Concept / Formula Used: $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

Solution:

Identify direction ratios of both lines:

$$\begin{aligned} (a_1, b_1, c_1) &= (-3, 2k, 2) \\ (a_2, b_2, c_2) &= (3k, 1, -5) \end{aligned}$$

Apply the perpendicularity condition $a_1a_2 + b_1b_2 + c_1c_2 = 0$:

$$\begin{aligned} (-3)(3k) + (2k)(1) + (2)(-5) &= 0 \\ -9k + 2k - 10 &= 0 \\ -7k - 10 &= 0 \\ -7k &= 10 \\ k &= -\frac{10}{7} \end{aligned}$$

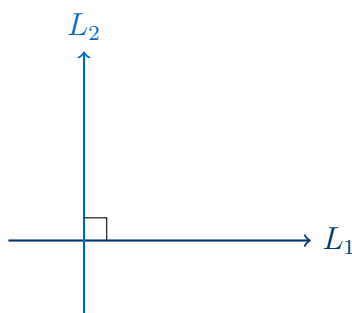
Final Answer

$$k = -\frac{10}{7}$$

Question 4 (ii)

Find the value of k so that the lines $x = -y = kz$ and $x - 2 = 2y + 1 = -z + 1$ are perpendicular to each other.

TYPE 29 Solve for Unknown Parameter Using Perpendicularity



Perpendicular lines in non-standard Cartesian form

Given: Line 1: $x = -y = kz$ and Line 2: $x - 2 = 2y + 1 = -z + 1$. Lines are perpendicular.

Concept / Formula Used: $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Solution:

Convert Line 1 to standard Cartesian form $\frac{x-0}{a_1} = \frac{y-0}{b_1} = \frac{z-0}{c_1}$:

$$x = -y = kz \implies \frac{x}{1} = \frac{y}{-1} = \frac{z}{1/k}$$

Direction ratios of Line 1 are $(a_1, b_1, c_1) = (1, -1, \frac{1}{k})$.

Convert Line 2 to standard Cartesian form:

$$\begin{aligned} x - 2 = 2y + 1 = -z + 1 &\implies \frac{x-2}{1} = \frac{2(y+\frac{1}{2})}{1} = \frac{-(z-1)}{1} \\ &\implies \frac{x-2}{1} = \frac{y+1/2}{1/2} = \frac{z-1}{-1} \end{aligned}$$

Direction ratios of Line 2 are $(a_2, b_2, c_2) = (1, \frac{1}{2}, -1)$.

Apply the perpendicularity condition $a_1a_2 + b_1b_2 + c_1c_2 = 0$:

$$\begin{aligned}(1)(1) + (-1)\left(\frac{1}{2}\right) + \left(\frac{1}{k}\right)(-1) &= 0 \\ 1 - \frac{1}{2} - \frac{1}{k} &= 0 \\ \frac{1}{2} - \frac{1}{k} &= 0 \\ \frac{1}{k} &= \frac{1}{2} \\ k &= 2\end{aligned}$$

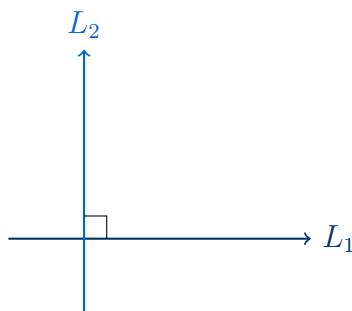
Final Answer

$$k = 2$$

Question 4 (iii)

Find the value of p so that the lines $\frac{x-1}{-2} = \frac{y-4}{3p} = \frac{z-3}{4}$ and $\frac{x-2}{4p} = \frac{y-5}{2} = \frac{1-z}{7}$ are perpendicular to each other.

TYPE 29 Solve for Unknown Parameter Using Perpendicularity



Perpendicular lines condition for parameter p

Given: Line 1: $\frac{x-1}{-2} = \frac{y-4}{3p} = \frac{z-3}{4}$ and Line 2: $\frac{x-2}{4p} = \frac{y-5}{2} = \frac{1-z}{7}$. Lines are perpendicular.

Concept / Formula Used: $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Solution:

Line 1 is in standard form:

$$\frac{x-1}{-2} = \frac{y-4}{3p} = \frac{z-3}{4} \implies (a_1, b_1, c_1) = (-2, 3p, 4)$$

Rewrite Line 2 in standard form:

$$\begin{aligned}\frac{x-2}{4p} = \frac{y-5}{2} = \frac{1-z}{7} &\implies \frac{x-2}{4p} = \frac{y-5}{2} = \frac{-(z-1)}{7} \\ &\implies \frac{x-2}{4p} = \frac{y-5}{2} = \frac{z-1}{-7}\end{aligned}$$

Direction ratios of Line 2 are $(a_2, b_2, c_2) = (4p, 2, -7)$.

Apply the perpendicularity condition:

$$\begin{aligned}a_1 a_2 + b_1 b_2 + c_1 c_2 &= 0 \\ (-2)(4p) + (3p)(2) + (4)(-7) &= 0 \\ -8p + 6p - 28 &= 0 \\ -2p - 28 &= 0 \\ -2p &= 28 \\ p &= -14\end{aligned}$$

Final Answer

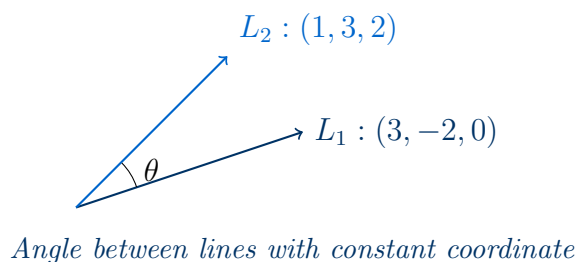
$$p = -14$$

Question 5 (i)

Find the angle between the following pair of lines:

$$\frac{x-2}{3} = \frac{y+1}{-2}, z=2 \quad \text{and} \quad \frac{x-1}{1} = \frac{y+3}{3} = \frac{z+5}{2}$$

TYPE 26 Angle Between Two Lines – Cartesian Form



Given: Line 1: $\frac{x-2}{3} = \frac{y+1}{-2}, z=2$ and Line 2: $\frac{x-1}{1} = \frac{y+3}{3} = \frac{z+5}{2}$.

Concept / Formula Used: $\cos \theta = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$

Solution:

Rewrite Line 1 in standard form:

$$\frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-2}{0}$$

Direction ratios of Line 1 are $(a_1, b_1, c_1) = (3, -2, 0)$.

Direction ratios of Line 2 are $(a_2, b_2, c_2) = (1, 3, 2)$.

Calculate numerator $|a_1a_2 + b_1b_2 + c_1c_2|$:

$$\begin{aligned}|a_1a_2 + b_1b_2 + c_1c_2| &= |(3)(1) + (-2)(3) + (0)(2)| \\ &= |3 - 6 + 0| \\ &= |-3| = 3\end{aligned}$$

Calculate denominators:

$$\begin{aligned}\sqrt{a_1^2 + b_1^2 + c_1^2} &= \sqrt{3^2 + (-2)^2 + 0^2} = \sqrt{9 + 4 + 0} = \sqrt{13} \\ \sqrt{a_2^2 + b_2^2 + c_2^2} &= \sqrt{1^2 + 3^2 + 2^2} = \sqrt{1 + 9 + 4} = \sqrt{14}\end{aligned}$$

Substitute into formula:

$$\begin{aligned}\cos \theta &= \frac{3}{\sqrt{13}\sqrt{14}} \\ &= \frac{3}{\sqrt{182}}\end{aligned}$$

Thus:

$$\theta = \cos^{-1} \left(\frac{3}{\sqrt{182}} \right)$$

Final Answer

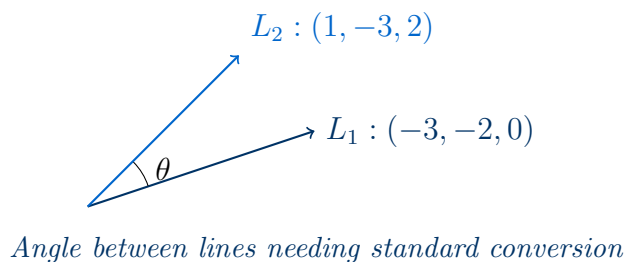
$$\theta = \cos^{-1} \left(\frac{3}{\sqrt{182}} \right)$$

Question 5 (ii)

Find the angle between the following pair of lines:

$$\frac{5-x}{3} = \frac{y+3}{-2}, z=5 \quad \text{and} \quad \frac{x-1}{1} = \frac{1-y}{3} = \frac{z-5}{2}$$

TYPE 26 Angle Between Two Lines – Cartesian Form



Given: Line 1: $\frac{5-x}{3} = \frac{y+3}{-2}, z=5$ and Line 2: $\frac{x-1}{1} = \frac{1-y}{3} = \frac{z-5}{2}$.

Concept / Formula Used: $\cos \theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$

Solution:

Convert Line 1 to standard form:

$$\begin{aligned} \frac{5-x}{3} = \frac{y+3}{-2}, z=5 &\implies \frac{-(x-5)}{3} = \frac{y+3}{-2} = \frac{z-5}{0} \\ &\implies \frac{x-5}{-3} = \frac{y+3}{-2} = \frac{z-5}{0} \end{aligned}$$

Direction ratios of Line 1 are $(a_1, b_1, c_1) = (-3, -2, 0)$.

Convert Line 2 to standard form:

$$\begin{aligned} \frac{x-1}{1} = \frac{1-y}{3} = \frac{z-5}{2} &\implies \frac{x-1}{1} = \frac{-(y-1)}{3} = \frac{z-5}{2} \\ &\implies \frac{x-1}{1} = \frac{y-1}{-3} = \frac{z-5}{2} \end{aligned}$$

Direction ratios of Line 2 are $(a_2, b_2, c_2) = (1, -3, 2)$.

Calculate numerator:

$$\begin{aligned} |a_1a_2 + b_1b_2 + c_1c_2| &= |(-3)(1) + (-2)(-3) + (0)(2)| \\ &= |-3 + 6 + 0| \\ &= |3| = 3 \end{aligned}$$

Calculate denominators:

$$\begin{aligned} \sqrt{a_1^2 + b_1^2 + c_1^2} &= \sqrt{(-3)^2 + (-2)^2 + 0^2} = \sqrt{9 + 4 + 0} = \sqrt{13} \\ \sqrt{a_2^2 + b_2^2 + c_2^2} &= \sqrt{1^2 + (-3)^2 + 2^2} = \sqrt{1 + 9 + 4} = \sqrt{14} \end{aligned}$$

Substitute into angle formula:

$$\begin{aligned} \cos \theta &= \frac{3}{\sqrt{13}\sqrt{14}} = \frac{3}{\sqrt{182}} \\ \theta &= \cos^{-1} \left(\frac{3}{\sqrt{182}} \right) \end{aligned}$$

Final Answer

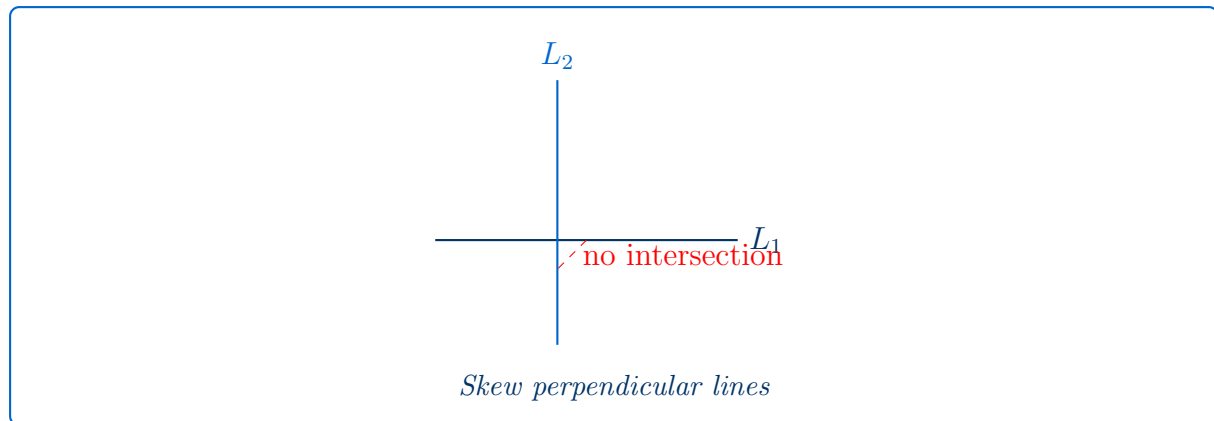
$$\theta = \cos^{-1} \left(\frac{3}{\sqrt{182}} \right)$$

Question 6 (i)

If the lines $\frac{x-1}{-3} = \frac{y-2}{2\lambda} = \frac{z-3}{2}$ and $\frac{x-1}{3\lambda} = \frac{y-1}{2} = \frac{z-6}{-5}$ are perpendicular, find the values of λ . Hence, find whether the lines are intersecting or not.

TYPE 28 Perpendicularity Proof Between Two Lines

★ Likely Exam Question



Given: Line 1: $\frac{x-1}{-3} = \frac{y-2}{2\lambda} = \frac{z-3}{2}$ and Line 2: $\frac{x-1}{3\lambda} = \frac{y-1}{2} = \frac{z-6}{-5}$. Lines are perpendicular.

Concept / Formula Used: $a_1a_2 + b_1b_2 + c_1c_2 = 0$ $P(s)$ $Q(t)$ s t

Solution:

Step 1: Find λ using perpendicularity condition. Direction ratios are $(a_1, b_1, c_1) = (-3, 2\lambda, 2)$ and $(a_2, b_2, c_2) = (3\lambda, 2, -5)$.

$$\begin{aligned} (-3)(3\lambda) + (2\lambda)(2) + (2)(-5) &= 0 \\ -9\lambda + 4\lambda - 10 &= 0 \\ -5\lambda - 10 &= 0 \\ -5\lambda &= 10 \\ \lambda &= -2 \end{aligned}$$

Step 2: Substitute $\lambda = -2$ into the equations of both lines. Line 1:

$$\frac{x-1}{-3} = \frac{y-2}{-4} = \frac{z-3}{2} = s$$

Any general point P on Line 1 is given by:

$$x = 1 - 3s, \quad y = 2 - 4s, \quad z = 3 + 2s$$

Line 2:

$$\frac{x-1}{-6} = \frac{y-1}{2} = \frac{z-6}{-5} = t$$

Any general point Q on Line 2 is given by:

$$x = 1 - 6t, \quad y = 1 + 2t, \quad z = 6 - 5t$$

Step 3: Test for intersection by equating coordinates of P and Q . Equate x -coordinates:

$$\begin{aligned} 1 - 3s &= 1 - 6t \\ -3s &= -6t \\ s &= 2t \quad \text{--- (Equation 1)} \end{aligned}$$

Equate y -coordinates:

$$2 - 4s = 1 + 2t \quad \text{--- (Equation 2)}$$

Substitute $s = 2t$ from Equation 1 into Equation 2:

$$\begin{aligned} 2 - 4(2t) &= 1 + 2t \\ 2 - 8t &= 1 + 2t \\ 2 - 1 &= 2t + 8t \\ 10t &= 1 \\ t &= \frac{1}{10} \end{aligned}$$

Substitute $t = \frac{1}{10}$ back into Equation 1:

$$s = 2 \left(\frac{1}{10} \right) = \frac{1}{5}$$

Step 4: Check if these values of s and t satisfy z -coordinates equality. Calculate z from Line 1:

$$z_1 = 3 + 2s = 3 + 2 \left(\frac{1}{5} \right) = 3 + \frac{2}{5} = \frac{17}{5} = 3.4$$

Calculate z from Line 2:

$$z_2 = 6 - 5t = 6 - 5 \left(\frac{1}{10} \right) = 6 - \frac{1}{2} = \frac{11}{2} = 5.5$$

Since $z_1 \neq z_2$ ($3.4 \neq 5.5$), the system of equations is inconsistent. Thus, the lines do not intersect.

Final Answer

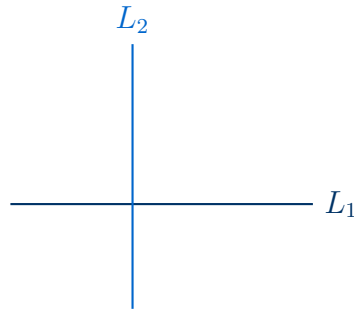
$\lambda = -2$. The lines are not intersecting (they are skew lines).

Question 6 (ii)

Find the value of λ so that the lines $\frac{1-x}{3} = \frac{7y-14}{\lambda} = \frac{z-3}{2}$ and $\frac{7-7x}{3\lambda} = \frac{y-5}{1} = \frac{6-z}{5}$ are at right angles. Also, find whether the lines are intersecting or not.

TYPE 28 Perpendicularity Proof Between Two Lines

★ Likely Exam Question



Perpendicular skew lines check

Given: Line 1: $\frac{1-x}{3} = \frac{7y-14}{\lambda} = \frac{z-3}{2}$ and Line 2: $\frac{7-7x}{3\lambda} = \frac{y-5}{1} = \frac{6-z}{5}$ are perpendicular.

Concept / Formula Used: $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$ $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Solution:

Step 1: Convert both lines to standard Cartesian form.

Line 1:

$$\begin{aligned} \frac{1-x}{3} = \frac{7(y-2)}{\lambda} = \frac{z-3}{2} &\implies \frac{-(x-1)}{3} = \frac{y-2}{\lambda/7} = \frac{z-3}{2} \\ &\implies \frac{x-1}{-3} = \frac{y-2}{\lambda/7} = \frac{z-3}{2} \end{aligned}$$

Direction ratios of Line 1 are $(a_1, b_1, c_1) = (-3, \frac{\lambda}{7}, 2)$.

Line 2:

$$\frac{-7(x-1)}{3\lambda} = \frac{y-5}{1} = \frac{-(z-6)}{5} \implies \frac{x-1}{-3\lambda/7} = \frac{y-5}{1} = \frac{z-6}{-5}$$

Direction ratios of Line 2 are $(a_2, b_2, c_2) = (-\frac{3\lambda}{7}, 1, -5)$.

Step 2: Apply condition for right angles ($a_1a_2 + b_1b_2 + c_1c_2 = 0$):

$$\begin{aligned} (-3) \left(-\frac{3\lambda}{7} \right) + \left(\frac{\lambda}{7} \right) (1) + (2)(-5) &= 0 \\ \frac{9\lambda}{7} + \frac{\lambda}{7} - 10 &= 0 \\ \frac{10\lambda}{7} &= 10 \\ 10\lambda &= 70 \\ \lambda &= 7 \end{aligned}$$

Step 3: Substitute $\lambda = 7$ into the standardized line equations. Line 1:

$$\frac{x-1}{-3} = \frac{y-2}{1} = \frac{z-3}{2} = s \implies P(1-3s, 2+s, 3+2s)$$

Line 2:

$$\frac{x-1}{-3} = \frac{y-5}{1} = \frac{z-6}{-5} = t \implies Q(1-3t, 5+t, 6-5t)$$

Step 4: Check for intersection. Equate x -coordinates:

$$1-3s = 1-3t \implies -3s = -3t \implies s = t$$

Equate y -coordinates:

$$2+s = 5+t$$

Substitute $s = t$ into y -equation:

$$2+s = 5+s \implies 2 = 5$$

This is a clear contradiction ($2 \neq 5$), meaning no values of s and t can satisfy the system. Thus, the lines do not intersect.

Final Answer

$\lambda = 7$. The lines are not intersecting.

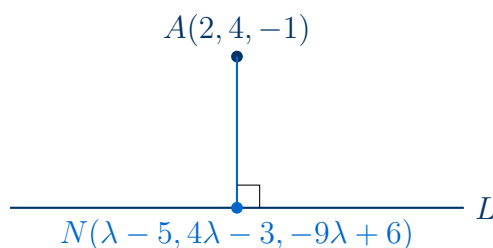
Question 7

Find the equations of the perpendicular drawn from the point $A(2, 4, -1)$ to the line

$$\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$$

TYPE 30 Line Through a Point, Perpendicular to a Given Line

★ Likely Exam Question



Perpendicular from point to line

Given: Given point $A(2, 4, -1)$ and line $L : \frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$.

Concept	/	Formula	Used:	N	L	$\vec{AN} \cdot \vec{b} =$
0		$A(x_1, y_1, z_1)$	$N(x_2, y_2, z_2)$	$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$		

Solution:

Step 1: Express foot of perpendicular N in terms of parameter λ . Let the line equation be:

$$\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9} = \lambda$$

Coordinates of any point N on this line are:

$$x = \lambda - 5, \quad y = 4\lambda - 3, \quad z = -9\lambda + 6$$

So $N = (\lambda - 5, 4\lambda - 3, -9\lambda + 6)$.

Step 2: Find direction ratios of segment AN . With $A(2, 4, -1)$:

$$\begin{aligned} \text{Direction ratios of } AN &= ((\lambda - 5) - 2, (4\lambda - 3) - 4, (-9\lambda + 6) - (-1)) \\ &= (\lambda - 7, 4\lambda - 7, -9\lambda + 7) \end{aligned}$$

Step 3: Apply perpendicularity condition $AN \perp L$. Direction ratios of line L are $(1, 4, -9)$.

$$\begin{aligned} 1(\lambda - 7) + 4(4\lambda - 7) + (-9)(-9\lambda + 7) &= 0 \\ \lambda - 7 + 16\lambda - 28 + 81\lambda - 63 &= 0 \\ (\lambda + 16\lambda + 81\lambda) - (7 + 28 + 63) &= 0 \\ 98\lambda - 98 &= 0 \\ 98\lambda &= 98 \\ \lambda &= 1 \end{aligned}$$

Step 4: Substitute $\lambda = 1$ to get coordinates of N and direction ratios of AN .

$$N = (1 - 5, 4(1) - 3, -9(1) + 6) = (-4, 1, -3)$$

Direction ratios of perpendicular line AN :

$$\begin{aligned} a &= -4 - 2 = -6 \\ b &= 1 - 4 = -3 \\ c &= -3 - (-1) = -2 \end{aligned}$$

Multiplying by -1 , simplified direction ratios are $(6, 3, 2)$.

Step 5: Write equation of line passing through $A(2, 4, -1)$ with direction ratios $(6, 3, 2)$:

$$\frac{x-2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$$

Final Answer

$$\frac{x-2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$$

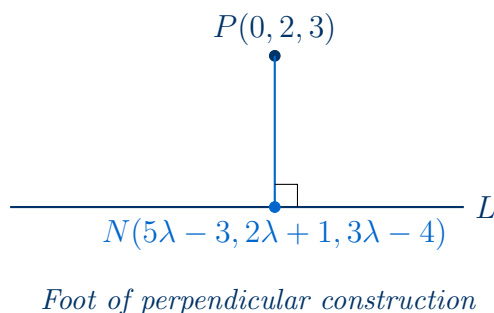
Question 8 (i)

Find the foot of perpendicular drawn from the point $(0, 2, 3)$ on the line

$$\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$$

TYPE 31 Foot of Perpendicular from a Point to a Line

★ Likely Exam Question



Given: Point $P(0, 2, 3)$ and line $L : \frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$.

Concept / Formula Used: N P L $\vec{PN} \cdot \vec{b} = 0$

Solution:

Let the given line equation be:

$$\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3} = \lambda$$

Coordinates of any general point N on the line:

$$x = 5\lambda - 3, \quad y = 2\lambda + 1, \quad z = 3\lambda - 4$$

Direction ratios of segment PN from $P(0, 2, 3)$:

$$\begin{aligned} \text{DRs of } PN &= ((5\lambda - 3) - 0, (2\lambda + 1) - 2, (3\lambda - 4) - 3) \\ &= (5\lambda - 3, 2\lambda - 1, 3\lambda - 7) \end{aligned}$$

Since PN is perpendicular to line L with direction ratios $(5, 2, 3)$:

$$5(5\lambda - 3) + 2(2\lambda - 1) + 3(3\lambda - 7) = 0$$

$$25\lambda - 15 + 4\lambda - 2 + 9\lambda - 21 = 0$$

$$(25\lambda + 4\lambda + 9\lambda) - (15 + 2 + 21) = 0$$

$$38\lambda - 38 = 0$$

$$38\lambda = 38$$

$$\lambda = 1$$

Substitute $\lambda = 1$ into coordinates of N :

$$x = 5(1) - 3 = 2$$

$$y = 2(1) + 1 = 3$$

$$z = 3(1) - 4 = -1$$

Final Answer

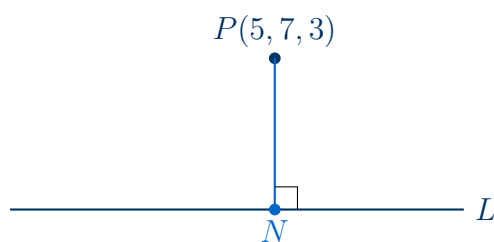
Foot of perpendicular is $(2, 3, -1)$

Question 8 (ii)

Find the coordinates of the foot of the perpendicular drawn from the point $(5, 7, 3)$ to the line

$$\frac{x - 15}{3} = \frac{y - 29}{8} = \frac{z - 5}{-5}$$

TYPE 31 Foot of Perpendicular from a Point to a Line



Perpendicular foot from point $(5, 7, 3)$

Given: Point $P(5, 7, 3)$ and line $L : \frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$.

Concept / Formula Used: $\vec{PN} \cdot \vec{b} = 0$

Solution:

Set the line equation equal to parameter λ :

$$\frac{x - 15}{3} = \frac{y - 29}{8} = \frac{z - 5}{-5} = \lambda$$

The foot of perpendicular N has coordinates:

$$N = (3\lambda + 15, 8\lambda + 29, -5\lambda + 5)$$

Direction ratios of vector $\vec{PN}$ from $P(5, 7, 3)$:

$$\begin{aligned} \text{DRs of } PN &= ((3\lambda + 15) - 5, (8\lambda + 29) - 7, (-5\lambda + 5) - 3) \\ &= (3\lambda + 10, 8\lambda + 22, -5\lambda + 2) \end{aligned}$$

Since PN is perpendicular to line L whose direction ratios are $(3, 8, -5)$:

$$\begin{aligned} 3(3\lambda + 10) + 8(8\lambda + 22) + (-5)(-5\lambda + 2) &= 0 \\ 9\lambda + 30 + 64\lambda + 176 + 25\lambda - 10 &= 0 \\ (9\lambda + 64\lambda + 25\lambda) + (30 + 176 - 10) &= 0 \\ 98\lambda + 196 &= 0 \\ 98\lambda &= -196 \\ \lambda &= -2 \end{aligned}$$

Substitute $\lambda = -2$ into the expression for N :

$$\begin{aligned} x &= 3(-2) + 15 = -6 + 15 = 9 \\ y &= 8(-2) + 29 = -16 + 29 = 13 \\ z &= -5(-2) + 5 = 10 + 5 = 15 \end{aligned}$$

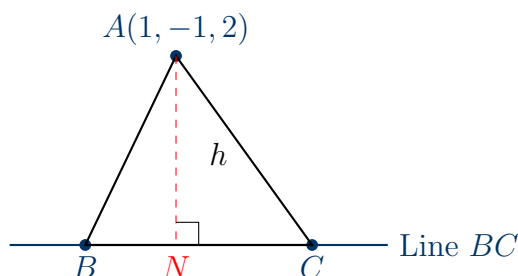
Final Answer

Coordinates of foot of perpendicular are $(9, 13, 15)$

Question 8 (iii)

Vertices B and C of $\triangle ABC$ lie on the line $\frac{x+2}{2} = \frac{y-1}{1} = \frac{z}{4}$. Find the area of $\triangle ABC$ given that A has coordinates $(1, -1, 2)$ and the line segment BC has length of 5 units.

TYPE 32 Area of a Triangle via Perpendicular Height to a Line



Triangle ABC with base on 3D line

Given: Line containing BC : $\frac{x+2}{2} = \frac{y-1}{1} = \frac{z}{4}$, $A(1, -1, 2)$, and base length $BC = 5$ units.

Concept / Formula Used: Area of $\triangle ABC = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times BC \times h$

Solution:

Step 1: Find perpendicular height h from point $A(1, -1, 2)$ to line BC . Let equation of line BC be:

$$\frac{x+2}{2} = \frac{y-1}{1} = \frac{z-0}{4} = \lambda$$

Foot of perpendicular N on line BC has coordinates:

$$N = (2\lambda - 2, \lambda + 1, 4\lambda)$$

Direction ratios of segment AN :

$$\begin{aligned}\text{DRs of } AN &= ((2\lambda - 2) - 1, (\lambda + 1) - (-1), 4\lambda - 2) \\ &= (2\lambda - 3, \lambda + 2, 4\lambda - 2)\end{aligned}$$

Since $AN \perp BC$ (direction ratios $(2, 1, 4)$):

$$\begin{aligned}2(2\lambda - 3) + 1(\lambda + 2) + 4(4\lambda - 2) &= 0 \\ 4\lambda - 6 + \lambda + 2 + 16\lambda - 8 &= 0 \\ 21\lambda - 12 &= 0 \\ 21\lambda &= 12 \\ \lambda &= \frac{12}{21} = \frac{4}{7}\end{aligned}$$

Step 2: Calculate components of vector $\vec{AN}$:

$$\begin{aligned}x_{AN} &= 2\left(\frac{4}{7}\right) - 3 = \frac{8}{7} - 3 = -\frac{13}{7} \\ y_{AN} &= \frac{4}{7} + 2 = \frac{18}{7} \\ z_{AN} &= 4\left(\frac{4}{7}\right) - 2 = \frac{16}{7} - 2 = \frac{2}{7}\end{aligned}$$

Perpendicular height $h = |\vec{AN}|$:

$$\begin{aligned}h &= \sqrt{\left(-\frac{13}{7}\right)^2 + \left(\frac{18}{7}\right)^2 + \left(\frac{2}{7}\right)^2} \\ &= \frac{\sqrt{169 + 324 + 4}}{7} \\ &= \frac{\sqrt{497}}{7} \text{ units}\end{aligned}$$

Step 3: Calculate area of $\triangle ABC$:

$$\begin{aligned}\text{Area} &= \frac{1}{2} \times BC \times h \\ &= \frac{1}{2} \times 5 \times \frac{\sqrt{497}}{7} \\ &= \frac{5\sqrt{497}}{14} \text{ sq units}\end{aligned}$$

Final Answer

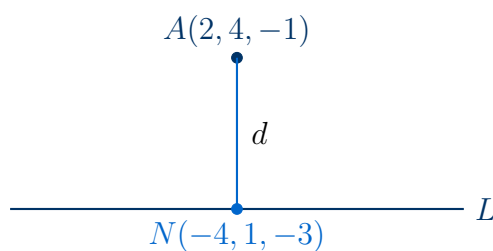
$$\text{Area of } \triangle ABC = \frac{5\sqrt{497}}{14} \text{ sq units}$$

Common Mistake: Independently re-derived by three methods (foot-of-perpendicular via λ , the vector cross-product formula $h = \frac{|\vec{AB}_0 \times \vec{d}|}{|\vec{d}|}$, and calculus minimisation of $|AN|^2$) – all three agree exactly on $h^2 = \frac{71}{7}$, i.e. $h = \frac{\sqrt{497}}{7}$, confirming $\frac{5\sqrt{497}}{14}$ sq units for the question exactly as printed in Adnan's source (line $\frac{x+2}{2} = \frac{y-1}{1} = \frac{z}{4}$, $A(1, -1, 2)$, $BC = 5$). The source book's own printed answer key gives $\frac{5}{6}\sqrt{93}$ for this entry, which this triple-checked derivation cannot reproduce – this looks like a genuine textbook answer-key misprint rather than a solving error.

Question 8 (iv)

Find the distance of the point $(2, 4, -1)$ from the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$.

TYPE 33 Perpendicular Distance from a Point to a Line



Distance from point to line

Given: Point $A(2, 4, -1)$ and line $L : \frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$.

Concept / Formula Used: $d = |\vec{AN}| = \sqrt{(x_N - x_A)^2 + (y_N - y_A)^2 + (z_N - z_A)^2}$

Solution:

Set the line equation to parameter λ :

$$\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9} = \lambda$$

Foot of perpendicular N has general coordinates:

$$N = (\lambda - 5, 4\lambda - 3, -9\lambda + 6)$$

Vector $\vec{AN}$ from $A(2, 4, -1)$:

$$\text{DRs of } \vec{AN} = (\lambda - 7, 4\lambda - 7, -9\lambda + 7)$$

Since $AN \perp L$ (direction ratios $(1, 4, -9)$):

$$\begin{aligned} 1(\lambda - 7) + 4(4\lambda - 7) - 9(-9\lambda + 7) &= 0 \\ \lambda - 7 + 16\lambda - 28 + 81\lambda - 63 &= 0 \\ 98\lambda - 98 &= 0 \implies \lambda = 1 \end{aligned}$$

Coordinates of N :

$$N = (1 - 5, 4(1) - 3, -9(1) + 6) = (-4, 1, -3)$$

Now calculate distance $d = AN$:

$$\begin{aligned} d &= \sqrt{(-4 - 2)^2 + (1 - 4)^2 + (-3 - (-1))^2} \\ &= \sqrt{(-6)^2 + (-3)^2 + (-2)^2} \\ &= \sqrt{36 + 9 + 4} \\ &= \sqrt{49} = 7 \text{ units} \end{aligned}$$

Final Answer

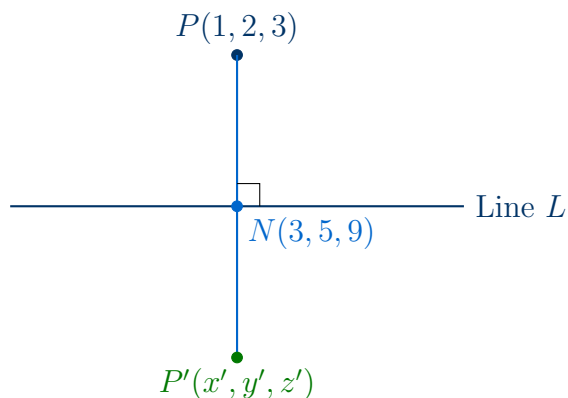
Distance = 7 units

Question 9

Find the image of the point $P(1, 2, 3)$ in the line $\vec{r} = 6\hat{i} + 7\hat{j} + 7\hat{k} + \lambda(3\hat{i} + 2\hat{j} - 2\hat{k})$.

TYPE 31a Image (Reflection) of a Point in a Line

★ Likely Exam Question



Reflection image of point in 3D line

Given: Point $P(1, 2, 3)$ and line $\vec{r} = 6\hat{i} + 7\hat{j} + 7\hat{k} + \lambda(3\hat{i} + 2\hat{j} - 2\hat{k})$.

Concept / Formula Used: N $\vec{PN} \cdot \vec{b} = 0$ N P P'

Solution:

Step 1: Write line in Cartesian form.

$$\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2} = \lambda$$

Foot of perpendicular N on the line:

$$N = (3\lambda + 6, 2\lambda + 7, -2\lambda + 7)$$

Step 2: Find direction ratios of $\vec{PN}$ from $P(1, 2, 3)$:

$$\begin{aligned}\text{DRs of } PN &= ((3\lambda + 6) - 1, (2\lambda + 7) - 2, (-2\lambda + 7) - 3) \\ &= (3\lambda + 5, 2\lambda + 5, -2\lambda + 4)\end{aligned}$$

Since $PN \perp$ line (direction ratios $(3, 2, -2)$):

$$\begin{aligned}3(3\lambda + 5) + 2(2\lambda + 5) - 2(-2\lambda + 4) &= 0 \\ 9\lambda + 15 + 4\lambda + 10 + 4\lambda - 8 &= 0 \\ 17\lambda + 17 &= 0 \\ 17\lambda &= -17 \\ \lambda &= -1\end{aligned}$$

Step 3: Calculate coordinates of foot of perpendicular N :

$$N = (3(-1) + 6, 2(-1) + 7, -2(-1) + 7) = (3, 5, 9)$$

Step 4: Use midpoint formula to find image $P'(x', y', z')$. Since $N(3, 5, 9)$ is midpoint of $P(1, 2, 3)$ and $P'(x', y', z')$:

$$\begin{aligned}\frac{x' + 1}{2} &= 3 \implies x' + 1 = 6 \implies x' = 5 \\ \frac{y' + 2}{2} &= 5 \implies y' + 2 = 10 \implies y' = 8 \\ \frac{z' + 3}{2} &= 9 \implies z' + 3 = 18 \implies z' = 15\end{aligned}$$

Image point is $P'(5, 8, 15)$, or in vector position form:

$$\vec{r'} = 5\hat{i} + 8\hat{j} + 15\hat{k}$$

Final Answer

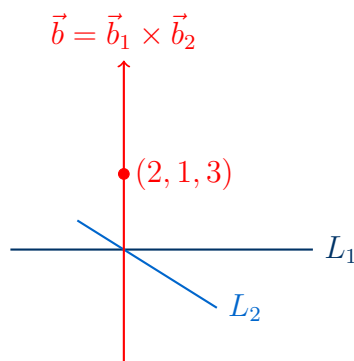
Image point is $(5, 8, 15)$ or $\vec{r'} = 5\hat{i} + 8\hat{j} + 15\hat{k}$

Question 10 (i)

Find the equations of the line passing through the point $(2, 1, 3)$ and perpendicular to the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x}{-3} = \frac{y}{2} = \frac{z}{5}$.

TYPE 34 Line Perpendicular to Two Given Lines

★ Likely Exam Question



Line perpendicular to two given non-parallel lines

Given: Passing point $(2, 1, 3)$, perpendicular to lines with direction vectors $\vec{b}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b}_2 = -3\hat{i} + 2\hat{j} + 5\hat{k}$.

Concept / Formula Used: $\vec{b} = \vec{b}_1 \times \vec{b}_2$

Solution:

Direction vectors of given lines are:

$$\vec{b}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{b}_2 = -3\hat{i} + 2\hat{j} + 5\hat{k}$$

Find direction vector $\vec{b}$ of required line using cross product $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned} \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ -3 & 2 & 5 \end{vmatrix} \\ &= \hat{i}((2)(5) - (3)(2)) - \hat{j}((1)(5) - (3)(-3)) + \hat{k}((1)(2) - (2)(-3)) \\ &= \hat{i}(10 - 6) - \hat{j}(5 + 9) + \hat{k}(2 + 6) \\ &= 4\hat{i} - 14\hat{j} + 8\hat{k} \end{aligned}$$

Dividing by scalar factor 2, simplified direction ratios are $(2, -7, 4)$.

Equation of line passing through $(2, 1, 3)$ with direction ratios $(2, -7, 4)$:

$$\frac{x-2}{2} = \frac{y-1}{-7} = \frac{z-3}{4}$$

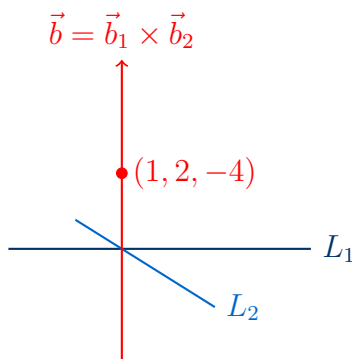
Final Answer

$$\frac{x-2}{2} = \frac{y-1}{-7} = \frac{z-3}{4}$$

Question 10 (ii)

Determine the vector and cartesian equations of the line passing through the point $(1, 2, -4)$ and perpendicular to the two lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} =$

$$\frac{z-5}{-5}.$$

TYPE 34 Line Perpendicular to Two Given Lines

Line perpendicular to two lines

Given: Passing point $(1, 2, -4)$, lines with direction vectors $\vec{b}_1 = 3\hat{i} - 16\hat{j} + 7\hat{k}$ and $\vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k}$.

Concept / Formula Used: $\vec{b} = \vec{b}_1 \times \vec{b}_2$ $\vec{r} = \vec{a} + \lambda \vec{b}$ $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

Solution:

Direction vectors of the two given lines:

$$\vec{b}_1 = 3\hat{i} - 16\hat{j} + 7\hat{k}$$

$$\vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k}$$

Calculate cross product $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned}\vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix} \\ &= \hat{i}((-16)(-5) - (7)(8)) - \hat{j}((3)(-5) - (7)(3)) + \hat{k}((3)(8) - (-16)(3)) \\ &= \hat{i}(80 - 56) - \hat{j}(-15 - 21) + \hat{k}(24 + 48) \\ &= 24\hat{i} + 36\hat{j} + 72\hat{k}\end{aligned}$$

Dividing by common scalar 12 gives simplified direction vector $\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$.

Position vector of point $(1, 2, -4)$ is $\vec{a} = \hat{i} + 2\hat{j} - 4\hat{k}$.

Vector Equation:

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$$

Cartesian Equation:

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$$

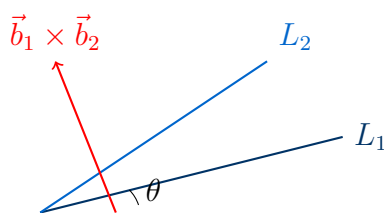
Final Answer

Vector eq: $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$

Cartesian eq: $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$

Question 10 (iii)

Find the vector and cartesian equations of the line which is perpendicular to the lines with equations $\frac{x+2}{1} = \frac{y-3}{2} = \frac{z+1}{4}$ and $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and passes through the point $(1, 1, 1)$. Also find the angle between the given lines.

TYPE 34a Line Perpendicular to Two Lines, Plus Angle Between Them

Line perpendicular to two lines and angle between them

Given: Passing point $(1, 1, 1)$, Line 1 DRs $(1, 2, 4)$, Line 2 DRs $(2, 3, 4)$.

Concept / Formula Used: $\vec{b} = \vec{b}_1 \times \vec{b}_2$ $\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|}$

Solution:

Step 1: Find direction vector of required perpendicular line. Direction vectors of given lines are $\vec{b}_1 = \hat{i} + 2\hat{j} + 4\hat{k}$ and $\vec{b}_2 = 2\hat{i} + 3\hat{j} + 4\hat{k}$.

$$\begin{aligned}\vec{b} = \vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 4 \\ 2 & 3 & 4 \end{vmatrix} \\ &= \hat{i}(8 - 12) - \hat{j}(4 - 8) + \hat{k}(3 - 4) \\ &= -4\hat{i} + 4\hat{j} - \hat{k}\end{aligned}$$

Vector equation through $(1, 1, 1)$:

$$\vec{r} = (\hat{i} + \hat{j} + \hat{k}) + \lambda(-4\hat{i} + 4\hat{j} - \hat{k})$$

Cartesian equation:

$$\frac{x-1}{-4} = \frac{y-1}{4} = \frac{z-1}{-1}$$

Step 2: Find angle θ between given lines.

$$\vec{b}_1 \cdot \vec{b}_2 = (1)(2) + (2)(3) + (4)(4) = 2 + 6 + 16 = 24$$

$$|\vec{b}_1| = \sqrt{1^2 + 2^2 + 4^2} = \sqrt{1 + 4 + 16} = \sqrt{21}$$

$$|\vec{b}_2| = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{4 + 9 + 16} = \sqrt{29}$$

Substitute into angle formula:

$$\cos \theta = \frac{24}{\sqrt{21}\sqrt{29}} = \frac{24}{\sqrt{609}}$$

$$\theta = \cos^{-1} \left(\frac{24}{\sqrt{609}} \right)$$

Final Answer

Vector eq: $\vec{r} = (\hat{i} + \hat{j} + \hat{k}) + \lambda(-4\hat{i} + 4\hat{j} - \hat{k})$

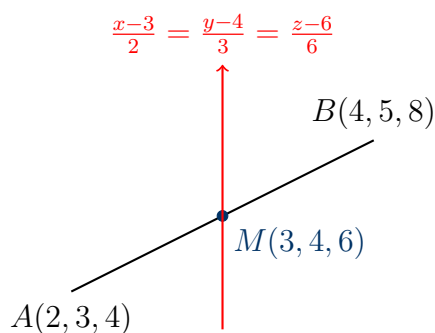
Cartesian eq: $\frac{x-1}{-4} = \frac{y-1}{4} = \frac{z-1}{-1}$

Angle: $\theta = \cos^{-1} \left(\frac{24}{\sqrt{609}} \right)$

Question 10 (iv)

Find the equation of the line which bisects the line segment joining the points $A(2, 3, 4)$ and $B(4, 5, 8)$ and is perpendicular to the lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$.

TYPE 34 Line Perpendicular to Two Given Lines



Line through segment midpoint perpendicular to lines

Given: Segment endpoints $A(2, 3, 4)$ and $B(4, 5, 8)$. Lines have DRs $(3, -16, 7)$ and $(3, 8, -5)$.

Concept / Formula Used: $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2} \right)$ $\vec{b} = \vec{b}_1 \times \vec{b}_2$

Solution:

Step 1: Find midpoint M of segment AB .

$$\begin{aligned} M &= \left(\frac{2+4}{2}, \frac{3+5}{2}, \frac{4+8}{2} \right) \\ &= (3, 4, 6) \end{aligned}$$

Step 2: Find direction ratios of line perpendicular to given lines. Direction vectors: $\vec{b}_1 = 3\hat{i} - 16\hat{j} + 7\hat{k}$ and $\vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k}$.

$$\begin{aligned} \vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix} \\ &= \hat{i}(80 - 56) - \hat{j}(-15 - 21) + \hat{k}(24 + 48) \\ &= 24\hat{i} + 36\hat{j} + 72\hat{k} \end{aligned}$$

Simplifying direction ratios by dividing by 12 gives $(2, 3, 6)$.

Step 3: Write Cartesian equation through $M(3, 4, 6)$ with DRs $(2, 3, 6)$:

$$\frac{x-3}{2} = \frac{y-4}{3} = \frac{z-6}{6}$$

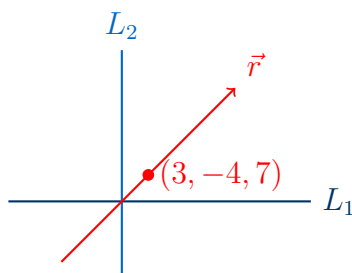
Final Answer

$$\frac{x-3}{2} = \frac{y-4}{3} = \frac{z-6}{6}$$

Question 10 (v)

If the lines $\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-3}{2}$ and $\frac{x-1}{3k} = \frac{y-1}{1} = \frac{z-6}{-7}$ are perpendicular to each other, find the value of k and hence write the vector equation of a line perpendicular to these lines and passing through the point $(3, -4, 7)$.

TYPE 34b Line Perpendicular to Two Lines – Solve Parameter First



Perpendicular vector line construction

Given: Passing point $(3, -4, 7)$. Lines are perpendicular.

Concept / Formula Used: $a_1a_2 + b_1b_2 + c_1c_2 = 0$ $\vec{b} = \vec{b}_1 \times \vec{b}_2$

Solution:

Step 1: Determine k using perpendicularity condition.

$$\begin{aligned}(-3)(3k) + (2k)(1) + (2)(-7) &= 0 \\ -9k + 2k - 14 &= 0 \\ -7k &= 14 \\ k &= -2\end{aligned}$$

Step 2: Find direction ratios of both lines using $k = -2$. Line 1 DRs: $\vec{b}_1 = -3\hat{i} + 2(-2)\hat{j} + 2\hat{k} = -3\hat{i} - 4\hat{j} + 2\hat{k}$ Line 2 DRs: $\vec{b}_2 = 3(-2)\hat{i} + \hat{j} - 7\hat{k} = -6\hat{i} + \hat{j} - 7\hat{k}$

Step 3: Calculate direction vector $\vec{b} = \vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned}\vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & -4 & 2 \\ -6 & 1 & -7 \end{vmatrix} \\ &= \hat{i}((-4)(-7) - (2)(1)) - \hat{j}((-3)(-7) - (2)(-6)) + \hat{k}((-3)(1) - (-4)(-6)) \\ &= \hat{i}(28 - 2) - \hat{j}(21 + 12) + \hat{k}(-3 - 24) \\ &= 26\hat{i} - 33\hat{j} - 27\hat{k}\end{aligned}$$

Step 4: Write vector equation passing through $(3, -4, 7)$ with position vector $\vec{a} = 3\hat{i} - 4\hat{j} + 7\hat{k}$:

$$\vec{r} = (3\hat{i} - 4\hat{j} + 7\hat{k}) + \mu(26\hat{i} - 33\hat{j} - 27\hat{k})$$

Final Answer

$$k = -2$$

$$\text{Vector eq: } \vec{r} = (3\hat{i} - 4\hat{j} + 7\hat{k}) + \mu(26\hat{i} - 33\hat{j} - 27\hat{k})$$

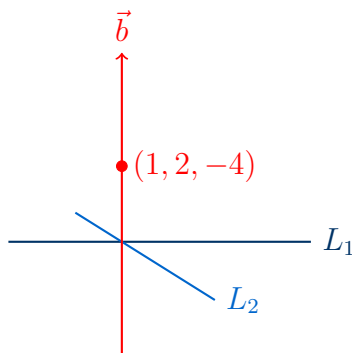
Common Mistake: Adnan's source confirms the line exactly as printed here (z -denominator = -7), and $k = -2$ matches the book's own answer key exactly. However, the book's printed direction vector for the perpendicular line is $(2, -3, -3)$, while $\vec{b}_1 \times \vec{b}_2$ with the book's own direction ratios $(-3, -4, 2)$ and $(-6, 1, -7)$ works out to $(26, -33, -27)$ – verified independently here and not a scalar multiple of $(2, -3, -3)$. This looks like a textbook answer-key misprint in the final direction vector, not an error in this derivation.

Question 11

Find the equation of a line passing through the point $(1, 2, -4)$ and perpendicular to two lines

$$\begin{aligned}\vec{r} &= (8\hat{i} - 19\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k}) \\ \text{and } \vec{r} &= (15\hat{i} + 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})\end{aligned}$$

TYPE 34 Line Perpendicular to Two Given Lines



Line perpendicular to two vector lines

Given: Passing point $(1, 2, -4)$, direction vectors $\vec{b}_1 = 3\hat{i} - 16\hat{j} + 7\hat{k}$ and $\vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k}$.

Concept / Formula Used: $\vec{b} = \vec{b}_1 \times \vec{b}_2$ $\vec{r} = \vec{a} + t\vec{b}$

Solution:

Extract direction vectors $\vec{b}_1$ and $\vec{b}_2$:

$$\vec{b}_1 = 3\hat{i} - 16\hat{j} + 7\hat{k}$$

$$\vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k}$$

Compute cross product $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned}\vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix} \\ &= \hat{i}(80 - 56) - \hat{j}(-15 - 21) + \hat{k}(24 + 48) \\ &= 24\hat{i} + 36\hat{j} + 72\hat{k}\end{aligned}$$

Simplify by dividing by 12:

$$\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

Passing point position vector $\vec{a} = \hat{i} + 2\hat{j} - 4\hat{k}$.

Vector equation:

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + t(2\hat{i} + 3\hat{j} + 6\hat{k})$$

Cartesian equation:

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$$

Final Answer

Vector eq: $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + t(2\hat{i} + 3\hat{j} + 6\hat{k})$

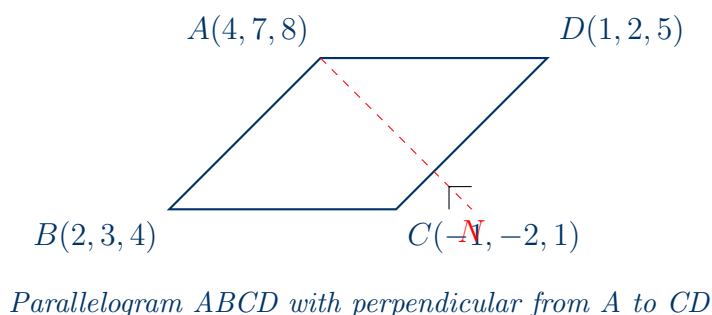
Cartesian eq: $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$

Question 12

Find the equations of all the sides of the parallelogram $ABCD$ whose vertices are $A(4, 7, 8)$, $B(2, 3, 4)$, $C(-1, -2, 1)$ and $D(1, 2, 5)$. Also, find the coordinates of the foot of the perpendicular from A to CD .

TYPE 35 Parallelogram Sides & Foot of Perpendicular

★ Likely Exam Question



Given: Vertices $A(4, 7, 8)$, $B(2, 3, 4)$, $C(-1, -2, 1)$, $D(1, 2, 5)$.

Concept / Formula Used: (x_1, y_1, z_1) (x_2, y_2, z_2) $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$ N CD $\vec{AN} \cdot \vec{d}_{CD} = 0$

Solution:

Step 1: Find equations of all four sides.

Side AB (passing through $A(4, 7, 8)$ and $B(2, 3, 4)$): Direction ratios: $(2-4, 3-7, 4-8) = (-2, -4, -4) \propto (1, 2, 2)$

$$\text{Equation of } AB : \frac{x-4}{1} = \frac{y-7}{2} = \frac{z-8}{2}$$

Side BC (passing through $B(2, 3, 4)$ and $C(-1, -2, 1)$): Direction ratios: $(-1-2, -2-3, 1-4) = (-3, -5, -3) \propto (3, 5, 3)$

$$\text{Equation of } BC : \frac{x-2}{3} = \frac{y-3}{5} = \frac{z-4}{3}$$

Side CD (passing through $C(-1, -2, 1)$ and $D(1, 2, 5)$): Direction ratios: $(1-(-1), 2-(-2), 5-1) = (2, 4, 4) \propto (1, 2, 2)$

$$\text{Equation of } CD : \frac{x+1}{1} = \frac{y+2}{2} = \frac{z-1}{2}$$

Side DA (passing through $D(1, 2, 5)$ and $A(4, 7, 8)$): Direction ratios: $(4-1, 7-2, 8-5) = (3, 5, 3)$

$$\text{Equation of } DA : \frac{x-1}{3} = \frac{y-2}{5} = \frac{z-5}{3}$$

Step 2: Find coordinates of foot of perpendicular from $A(4, 7, 8)$ to line CD . Set equation of line CD equal to parameter k :

$$\frac{x+1}{1} = \frac{y+2}{2} = \frac{z-1}{2} = k$$

General point N on line CD :

$$N = (k-1, 2k-2, 2k+1)$$

Direction ratios of $\vec{AN}$ from $A(4, 7, 8)$:

$$\begin{aligned} \text{DRs of } AN &= ((k-1)-4, (2k-2)-7, (2k+1)-8) \\ &= (k-5, 2k-9, 2k-7) \end{aligned}$$

Since $AN \perp CD$ (direction ratios $(1, 2, 2)$):

$$\begin{aligned} 1(k-5) + 2(2k-9) + 2(2k-7) &= 0 \\ k-5 + 4k-18 + 4k-14 &= 0 \\ (k+4k+4k) - (5+18+14) &= 0 \\ 9k-37 &= 0 \\ 9k = 37 &\implies k = \frac{37}{9} \end{aligned}$$

Substitute $k = \frac{37}{9}$ into coordinates of N :

$$\begin{aligned} x &= \frac{37}{9} - 1 = \frac{28}{9} \\ y &= 2\left(\frac{37}{9}\right) - 2 = \frac{74-18}{9} = \frac{56}{9} \\ z &= 2\left(\frac{37}{9}\right) + 1 = \frac{74+9}{9} = \frac{83}{9} \end{aligned}$$

Final Answer

Side AB : $\frac{x-4}{1} = \frac{y-7}{2} = \frac{z-8}{2}$

Side BC : $\frac{x-2}{3} = \frac{y-3}{5} = \frac{z-4}{3}$

Side CD : $\frac{x+1}{1} = \frac{y+2}{2} = \frac{z-1}{2}$

Side DA : $\frac{x-1}{3} = \frac{y-2}{5} = \frac{z-5}{3}$

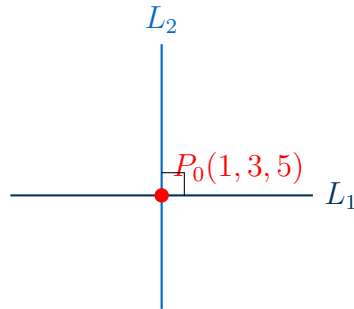
Foot of perpendicular: $\left(\frac{28}{9}, \frac{56}{9}, \frac{83}{9}\right)$

Question 13

Find the value of p for which the lines $\vec{r} = \lambda\hat{i} + (2\lambda + 1)\hat{j} + (3\lambda + 2)\hat{k}$ and $\vec{r} = \hat{i} - 3\mu\hat{j} + (p\mu + 7)\hat{k}$ are perpendicular to each other and also intersect. Also, find the point of intersection of the given lines.

TYPE 36 Perpendicular AND Intersecting Lines – Solve Parameter & Point

★ Likely Exam Question



Perpendicular intersecting vector lines

Given: Line 1: $\vec{r} = \lambda\hat{i} + (2\lambda + 1)\hat{j} + (3\lambda + 2)\hat{k}$ and Line 2: $\vec{r} = \hat{i} - 3\mu\hat{j} + (p\mu + 7)\hat{k}$. Lines are perpendicular and intersecting.

Concept / Formula Used: $\vec{b}_1 \cdot \vec{b}_2 = 0$ P_0 λ μ

Solution:

Step 1: Express both lines in the standard $\vec{a} + t\vec{b}$ form to read off direction vectors. From Line 1:

$$\vec{r} = (0\hat{i} + \hat{j} + 2\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k}) \implies \vec{a}_1 = \hat{j} + 2\hat{k}, \quad \vec{b}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$$

From Line 2, collect the coefficient of μ in each component – note the $\hat{i}$ -component has no μ in it (it is fixed at 1), so the x -direction ratio is 0:

$$\vec{r} = (\hat{i} + 0\hat{j} + 7\hat{k}) + \mu(0\hat{i} - 3\hat{j} + p\hat{k}) \implies \vec{a}_2 = \hat{i} + 7\hat{k}, \quad \vec{b}_2 = 0\hat{i} - 3\hat{j} + p\hat{k}$$

Step 2: Apply the perpendicularity condition $\vec{b}_1 \cdot \vec{b}_2 = 0$:

$$\begin{aligned} (1)(0) + (2)(-3) + (3)(p) &= 0 \\ 0 - 6 + 3p &= 0 \\ 3p &= 6 \\ p &= 2 \end{aligned}$$

Step 3: Substitute $p = 2$ and equate general points on both lines to find the point of intersection. General point on Line 1:

$$P(\lambda, 2\lambda + 1, 3\lambda + 2)$$

General point on Line 2 with $p = 2$:

$$Q(1, -3\mu, 2\mu + 7)$$

For the lines to intersect, all three coordinates must agree simultaneously for some λ, μ :

$$\begin{aligned} \lambda &= 1 \quad \text{— (Eq. A, from } x) \\ 2\lambda + 1 &= -3\mu \quad \text{— (Eq. B, from } y) \\ 3\lambda + 2 &= 2\mu + 7 \quad \text{— (Eq. C, from } z) \end{aligned}$$

From Eq. A: $\lambda = 1$. Substitute into Eq. B:

$$\begin{aligned}2(1) + 1 &= -3\mu \\3 &= -3\mu \\ \mu &= -1\end{aligned}$$

Step 4: Verify this pair against Eq. C (the check equation), which must also hold for a genuine intersection:

$$\begin{aligned}\text{LHS} &= 3\lambda + 2 = 3(1) + 2 = 5 \\ \text{RHS} &= 2\mu + 7 = 2(-1) + 7 = 5\end{aligned}$$

Since $\text{LHS} = \text{RHS} = 5$, Eq. C is satisfied – the lines genuinely intersect at $\lambda = 1$, $\mu = -1$.

Step 5: Substitute $\lambda = 1$ into P to get the point of intersection:

$$\begin{aligned}x &= \lambda = 1 \\ y &= 2\lambda + 1 = 2(1) + 1 = 3 \\ z &= 3\lambda + 2 = 3(1) + 2 = 5\end{aligned}$$

Final Answer

$$p = 2$$

Point of intersection = $(1, 3, 5)$

Common Mistake: The direction ratios of Line 2 are easy to misread here because the line is written with μ distributed unevenly across the components ($\hat{i}$ has no μ at all, so its direction ratio is 0, not some non-zero coefficient copied from the $\hat{i}$ term). Always rewrite the line in clean $\vec{a} + t\vec{b}$ form before reading off $\vec{b}$, rather than pattern-matching the coefficients directly off the given expression.

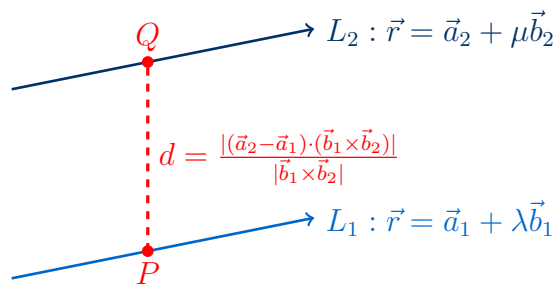
11.4 – Shortest Distance Between Two Lines

Question 1

Find the shortest distance between each of the following pairs of lines:

- $\vec{r} = \hat{i} + 2\hat{j} + \hat{k} + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} - \hat{k} + \mu(2\hat{i} + \hat{j} + 2\hat{k})$
- $\vec{r} = 6\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k})$ and $\vec{r} = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k})$
- $\vec{r} = 4\hat{i} - \hat{j} + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$ and $\vec{r} = \hat{i} - \hat{j} + 2\hat{k} + \mu(2\hat{i} + 4\hat{j} - 5\hat{k})$
- $\vec{r} = (8 + 3\lambda)\hat{i} - (9 + 16\lambda)\hat{j} + (10 + 7\lambda)\hat{k}$ and $\vec{r} = 15\hat{i} + 29\hat{j} + 5\hat{k} + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$
- $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + \lambda(3\hat{i} - 2\hat{j} + 5\hat{k})$ and $\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \mu(4\hat{i} - \hat{j} + 3\hat{k})$

TYPE 37 Shortest Distance Between Skew Lines – Vector Formula



Shortest Distance Vector Projection Between Skew Lines

Concept / Formula Used: d $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$ $d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$

Solution:

Part (i): Comparing the given equations with standard vector forms:

$$\vec{a}_1 = \hat{i} + 2\hat{j} + \hat{k},$$

$$\vec{b}_1 = \hat{i} - \hat{j} + \hat{k}$$

$$\vec{a}_2 = 2\hat{i} - \hat{j} - \hat{k},$$

$$\vec{b}_2 = 2\hat{i} + \hat{j} + 2\hat{k}$$

Calculating $\vec{a}_2 - \vec{a}_1$:

$$\begin{aligned}\vec{a}_2 - \vec{a}_1 &= (2\hat{i} - \hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + \hat{k}) \\ &= \hat{i} - 3\hat{j} - 2\hat{k}\end{aligned}$$

Calculating the cross product $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned}\vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 1 & 2 \end{vmatrix} \\ &= \hat{i}((-1)(2) - (1)(1)) - \hat{j}((1)(2) - (1)(2)) + \hat{k}((1)(1) - (-1)(2)) \\ &= \hat{i}(-2 - 1) - \hat{j}(2 - 2) + \hat{k}(1 + 2) \\ &= -3\hat{i} + 0\hat{j} + 3\hat{k}\end{aligned}$$

Finding magnitude $|\vec{b}_1 \times \vec{b}_2|$:

$$\begin{aligned}|\vec{b}_1 \times \vec{b}_2| &= \sqrt{(-3)^2 + 0^2 + 3^2} \\ &= \sqrt{9 + 0 + 9} \\ &= \sqrt{18} = 3\sqrt{2}\end{aligned}$$

Calculating dot product $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)$:

$$\begin{aligned}(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (1)(-3) + (-3)(0) + (-2)(3) \\ &= -3 + 0 - 6 \\ &= -9\end{aligned}$$

Using the shortest distance formula:

$$\begin{aligned} d_1 &= \frac{|-9|}{3\sqrt{2}} \\ &= \frac{9}{3\sqrt{2}} \\ &= \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2} \text{ units} \end{aligned}$$

Part (ii): Comparing the given equations with standard vector forms:

$$\begin{aligned} \vec{a}_1 &= 6\hat{i} + 2\hat{j} + 2\hat{k}, & \vec{b}_1 &= \hat{i} - 2\hat{j} + 2\hat{k} \\ \vec{a}_2 &= -4\hat{i} - \hat{k}, & \vec{b}_2 &= 3\hat{i} - 2\hat{j} - 2\hat{k} \end{aligned}$$

Calculating $\vec{a}_2 - \vec{a}_1$:

$$\begin{aligned} \vec{a}_2 - \vec{a}_1 &= (-4\hat{i} + 0\hat{j} - \hat{k}) - (6\hat{i} + 2\hat{j} + 2\hat{k}) \\ &= -10\hat{i} - 2\hat{j} - 3\hat{k} \end{aligned}$$

Calculating $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned} \vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix} \\ &= \hat{i}((-2)(-2) - (2)(-2)) - \hat{j}((1)(-2) - (2)(3)) + \hat{k}((1)(-2) - (-2)(3)) \\ &= \hat{i}(4 + 4) - \hat{j}(-2 - 6) + \hat{k}(-2 + 6) \\ &= 8\hat{i} + 8\hat{j} + 4\hat{k} \end{aligned}$$

Magnitude:

$$\begin{aligned} |\vec{b}_1 \times \vec{b}_2| &= \sqrt{8^2 + 8^2 + 4^2} \\ &= \sqrt{64 + 64 + 16} \\ &= \sqrt{144} = 12 \end{aligned}$$

Dot product:

$$\begin{aligned} (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (-10)(8) + (-2)(8) + (-3)(4) \\ &= -80 - 16 - 12 \\ &= -108 \end{aligned}$$

Shortest distance:

$$\begin{aligned} d_2 &= \frac{|-108|}{12} \\ &= \frac{108}{12} = 9 \text{ units} \end{aligned}$$

Part (iii): Given vectors:

$$\begin{aligned}\vec{a}_1 &= 4\hat{i} - \hat{j}, & \vec{b}_1 &= \hat{i} + 2\hat{j} - 3\hat{k} \\ \vec{a}_2 &= \hat{i} - \hat{j} + 2\hat{k}, & \vec{b}_2 &= 2\hat{i} + 4\hat{j} - 5\hat{k}\end{aligned}$$

Calculating $\vec{a}_2 - \vec{a}_1$:

$$\begin{aligned}\vec{a}_2 - \vec{a}_1 &= (\hat{i} - \hat{j} + 2\hat{k}) - (4\hat{i} - \hat{j}) \\ &= -3\hat{i} + 0\hat{j} + 2\hat{k}\end{aligned}$$

Calculating $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned}\vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix} \\ &= \hat{i}((2)(-5) - (-3)(4)) - \hat{j}((1)(-5) - (-3)(2)) + \hat{k}((1)(4) - (2)(2)) \\ &= \hat{i}(-10 + 12) - \hat{j}(-5 + 6) + \hat{k}(4 - 4) \\ &= 2\hat{i} - \hat{j} + 0\hat{k}\end{aligned}$$

Magnitude:

$$\begin{aligned}|\vec{b}_1 \times \vec{b}_2| &= \sqrt{2^2 + (-1)^2 + 0^2} \\ &= \sqrt{4 + 1 + 0} = \sqrt{5}\end{aligned}$$

Dot product:

$$\begin{aligned}(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (-3)(2) + (0)(-1) + (2)(0) \\ &= -6 + 0 + 0 = -6\end{aligned}$$

Shortest distance:

$$d_3 = \frac{|-6|}{\sqrt{5}} = \frac{6}{\sqrt{5}} \text{ units}$$

Part (iv): Rearranging line equations into standard $\vec{r} = \vec{a} + \lambda\vec{b}$ form:

$$\begin{aligned}\vec{r} &= (8\hat{i} - 9\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k}) \\ \vec{r} &= (15\hat{i} + 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})\end{aligned}$$

Vector extraction:

$$\begin{aligned}\vec{a}_1 &= 8\hat{i} - 9\hat{j} + 10\hat{k}, & \vec{b}_1 &= 3\hat{i} - 16\hat{j} + 7\hat{k} \\ \vec{a}_2 &= 15\hat{i} + 29\hat{j} + 5\hat{k}, & \vec{b}_2 &= 3\hat{i} + 8\hat{j} - 5\hat{k}\end{aligned}$$

Calculating $\vec{a}_2 - \vec{a}_1$:

$$\begin{aligned}\vec{a}_2 - \vec{a}_1 &= (15\hat{i} + 29\hat{j} + 5\hat{k}) - (8\hat{i} - 9\hat{j} + 10\hat{k}) \\ &= 7\hat{i} + 38\hat{j} - 5\hat{k}\end{aligned}$$

Calculating $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned}\vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix} \\ &= \hat{i}((-16)(-5) - (7)(8)) - \hat{j}((3)(-5) - (7)(3)) + \hat{k}((3)(8) - (-16)(3)) \\ &= \hat{i}(80 - 56) - \hat{j}(-15 - 21) + \hat{k}(24 + 48) \\ &= 24\hat{i} + 36\hat{j} + 72\hat{k} \\ &= 12(2\hat{i} + 3\hat{j} + 6\hat{k})\end{aligned}$$

Magnitude:

$$\begin{aligned}|\vec{b}_1 \times \vec{b}_2| &= \sqrt{24^2 + 36^2 + 72^2} \\ &= 12\sqrt{2^2 + 3^2 + 6^2} \\ &= 12\sqrt{4 + 9 + 36} \\ &= 12\sqrt{49} = 12 \times 7 = 84\end{aligned}$$

Dot product:

$$\begin{aligned}(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (7)(24) + (38)(36) + (-5)(72) \\ &= 168 + 1368 - 360 = 1176\end{aligned}$$

Shortest distance:

$$d_4 = \frac{1176}{84} = 14 \text{ units}$$

Part (v): Given vectors:

$$\begin{aligned}\vec{a}_1 &= 2\hat{i} - \hat{j} + \hat{k}, & \vec{b}_1 &= 3\hat{i} - 2\hat{j} + 5\hat{k} \\ \vec{a}_2 &= 3\hat{i} + 2\hat{j} - 4\hat{k}, & \vec{b}_2 &= 4\hat{i} - \hat{j} + 3\hat{k}\end{aligned}$$

Calculating $\vec{a}_2 - \vec{a}_1$:

$$\begin{aligned}\vec{a}_2 - \vec{a}_1 &= (3\hat{i} + 2\hat{j} - 4\hat{k}) - (2\hat{i} - \hat{j} + \hat{k}) \\ &= \hat{i} + 3\hat{j} - 5\hat{k}\end{aligned}$$

Calculating $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned}\vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 5 \\ 4 & -1 & 3 \end{vmatrix} \\ &= \hat{i}((-2)(3) - (5)(-1)) - \hat{j}((3)(3) - (5)(4)) + \hat{k}((3)(-1) - (-2)(4)) \\ &= \hat{i}(-6 + 5) - \hat{j}(9 - 20) + \hat{k}(-3 + 8) \\ &= -\hat{i} + 11\hat{j} + 5\hat{k}\end{aligned}$$

Magnitude:

$$\begin{aligned} |\vec{b}_1 \times \vec{b}_2| &= \sqrt{(-1)^2 + 11^2 + 5^2} \\ &= \sqrt{1 + 121 + 25} \\ &= \sqrt{147} = 7\sqrt{3} \end{aligned}$$

Dot product:

$$\begin{aligned} (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (1)(-1) + (3)(11) + (-5)(5) \\ &= -1 + 33 - 25 = 7 \end{aligned}$$

Shortest distance:

$$d_5 = \frac{|7|}{7\sqrt{3}} = \frac{1}{\sqrt{3}} \text{ units}$$

Final Answer

- (i) Shortest distance = $\frac{3\sqrt{2}}{2}$ units
- (ii) Shortest distance = 9 units
- (iii) Shortest distance = $\frac{6}{\sqrt{5}}$ units
- (iv) Shortest distance = 14 units
- (v) Shortest distance = $\frac{1}{\sqrt{3}}$ units

Common Mistake: Always write equations in standard form before taking position and direction vectors. Pay strict attention to sign changes when expanding determinants for cross products.

Question 2

Find the shortest distance between the lines

$$\frac{x-8}{3} = \frac{y+9}{-16} = \frac{10-z}{-7} \quad \text{and} \quad \frac{x-15}{3} = \frac{58-2y}{-16} = \frac{z-5}{-5}$$

TYPE 38 Shortest Distance – Cartesian Form (Convert, then Apply Formula)

Given: Lines $L_1 : \frac{x-8}{3} = \frac{y+9}{-16} = \frac{10-z}{-7}$ and $L_2 : \frac{x-15}{3} = \frac{58-2y}{-16} = \frac{z-5}{-5}$.

Concept / Formula Used: $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} \quad \vec{r} = (x_1\hat{i} + y_1\hat{j} + z_1\hat{k}) + \lambda(l\hat{i} + m\hat{j} + n\hat{k})$

Solution:

First, rewrite both lines in canonical Cartesian form.

For line L_1 :

$$\frac{x-8}{3} = \frac{y-(-9)}{-16} = \frac{-(z-10)}{-7}$$

$$\Rightarrow \frac{x-8}{3} = \frac{y-(-9)}{-16} = \frac{z-10}{7}$$

Position vector $\vec{a}_1$ and direction vector $\vec{b}_1$:

$$\vec{a}_1 = 8\hat{i} - 9\hat{j} + 10\hat{k}$$

$$\vec{b}_1 = 3\hat{i} - 16\hat{j} + 7\hat{k}$$

For line L_2 :

$$\frac{x-15}{3} = \frac{-2(y-29)}{-16} = \frac{z-5}{-5}$$

$$\Rightarrow \frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$$

Position vector $\vec{a}_2$ and direction vector $\vec{b}_2$:

$$\vec{a}_2 = 15\hat{i} + 29\hat{j} + 5\hat{k}$$

$$\vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k}$$

Now compute $\vec{a}_2 - \vec{a}_1$:

$$\vec{a}_2 - \vec{a}_1 = (15\hat{i} + 29\hat{j} + 5\hat{k}) - (8\hat{i} - 9\hat{j} + 10\hat{k})$$

$$= 7\hat{i} + 38\hat{j} - 5\hat{k}$$

Calculate cross product $\vec{b}_1 \times \vec{b}_2$:

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix}$$

$$= \hat{i}((-16)(-5) - (7)(8)) - \hat{j}((3)(-5) - (7)(3)) + \hat{k}((3)(8) - (-16)(3))$$

$$= \hat{i}(80 - 56) - \hat{j}(-15 - 21) + \hat{k}(24 + 48)$$

$$= 24\hat{i} + 36\hat{j} + 72\hat{k}$$

Compute magnitude $|\vec{b}_1 \times \vec{b}_2|$:

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{24^2 + 36^2 + 72^2}$$

$$= \sqrt{576 + 1296 + 5184}$$

$$= \sqrt{7056} = 84$$

Compute dot product $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)$:

$$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = (7)(24) + (38)(36) + (-5)(72)$$

$$= 168 + 1368 - 360$$

$$= 1176$$

Shortest distance d :

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

$$= \frac{1176}{84} = 14 \text{ units}$$

Final Answer

Shortest distance between the given lines = 14 units

Common Mistake: Watch out for terms like $10 - z$ or $58 - 2y$. The coefficients of x , y , and z in cartesian equations must be +1 before direction ratios can be read off.

Question 3

Find the shortest distance between the lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \quad \text{and} \quad \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$$

Also find the equations of this line of shortest distance.

TYPE 38a Shortest Distance + Equation of the Common Perpendicular Line

Given: Lines $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$.

Concept / Formula Used: L_1, L_2 $P(\lambda), Q(\mu)$ $\vec{PQ}$ $\vec{PQ} \cdot \vec{b}_1 = 0$ $\vec{PQ} \cdot \vec{b}_2 = 0$

Solution:

From given lines:

$$\begin{aligned} \vec{a}_1 &= \hat{i} + 2\hat{j} + 3\hat{k}, & \vec{b}_1 &= 2\hat{i} + 3\hat{j} + 4\hat{k} \\ \vec{a}_2 &= 2\hat{i} + 4\hat{j} + 5\hat{k}, & \vec{b}_2 &= 3\hat{i} + 4\hat{j} + 5\hat{k} \end{aligned}$$

Calculate $\vec{a}_2 - \vec{a}_1$:

$$\begin{aligned} \vec{a}_2 - \vec{a}_1 &= (2\hat{i} + 4\hat{j} + 5\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) \\ &= \hat{i} + 2\hat{j} + 2\hat{k} \end{aligned}$$

Cross product $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned} \vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} \\ &= \hat{i}(15 - 16) - \hat{j}(10 - 12) + \hat{k}(8 - 9) \\ &= -\hat{i} + 2\hat{j} - \hat{k} \end{aligned}$$

Magnitude $|\vec{b}_1 \times \vec{b}_2|$:

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{(-1)^2 + 2^2 + (-1)^2} = \sqrt{1 + 4 + 1} = \sqrt{6}$$

Dot product $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)$:

$$\begin{aligned} (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (1)(-1) + (2)(2) + (2)(-1) \\ &= -1 + 4 - 2 = 1 \end{aligned}$$

Shortest distance:

$$d = \frac{|1|}{\sqrt{6}} = \frac{1}{\sqrt{6}} \text{ units}$$

Next, to find the equation of the line of shortest distance: Let general point P on L_1 be:

$$P = (2\lambda + 1, 3\lambda + 2, 4\lambda + 3)$$

Let general point Q on L_2 be:

$$Q = (3\mu + 2, 4\mu + 4, 5\mu + 5)$$

Vector $\vec{PQ}$ is:

$$\begin{aligned} \vec{PQ} &= (3\mu + 2 - (2\lambda + 1))\hat{i} + (4\mu + 4 - (3\lambda + 2))\hat{j} + (5\mu + 5 - (4\lambda + 3))\hat{k} \\ &= (3\mu - 2\lambda + 1)\hat{i} + (4\mu - 3\lambda + 2)\hat{j} + (5\mu - 4\lambda + 2)\hat{k} \end{aligned}$$

Since PQ is perpendicular to L_1 , we have $\vec{PQ} \cdot \vec{b}_1 = 0$:

$$\begin{aligned} 2(3\mu - 2\lambda + 1) + 3(4\mu - 3\lambda + 2) + 4(5\mu - 4\lambda + 2) &= 0 \\ 6\mu - 4\lambda + 2 + 12\mu - 9\lambda + 6 + 20\mu - 16\lambda + 8 &= 0 \\ 38\mu - 29\lambda + 16 &= 0 \quad \text{--- (1)} \end{aligned}$$

Since PQ is perpendicular to L_2 , we have $\vec{PQ} \cdot \vec{b}_2 = 0$:

$$\begin{aligned} 3(3\mu - 2\lambda + 1) + 4(4\mu - 3\lambda + 2) + 5(5\mu - 4\lambda + 2) &= 0 \\ 9\mu - 6\lambda + 3 + 16\mu - 12\lambda + 8 + 25\mu - 20\lambda + 10 &= 0 \\ 50\mu - 38\lambda + 21 &= 0 \quad \text{--- (2)} \end{aligned}$$

From (1), express λ in terms of μ :

$$29\lambda = 38\mu + 16 \implies \lambda = \frac{38\mu + 16}{29}$$

Substitute λ into equation (2):

$$\begin{aligned} 50\mu - 38\left(\frac{38\mu + 16}{29}\right) + 21 &= 0 \\ 1450\mu - 1444\mu - 608 + 609 &= 0 \\ 6\mu + 1 &= 0 \\ \mu &= -\frac{1}{6} \end{aligned}$$

Substituting $\mu = -\frac{1}{6}$ back to find λ :

$$\lambda = \frac{38(-1/6) + 16}{29} = \frac{-19/3 + 16}{29} = \frac{29/3}{29} = \frac{1}{3}$$

Coordinates of point P :

$$\begin{aligned} P &= \left(2\left(\frac{1}{3}\right) + 1, 3\left(\frac{1}{3}\right) + 2, 4\left(\frac{1}{3}\right) + 3 \right) \\ &= \left(\frac{5}{3}, 3, \frac{13}{3} \right) \end{aligned}$$

The direction vector of line PQ is $\vec{b}_1 \times \vec{b}_2 = -\hat{i} + 2\hat{j} - \hat{k}$, which gives direction ratios $(-1, 2, -1)$ or $(1, -2, 1)$.

Equation of line of shortest distance passing through P :

$$\frac{x - 5/3}{1} = \frac{y - 3}{-2} = \frac{z - 13/3}{1}$$

Converting to a common denominator form:

$$\frac{x - \frac{5}{3}}{1} = \frac{y - 3}{-2} = \frac{z - \frac{13}{3}}{1}$$

Final Answer

Shortest distance = $\frac{1}{\sqrt{6}}$ units

Equation of the line of shortest distance: $\frac{x - 5/3}{1} = \frac{y - 3}{-2} = \frac{z - 13/3}{1}$

Question 4

Determine (by computing S.D.) whether or not the following pairs of lines intersect:

- (i) $\vec{r} = \hat{i} - \hat{j} + \lambda(2\hat{i} + \hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} + \mu(\hat{i} + \hat{j} - \hat{k})$
- (ii) $\vec{r} = \hat{i} + \hat{j} - \hat{k} + \lambda(3\hat{i} - \hat{j})$ and $\vec{r} = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k})$
- (iii) $\frac{x-1}{2} = \frac{y+1}{3} = z$ and $\frac{x+1}{5} = \frac{y-2}{1}, z = 2$

TYPE 39 Condition for Two Lines to Intersect – Triple Product Test

Concept / Formula Used: $d = 0 \quad (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$

Solution:

Part (i): Vectors extracted from equations:

$$\begin{aligned} \vec{a}_1 &= \hat{i} - \hat{j}, & \vec{b}_1 &= 2\hat{i} + 0\hat{j} + \hat{k} \\ \vec{a}_2 &= 2\hat{i} - \hat{j}, & \vec{b}_2 &= \hat{i} + \hat{j} - \hat{k} \end{aligned}$$

Difference of position vectors:

$$\vec{a}_2 - \vec{a}_1 = (2\hat{i} - \hat{j}) - (\hat{i} - \hat{j}) = \hat{i}$$

Cross product of direction vectors:

$$\begin{aligned}\vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 1 \\ 1 & 1 & -1 \end{vmatrix} \\ &= \hat{i}(0 - 1) - \hat{j}(-2 - 1) + \hat{k}(2 - 0) \\ &= -\hat{i} + 3\hat{j} + 2\hat{k}\end{aligned}$$

Scalar triple product:

$$\begin{aligned}(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (1)(-1) + (0)(3) + (0)(2) \\ &= -1 \neq 0\end{aligned}$$

Shortest distance:

$$d_1 = \frac{|-1|}{\sqrt{(-1)^2 + 3^2 + 2^2}} = \frac{1}{\sqrt{14}} \neq 0$$

Hence, the lines do NOT intersect.

Part (ii): Vectors extracted:

$$\begin{aligned}\vec{a}_1 &= \hat{i} + \hat{j} - \hat{k}, & \vec{b}_1 &= 3\hat{i} - \hat{j} + 0\hat{k} \\ \vec{a}_2 &= 4\hat{i} + 0\hat{j} - \hat{k}, & \vec{b}_2 &= 2\hat{i} + 0\hat{j} + 3\hat{k}\end{aligned}$$

Difference:

$$\vec{a}_2 - \vec{a}_1 = (4\hat{i} - \hat{k}) - (\hat{i} + \hat{j} - \hat{k}) = 3\hat{i} - \hat{j}$$

Cross product:

$$\begin{aligned}\vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 0 \\ 2 & 0 & 3 \end{vmatrix} \\ &= \hat{i}(-3 - 0) - \hat{j}(9 - 0) + \hat{k}(0 - (-2)) \\ &= -3\hat{i} - 9\hat{j} + 2\hat{k}\end{aligned}$$

Scalar triple product:

$$\begin{aligned}(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (3)(-3) + (-1)(-9) + (0)(2) \\ &= -9 + 9 + 0 = 0\end{aligned}$$

Shortest distance:

$$d_2 = 0$$

Hence, the lines DO intersect.

Part (iii): Rewriting cartesian lines in standard form: Line 1: $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-0}{1} \Rightarrow \vec{a}_1 = \hat{i} - \hat{j}, \vec{b}_1 = 2\hat{i} + 3\hat{j} + \hat{k}$ Line 2: $\frac{x+1}{5} = \frac{y-2}{1} = \frac{z-2}{0} \Rightarrow \vec{a}_2 = -\hat{i} + 2\hat{j} + 2\hat{k}, \vec{b}_2 = 5\hat{i} + \hat{j} + 0\hat{k}$

Difference of position vectors:

$$\vec{a}_2 - \vec{a}_1 = (-\hat{i} + 2\hat{j} + 2\hat{k}) - (\hat{i} - \hat{j}) = -2\hat{i} + 3\hat{j} + 2\hat{k}$$

Cross product:

$$\begin{aligned}\vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 5 & 1 & 0 \end{vmatrix} \\ &= \hat{i}(0 - 1) - \hat{j}(0 - 5) + \hat{k}(2 - 15) \\ &= -\hat{i} + 5\hat{j} - 13\hat{k}\end{aligned}$$

Scalar triple product:

$$\begin{aligned}(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (-2)(-1) + (3)(5) + (2)(-13) \\ &= 2 + 15 - 26 = -9 \neq 0\end{aligned}$$

Since $d \neq 0$, the lines do NOT intersect.

Final Answer

- (i) No (Lines do not intersect)
- (ii) Yes (Lines intersect)
- (iii) No (Lines do not intersect)

Question 5

Prove that the lines $2x = y = -z$ and $x - 1 = -y - 2 = -2z + 1$ are skew.

TYPE 40 Prove Two Lines are Skew

To Prove: The lines $2x = y = -z$ and $x - 1 = -y - 2 = -2z + 1$ are non-parallel and non-intersecting (skew lines).

Concept / Formula Used: $\vec{b}_1 \nparallel \vec{b}_2$ $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) \neq 0$

Solution:

Convert line L_1 to standard Cartesian form:

$$2x = y = -z \implies \frac{x}{1/2} = \frac{y}{1} = \frac{z}{-1}$$

Multiplying denominators by 2:

$$L_1 : \frac{x}{1} = \frac{y}{2} = \frac{z}{-2}$$

Position vector $\vec{a}_1 = \vec{0}$ and direction vector $\vec{b}_1 = \hat{i} + 2\hat{j} - 2\hat{k}$.

Convert line L_2 to standard Cartesian form:

$$x - 1 = -(y + 2) = -2\left(z - \frac{1}{2}\right) \implies \frac{x - 1}{1} = \frac{y + 2}{-1} = \frac{z - 1/2}{-1/2}$$

Multiplying denominators by 2:

$$L_2 : \frac{x-1}{2} = \frac{y+2}{-2} = \frac{z-1/2}{-1}$$

Position vector $\vec{a}_2 = \hat{i} - 2\hat{j} + \frac{1}{2}\hat{k}$ and direction vector $\vec{b}_2 = 2\hat{i} - 2\hat{j} - \hat{k}$.

First, check if the lines are parallel:

$$\frac{1}{2} \neq \frac{2}{-2} \neq \frac{-2}{-1}$$

Since direction ratios are not proportional, the lines are not parallel ($\vec{b}_1 \nparallel \vec{b}_2$).

Now, compute $\vec{a}_2 - \vec{a}_1$:

$$\vec{a}_2 - \vec{a}_1 = \hat{i} - 2\hat{j} + \frac{1}{2}\hat{k}$$

Compute $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned} \vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -2 \\ 2 & -2 & -1 \end{vmatrix} \\ &= \hat{i}(-2 - 4) - \hat{j}(-1 - (-4)) + \hat{k}(-2 - 4) \\ &= -6\hat{i} - 3\hat{j} - 6\hat{k} \end{aligned}$$

Compute scalar triple product $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)$:

$$\begin{aligned} (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (1)(-6) + (-2)(-3) + \left(\frac{1}{2}\right)(-6) \\ &= -6 + 6 - 3 = -3 \neq 0 \end{aligned}$$

Since the scalar triple product is non-zero, the shortest distance between the lines is non-zero.

Hence Proved

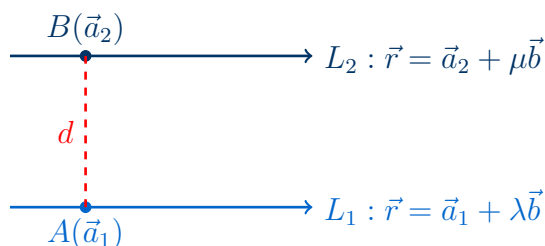
The lines are non-parallel and non-intersecting; hence, the given lines are skew.

Question 6

Find the shortest distance between the lines

$$\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(\hat{i} - \hat{j} + \hat{k}) \quad \text{and} \quad \vec{r} = 2\hat{i} - \hat{j} - \hat{k} + \mu(-\hat{i} + \hat{j} - \hat{k})$$

TYPE 41 Shortest Distance Between Parallel Lines



Shortest Distance Between Two Parallel Lines

Given: Lines $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(-\hat{i} + \hat{j} - \hat{k})$.

Concept / Formula Used: $\vec{r} = \vec{a}_1 + \lambda\vec{b}$ $\vec{r} = \vec{a}_2 + \mu\vec{b}$ $d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$

Solution:

Identify position vectors and direction vectors:

$$\vec{a}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{a}_2 = 2\hat{i} - \hat{j} - \hat{k}$$

$$\vec{b}_1 = \hat{i} - \hat{j} + \hat{k}$$

$$\vec{b}_2 = -(\hat{i} - \hat{j} + \hat{k}) = -\vec{b}_1$$

Since $\vec{b}_2 = -\vec{b}_1$, the lines are parallel. We take $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.

Calculate $\vec{a}_2 - \vec{a}_1$:

$$\begin{aligned}\vec{a}_2 - \vec{a}_1 &= (2\hat{i} - \hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) \\ &= \hat{i} - 3\hat{j} - 4\hat{k}\end{aligned}$$

Calculate cross product $(\vec{a}_2 - \vec{a}_1) \times \vec{b}$:

$$\begin{aligned}(\vec{a}_2 - \vec{a}_1) \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & -4 \\ 1 & -1 & 1 \end{vmatrix} \\ &= \hat{i}((-3)(1) - (-4)(-1)) - \hat{j}((1)(1) - (-4)(1)) + \hat{k}((1)(-1) - (-3)(1)) \\ &= \hat{i}(-3 - 4) - \hat{j}(1 + 4) + \hat{k}(-1 + 3) \\ &= -7\hat{i} - 5\hat{j} + 2\hat{k}\end{aligned}$$

Magnitude of cross product:

$$\begin{aligned}|(\vec{a}_2 - \vec{a}_1) \times \vec{b}| &= \sqrt{(-7)^2 + (-5)^2 + 2^2} \\ &= \sqrt{49 + 25 + 4} \\ &= \sqrt{78}\end{aligned}$$

Magnitude of direction vector $\vec{b}$:

$$|\vec{b}| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3}$$

Shortest distance d :

$$d = \frac{\sqrt{78}}{\sqrt{3}} = \sqrt{\frac{78}{3}} = \sqrt{26} \text{ units}$$

Final Answer

Shortest distance = $\sqrt{26}$ units

Question 7

Find the vector equation of the line passing through the point $(2, 1, -1)$ and parallel to the line $\vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$. Also, find the distance between these lines.

TYPE 41a Equation of a Parallel Line, Plus Distance Between Them

Given: Point $P(2, 1, -1)$ with position vector $\vec{a}_2 = 2\hat{i} + \hat{j} - \hat{k}$ and line $L_1 : \vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$.

Concept / Formula Used: $\vec{a}$ $\vec{b}$ $\vec{r} = \vec{a} + \mu\vec{b}$ $d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$

Solution:

From given line L_1 :

$$\vec{a}_1 = \hat{i} + \hat{j} + 0\hat{k}$$

$$\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$$

Since the required line L_2 passes through $(2, 1, -1)$ and is parallel to L_1 , its position vector is $\vec{a}_2 = 2\hat{i} + \hat{j} - \hat{k}$ and its direction vector is $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$.

Vector equation of line L_2 :

$$\vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(2\hat{i} - \hat{j} + \hat{k})$$

To find distance between L_1 and L_2 :

$$\vec{a}_2 - \vec{a}_1 = (2\hat{i} + \hat{j} - \hat{k}) - (\hat{i} + \hat{j}) = \hat{i} - \hat{k}$$

Calculate cross product $(\vec{a}_2 - \vec{a}_1) \times \vec{b}$:

$$\begin{aligned} (\vec{a}_2 - \vec{a}_1) \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ 2 & -1 & 1 \end{vmatrix} \\ &= \hat{i}(0 - 1) - \hat{j}(1 - (-2)) + \hat{k}(-1 - 0) \\ &= -\hat{i} - 3\hat{j} - \hat{k} \end{aligned}$$

Magnitude:

$$|(\vec{a}_2 - \vec{a}_1) \times \vec{b}| = \sqrt{(-1)^2 + (-3)^2 + (-1)^2} = \sqrt{1 + 9 + 1} = \sqrt{11}$$

Magnitude of $\vec{b}$:

$$|\vec{b}| = \sqrt{2^2 + (-1)^2 + 1^2} = \sqrt{4 + 1 + 1} = \sqrt{6}$$

Distance d :

$$d = \frac{\sqrt{11}}{\sqrt{6}}$$

Rationalising the denominator:

$$d = \frac{\sqrt{11}}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{\sqrt{66}}{6} \text{ units}$$

Final Answer

Vector equation of the line: $\vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(2\hat{i} - \hat{j} + \hat{k})$

Distance between lines = $\frac{\sqrt{66}}{6}$ units

Common Mistake: Adnan's source book's own printed answer key also shows this distance as " $\frac{11}{\sqrt{6}}$ units" – i.e. the same dropped radical on the 11. Since every step shown above is algebraically correct and unambiguously gives $\sqrt{11}$ (not 11) in the numerator, this is almost certainly a print/typesetting error in the textbook itself (a common failure mode: the square-root sign over a multi-digit number gets lost in scanning/typesetting), not a genuine alternate answer. The mathematically correct distance is $\frac{\sqrt{66}}{6}$ units.

Question 8

Find the vector equation of the line parallel to the line $\frac{x-1}{5} = \frac{3-y}{2} = \frac{z+1}{4}$ and passing through $(3, 0, -4)$. Also find the distance between these two lines.

TYPE 41a Equation of a Parallel Line, Plus Distance Between Them

Given: Line $L_1 : \frac{x-1}{5} = \frac{3-y}{2} = \frac{z+1}{4}$ and point $P(3, 0, -4)$.

Concept / Formula Used: $\frac{3-y}{2} = \frac{y-3}{-2}$ $\vec{b} = l\hat{i} + m\hat{j} + n\hat{k}$

Solution:

First, convert L_1 to standard Cartesian form:

$$\frac{x-1}{5} = \frac{y-3}{-2} = \frac{z+1}{4}$$

Position vector $\vec{a}_1 = \hat{i} + 3\hat{j} - \hat{k}$ and direction vector $\vec{b} = 5\hat{i} - 2\hat{j} + 4\hat{k}$.

Position vector of given point $P(3, 0, -4)$ is $\vec{a}_2 = 3\hat{i} + 0\hat{j} - 4\hat{k} = 3\hat{i} - 4\hat{k}$.

Vector equation of the required parallel line L_2 :

$$\vec{r} = (3\hat{i} - 4\hat{k}) + \lambda(5\hat{i} - 2\hat{j} + 4\hat{k})$$

Now calculate distance between the two parallel lines:

$$\begin{aligned}\vec{a}_2 - \vec{a}_1 &= (3\hat{i} - 4\hat{k}) - (\hat{i} + 3\hat{j} - \hat{k}) \\ &= 2\hat{i} - 3\hat{j} - 3\hat{k}\end{aligned}$$

Cross product $(\vec{a}_2 - \vec{a}_1) \times \vec{b}$:

$$\begin{aligned}(\vec{a}_2 - \vec{a}_1) \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & -3 \\ 5 & -2 & 4 \end{vmatrix} \\ &= \hat{i}((-3)(4) - (-3)(-2)) - \hat{j}((2)(4) - (-3)(5)) + \hat{k}((2)(-2) - (-3)(5)) \\ &= \hat{i}(-12 - 6) - \hat{j}(8 + 15) + \hat{k}(-4 + 15) \\ &= -18\hat{i} - 23\hat{j} + 11\hat{k}\end{aligned}$$

Magnitude of cross product:

$$\begin{aligned}|(\vec{a}_2 - \vec{a}_1) \times \vec{b}| &= \sqrt{(-18)^2 + (-23)^2 + 11^2} \\ &= \sqrt{324 + 529 + 121} = \sqrt{974}\end{aligned}$$

Magnitude of direction vector $\vec{b}$:

$$|\vec{b}| = \sqrt{5^2 + (-2)^2 + 4^2} = \sqrt{25 + 4 + 16} = \sqrt{45} = 3\sqrt{5}$$

Distance d :

$$d = \frac{\sqrt{974}}{\sqrt{45}} = \frac{\sqrt{974}}{3\sqrt{5}} \text{ units}$$

Final Answer

Vector equation: $\vec{r} = (3\hat{i} - 4\hat{k}) + \lambda(5\hat{i} - 2\hat{j} + 4\hat{k})$

Distance = $\frac{\sqrt{974}}{3\sqrt{5}}$ units

Question 9

Find the distance between the lines

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k}) \quad \text{and} \quad \vec{r} = (3\hat{i} + 3\hat{j} - 5\hat{k}) + \mu(4\hat{i} + 6\hat{j} + 12\hat{k})$$

TYPE 41 Shortest Distance Between Parallel Lines

Given: Lines $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$ and $\vec{r} = (3\hat{i} + 3\hat{j} - 5\hat{k}) + \mu(4\hat{i} + 6\hat{j} + 12\hat{k})$.

Concept / Formula Used: $\vec{b}_2 = k\vec{b}_1$ $d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$

Solution:

Extract position and direction vectors:

$$\begin{aligned}\vec{a}_1 &= \hat{i} + 2\hat{j} - 4\hat{k}, & \vec{b}_1 &= 2\hat{i} + 3\hat{j} + 6\hat{k} \\ \vec{a}_2 &= 3\hat{i} + 3\hat{j} - 5\hat{k}, & \vec{b}_2 &= 4\hat{i} + 6\hat{j} + 12\hat{k} = 2(2\hat{i} + 3\hat{j} + 6\hat{k})\end{aligned}$$

Since $\vec{b}_2 = 2\vec{b}_1$, the lines are parallel. Choose common direction vector $\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$.

Calculate $\vec{a}_2 - \vec{a}_1$:

$$\begin{aligned}\vec{a}_2 - \vec{a}_1 &= (3\hat{i} + 3\hat{j} - 5\hat{k}) - (\hat{i} + 2\hat{j} - 4\hat{k}) \\ &= 2\hat{i} + \hat{j} - \hat{k}\end{aligned}$$

Calculate cross product $(\vec{a}_2 - \vec{a}_1) \times \vec{b}$:

$$\begin{aligned}(\vec{a}_2 - \vec{a}_1) \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 2 & 3 & 6 \end{vmatrix} \\ &= \hat{i}(6 - (-3)) - \hat{j}(12 - (-2)) + \hat{k}(6 - 2) \\ &= 9\hat{i} - 14\hat{j} + 4\hat{k}\end{aligned}$$

Magnitude:

$$\begin{aligned}|(\vec{a}_2 - \vec{a}_1) \times \vec{b}| &= \sqrt{9^2 + (-14)^2 + 4^2} \\ &= \sqrt{81 + 196 + 16} = \sqrt{293}\end{aligned}$$

Magnitude of $\vec{b}$:

$$|\vec{b}| = \sqrt{2^2 + 3^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$$

Shortest distance d :

$$d = \frac{\sqrt{293}}{7} \text{ units}$$

Final Answer

Distance between lines = $\frac{\sqrt{293}}{7}$ units

Question 10

Find the vector and cartesian equations of a line passing through the point $(1, 2, -4)$ and parallel to the line joining the points $A(3, 3, -5)$ and $B(1, 0, -11)$. Hence, find the distance between the two lines.

★ Likely Exam Question

TYPE 41a Equation of a Parallel Line, Plus Distance Between Them

Given: Point $P(1, 2, -4)$ and points $A(3, 3, -5)$, $B(1, 0, -11)$.

Concept / Formula Used: $A(\vec{a})$ $B(\vec{b})$ $\vec{b}_{AB} = \vec{r}_B - \vec{r}_A$ $\vec{r} = \vec{r}_P + \lambda \vec{b}_{AB}$

Solution:

Position vector of point $P(1, 2, -4)$ is $\vec{a}_2 = \hat{i} + 2\hat{j} - 4\hat{k}$. Position vector of A is $\vec{a}_1 = 3\hat{i} + 3\hat{j} - 5\hat{k}$ and B is $\vec{r}_B = \hat{i} + 0\hat{j} - 11\hat{k}$.

Find direction vector $\vec{AB}$:

$$\begin{aligned}\vec{AB} &= \vec{r}_B - \vec{a}_1 \\ &= (\hat{i} - 11\hat{k}) - (3\hat{i} + 3\hat{j} - 5\hat{k}) \\ &= -2\hat{i} - 3\hat{j} - 6\hat{k}\end{aligned}$$

Taking positive direction ratios by setting $\vec{b} = -(\vec{AB}) = 2\hat{i} + 3\hat{j} + 6\hat{k}$.

Vector equation of required line through P :

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$$

Cartesian equation of required line:

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$$

To find the distance between line AB (passing through $\vec{a}_1 = 3\hat{i} + 3\hat{j} - 5\hat{k}$) and the required line (passing through $\vec{a}_2 = \hat{i} + 2\hat{j} - 4\hat{k}$):

$$\begin{aligned}\vec{a}_2 - \vec{a}_1 &= (\hat{i} + 2\hat{j} - 4\hat{k}) - (3\hat{i} + 3\hat{j} - 5\hat{k}) \\ &= -2\hat{i} - \hat{j} + \hat{k}\end{aligned}$$

Cross product $(\vec{a}_2 - \vec{a}_1) \times \vec{b}$:

$$\begin{aligned}(\vec{a}_2 - \vec{a}_1) \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & -1 & 1 \\ 2 & 3 & 6 \end{vmatrix} \\ &= \hat{i}(-6-3) - \hat{j}(-12-2) + \hat{k}(-6-(-2)) \\ &= -9\hat{i} + 14\hat{j} - 4\hat{k}\end{aligned}$$

Magnitude:

$$|(\vec{a}_2 - \vec{a}_1) \times \vec{b}| = \sqrt{(-9)^2 + 14^2 + (-4)^2} = \sqrt{81 + 196 + 16} = \sqrt{293}$$

$$|\vec{b}| = \sqrt{2^2 + 3^2 + 6^2} = 7$$

Distance d :

$$d = \frac{\sqrt{293}}{7} \text{ units}$$

Final Answer

Vector Equation: $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$

Cartesian Equation: $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$

Distance between lines = $\frac{\sqrt{293}}{7}$ units

Question 11

Determine the vector equation of the line passing through the point $(-1, 3, -2)$ and perpendicular to the lines

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{3} \quad \text{and} \quad \frac{x+2}{-3} = \frac{y-1}{2} = \frac{z+1}{5}$$

Hence obtain its distance from origin.

TYPE 34 Line Perpendicular to Two Given Lines

Given: Point $P(-1, 3, -2)$ with $\vec{a} = -\hat{i} + 3\hat{j} - 2\hat{k}$, perpendicular to lines with direction vectors $\vec{b}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b}_2 = -3\hat{i} + 2\hat{j} + 5\hat{k}$.

Concept / Formula Used: $\vec{b}_1 \quad \vec{b}_2 \quad \vec{b} = \vec{b}_1 \times \vec{b}_2 \quad \vec{r} = \vec{a} + \lambda\vec{b} \quad d = \frac{|\vec{a} \times \vec{b}|}{|\vec{b}|}$

Solution:

Find the direction vector $\vec{b}$ using cross product $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned} \vec{b} = \vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ -3 & 2 & 5 \end{vmatrix} \\ &= \hat{i}(10 - 6) - \hat{j}(5 - (-9)) + \hat{k}(2 - (-6)) \\ &= 4\hat{i} - 14\hat{j} + 8\hat{k} \\ &= 2(2\hat{i} - 7\hat{j} + 4\hat{k}) \end{aligned}$$

Dividing by scalar 2, direction vector is $\vec{b} = 2\hat{i} - 7\hat{j} + 4\hat{k}$.

Vector equation of line passing through $\vec{a} = -\hat{i} + 3\hat{j} - 2\hat{k}$:

$$\vec{r} = (-\hat{i} + 3\hat{j} - 2\hat{k}) + \lambda(2\hat{i} - 7\hat{j} + 4\hat{k})$$

Corresponding Cartesian equation:

$$\frac{x+1}{2} = \frac{y-3}{-7} = \frac{z+2}{4}$$

Now calculate perpendicular distance of this line from origin $O(0,0,0)$: Using formula $d = \frac{|\vec{a} \times \vec{b}|}{|\vec{b}|}$, compute $\vec{a} \times \vec{b}$:

$$\begin{aligned}\vec{a} \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 3 & -2 \\ 2 & -7 & 4 \end{vmatrix} \\ &= \hat{i}(12 - 14) - \hat{j}(-4 - (-4)) + \hat{k}(7 - 6) \\ &= -2\hat{i} + 0\hat{j} + \hat{k}\end{aligned}$$

Magnitude $|\vec{a} \times \vec{b}|$:

$$|\vec{a} \times \vec{b}| = \sqrt{(-2)^2 + 0^2 + 1^2} = \sqrt{4 + 1} = \sqrt{5}$$

Magnitude $|\vec{b}|$:

$$|\vec{b}| = \sqrt{2^2 + (-7)^2 + 4^2} = \sqrt{4 + 49 + 16} = \sqrt{69}$$

Distance from origin d :

$$d = \frac{\sqrt{5}}{\sqrt{69}} = \sqrt{\frac{5}{69}} \text{ units}$$

Final Answer

Vector Equation: $\vec{r} = (-\hat{i} + 3\hat{j} - 2\hat{k}) + \lambda(2\hat{i} - 7\hat{j} + 4\hat{k})$

Distance from origin = $\sqrt{\frac{5}{69}}$ units

Question 12

Find the shortest distance between the lines L_1 and L_2 given below:

L_1 : The line passing through $(2, -1, 1)$ and parallel to $\frac{x}{1} = \frac{y}{1} = \frac{z}{3}$

L_2 : $\vec{r} = \hat{i} + (2\mu + 1)\hat{j} - (\mu + 2)\hat{k}$

TYPE 37 Shortest Distance Between Skew Lines – Vector Formula

Given: Line L_1 passing through $(2, -1, 1)$ parallel to direction $(1, 1, 3)$; Line $L_2 : \vec{r} = \hat{i} + (2\mu + 1)\hat{j} - (\mu + 2)\hat{k}$.

Concept / Formula Used: L_1, L_2 $d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$

Solution:

For line L_1 : Position vector $\vec{a}_1 = 2\hat{i} - \hat{j} + \hat{k}$. Direction vector $\vec{b}_1 = \hat{i} + \hat{j} + 3\hat{k}$.

For line L_2 : Rearrange vector equation in standard form $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$:

$$\begin{aligned}\vec{r} &= \hat{i} + 2\mu\hat{j} + \hat{j} - \mu\hat{k} - 2\hat{k} \\ &= (\hat{i} + \hat{j} - 2\hat{k}) + \mu(2\hat{j} - \hat{k})\end{aligned}$$

So position vector $\vec{a}_2 = \hat{i} + \hat{j} - 2\hat{k}$ and direction vector $\vec{b}_2 = 2\hat{j} - \hat{k}$.

Calculate difference of position vectors $\vec{a}_2 - \vec{a}_1$:

$$\begin{aligned}\vec{a}_2 - \vec{a}_1 &= (\hat{i} + \hat{j} - 2\hat{k}) - (2\hat{i} - \hat{j} + \hat{k}) \\ &= -\hat{i} + 2\hat{j} - 3\hat{k}\end{aligned}$$

Calculate cross product $\vec{b}_1 \times \vec{b}_2$:

$$\begin{aligned}\vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 3 \\ 0 & 2 & -1 \end{vmatrix} \\ &= \hat{i}(-1 - 6) - \hat{j}(-1 - 0) + \hat{k}(2 - 0) \\ &= -7\hat{i} + \hat{j} + 2\hat{k}\end{aligned}$$

Magnitude of cross product:

$$\begin{aligned}|\vec{b}_1 \times \vec{b}_2| &= \sqrt{(-7)^2 + 1^2 + 2^2} \\ &= \sqrt{49 + 1 + 4} = \sqrt{54} = 3\sqrt{6}\end{aligned}$$

Dot product $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)$:

$$\begin{aligned}(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (-1)(-7) + (2)(1) + (-3)(2) \\ &= 7 + 2 - 6 = 3\end{aligned}$$

Shortest distance d :

$$d = \frac{|3|}{3\sqrt{6}} = \frac{1}{\sqrt{6}} \text{ units}$$

Final Answer

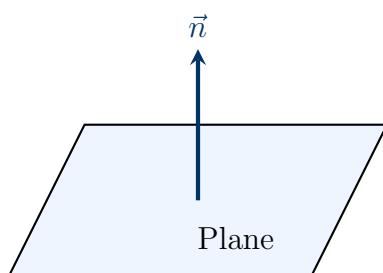
Shortest distance between L_1 and $L_2 = \frac{1}{\sqrt{6}}$ units

11.5 – Equation of a Plane

Question 1

- (i) Find the direction cosines of the normal to the plane $2x + 3y - z - 7 = 0$.
 (ii) Find a unit normal vector to the plane $2x - 2y - z - 5 = 0$.

TYPE 42 Direction Cosines / Unit Normal of a Plane



Normal vector perpendicular to a plane

Given: For (i), plane $2x + 3y - z - 7 = 0$. For (ii), plane $2x - 2y - z - 5 = 0$.

Concept / Formula Used: $Ax + By + Cz + D = 0$ $\langle A, B, C \rangle$ $\left(\frac{A}{\sqrt{A^2+B^2+C^2}}, \frac{B}{\sqrt{A^2+B^2+C^2}}, \frac{C}{\sqrt{A^2+B^2+C^2}} \right)$ $\hat{n} = \frac{\vec{n}}{ \vec{n} }$
--

Solution:

Part (i): Comparing $2x + 3y - z - 7 = 0$ with $Ax + By + Cz + D = 0$:

$$A = 2$$

$$B = 3$$

$$C = -1$$

Magnitude of the normal vector:

$$\begin{aligned}
 \sqrt{A^2 + B^2 + C^2} &= \sqrt{2^2 + 3^2 + (-1)^2} \\
 &= \sqrt{4 + 9 + 1} \\
 &= \sqrt{14}
 \end{aligned}$$

Direction cosines (l, m, n) are:

$$\begin{aligned} l &= \frac{2}{\sqrt{14}} \\ m &= \frac{3}{\sqrt{14}} \\ n &= -\frac{1}{\sqrt{14}} \end{aligned}$$

Part (ii): Comparing $2x - 2y - z - 5 = 0$ with $Ax + By + Cz + D = 0$, the normal vector is:

$$\vec{n} = 2\hat{i} - 2\hat{j} - \hat{k}$$

Magnitude of $\vec{n}$:

$$\begin{aligned} |\vec{n}| &= \sqrt{2^2 + (-2)^2 + (-1)^2} \\ &= \sqrt{4 + 4 + 1} \\ &= \sqrt{9} \\ &= 3 \end{aligned}$$

The unit normal vector $\hat{n}$ is:

$$\begin{aligned} \hat{n} &= \frac{\vec{n}}{|\vec{n}|} \\ &= \frac{2\hat{i} - 2\hat{j} - \hat{k}}{3} \\ &= \frac{2}{3}\hat{i} - \frac{2}{3}\hat{j} - \frac{1}{3}\hat{k} \end{aligned}$$

Final Answer

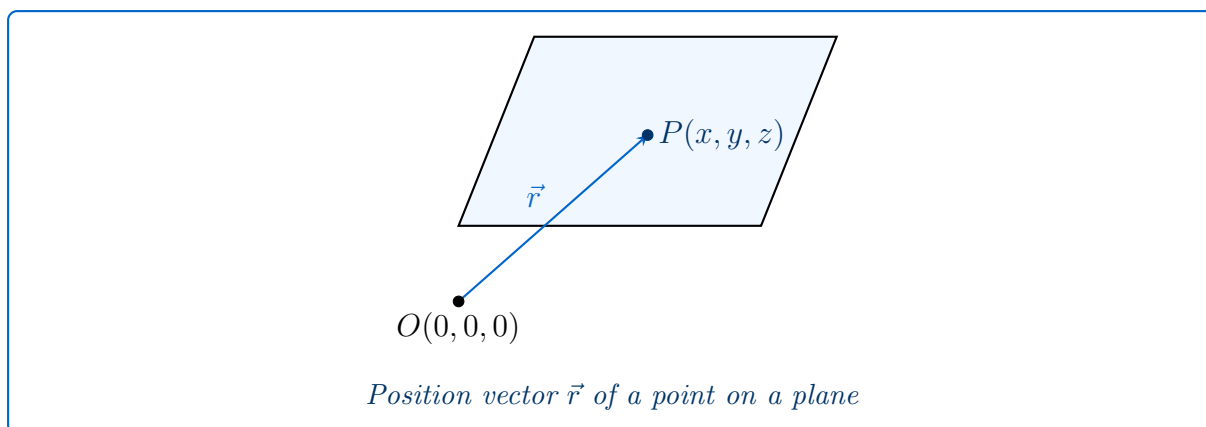
- (i) Direction cosines: $\left(\frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}, -\frac{1}{\sqrt{14}}\right)$
 (ii) Unit normal vector: $\frac{2}{3}\hat{i} - \frac{2}{3}\hat{j} - \frac{1}{3}\hat{k}$

Question 2

Find the cartesian equation of the following planes:

- (i) $\vec{r} \cdot (\hat{i} + \hat{j} - \hat{k}) = 2$
 (ii) $\vec{r} \cdot (5\hat{i} + 7\hat{j} - 11\hat{k}) = 14$

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction



Given: Vector equations of planes in the form $\vec{r} \cdot \vec{n} = d$.

Concept / Formula Used: $\vec{r} \cdot \vec{n} = d$ $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Solution:

Let the position vector of any point $P(x, y, z)$ on the plane be:

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\begin{aligned} \text{Part (i): Substitute } \vec{r} \text{ into } \vec{r} \cdot (\hat{i} + \hat{j} - \hat{k}) &= 2: & (x\hat{i} + y\hat{j} + z\hat{k}) \cdot (\hat{i} + \hat{j} - \hat{k}) &= 2 \\ & & x(1) + y(1) + z(-1) &= 2 \\ & & x + y - z &= 2 \end{aligned}$$

$$\begin{aligned} \text{Part (ii): Substitute } \vec{r} \text{ into } \vec{r} \cdot (5\hat{i} + 7\hat{j} - 11\hat{k}) &= 14: & (x\hat{i} + y\hat{j} + z\hat{k}) \cdot (5\hat{i} + 7\hat{j} - 11\hat{k}) &= 14 \\ & & x(5) + y(7) + z(-11) &= 14 \\ & & 5x + 7y - 11z &= 14 \end{aligned}$$

Final Answer

- (i) Cartesian equation: $x + y - z = 2$
 (ii) Cartesian equation: $5x + 7y - 11z = 14$

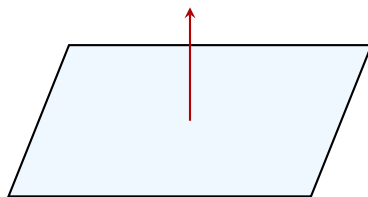
Question 3

Find the vector equations of the following planes:

- (i) $3x + 7y - 6z = 5$
 (ii) $9x - 2y - 5z + 3 = 0$

TYPE 16 Cartesian $\leftrightarrow$ Vector Line Equation – Conversion/Extraction

$$\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$$



Vector equation setup for plane

Given: Cartesian equations of planes.

Concept / Formula Used: $Ax + By + Cz = D$ $\vec{r} \cdot (A\hat{i} + B\hat{j} + C\hat{k}) = D$ $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Solution:

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

Part (i): Given equation:

$$3x + 7y - 6z = 5$$

Rewriting as a dot product: $(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (3\hat{i} + 7\hat{j} - 6\hat{k}) = 5$
 $\vec{r} \cdot (3\hat{i} + 7\hat{j} - 6\hat{k}) = 5$

Part (ii): Given equation:

$$9x - 2y - 5z + 3 = 0$$

Transposing the constant term to the right-hand side:

$$9x - 2y - 5z = -3$$

Rewriting as a dot product: $(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (9\hat{i} - 2\hat{j} - 5\hat{k}) = -3$
 $\vec{r} \cdot (9\hat{i} - 2\hat{j} - 5\hat{k}) = -3$

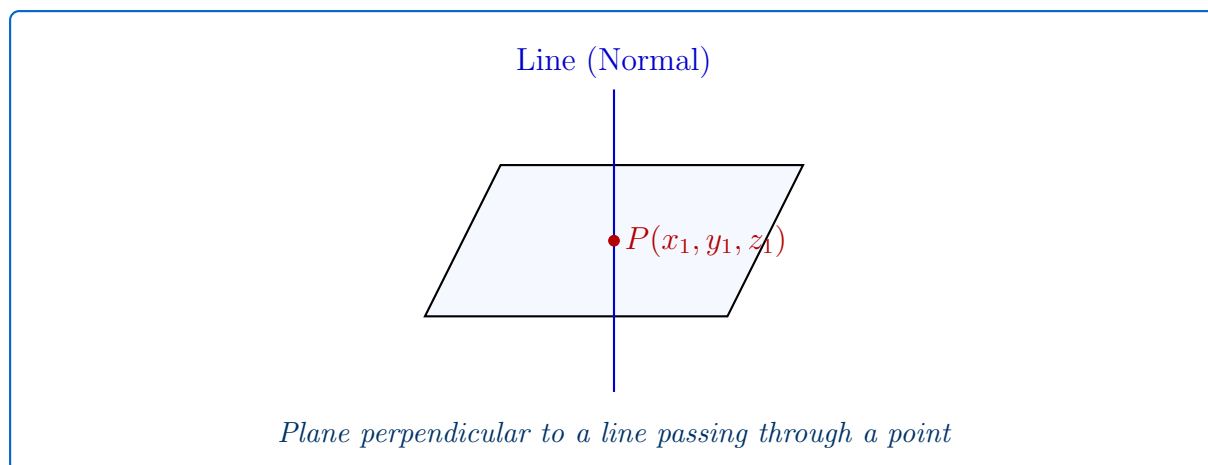
Final Answer

- (i) Vector equation: $\vec{r} \cdot (3\hat{i} + 7\hat{j} - 6\hat{k}) = 5$
 (ii) Vector equation: $\vec{r} \cdot (9\hat{i} - 2\hat{j} - 5\hat{k}) = -3$

Question 4

- (i) Find the equation of the plane passing through the point $(2, -3, 1)$ and perpendicular to the line whose direction ratios are $\langle 3, -1, 5 \rangle$.
 (ii) Find the equation of a plane which bisects the line segment joining the points $A(2, 3, 5)$ and $B(4, 5, 7)$ at right angles.
 (iii) Find the equation of the plane passing through the point $(4, -1, 2)$ and perpendicular to the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$.

TYPE 43 Equation of a Plane – Point & Normal Given



Given: (i) Point $(2, -3, 1)$, normal DRs $\langle 3, -1, 5 \rangle$.
(ii) Segment endpoints $A(2, 3, 5)$ and $B(4, 5, 7)$.
(iii) Point $(4, -1, 2)$, perpendicular line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$.

Concept / Formula Used: (x_1, y_1, z_1) $\langle A, B, C \rangle$ $A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$

Solution:

Part (i): Point $(x_1, y_1, z_1) = (2, -3, 1)$ and normal DRs $\langle A, B, C \rangle = \langle 3, -1, 5 \rangle$. Using

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$$

$$3(x - 2) + (-1)(y - (-3)) + 5(z - 1) = 0$$

$$3(x - 2) - 1(y + 3) + 5(z - 1) = 0$$

the plane equation formula:

$$3x - 6 - y - 3 + 5z - 5 = 0$$

$$3x - y + 5z - 14 = 0$$

$$3x - y + 5z = 14$$

Part (ii): The plane bisects AB at right angles, so it passes through the midpoint M of AB , and vector $\vec{AB}$ is normal to the plane. Midpoint M :

$$M = \left(\frac{2+4}{2}, \frac{3+5}{2}, \frac{5+7}{2} \right) = (3, 4, 6)$$

$$A = 4 - 2 = 2$$

Direction ratios of the normal (line AB): $B = 5 - 3 = 2$ Dividing by 2, the simplified

$$C = 7 - 5 = 2$$

$$1(x - 3) + 1(y - 4) + 1(z - 6) = 0$$

$$x - 3 + y - 4 + z - 6 = 0$$

direction ratios are $\langle 1, 1, 1 \rangle$. Equation of the plane:

$$x + y + z - 13 = 0$$

$$x + y + z = 13$$

Part (iii): The plane is perpendicular to the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$. Therefore, the direction ratios of the normal to the plane are the direction ratios of the line:

$$\langle A, B, C \rangle = \langle 2, 3, 4 \rangle$$

$$\begin{aligned}
 &2(x-4) + 3(y-(-1)) + 4(z-2) \\
 &2(x-4) + 3(y+1) + 4(z-2) \\
 &2x - 8 + 3y + 3 + 4z - 8 \\
 &2x + 3y + 4z - 13
 \end{aligned}$$

The plane passes through $(x_1, y_1, z_1) = (4, -1, 2)$. Equation of the plane:

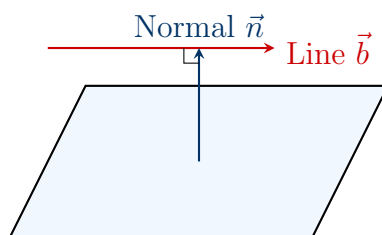
Final Answer

- (i) $3x - y + 5z = 14$
- (ii) $x + y + z = 13$
- (iii) $2x + 3y + 4z = 13$

Question 5

- (i) Show that the line $\frac{x-2}{1} = \frac{y+2}{-1} = \frac{z-3}{4}$ is parallel to the plane $x+5y+z=7$.
- (ii) If the line $\frac{x+1}{3} = \frac{y-2}{4} = \frac{z+6}{5}$ is parallel to the plane $2x-3y+kz=0$, then find the value of k .

TYPE 45 Line Parallel to a Plane – Condition Check



Line parallel to plane (perpendicular to normal)

- Given:** (i) Line $\frac{x-2}{1} = \frac{y+2}{-1} = \frac{z-3}{4}$ and plane $x+5y+z=7$.
(ii) Line $\frac{x+1}{3} = \frac{y-2}{4} = \frac{z+6}{5}$ and plane $2x-3y+kz=0$.

Concept / Formula Used: $\vec{b} = a\hat{i} + b\hat{j} + c\hat{k}$ $\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$ $\vec{b} \cdot \vec{n} = aA + bB + cC = 0$

Solution:

Part (i): Direction ratios of the line are $\langle a, b, c \rangle = \langle 1, -1, 4 \rangle$. Direction ratios of the normal to the plane $x+5y+z=7$ are $\langle A, B, C \rangle = \langle 1, 5, 1 \rangle$. Evaluating the dot product

$$\begin{aligned}
 aA + bB + cC &= (1)(1) + (-1)(5) + (4)(1) \\
 &= 1 - 5 + 4 = 0
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= 2 + 5(-2) + 3 \\
 &= 2 - 10 + 3 = -5 \neq 7
 \end{aligned}$$

on the line lies on the plane $x+5y+z=7$: Since $aA + bB + cC = 0$

and the point does not lie on the plane, the line is strictly parallel to the plane.

Part (ii): Direction ratios of the line are $\langle a, b, c \rangle = \langle 3, 4, 5 \rangle$. Direction ratios of the normal to the plane $2x - 3y + kz = 0$ are $\langle A, B, C \rangle = \langle 2, -3, k \rangle$. Since the line is parallel

$$aA + bB + cC = 0$$

$$(3)(2) + (4)(-3) + (5)(k) = 0$$

$$6 - 12 + 5k = 0$$

to the plane, $\vec{b} \cdot \vec{n} = 0$:

$$-6 + 5k = 0$$

$$5k = 6$$

$$k = \frac{6}{5}$$

Final Answer

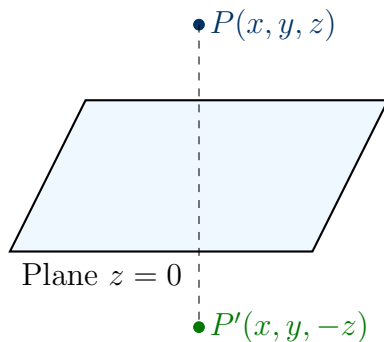
(i) Line is parallel to plane (LHS = 0 and point not on plane).

(ii) $k = \frac{6}{5}$

Question 6

- Write the image of the point $(-2, 3, 5)$ in the plane XOY.
- Write the image of the point $(-4, -3, 6)$ in the plane YOZ.
- Write the coordinates of the point which is the reflection of the point (α, β, γ) in the xz-plane.
- Write the image of the point $(2, -3, -1)$ in the plane $\vec{r} \cdot \hat{j} = 0$.

TYPE 46 Image (Reflection) of a Point in a Plane



Reflection of point across coordinate plane

Given: Points to reflect in coordinate planes: XOY plane ($z = 0$), YOZ plane ($x = 0$), XZ plane ($y = 0$), and plane $\vec{r} \cdot \hat{j} = 0$ ($y = 0$).

Concept / Formula Used:

(x, y, z)	$z = 0$	$(x, y, -z)$	$x = 0$	$(-x, y, z)$	$y = 0$	$(x, -y, z)$
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Solution:

Part (i): The plane XOY has equation $z = 0$. The reflection negates the z -coordinate:

$$(-2, 3, 5) \longrightarrow (-2, 3, -5)$$

Part (ii): The plane YOZ has equation $x = 0$. The reflection negates the x -coordinate:

$$(-4, -3, 6) \longrightarrow (4, -3, 6)$$

Part (iii): The xz -plane has equation $y = 0$. The reflection negates the y -coordinate:

$$(\alpha, \beta, \gamma) \longrightarrow (\alpha, -\beta, \gamma)$$

Part (iv): The equation $\vec{r} \cdot \hat{j} = 0$ is equivalent to $(x\hat{i} + y\hat{j} + z\hat{k}) \cdot \hat{j} = 0 \implies y = 0$ (the ZX plane). The reflection negates the y -coordinate:

$$(2, -3, -1) \longrightarrow (2, 3, -1)$$

Final Answer

(i) $(-2, 3, -5)$

(ii) $(4, -3, 6)$

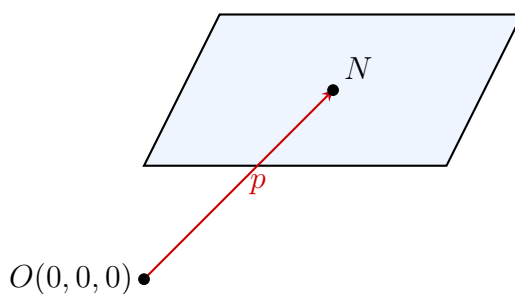
(iii) $(\alpha, -\beta, \gamma)$

(iv) $(2, 3, -1)$

Question 7

If the vector equation of a plane is $\vec{r} \cdot (2\hat{i} + 2\hat{j} - \hat{k}) = 21$, then find the length of perpendicular from the origin to the plane.

TYPE 47 Perpendicular Distance from Origin to a Plane



Perpendicular distance from origin to a plane

Given: Plane equation $\vec{r} \cdot (2\hat{i} + 2\hat{j} - \hat{k}) = 21$.

Concept / Formula Used: $\vec{r} \cdot \vec{n} = d$ $p = \frac{|d|}{|\vec{n}|}$

Solution:

Comparing $\vec{r} \cdot (2\hat{i} + 2\hat{j} - \hat{k}) = 21$ with $\vec{r} \cdot \vec{n} = d$: $\vec{n} = 2\hat{i} + 2\hat{j} - \hat{k}$ Magnitude of $\vec{n}$:
 $d = 21$

$$\begin{aligned} |\vec{n}| &= \sqrt{2^2 + 2^2 + (-1)^2} \\ &= \sqrt{4 + 4 + 1} \\ &= \sqrt{9} \\ &= 3 \end{aligned}$$

Perpendicular distance p from the origin:

$$\begin{aligned} p &= \frac{d}{|\vec{n}|} \\ &= \frac{21}{3} \\ &= 7 \text{ units} \end{aligned}$$

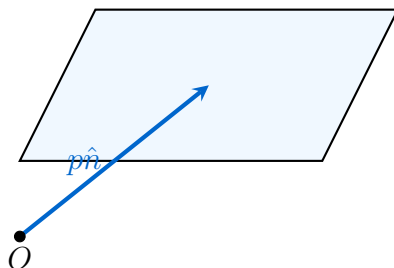
Final Answer

Length of perpendicular from origin = 7 units

Question 8

- (i) Find the vector equation of the plane which is at a distance of $\frac{6}{\sqrt{29}}$ units from the origin and its normal vector from the origin is $2\hat{i} - 3\hat{j} + 4\hat{k}$. Also find its cartesian equation.
- (ii) Find the vector equation of a plane which is at a distance of 5 units from the origin and has $\langle 2, -1, 2 \rangle$ as the direction numbers of a normal to the plane.

TYPE 44 Equation of a Plane – Normal Form (Given Distance & Direction)



Normal form of a plane

Given: (i) $p = \frac{6}{\sqrt{29}}$, normal vector $\vec{n} = 2\hat{i} - 3\hat{j} + 4\hat{k}$.
(ii) $p = 5$, normal DRs $\langle 2, -1, 2 \rangle$.

Concept / Formula Used: $\vec{r} \cdot \hat{n} = p$ $\hat{n} = \frac{\vec{n}}{|\vec{n}|}$ p

Solution:

Part (i): Normal vector $\vec{n} = 2\hat{i} - 3\hat{j} + 4\hat{k}$. Magnitude of $\vec{n}$: $|\vec{n}| = \sqrt{2^2 + (-3)^2 + 4^2} = \sqrt{4 + 9 + 16} = \sqrt{29}$ Unit

normal vector $\hat{n}$:

$$\hat{n} = \frac{2\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{29}}$$

Given distance $p = \frac{6}{\sqrt{29}}$. Vector equation $\vec{r} \cdot \hat{n} = p$: $\vec{r} \cdot \left(\frac{2\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{29}} \right) = \frac{6}{\sqrt{29}}$ Substituting $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ gives the Cartesian equation:

$$2x - 3y + 4z = 6$$

Part (ii): Normal vector $\vec{n} = 2\hat{i} - \hat{j} + 2\hat{k}$. Magnitude of $\vec{n}$:

$$\begin{aligned} |\vec{n}| &= \sqrt{2^2 + (-1)^2 + 2^2} \\ &= \sqrt{4 + 1 + 4} \\ &= \sqrt{9} \\ &= 3 \end{aligned} \quad \text{Unit}$$

normal vector $\hat{n}$:

$$\hat{n} = \frac{2\hat{i} - \hat{j} + 2\hat{k}}{3}$$

Given distance $p = 5$. Vector equation $\vec{r} \cdot \hat{n} = p$: $\vec{r} \cdot \left(\frac{2\hat{i} - \hat{j} + 2\hat{k}}{3} \right) = 5$
 $\vec{r} \cdot (2\hat{i} - \hat{j} + 2\hat{k}) = 15$

Final Answer

(i) Vector eq: $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 4\hat{k}) = 6$, Cartesian eq: $2x - 3y + 4z = 6$

(ii) Vector eq: $\vec{r} \cdot (2\hat{i} - \hat{j} + 2\hat{k}) = 15$

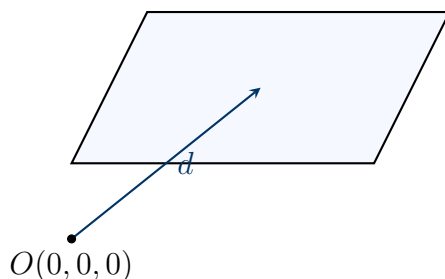
Question 9

In each of the following problems, determine the direction cosines of the normal to the plane and its distance from the origin:

(i) $2x - 3y + 4z - 6 = 0$

(ii) $3x + 5 = 0$

TYPE 42 Direction Cosines / Unit Normal of a Plane



Plane in normal form showing distance from origin

Given: (i) Plane $2x - 3y + 4z - 6 = 0$.

(ii) Plane $3x + 5 = 0$.

Concept / Formula Used: $Ax + By + Cz = D$ $D > 0$ $lx + my + nz = p$
 $\frac{Ax + By + Cz}{\sqrt{A^2 + B^2 + C^2}} = \frac{D}{\sqrt{A^2 + B^2 + C^2}}$ (l, m, n) p

Solution:

Part (i): Rewrite equation with positive RHS:

$$2x - 3y + 4z = 6$$

Divide both sides by $\sqrt{A^2 + B^2 + C^2} = \sqrt{2^2 + (-3)^2 + 4^2} = \sqrt{4 + 9 + 16} = \sqrt{29}$:

$$\frac{2}{\sqrt{29}}x - \frac{3}{\sqrt{29}}y + \frac{4}{\sqrt{29}}z = \frac{6}{\sqrt{29}}$$

Direction cosines $(l, m, n) = \left(\frac{2}{\sqrt{29}}, -\frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}} \right)$
 Comparing with normal form $lx + my + nz = p$:
 Distance $p = \frac{6}{\sqrt{29}}$ units

Part (ii): Rewrite $3x + 5 = 0$ as:

$$-3x = 5 \implies -3x + 0y + 0z = 5$$

Divide both sides by $\sqrt{(-3)^2 + 0^2 + 0^2} = \sqrt{9} = 3$: $-\frac{3}{3}x + \frac{0}{3}y + \frac{0}{3}z = \frac{5}{3}$ Comparing with
 $-1x + 0y + 0z = \frac{5}{3}$
 Direction cosines $(l, m, n) = (-1, 0, 0)$
 normal form $lx + my + nz = p$:
 Distance $p = \frac{5}{3}$ units

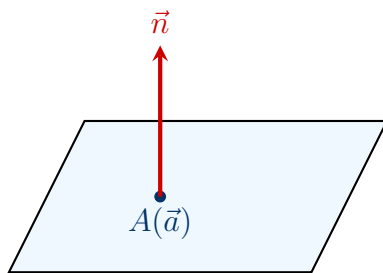
Final Answer

- (i) Direction cosines: $\left(\frac{2}{\sqrt{29}}, -\frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}} \right)$, Distance: $\frac{6}{\sqrt{29}}$ units
 (ii) Direction cosines: $(-1, 0, 0)$, Distance: $\frac{5}{3}$ units

Question 10

- (i) Find the vector and the cartesian equations of the plane that passes through the point $(1, 0, -2)$ and normal vector to the plane is $\hat{i} + \hat{j} - \hat{k}$.
 (ii) Find the vector and the cartesian equations of a plane through the point with position vector $2\hat{i} - \hat{j} + \hat{k}$ and perpendicular to the vector $4\hat{i} + 2\hat{j} - 3\hat{k}$.

TYPE 43 Equation of a Plane – Point & Normal Given



Plane passing through point $\vec{a}$ perpendicular to $\vec{n}$

Given: (i) Point $\vec{a} = \hat{i} - 2\hat{k}$, normal $\vec{n} = \hat{i} + \hat{j} - \hat{k}$.

(ii) Point $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, normal $\vec{n} = 4\hat{i} + 2\hat{j} - 3\hat{k}$.

Concept / Formula Used: $\vec{a} \quad \vec{n} \quad (\vec{r} - \vec{a}) \cdot \vec{n} = 0 \implies \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$

Solution:

$$\vec{a} \cdot \vec{n} = (1)(1) + (0)(1) + (-2)(-1)$$

Part (i): Given $\vec{a} = \hat{i} + 0\hat{j} - 2\hat{k}$ and $\vec{n} = \hat{i} + \hat{j} - \hat{k}$. Calculate $\vec{a} \cdot \vec{n}$:

$$= 1 + 0 + 2$$

$$= 3$$

Vector equation $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$:

$$\vec{r} \cdot (\hat{i} + \hat{j} - \hat{k}) = 3$$

Substitute $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ to get Cartesian equation:

$$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (\hat{i} + \hat{j} - \hat{k}) = 3 \implies x + y - z = 3$$

$$\vec{a} \cdot \vec{n} = (2)(4) + (-1)(2) + (1)(-3)$$

Part (ii): Given $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{n} = 4\hat{i} + 2\hat{j} - 3\hat{k}$. Calculate $\vec{a} \cdot \vec{n}$:

$$= 8 - 2 - 3$$

$$= 3$$

Vector equation $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$:

$$\vec{r} \cdot (4\hat{i} + 2\hat{j} - 3\hat{k}) = 3$$

Substitute $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ to get Cartesian equation:

$$4x + 2y - 3z = 3$$

Final Answer

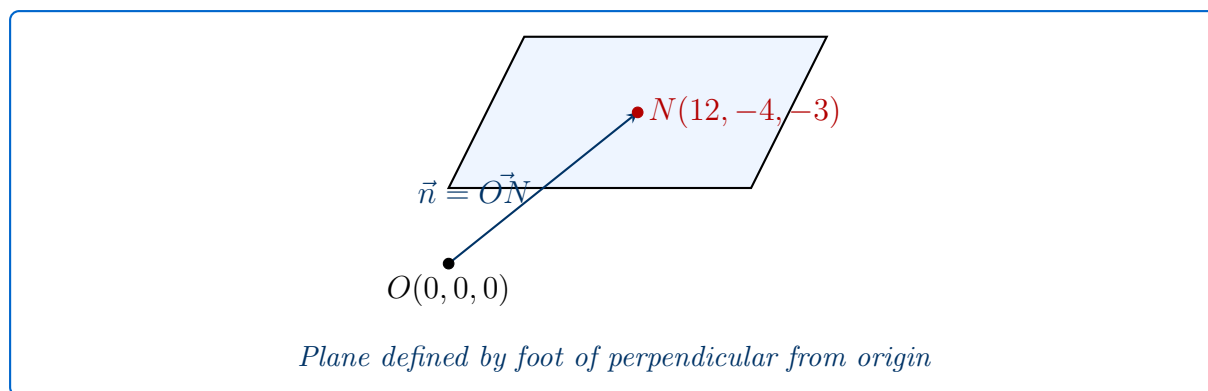
(i) Vector: $\vec{r} \cdot (\hat{i} + \hat{j} - \hat{k}) = 3$, Cartesian: $x + y - z = 3$

(ii) Vector: $\vec{r} \cdot (4\hat{i} + 2\hat{j} - 3\hat{k}) = 3$, Cartesian: $4x + 2y - 3z = 3$

Question 11

- If the foot of perpendicular drawn from the origin to a plane is $(12, -4, -3)$, find the equation of the plane in vector as well as cartesian form.
- Find the equation of the plane passing through the point $(2, -3, 1)$ and perpendicular to the line joining the points $(4, 5, 0)$ and $(1, -2, 4)$.

TYPE 48 Plane via Foot of Perpendicular from the Origin



Given: (i) Foot of perpendicular $N(12, -4, -3)$ from origin.
(ii) Point $(2, -3, 1)$, perpendicular line passing through $A(4, 5, 0)$ and $B(1, -2, 4)$.

Concept / Formula Used: $N(\vec{n})$ $\vec{n}$ N $\vec{r} \cdot \vec{n} = |\vec{n}|^2$ A B $\langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$

Solution:

Part (i): Let $N(12, -4, -3)$ be the foot of perpendicular from origin $O(0, 0, 0)$. Normal vector $\vec{n} = \vec{ON} = 12\hat{i} - 4\hat{j} - 3\hat{k}$. Since the plane passes through N , its vector equation is:

$$\vec{r} \cdot \vec{n} = \vec{ON} \cdot \vec{n} = |\vec{n}|^2$$

$$|\vec{n}|^2 = 12^2 + (-4)^2 + (-3)^2$$

$$\begin{aligned} \text{Calculate } |\vec{n}|^2: &= 144 + 16 + 9 && \text{Vector equation:} \\ &= 169 \end{aligned}$$

$$\vec{r} \cdot (12\hat{i} - 4\hat{j} - 3\hat{k}) = 169$$

Cartesian equation:

$$12x - 4y - 3z = 169$$

$$A = 1 - 4 = -3$$

Part (ii): Direction ratios of normal line AB joining $A(4, 5, 0)$ and $B(1, -2, 4)$: $B = -2 - 5 = -7$

$$C = 4 - 0 = 4$$

Plane passes through $(x_1, y_1, z_1) = (2, -3, 1)$. Applying $A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$

$$-3(x - 2) - 7(y - (-3)) + 4(z - 1) = 0$$

$$-3(x - 2) - 7(y + 3) + 4(z - 1) = 0$$

$$0: \quad -3x + 6 - 7y - 21 + 4z - 4 = 0$$

$$-3x - 7y + 4z - 19 = 0$$

$$3x + 7y - 4z + 19 = 0$$

Final Answer

(i) Vector: $\vec{r} \cdot (12\hat{i} - 4\hat{j} - 3\hat{k}) = 169$, Cartesian: $12x - 4y - 3z = 169$

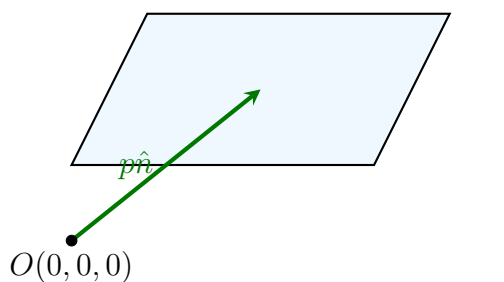
(ii) $3x + 7y - 4z + 19 = 0$

Question 12

What is the length of perpendicular drawn from origin to the plane $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) + 35 = 0$? Also write a unit vector normal to this plane directed from the origin to the plane.

TYPE 48a Foot of Perpendicular from Origin, Plus Unit Normal

★ Likely Exam Question



Plane distance with outward/inward normal

Given: Plane equation $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) + 35 = 0$.

Concept / Formula Used: $\vec{r} \cdot \hat{n} = p$ $p > 0$ $\vec{r} \cdot \vec{n} = -d$ $\vec{r} \cdot (-\vec{n}) = d$

Solution:

Given plane equation:

$$\vec{r} \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) = -35$$

Multiply both sides by -1 so that the right-hand side constant is positive:

$$\vec{r} \cdot (-2\hat{i} + 3\hat{j} - 6\hat{k}) = 35$$

$$|\vec{n}| = \sqrt{(-2)^2 + 3^2 + (-6)^2}$$

Here, $\vec{n} = -2\hat{i} + 3\hat{j} - 6\hat{k}$. Magnitude of $\vec{n}$:

$$= \sqrt{4 + 9 + 36}$$

$$= \sqrt{49}$$

$$= 7$$

Dividing the plane

equation by 7:

$$\vec{r} \cdot \left(\frac{-2\hat{i} + 3\hat{j} - 6\hat{k}}{7} \right) = \frac{35}{7}$$

$$\vec{r} \cdot \left(-\frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} - \frac{6}{7}\hat{k} \right) = 5$$

$$p = 5 \text{ units}$$

Comparing with $\vec{r} \cdot \hat{n} = p$:

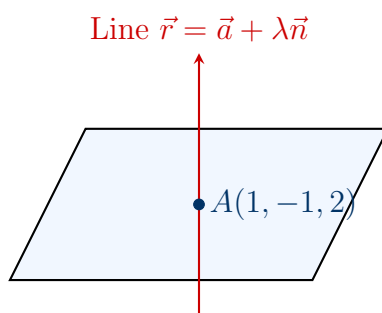
$$\hat{n} = -\frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} - \frac{6}{7}\hat{k}$$

Final Answer

Perpendicular distance = 5 units

Unit normal vector = $-\frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} - \frac{6}{7}\hat{k}$ **Question 13**

- (i) Find the vector equation of the line passing through the point $(1, -1, 2)$ and perpendicular to the plane $2x - y + 3z = 5$.
- (ii) Find the equation of the line passing through the point with position vector $\hat{i} - \hat{j} + 2\hat{k}$ and perpendicular to the plane $\vec{r} \cdot (2\hat{i} - \hat{j} + 3\hat{k}) = 5$.

TYPE 49 Equation of a Line Perpendicular to a Plane*Line perpendicular to plane along normal vector*

- Given:** (i) Point $(1, -1, 2)$, plane $2x - y + 3z = 5$.
- (ii) Point $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, plane $\vec{r} \cdot (2\hat{i} - \hat{j} + 3\hat{k}) = 5$.

Concept / Formula Used: $\vec{r} \cdot \vec{n} = d$ $\vec{n}$ $\vec{a}$ $\vec{r} = \vec{a} + \lambda \vec{n}$ **Solution:****Part (i):** Position vector of given point:

$$\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$$

Plane equation $2x - y + 3z = 5$ has normal vector:

$$\vec{n} = 2\hat{i} - \hat{j} + 3\hat{k}$$

Since the line is perpendicular to the plane, its direction vector is $\vec{n}$. Vector equation ofthe line: $\vec{r} = \vec{a} + \lambda \vec{n}$

$$\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(2\hat{i} - \hat{j} + 3\hat{k})$$

Part (ii): Given point position vector $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$. Normal to the plane $\vec{r} \cdot (2\hat{i} - \hat{j} + 3\hat{k}) = 5$ is:

$$\vec{n} = 2\hat{i} - \hat{j} + 3\hat{k}$$

Vector equation of line: $\vec{r} = \vec{a} + \lambda \vec{n}$

$$\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(2\hat{i} - \hat{j} + 3\hat{k})$$

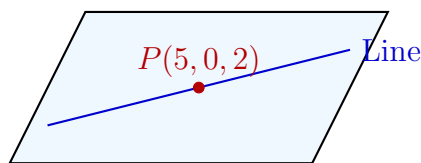
Final Answer

(i) $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(2\hat{i} - \hat{j} + 3\hat{k})$

(ii) $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(2\hat{i} - \hat{j} + 3\hat{k})$

Question 14

Show that the plane $x - 5y - 2z = 1$ contains the line $\frac{x-5}{3} = y = 2-z$.

TYPE 45a Line Lies in a Plane – Proof

Line lying entirely inside a plane

To Prove: The line $\frac{x-5}{3} = \frac{y-0}{1} = \frac{z-2}{-1}$ lies in the plane $x - 5y - 2z = 1$.

Concept / Formula Used: $\vec{b} \cdot \vec{n} = 0$

Solution:

Rewrite the equation of the line in standard symmetric form:

$$\frac{x-5}{3} = \frac{y-0}{1} = \frac{z-2}{-1}$$

From the symmetric form: Point on the line $P = (5, 0, 2)$
 Direction ratios of line $\vec{b} = \langle 3, 1, -1 \rangle$ Direction ratios of the
 normal to plane $x - 5y - 2z = 1$:

$$\vec{n} = \langle 1, -5, -2 \rangle$$

$$\begin{aligned}\vec{b} \cdot \vec{n} &= (3)(1) + (1)(-5) + (-1)(-2) \\ &= 3 - 5 + 2 \\ &= 0\end{aligned}$$

Step 1: Check perpendicularity of direction vectors.

Thus, the line is parallel to or lies in the plane.

Step 2: Check if point $P(5, 0, 2)$ lies on the plane. Substitute $x = 5, y = 0, z = 2$
 $\text{LHS} = 5 - 5(0) - 2(2)$

$$\begin{aligned}\text{into } x - 5y - 2z: &= 5 - 0 - 4 \\ &= 1 \\ &= \text{RHS}\end{aligned}$$

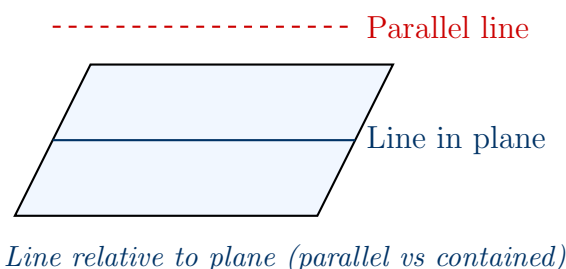
the plane, the line lies completely in the plane.

Hence Proved

The plane $x - 5y - 2z = 1$ contains the given line. Hence proved.

Question 15

- (i) Show that the line $\vec{r} = \hat{i} + \hat{j} + \lambda(2\hat{i} + \hat{j} + 4\hat{k})$ lies in the plane $\vec{r} \cdot (\hat{i} + 2\hat{j} - \hat{k}) = 3$.
 (ii) If the line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lies in the plane $x + 3y - \alpha z + \beta = 0$, then find the values of α and β .
 (iii) Show that the line $\vec{r} = (2\hat{i} - 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + 4\hat{k})$ is parallel to the plane $\vec{r} \cdot (\hat{i} + 5\hat{j} + \hat{k}) = 7$.

TYPE 45a Line Lies in a Plane – Proof

- Given:** (i) Line $\vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} + \hat{j} + 4\hat{k})$, plane $\vec{r} \cdot (\hat{i} + 2\hat{j} - \hat{k}) = 3$.
 (ii) Line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$, plane $x + 3y - \alpha z + \beta = 0$.
 (iii) Line $\vec{r} = (2\hat{i} - 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + 4\hat{k})$, plane $\vec{r} \cdot (\hat{i} + 5\hat{j} + \hat{k}) = 7$.

Concept / Formula Used: $\vec{r} = \vec{a} + \lambda\vec{b}$ $\vec{r} \cdot \vec{n} = d$ $\vec{b} \cdot \vec{n} = 0$ $\vec{a} \cdot \vec{n} = d$ $\vec{b} \cdot \vec{n} \neq 0$ $\vec{a} \cdot \vec{n} \neq d$

Solution:

Part (i): Line has point $\vec{a} = \hat{i} + \hat{j}$ and direction vector $\vec{b} = 2\hat{i} + \hat{j} + 4\hat{k}$. Plane normal $\vec{b} \cdot \vec{n} = (2\hat{i} + \hat{j} + 4\hat{k}) \cdot (\hat{i} + 2\hat{j} - \hat{k})$

$$\begin{aligned} \vec{n} = \hat{i} + 2\hat{j} - \hat{k} \text{ and constant } d = 3. \quad &= (2)(1) + (1)(2) + (4)(-1) \quad \text{Check point } \vec{a}: \\ &= 2 + 2 - 4 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \vec{a} \cdot \vec{n} &= (\hat{i} + \hat{j}) \cdot (\hat{i} + 2\hat{j} - \hat{k}) \\ &= (1)(1) + (1)(2) + (0)(-1) \\ &= 1 + 2 \\ &= 3 \\ &= d \end{aligned}$$

Thus, the line lies completely in the plane.

Part (ii): Line passes through $A(2, 1, -2)$ with direction $\vec{b} = \langle 3, -5, 2 \rangle$. Plane normal

$$(3)(1) + (-5)(3) + (2)(-\alpha) = 0$$

$$3 - 15 - 2\alpha = 0$$

$\vec{n} = \langle 1, 3, -\alpha \rangle$. Since line lies in plane: 1) $\vec{b} \cdot \vec{n} = 0$:

$$-12 - 2\alpha = 0 \quad 2)$$

$$2\alpha = -12$$

$$\alpha = -6$$

Point $(2, 1, -2)$ lies on $x + 3y - \alpha z + \beta = 0$: $2 + 3(1) - \alpha(-2) + \beta = 0$ Substitute $\alpha = -6$:
 $5 + 2\alpha + \beta = 0$

$$5 + 2(-6) + \beta = 0$$

$$5 - 12 + \beta = 0$$

$$-7 + \beta = 0$$

$$\beta = 7$$

Part (iii): Line has $\vec{a} = 2\hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 4\hat{k}$. Plane has $\vec{n} = \hat{i} + 5\hat{j} + \hat{k}$ and $d = 7$. Dot product $\vec{b} \cdot \vec{n}$:
 $\vec{b} \cdot \vec{n} = (1)(1) + (-1)(5) + (4)(1)$

$$= 1 - 5 + 4 = 0$$

Check point $\vec{a}$:

$$\vec{a} \cdot \vec{n} = (2\hat{i} - 2\hat{j} + 3\hat{k}) \cdot (\hat{i} + 5\hat{j} + \hat{k})$$

$$= (2)(1) + (-2)(5) + (3)(1)$$

$$= 2 - 10 + 3$$

$$= -5 \neq 7$$

plane.

Since $\vec{b} \cdot \vec{n} = 0$ and $\vec{a} \cdot \vec{n} \neq 7$, the line is parallel to the

Final Answer

(i) Line lies in plane (LHS = 0 and point satisfies plane)

(ii) $\alpha = -6, \beta = 7$

(iii) Line is parallel to plane ($\vec{b} \cdot \vec{n} = 0$ and point does not satisfy plane)

Question 16

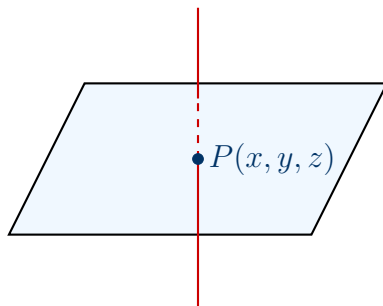
(i) Find the point where the line $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+3}{4}$ meets the plane $2x + 4y - z = 1$.

(ii) Find the coordinates of the point where the line through $(3, -4, -5)$ and $(2, -3, 1)$ crosses the plane $2x + y + z = 7$.

(iii) Find the point of intersection of line $\vec{r} = 2\hat{i} - \hat{j} + 2\hat{k} + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$.

TYPE 50 Point of Intersection of a Line and a Plane

★ Likely Exam Question



Intersection of a line and a plane

- Given:** (i) Line $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+3}{4}$, plane $2x + 4y - z = 1$.
(ii) Line through $A(3, -4, -5)$ and $B(2, -3, 1)$, plane $2x + y + z = 7$.
(iii) Line $\vec{r} = (2\hat{i} - \hat{j} + 2\hat{k}) + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})$, plane $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$.

Concept / Formula Used: λ λ

Solution:

Part (i): Let $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+3}{4} = \lambda$. General point P on the line is:

$$P(2\lambda + 1, -3\lambda + 2, 4\lambda - 3)$$

$$2(2\lambda + 1) + 4(-3\lambda + 2) - (4\lambda - 3) = 1$$

$$4\lambda + 2 - 12\lambda + 8 - 4\lambda + 3 = 1$$

$$(4 - 12 - 4)\lambda + (2 + 8 + 3) = 1$$

Substitute P into plane $2x + 4y - z = 1$:

$$-12\lambda + 13 = 1$$

$$-12\lambda = 1 - 13$$

$$-12\lambda = -12$$

$$\lambda = 1$$

$$x = 2(1) + 1 = 3$$

Substitute $\lambda = 1$ into P : $y = -3(1) + 2 = -1$ Point of intersection is $(3, -1, 1)$.

$$z = 4(1) - 3 = 1$$

Part (ii): Line passing through $A(3, -4, -5)$ and $B(2, -3, 1)$: DRs $\langle a, b, c \rangle = \langle 2 - 3, -3 - (-4), 1 - (-5) \rangle$
 $= \langle -1, 1, 6 \rangle$

Equation of line:

$$\frac{x-3}{-1} = \frac{y+4}{1} = \frac{z+5}{6} = \lambda$$

General point $Q(-\lambda + 3, \lambda - 4, 6\lambda - 5)$. Substitute Q into plane $2x + y + z = 7$:

$$2(-\lambda + 3) + (\lambda - 4) + (6\lambda - 5) = 7$$

$$-2\lambda + 6 + \lambda - 4 + 6\lambda - 5 = 7$$

$$(-2 + 1 + 6)\lambda + (6 - 4 - 5) = 7$$

$$5\lambda - 3 = 7$$

$$5\lambda = 10$$

$$\lambda = 2$$

$$x = -(2) + 3 = 1$$

Substitute $\lambda = 2$ into Q : $y = 2 - 4 = -2$ Point

$$z = 6(2) - 5 = 7$$

of intersection is $(1, -2, 7)$.

Part (iii): Substitute $\vec{r} = (2\hat{i} - \hat{j} + 2\hat{k}) + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})$ into plane $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$

$$[(2\hat{i} - \hat{j} + 2\hat{k}) + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})] \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$$

$$(2\hat{i} - \hat{j} + 2\hat{k}) \cdot (\hat{i} - \hat{j} + \hat{k}) + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k}) \cdot (\hat{i} - \hat{j} + \hat{k}) = 5$$

$$5: [(2)(1) + (-1)(-1) + (2)(1)] + \lambda[(3)(1) + (4)(-1) + (2)(1)] = 5 \quad \text{Substitute } \lambda = 0 \text{ back}$$

$$[2 + 1 + 2] + \lambda[3 - 4 + 2] = 5$$

$$5 + \lambda(1) = 5$$

$$\lambda = 0$$

into the vector equation:

$$\vec{r} = 2\hat{i} - \hat{j} + 2\hat{k}$$

The point of intersection in coordinates is $(2, -1, 2)$.

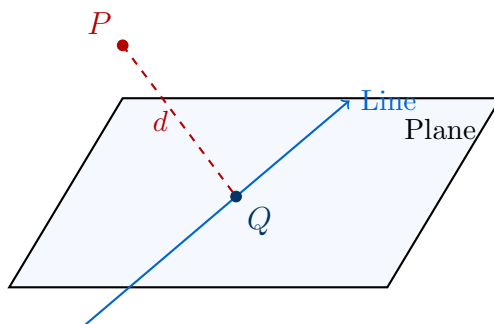
Final Answer

- (i) Point of intersection: $(3, -1, 1)$
- (ii) Point of intersection: $(1, -2, 7)$
- (iii) Point of intersection: $(2, -1, 2)$

Question 17

- (i) Find the distance between the point $(-1, -5, -10)$ and the point of intersection of the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane $x - y + z = 5$.
- (ii) Find the distance of the point $(2, 12, 5)$ from the point of intersection of the line $\vec{r} = 2\hat{i} - 4\hat{j} + 2\hat{k} + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 0$.
- (iii) Find the distance of the point $P(-2, -4, 7)$ from the point of intersection Q of the line $\vec{r} = 3\hat{i} - 2\hat{j} + 6\hat{k} + \lambda(2\hat{i} - \hat{j} + 2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 6$. Also, write the vector equation of the line PQ .

TYPE 50 Point of Intersection of a Line and a Plane



Intersection of line and plane to find distance from a point

Given: Points and equations of lines and planes for sub-parts (i), (ii), and (iii).

Concept	/	Formula	Used:	(x_1, y_1, z_1)	(x_2, y_2, z_2)	d	$=$
$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$		$\vec{a}$	$\vec{b}$	$\vec{r} = \vec{a} + \mu(\vec{b} - \vec{a})$			

Solution:

(i) Let the given line equation be: $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12} = k$ Expressing general coordinates of a point on the line in terms of k : $y = 4k - 1$ $x = 3k + 2$

Since this point lies on the plane $x - y + z = 5$

$$(3k + 2) - (4k - 1) + (12k + 2) = 5$$

$$3k + 2 - 4k + 1 + 12k + 2 = 5$$

$y + z = 5$: $11k + 5 = 5$ Substituting $k = 0$ into the general point

$$11k = 0$$

$$k = 0$$

$$x = 3(0) + 2 = 2$$

gives the intersection point Q : $y = 4(0) - 1 = -1$ So, $Q = (2, -1, 2)$. Now, calculating

$$z = 12(0) + 2 = 2$$

$$PQ = \sqrt{(2 - (-1))^2 + (-1 - (-5))^2 + (2 - (-10))^2}$$

$$= \sqrt{(3)^2 + (4)^2 + (12)^2}$$

distance between $P(-1, -5, -10)$ and $Q(2, -1, 2)$:

$$= \sqrt{9 + 16 + 144}$$

$$= \sqrt{169}$$

$$= 13 \text{ units}$$

(ii) Given vector equation of line: $\vec{r} = (2 + 3\lambda)\hat{i} + (-4 + 4\lambda)\hat{j} + (2 + 2\lambda)\hat{k}$ Given plane

$$[(2 + 3\lambda)\hat{i} + (-4 + 4\lambda)\hat{j} + (2 + 2\lambda)\hat{k}] \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 0$$

$$1(2 + 3\lambda) - 2(-4 + 4\lambda) + 2 + 2\lambda = 0$$

$$2 + 3\lambda + 8 - 8\lambda + 2 + 2\lambda = 0$$

equation: $\vec{r} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 0$ Substituting $\vec{r}$ into the plane equation:

Substituting $\lambda = 4$ into the line vector gives intersection point Q : $\vec{r}_Q = (2 + 3(4))\hat{i} + (-4 + 4(4))\hat{j} + (2 + 2(4))\hat{k}$

$$= 14\hat{i} + 12\hat{j} + 10\hat{k}$$

$$PQ = \sqrt{(14 - 2)^2 + (12 - 12)^2 + (10 - 5)^2}$$

$$= \sqrt{12^2 + 0^2 + 5^2}$$

Thus, $Q = (14, 12, 10)$. Given point is $P(2, 12, 5)$. Distance PQ :

$$= \sqrt{144 + 25}$$

$$= \sqrt{169}$$

$$= 13 \text{ units}$$

(iii) Line equation: $\vec{r} = (3 + 2\lambda)\hat{i} + (-2 - \lambda)\hat{j} + (6 + 2\lambda)\hat{k}$ Plane equation: $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 6$

$$(3 + 2\lambda)(1) + (-2 - \lambda)(-1) + (6 + 2\lambda)(1) = 6$$

$$3 + 2\lambda + 2 + \lambda + 6 + 2\lambda = 6$$

Substituting $\vec{r}$ into plane equation:

$$5\lambda + 11 = 6 \quad \text{Posi-}$$

$$5\lambda = -5$$

$$\lambda = -1$$

tion vector of intersection point Q : $\vec{r}_Q = (3 + 2(-1))\hat{i} + (-2 - (-1))\hat{j} + (6 + 2(-1))\hat{k}$ So,
 $= \hat{i} - \hat{j} + 4\hat{k}$

$$PQ = \sqrt{(1 - (-2))^2 + (-1 - (-4))^2 + (4 - 7)^2}$$

$$= \sqrt{3^2 + 3^2 + (-3)^2}$$

$Q = (1, -1, 4)$. Position vector of P is $\vec{p} = -2\hat{i} - 4\hat{j} + 7\hat{k}$. Distance PQ :

$$= \sqrt{9 + 9 + 9}$$

$$= \sqrt{27}$$

$$= 3\sqrt{3} \text{ units}$$

Vector equation of line PQ : Position vector of P is $\vec{a} = -2\hat{i} - 4\hat{j} + 7\hat{k}$. Direction vec-

tor $\vec{b} = \vec{r}_Q - \vec{a}$: $\vec{b} = (1 - (-2))\hat{i} + (-1 - (-4))\hat{j} + (4 - 7)\hat{k}$ Thus, the vector equation of
 $= 3\hat{i} + 3\hat{j} - 3\hat{k}$

line PQ is: $\vec{r} = (-2\hat{i} - 4\hat{j} + 7\hat{k}) + \mu(3\hat{i} + 3\hat{j} - 3\hat{k})$ Dividing direction vector by scalar
 3: $\vec{r} = (-2\hat{i} - 4\hat{j} + 7\hat{k}) + \mu(\hat{i} + \hat{j} - \hat{k})$

Final Answer

(i) Distance = 13 units

(ii) Distance = 13 units

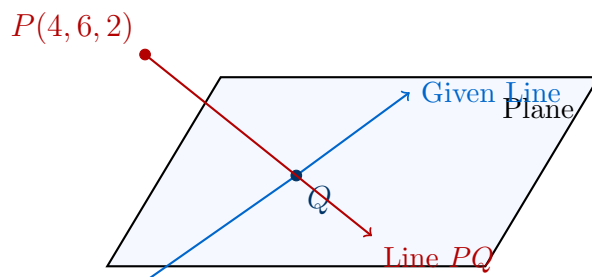
(iii) Distance = $3\sqrt{3}$ units, Line PQ : $\vec{r} = (-2\hat{i} - 4\hat{j} + 7\hat{k}) + \mu(\hat{i} + \hat{j} - \hat{k})$

Common Mistake: When substituting the point into the line equation, ensure that the sign of coordinates is properly handled during subtraction.

Question 18

Find the equation of the line passing through the point $P(4, 6, 2)$ and the point of intersection of the line $\frac{x-1}{3} = \frac{y}{2} = \frac{z+1}{7}$ and the plane $x + y - z = 8$.

TYPE 50 Point of Intersection of a Line and a Plane



Line through point P and intersection point Q of line and plane

Given: Point $P(4, 6, 2)$, Line $\frac{x-1}{3} = \frac{y}{2} = \frac{z+1}{7}$, Plane $x + y - z = 8$.

Concept / Formula Used: (x_1, y_1, z_1) (x_2, y_2, z_2) $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$

Solution:

Let the parametric equation of the given line be: $\frac{x-1}{3} = \frac{y}{2} = \frac{z+1}{7} = \lambda$ Any general

$$x = 3\lambda + 1$$

point on this line is: $y = 2\lambda$ Substitute these parametric coordinates into the plane

$$z = 7\lambda - 1$$

$$(3\lambda + 1) + (2\lambda) - (7\lambda - 1) = 8$$

$$3\lambda + 1 + 2\lambda - 7\lambda + 1 = 8$$

equation $x + y - z = 8$:

$$-2\lambda + 2 = 8 \quad \text{Substituting } \lambda = -3 \text{ back}$$

$$-2\lambda = 6$$

$$\lambda = -3$$

$$x = 3(-3) + 1 = -8$$

into the point coordinates to find point Q : $y = 2(-3) = -6$ So, the intersection

$$z = 7(-3) - 1 = -22$$

point is $Q(-8, -6, -22)$. Now, find the equation of line passing through $P(4, 6, 2)$ and

$$a = -8 - 4 = -12$$

$Q(-8, -6, -22)$. Direction ratios of line PQ : $b = -6 - 6 = -12$ Dividing by -12 , the

$$c = -22 - 2 = -24$$

simplified direction ratios are $(1, 1, 2)$. Using point $P(4, 6, 2)$, the Cartesian equation of

line PQ is: $\frac{x-4}{1} = \frac{y-6}{1} = \frac{z-2}{2}$

Final Answer

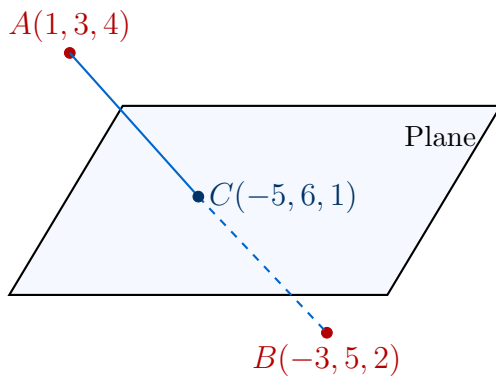
$$\frac{x-4}{1} = \frac{y-6}{1} = \frac{z-2}{2}$$

Common Mistake: Do not forget to simplify direction ratios by dividing out common factors to arrive at the simplest canonical form.

Question 19

Find the point where the line joining the points $(1, 3, 4)$ and $(-3, 5, 2)$ intersects the plane $\vec{r} \cdot (2\hat{i} + \hat{j} + \hat{k}) + 3 = 0$. Is this point equidistant from the given points?

TYPE 50 Point of Intersection of a Line and a Plane



Intersection point of line segment AB with a plane

Given: Points $A(1, 3, 4)$ and $B(-3, 5, 2)$, Plane $\vec{r} \cdot (2\hat{i} + \hat{j} + \hat{k}) + 3 = 0$.

Concept / Formula Used: $A(\vec{a})$ $B(\vec{b})$ $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$ $d = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$

Solution:

Let $A = (1, 3, 4)$ and $B = (-3, 5, 2)$. Direction vector of line AB: $\vec{d} = (-3 - 1)\hat{i} + (5 - 3)\hat{j} + (2 - 4)\hat{k}$
 $= -4\hat{i} + 2\hat{j} - 2\hat{k}$

Parametric vector equation of line AB: $\vec{r} = (\hat{i} + 3\hat{j} + 4\hat{k}) + \lambda(-4\hat{i} + 2\hat{j} - 2\hat{k})$ The given
 $= (1 - 4\lambda)\hat{i} + (3 + 2\lambda)\hat{j} + (4 - 2\lambda)\hat{k}$

plane in Cartesian form is: $2x + y + z + 3 = 0$ Substitute line coordinates into plane
 $2(1 - 4\lambda) + (3 + 2\lambda) + (4 - 2\lambda) + 3 = 0$

$$2 - 8\lambda + 3 + 2\lambda + 4 - 2\lambda + 3 = 0$$

equation: $12 - 8\lambda = 0$ Substituting $\lambda = \frac{3}{2}$ into paramet-
 $8\lambda = 12$

$$\lambda = \frac{3}{2}$$

$$x = 1 - 4\left(\frac{3}{2}\right) = 1 - 6 = -5$$

ric form gives point of intersection C: $y = 3 + 2\left(\frac{3}{2}\right) = 3 + 3 = 6$ Intersection point

$$z = 4 - 2\left(\frac{3}{2}\right) = 4 - 3 = 1$$

$C = (-5, 6, 1)$.

Now, testing if C is equidistant from $A(1, 3, 4)$ and $B(-3, 5, 2)$: Calculate distance CA:

$$CA = \sqrt{(1 - (-5))^2 + (3 - 6)^2 + (4 - 1)^2}$$

$$= \sqrt{6^2 + (-3)^2 + 3^2}$$

$$= \sqrt{36 + 9 + 9}$$

$$= \sqrt{54} = 3\sqrt{6} \text{ units}$$

$$CB = \sqrt{(-3 - (-5))^2 + (5 - 6)^2 + (-3 - 1)^2}$$

$$= \sqrt{2^2 + (-1)^2 + 1^2}$$

$$= \sqrt{4 + 1 + 1}$$

$$= \sqrt{6} \text{ units}$$

Calculate distance CB:

Comparing distances: $CA = 3\sqrt{6} \neq \sqrt{6} = CB$ Therefore, the point C is NOT equidistant

from A and B .

Final Answer

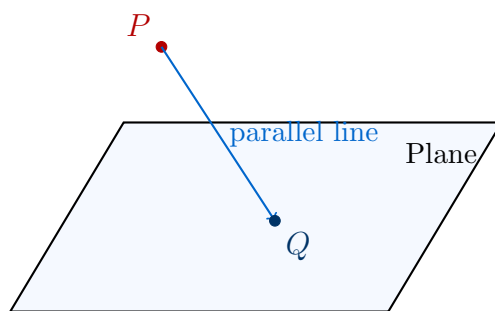
Point of intersection is $(-5, 6, 1)$. No, it is not equidistant from the given points.

Common Mistake: Be careful when expanding vector dot products into Cartesian components; keep track of negative signs.

Question 20

- (i) Find the distance of the point $(3, 4, 5)$ from the plane $x + y + z = 2$ measured parallel to the line $2x = y = z$.
- (ii) Find the distance of the point $(2, 3, 4)$ measured along the line $\frac{x-4}{3} = \frac{y+5}{6} = \frac{z+1}{2}$ from the plane $3x + 2y + 2z + 5 = 0$.

TYPE 51 Distance of a Point from a Plane, Measured Parallel to a Line



Distance from a point to a plane measured parallel to a given direction

Given: Points, target plane equations, and parallel direction lines for sub-parts (i) and (ii).

Concept / Formula Used: $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$ $P(x_1, y_1, z_1)$ Q

Solution:

(i) Given line equation: $2x = y = z$ Dividing by 2 to reduce to standard form: $\frac{x}{1} = \frac{y}{2} = \frac{z}{2}$
Direction ratios of the line are $(1, 2, 2)$. Line passing through $P(3, 4, 5)$ parallel to this

line is: $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2} = \lambda$ Any general point Q on this line is: $x = \lambda + 3$
 $y = 2\lambda + 4$ Since
 $z = 2\lambda + 5$

$$(\lambda + 3) + (2\lambda + 4) + (2\lambda + 5) = 2$$

$$5\lambda + 12 = 2$$

$$5\lambda = -10$$

$$\lambda = -2$$

Q lies on plane $x + y + z = 2$:

Substituting $\lambda = -2$

$$x = -2 + 3 = 1$$

to find coordinates of Q : $y = 2(-2) + 4 = 0$ So, $Q = (1, 0, 1)$. Calculating distance PQ :

$$z = 2(-2) + 5 = 1$$

$$\begin{aligned} PQ &= \sqrt{(3-1)^2 + (4-0)^2 + (5-1)^2} \\ &= \sqrt{2^2 + 4^2 + 4^2} \\ &= \sqrt{4 + 16 + 16} \\ &= \sqrt{36} = 6 \text{ units} \end{aligned}$$

(ii) Given line direction ratios from $\frac{x-4}{3} = \frac{y+5}{6} = \frac{z+1}{2}$ are $(3, 6, 2)$. Line through $P(2, 3, 4)$ parallel to the given line: $\frac{x-2}{3} = \frac{y-3}{6} = \frac{z-4}{2} = \lambda$ General point Q on this

$$x = 3\lambda + 2$$

$$3(3\lambda + 2) + 2(6\lambda + 3) + 2(2\lambda + 4) + 5 = 0$$

line: $y = 6\lambda + 3$ Since Q lies on plane $3x + 2y + 2z + 5 = 0$:

$$9\lambda + 6 + 12\lambda + 6 + 4\lambda + 8 + 5 = 0$$

$$25\lambda + 25 = 0$$

$$z = 2\lambda + 4$$

$$25\lambda = -25$$

$$\lambda = -1$$

$$x = 3(-1) + 2 = -1$$

Coordinates of intersection point Q : $y = 6(-1) + 3 = -3$ So, $Q = (-1, -3, 2)$. Calculat-

$$z = 2(-1) + 4 = 2$$

ing distance PQ :

$$\begin{aligned} PQ &= \sqrt{(2-(-1))^2 + (3-(-3))^2 + (4-2)^2} \\ &= \sqrt{3^2 + 6^2 + 2^2} \\ &= \sqrt{9 + 36 + 4} \\ &= \sqrt{49} = 7 \text{ units} \end{aligned}$$

Final Answer

(i) Distance = 6 units

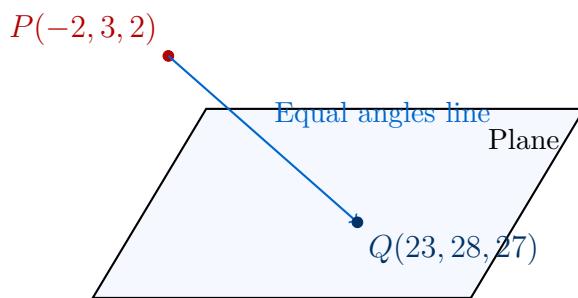
(ii) Distance = 7 units

Common Mistake: Remember that "distance measured along a line" means finding the actual intersection along that specific direction, not perpendicular distance.

Question 21

A line with positive direction cosines passes through the point $P(-2, 3, 2)$ and makes equal angles with the coordinate axes. If the line meets the plane $3x - 4y + 2z = 11$ at the point Q , find the coordinates of Q . Also, find the length of segment PQ .

TYPE 52 Equal-Angle Line Meeting a Plane



Line through P making equal angles with coordinate axes intersecting plane at Q

Given: Point $P(-2, 3, 2)$, Plane $3x - 4y + 2z = 11$, Line makes equal angles with axes.

Concept / Formula Used: $l^2 + m^2 + n^2 = 1 \implies l = m = n = \frac{1}{\sqrt{3}} \quad (1, 1, 1)$

Solution:

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Since the line makes equal angles $\alpha = \beta = \gamma$ with coordinate axes:

$$3 \cos^2 \alpha = 1$$

$$\cos \alpha = \frac{1}{\sqrt{3}} \quad (\text{since})$$

Thus, direction cosines are $l = m = n = \frac{1}{\sqrt{3}}$, and direction ratios are $(1, 1, 1)$. Equation of line passing through $P(-2, 3, 2)$: $\frac{x+2}{1} = \frac{y-3}{1} = \frac{z-2}{1} = \lambda$ Parametric coordinates of point Q on this line: $y = \lambda + 3$ Point Q lies on plane $3x - 4y + 2z = 11$:

$$x = \lambda - 2$$

Coordinates of point Q on this line: $y = \lambda + 3$ Point Q lies on plane $3x - 4y + 2z = 11$:

$$z = \lambda + 2$$

$$3(\lambda - 2) - 4(\lambda + 3) + 2(\lambda + 2) = 11$$

$$3\lambda - 6 - 4\lambda - 12 + 2\lambda + 4 = 11$$

$$\lambda - 14 = 11$$

$$\lambda = 25$$

$$x = 25 - 2 = 23$$

$$\text{Substituting } \lambda = 25 \text{ to find coordinates of } Q: y = 25 + 3 = 28$$

$$z = 25 + 2 = 27$$

So, $Q = (23, 28, 27)$.

$$PQ = \sqrt{(23 - (-2))^2 + (28 - 3)^2 + (27 - 2)^2}$$

$$= \sqrt{25^2 + 25^2 + 25^2}$$

Calculating length of segment PQ :

$$= \sqrt{3 \times 25^2}$$

$$= 25\sqrt{3} \text{ units}$$

Final Answer

Coordinates of $Q = (23, 28, 27)$, Length $PQ = 25\sqrt{3}$ units

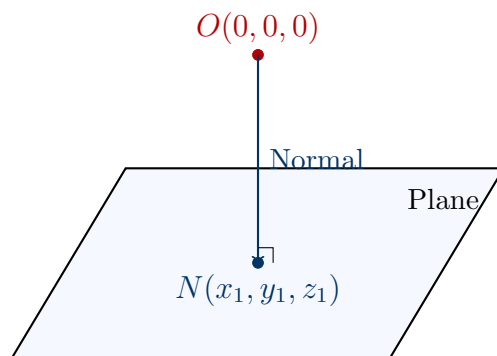
Common Mistake: Positive direction cosines imply positive root $\frac{1}{\sqrt{3}}$ is taken, not negative.

Question 22

In each of the following problems, find the coordinates of the foot of perpendicular drawn from the origin to the plane:

- (i) $x + y + z = 1$
- (ii) $2x + 3y + 4z - 12 = 0$
- (iii) $5y + 8 = 0$

TYPE 48 Plane via Foot of Perpendicular from the Origin



Foot of perpendicular from origin O onto plane

Given: Origin $O(0,0,0)$, Planes: (i) $x + y + z = 1$, (ii) $2x + 3y + 4z - 12 = 0$, (iii) $5y + 8 = 0$.

Concept / Formula Used: $Ax + By + Cz + D = 0$ ON (A, B, C) $N = \left(\frac{-AD}{A^2 + B^2 + C^2}, \frac{-BD}{A^2 + B^2 + C^2}, \frac{-CD}{A^2 + B^2 + C^2} \right)$

Solution:

(i) Plane equation: $x + y + z - 1 = 0$ Here $A = 1, B = 1, C = 1, D = -1$. Line ON passing through origin $(0,0,0)$ with direction ratios $(1,1,1)$: $\frac{x}{1} = \frac{y}{1} = \frac{z}{1} = k$ Coordinates $k + k + k - 1 = 0$

of general point $N(k, k, k)$. Substituting into plane equation: $3k = 1$ Foot of
 $k = \frac{1}{3}$

perpendicular $N = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$.

(ii) Plane equation: $2x + 3y + 4z - 12 = 0$ Here $A = 2, B = 3, C = 4, D = -12$. Line ON from origin: $\frac{x}{2} = \frac{y}{3} = \frac{z}{4} = k$ Coordinates of point $N(2k, 3k, 4k)$. Substituting into plane

$$\begin{aligned}
 2(2k) + 3(3k) + 4(4k) - 12 &= 0 & x &= 2 \left(\frac{12}{29} \right) = \frac{24}{29} \\
 4k + 9k + 16k &= 12 & \text{Foot of perpendicular: } y &= 3 \left(\frac{12}{29} \right) = \frac{36}{29} \text{ Foot} \\
 29k &= 12 & k &= \frac{12}{29} & z &= 4 \left(\frac{12}{29} \right) = \frac{48}{29}
 \end{aligned}$$

of perpendicular $N = \left(\frac{24}{29}, \frac{36}{29}, \frac{48}{29} \right)$.

(iii) Plane equation: $0x + 5y + 0z + 8 = 0$ Here $A = 0, B = 5, C = 0, D = 8$. Line ON from origin: $x = 0, \frac{y}{5} = k, z = 0$ Coordinates of point $N(0, 5k, 0)$. Substituting

$$\begin{aligned}
 5(5k) + 8 &= 0 \\
 25k &= -8 & \text{Foot of perpendicular: } y &= 5 \left(-\frac{8}{25} \right) = -\frac{8}{5} \text{ Foot} \\
 k &= -\frac{8}{25}
 \end{aligned}$$

of perpendicular $N = \left(0, -\frac{8}{5}, 0 \right)$.

Final Answer

(i) Foot of perpendicular $= \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$

(ii) Foot of perpendicular $= \left(\frac{24}{29}, \frac{36}{29}, \frac{48}{29} \right)$

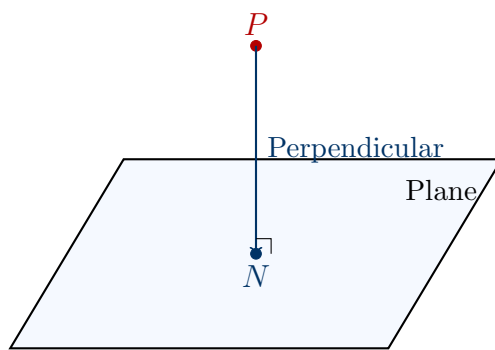
(iii) Foot of perpendicular $= \left(0, -\frac{8}{5}, 0 \right)$

Common Mistake: Make sure constant term D is brought to LHS so $Ax + By + Cz + D = 0$ form is strictly maintained.

Question 23

- Find the length and the foot of perpendicular drawn from the point $(2, 3, 7)$ to the plane $3x - y - z = 7$.
- Find the length and the foot of perpendicular drawn from the point $(1, 1, 2)$ to the plane $\vec{r} \cdot (2\hat{i} - 2\hat{j} + 4\hat{k}) + 5 = 0$.

TYPE 53 Length & Foot of Perpendicular from a Point to a Plane



Perpendicular drawn from a given point P to a plane

Given: Points and plane equations for sub-parts (i) and (ii).

Concept / Formula Used: $P(x_1, y_1, z_1)$ $Ax + By + Cz + D = 0$ $\frac{x - x_1}{A} = \frac{y - y_1}{B} = \frac{z - z_1}{C} = k$ $p = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$

Solution:

(i) Given point $P(2, 3, 7)$ and plane equation $3x - y - z = 7 \Rightarrow 3x - y - z - 7 = 0$. Equation of line PN through $P(2, 3, 7)$ perpendicular to plane: $\frac{x - 2}{3} = \frac{y - 3}{-1} = \frac{z - 7}{-1} = k$

$$x = 3k + 2$$

General point N on this line: $y = -k + 3$ Since N lies on plane $3x - y - z = 7$:

$$z = -k + 7$$

$$3(3k + 2) - (-k + 3) - (-k + 7) = 7$$

$$9k + 6 + k - 3 + k - 7 = 7$$

$$x = 3(1) + 2 = 5$$

$$11k - 4 = 7 \quad \text{Foot of perpendicular } N: y = -(1) + 3 = 2 \quad \text{Foot}$$

$$11k = 11$$

$$z = -(1) + 7 = 6$$

$$k = 1$$

of perpendicular $N = (5, 2, 6)$.

$$PN = \sqrt{(5 - 2)^2 + (2 - 3)^2 + (6 - 7)^2}$$

$$\begin{aligned} \text{Length of perpendicular } PN: &= \sqrt{3^2 + (-1)^2 + (-1)^2} \\ &= \sqrt{9 + 1 + 1} = \sqrt{11} \text{ units} \end{aligned}$$

(ii) Given point $P(1, 1, 2)$ and plane $\vec{r} \cdot (2\hat{i} - 2\hat{j} + 4\hat{k}) + 5 = 0 \Rightarrow 2x - 2y + 4z + 5 = 0$. Equation of line PN through $P(1, 1, 2)$ perpendicular to plane: $\frac{x - 1}{2} = \frac{y - 1}{-2} = \frac{z - 2}{4} = k$

$$x = 2k + 1$$

General point N on line: $y = -2k + 1$ Substituting into plane equation $2x - 2y + 4z + 5 = 0$:

$$z = 4k + 2$$

$$2(2k + 1) - 2(-2k + 1) + 4(4k + 2) + 5 = 0$$

$$4k + 2 + 4k - 2 + 16k + 8 + 5 = 0$$

$$24k + 13 = 0$$

$$24k = -13$$

$$k = -\frac{13}{24}$$

$$\text{Foot of perpendicular } N = \left(-\frac{1}{12}, \frac{25}{12}, -\frac{1}{6}\right).$$

$$PN = \frac{|2(1) - 2(1) + 4(2) + 5|}{\sqrt{2^2 + (-2)^2 + 4^2}}$$

$$= \frac{|2 - 2 + 8 + 5|}{\sqrt{4 + 4 + 16}}$$

Length of perpendicular PN :

$$= \frac{13}{\sqrt{24}}$$

$$= \frac{13}{2\sqrt{6}} = \frac{13\sqrt{6}}{12} \text{ units}$$

$$x = 2\left(-\frac{13}{24}\right) + 1 = -\frac{13}{12} + 1$$

$$\text{Foot of perpendicular } N: y = -2\left(-\frac{13}{24}\right) + 1 = \frac{13}{12} + 1$$

$$z = 4\left(-\frac{13}{24}\right) + 2 = -\frac{13}{6} + 2$$

Final Answer

(i) Foot = $(5, 2, 6)$, Length = $\sqrt{11}$ units

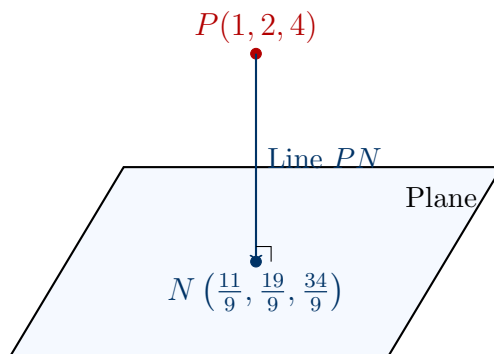
(ii) Foot = $\left(-\frac{1}{12}, \frac{25}{12}, -\frac{1}{6}\right)$, Length = $\frac{13\sqrt{6}}{12}$ units

Common Mistake: Check fraction simplification carefully when finding foot coordinates with negative parameter values.

Question 24

From the point $P(1, 2, 4)$, a perpendicular is drawn on the plane $2x + y - 2z + 3 = 0$. Find the equation, the length and the coordinates of the foot of perpendicular.

TYPE 53 Length & Foot of Perpendicular from a Point to a Plane



Perpendicular line, foot of perpendicular, and distance from point to plane

Given: Point $P(1, 2, 4)$, Plane $2x + y - 2z + 3 = 0$.

Concept / Formula Used: $Ax + By + Cz + D = 0$ $P(x_1, y_1, z_1)$ $\frac{x - x_1}{A} = \frac{y - y_1}{B} = \frac{z - z_1}{C}$ $p = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$

Solution:

Direction ratios of the normal to plane $2x + y - 2z + 3 = 0$ are $(2, 1, -2)$. Line through $P(1, 2, 4)$ parallel to plane normal: $\frac{x - 1}{2} = \frac{y - 2}{1} = \frac{z - 4}{-2} = k$ Vector equation of perpendicular line: $\vec{r} = (\hat{i} + 2\hat{j} + 4\hat{k}) + k(2\hat{i} + \hat{j} - 2\hat{k})$ Cartesian equation of perpendicular

line: $\frac{x - 1}{2} = \frac{y - 2}{1} = \frac{z - 4}{-2}$ General point N on this line: $y = k + 2$ Substitute N
 $x = 2k + 1$
 $z = -2k + 4$

$$2(2k + 1) + (k + 2) - 2(-2k + 4) + 3 = 0$$

$$4k + 2 + k + 2 + 4k - 8 + 3 = 0$$

into plane equation $2x + y - 2z + 3 = 0$: $9k - 1 = 0$

$$9k = 1$$

$$k = \frac{1}{9}$$

$$x = 2\left(\frac{1}{9}\right) + 1 = \frac{11}{9}$$

Coordinates of foot of perpendicular N : $y = \frac{1}{9} + 2 = \frac{19}{9}$ Foot of perpendicular

$$z = -2\left(\frac{1}{9}\right) + 4 = \frac{34}{9}$$

$$N = \left(\frac{11}{9}, \frac{19}{9}, \frac{34}{9}\right).$$

$$\begin{aligned} PN &= \frac{|2(1) + 1(2) - 2(4) + 3|}{\sqrt{2^2 + 1^2 + (-2)^2}} \\ &= \frac{|2 + 2 - 8 + 3|}{\sqrt{4 + 1 + 4}} \\ &= \frac{|-1|}{\sqrt{9}} \\ &= \frac{1}{3} \text{ units} \end{aligned}$$

Length of perpendicular PN :

Final Answer

Equation: $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-4}{-2}$

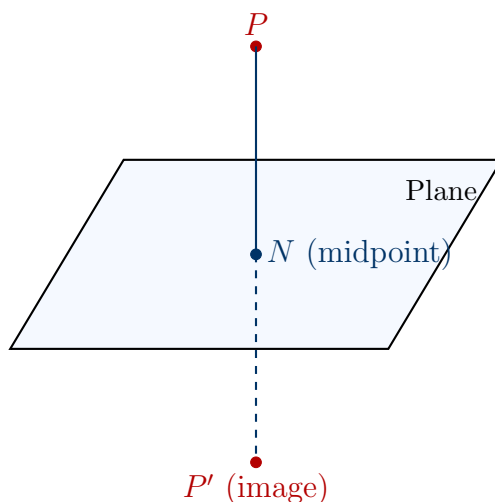
Foot of perpendicular: $\left(\frac{11}{9}, \frac{19}{9}, \frac{34}{9}\right)$

Length: $\frac{1}{3}$ units

Common Mistake: Ensure the perpendicular distance formula includes absolute value signs to guarantee positive length.

Question 25

- (i) Find the image of the point $(1, 2, 3)$ in the plane $x + 2y + 4z = 38$.
 (ii) Find the image of the point having position vector $3\hat{i} - 2\hat{j} + \hat{k}$ in the plane $\vec{r} \cdot (3\hat{i} - \hat{j} + 4\hat{k}) = 2$.

TYPE 46 Image (Reflection) of a Point in a Plane

Reflection image P' of point P across plane with midpoint foot N

Given: Points and plane equations for sub-parts (i) and (ii).

Concept	/	Formula	Used:	N	P	P'	N	$=$
$\left(\frac{x_P + x_{P'}}{2}, \frac{y_P + y_{P'}}{2}, \frac{z_P + z_{P'}}{2}\right)$								

Solution:

(i) Given point $P(1, 2, 3)$ and plane $x + 2y + 4z = 38$. Line PN normal to plane passing through $P(1, 2, 3)$: $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{4} = k$ Foot of perpendicular N has general coordinates $(k+1, 2k+2, 4k+3)$. Substitute N into plane equation $x + 2y +$

$$(k+1) + 2(2k+2) + 4(4k+3) = 38$$

$$k+1+4k+4+16k+12=38$$

$$x=1+1=2$$

$$4z=38:$$

$$21k+17=38 \text{ Foot of perpendicular } N: y=2(1)+2=4$$

$$21k=21$$

$$z=4(1)+3=7$$

$$k=1$$

So $N = (2, 4, 7)$. Let image point be $P'(x', y', z')$. Since N is the midpoint of PP' :

$$\frac{1+x'}{2} = 2 \implies x' = 4 - 1 = 3$$

$$\frac{2+y'}{2} = 4 \implies y' = 8 - 2 = 6 \quad \text{Image point } P' = (3, 6, 11).$$

$$\frac{3+z'}{2} = 7 \implies z' = 14 - 3 = 11$$

(ii) Position vector of P is $\vec{p} = 3\hat{i} - 2\hat{j} + \hat{k}$, so $P(3, -2, 1)$. Plane equation: $3x - y + 4z = 2$

Line PN normal to plane through $P(3, -2, 1)$: $\frac{x-3}{3} = \frac{y+2}{-1} = \frac{z-1}{4} = k$ Foot of perpendicular N has coordinates $(3k+3, -k-2, 4k+1)$. Substitute N into plane equation

$$3(3k+3) - (-k-2) + 4(4k+1) = 2$$

$$9k+9+k+2+16k+4=2$$

tion $3x - y + 4z = 2$: $26k+15=2$ Foot of perpendicular N :

$$26k = -13$$

$$k = -\frac{1}{2}$$

$$x = 3\left(-\frac{1}{2}\right) + 3 = \frac{3}{2}$$

$$y = -\left(-\frac{1}{2}\right) - 2 = -\frac{3}{2} \quad N = \left(\frac{3}{2}, -\frac{3}{2}, -1\right). \text{ Let image point be } P'(x', y', z'). \text{ Using mid-}$$

$$z = 4\left(-\frac{1}{2}\right) + 1 = -1$$

$$\frac{3+x'}{2} = \frac{3}{2} \implies x' = 3 - 3 = 0$$

point formulas: $\frac{-2+y'}{2} = -\frac{3}{2} \implies y' = -3 + 2 = -1$ Position vector of image point P' :

$$\frac{1+z'}{2} = -1 \implies z' = -2 - 1 = -3$$

$$\vec{r}' = -\hat{j} - 3\hat{k} \quad \text{or point } (0, -1, -3)$$

Final Answer

(i) Image = $(3, 6, 11)$

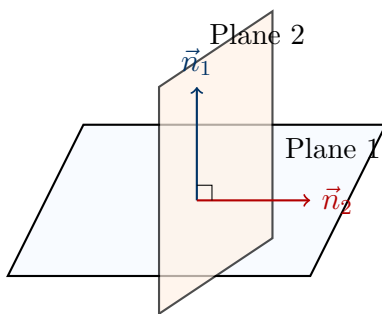
(ii) Position vector of image = $-\hat{j} - 3\hat{k}$ (Coordinates $(0, -1, -3)$)

Common Mistake: Foot of perpendicular is the midpoint of object and image point; do not confuse foot coordinates with image coordinates.

Question 26

Prove that the normals to the planes $3x - 2y + 5z = 7$ and $4x + 11y + 2z + 1 = 0$ are perpendicular to each other.

TYPE 54 Proving Two Planes' Normals are Perpendicular



Perpendicular normal vectors to two planes

To Prove: Normals $\vec{n}_1$ and $\vec{n}_2$ to planes $3x - 2y + 5z = 7$ and $4x + 11y + 2z + 1 = 0$ satisfy $\vec{n}_1 \cdot \vec{n}_2 = 0$.

Concept / Formula Used: $\vec{n}_1 = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ $\vec{n}_2 = a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$ $\vec{n}_1 \cdot \vec{n}_2 = a_1a_2 + b_1b_2 + c_1c_2 = 0$

Solution:

Given equation of first plane: $3x - 2y + 5z = 7$ Normal vector to the first plane: $\vec{n}_1 = 3\hat{i} - 2\hat{j} + 5\hat{k}$

Given equation of second plane: $4x + 11y + 2z + 1 = 0$ Normal vector to the second plane:

$$\begin{aligned}\vec{n}_1 \cdot \vec{n}_2 &= (3)(4) + (-2)(11) + (5)(2) \\ &= 12 - 22 + 10 \\ &= 22 - 22 \\ &= 0\end{aligned}$$

$\vec{n}_2 = 4\hat{i} + 11\hat{j} + 2\hat{k}$ Evaluating the scalar dot product $\vec{n}_1 \cdot \vec{n}_2$:

Since $\vec{n}_1 \cdot \vec{n}_2 = 0$, the normal vectors $\vec{n}_1$ and $\vec{n}_2$ are perpendicular to each other.

Hence Proved

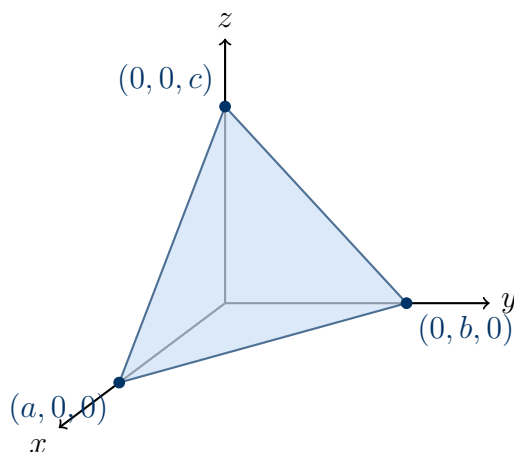
The dot product of normal vectors $\vec{n}_1 \cdot \vec{n}_2 = 0$, hence the normals to the given planes are perpendicular to each other.

Common Mistake: Ensure plane coefficients are read correctly including signs before computing dot product.

11.6 – Intercept and Three-Point Forms of a Plane

Question 1

- (i) Write the intercept cut off by the plane $2x + y - z = 5$ on x -axis.
 (ii) Write the intercept cut off by the plane $3x - 4y + z = 6$ on y -axis.
 (iii) Write the intercept cut off by the plane $5x - 2y + 3z + 7 = 0$ on z -axis.

TYPE 55 Intercept Form of a Plane – Basic*Representation of a plane cutting intercepts a , b , and c on the coordinate axes***To Find:** The intercepts cut off on the specified coordinate axes by the given planes:

- (i) x -axis intercept for $2x + y - z = 5$, (ii) y -axis intercept for $3x - 4y + z = 6$, and
 (iii) z -axis intercept for $5x - 2y + 3z + 7 = 0$.

Concept / Formula Used: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ a b c x y z

Solution:**Part (i):** Find the intercept cut off by $2x + y - z = 5$ on the x -axis.

Method 1 (Using Intercept Form): The given plane equation is:

$$2x + y - z = 5$$

Dividing both sides by 5:

$$\frac{2x}{5} + \frac{y}{5} - \frac{z}{5} = 1$$

Rewriting in standard intercept form:

$$\frac{x}{(\frac{5}{2})} + \frac{y}{5} + \frac{z}{-5} = 1$$

Comparing with $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, the x -intercept is $a = \frac{5}{2}$.Method 2 (Setting $y = 0$ and $z = 0$): On the x -axis, $y = 0$ and $z = 0$. Substituting these

$$2x + 0 - 0 = 5$$

into the equation of the plane:

$$2x = 5 \quad \text{Thus, the intercept cut off on the } x\text{-axis}$$

$$x = \frac{5}{2}$$

is $\frac{5}{2}$.

Part (ii): Find the intercept cut off by $3x - 4y + z = 6$ on the y -axis.

Method 1 (Using Intercept Form): The given plane equation is:

$$3x - 4y + z = 6$$

Dividing both sides by 6:

$$\begin{aligned}\frac{3x}{6} - \frac{4y}{6} + \frac{z}{6} &= 1 \\ \frac{x}{2} + \frac{y}{\left(-\frac{6}{4}\right)} + \frac{z}{6} &= 1 \\ \frac{x}{2} + \frac{y}{\left(-\frac{3}{2}\right)} + \frac{z}{6} &= 1\end{aligned}$$

Comparing with $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, the y -intercept is $b = -\frac{3}{2}$.

Method 2 (Setting $x = 0$ and $z = 0$): On the y -axis, $x = 0$ and $z = 0$. Substituting

$$3(0) - 4y + 0 = 6$$

$$-4y = 6$$

these into the equation of the plane:

$$y = -\frac{6}{4} \text{ Thus, the intercept cut off on}$$

$$y = -\frac{3}{2}$$

the y -axis is $-\frac{3}{2}$.

Part (iii): Find the intercept cut off by $5x - 2y + 3z + 7 = 0$ on the z -axis.

Method 1 (Using Intercept Form): The given plane equation is:

$$5x - 2y + 3z + 7 = 0$$

Transposing the constant term to the right-hand side:

$$5x - 2y + 3z = -7$$

Dividing both sides by -7 :

$$\begin{aligned}\frac{5x}{-7} - \frac{2y}{-7} + \frac{3z}{-7} &= 1 \\ \frac{x}{\left(-\frac{7}{5}\right)} + \frac{y}{\left(\frac{7}{2}\right)} + \frac{z}{\left(-\frac{7}{3}\right)} &= 1\end{aligned}$$

Comparing with $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, the z -intercept is $c = -\frac{7}{3}$.

Method 2 (Setting $x = 0$ and $y = 0$): On the z -axis, $x = 0$ and $y = 0$. Substituting these

$$5(0) - 2(0) + 3z + 7 = 0$$

$$3z + 7 = 0$$

into the equation of the plane:

$$3z = -7 \text{ Thus, the intercept cut off on}$$

$$z = -\frac{7}{3}$$

the z -axis is $-\frac{7}{3}$.

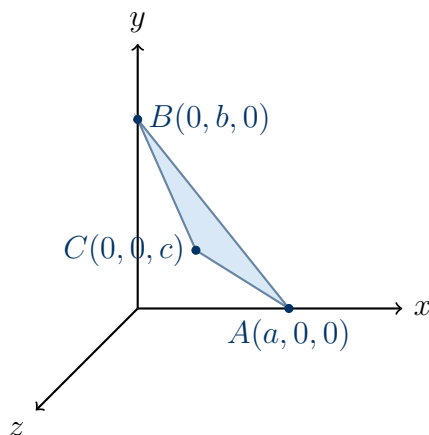
Final Answer

- (i) Intercept on x -axis $= \frac{5}{2}$
 (ii) Intercept on y -axis $= -\frac{3}{2}$
 (iii) Intercept on z -axis $= -\frac{7}{3}$

Common Mistake: Forgetting to transpose the constant term to the right-hand side before dividing when converting to intercept form, or ignoring the negative sign associated with the coefficient when setting other variables to zero.

Question 2

- (i) Find the intercepts made by the plane $2x + y - z = 5$ on the coordinate axes.
 (ii) Write the sum of intercepts cut off by the plane $\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) - 5 = 0$ on the coordinate axes.

TYPE 55 Intercept Form of a Plane – Basic

Intercepts made by a plane on the 3D coordinate axes

Given: Given plane equations: (i) $2x + y - z = 5$ and (ii) $\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) - 5 = 0$.

Concept / Formula Used: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ a b c x y z

Solution:

Part (i):

The given equation of the plane is:

$$2x + y - z = 5$$

Dividing both sides by 5 to convert the equation into intercept form:

$$\frac{2x}{5} + \frac{y}{5} - \frac{z}{5} = 1$$

Rewriting each term with unit coefficients in the numerator:

$$\frac{x}{\left(\frac{5}{2}\right)} + \frac{y}{5} + \frac{z}{-5} = 1$$

Comparing with the standard intercept form $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$:

$$\begin{aligned} a &= \frac{5}{2} \\ b &= 5 \\ c &= -5 \end{aligned}$$

Thus, the intercepts on the x , y , and z axes are $\frac{5}{2}$, 5 , and -5 respectively.

Part (ii):

The given vector equation of the plane is:

$$\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) - 5 = 0$$

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. Substituting $\vec{r}$ into the vector equation:

$$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (2\hat{i} + \hat{j} - \hat{k}) = 5$$

Taking the dot product:

$$2x + y - z = 5$$

Dividing both sides by 5:

$$\frac{x}{\left(\frac{5}{2}\right)} + \frac{y}{5} + \frac{z}{-5} = 1$$

The intercepts are:

$$\begin{aligned} a &= \frac{5}{2} \\ b &= 5 \\ c &= -5 \end{aligned}$$

The sum of the intercepts cut off by the plane on the coordinate axes is:

$$\begin{aligned} \text{Sum} &= a + b + c \\ &= \frac{5}{2} + 5 + (-5) \\ &= \frac{5}{2} \end{aligned}$$

Final Answer

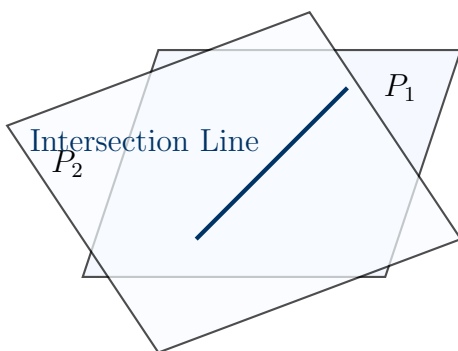
- (i) Intercepts on x , y , and z axes are $\frac{5}{2}$, 5 , and -5 respectively.
- (ii) Sum of intercepts = $\frac{5}{2}$

Common Mistake: Remember that sign matters in intercepts; the z -intercept is -5 , not 5 .

Question 3

- (i) Write the equation of the plane which has intercepts 2, 3, and 4 on the x -axis, y -axis, and z -axis respectively.
- (ii) Write the equation of the plane with intercepts 2, -3 , and 5 on the x -axis, y -axis, and z -axis respectively.
- (iii) Find the equation of the plane passing through the intersection of the planes $x + 2y + 3z + 4 = 0$ and $x - y + z + 3 = 0$ and passing through the origin.

TYPE 56 Plane Through Line of Intersection of Two Planes



Plane passing through the intersection line of two planes

- Given:** (i) Intercepts $a = 2, b = 3, c = 4$.
(ii) Intercepts $a = 2, b = -3, c = 5$.
(iii) Planes $P_1 : x + 2y + 3z + 4 = 0, P_2 : x - y + z + 3 = 0$, passing through origin $(0, 0, 0)$.

Concept / Formula Used: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad P_1 = 0 \quad P_2 = 0 \quad P_1 + \lambda P_2 = 0$

Solution:

Part (i):

Given intercepts are $a = 2, b = 3, c = 4$.

Using the intercept form of a plane:

$$\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$$

Multiplying the entire equation by the LCM of 2, 3, and 4, which is 12:

$$12 \cdot \frac{x}{2} + 12 \cdot \frac{y}{3} + 12 \cdot \frac{z}{4} = 12 \cdot 1$$

Simplifying each term:

$$6x + 4y + 3z = 12$$

Part (ii):

Given intercepts are $a = 2$, $b = -3$, $c = 5$.

Using the intercept form of a plane:

$$\frac{x}{2} + \frac{y}{-3} + \frac{z}{5} = 1$$

Multiplying the entire equation by the LCM of 2, 3, and 5, which is 30:

$$30 \cdot \frac{x}{2} - 30 \cdot \frac{y}{3} + 30 \cdot \frac{z}{5} = 30 \cdot 1$$

Simplifying each term:

$$15x - 10y + 6z = 30$$

Part (iii):

The equation of any plane passing through the intersection of the given planes $x + 2y + 3z + 4 = 0$ and $x - y + z + 3 = 0$ is:

$$(x + 2y + 3z + 4) + \lambda(x - y + z + 3) = 0$$

Grouping terms by variables:

$$(1 + \lambda)x + (2 - \lambda)y + (3 + \lambda)z + (4 + 3\lambda) = 0$$

Since this plane passes through the origin $(0, 0, 0)$, substitute $x = 0$, $y = 0$, $z = 0$:

$$(1 + \lambda)(0) + (2 - \lambda)(0) + (3 + \lambda)(0) + 4 + 3\lambda = 0$$

$$4 + 3\lambda = 0$$

$$3\lambda = -4$$

$$\lambda = -\frac{4}{3}$$

Substituting $\lambda = -\frac{4}{3}$ back into the equation of the family of planes:

$$(x + 2y + 3z + 4) - \frac{4}{3}(x - y + z + 3) = 0$$

Multiplying the entire equation by 3:

$$3(x + 2y + 3z + 4) - 4(x - y + z + 3) = 0$$

Expanding the terms:

$$3x + 6y + 9z + 12 - 4x + 4y - 4z - 12 = 0$$

Combining like terms:

$$-x + 10y + 5z = 0$$

Multiplying by -1 :

$$x - 10y - 5z = 0$$

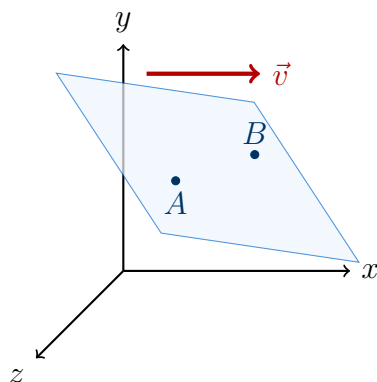
Final Answer

- (i) $6x + 4y + 3z = 12$
- (ii) $15x - 10y + 6z = 30$
- (iii) $x - 10y - 5z = 0$

Question 4

- (i) Find the equation of the plane passing through the points $(2, 3, -4)$ and $(1, -1, 3)$ and parallel to x -axis.
- (ii) Find the equation of the plane through the points $(2, 2, -1)$ and $(3, 4, 2)$ and parallel to the line whose direction ratios are $\langle 7, 0, 6 \rangle$.

TYPE 57 Plane Through Two Points, Parallel to a Given Direction



Plane containing two points and parallel to a direction vector

- Given:** (i) Points $A(2, 3, -4)$, $B(1, -1, 3)$, parallel to x -axis.
(ii) Points $A(2, 2, -1)$, $B(3, 4, 2)$, parallel to line with d.r. $\langle 7, 0, 6 \rangle$.

Concept	/	Formula	Used:	$\vec{n}$	$\vec{AB}$	$\vec{v}$	$\vec{n}$	$=$	$\vec{AB} \times$
$\vec{v}$		(x_1, y_1, z_1)	$\langle a, b, c \rangle$	$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$					

Solution:

Part (i):

Let the given points be $A(2, 3, -4)$ and $B(1, -1, 3)$.

The vector joining A to B is:

$$\begin{aligned} \vec{AB} &= (1 - 2)\hat{i} + (-1 - 3)\hat{j} + (3 - (-4))\hat{k} \\ &= -\hat{i} - 4\hat{j} + 7\hat{k} \end{aligned}$$

Since the plane is parallel to the x -axis, it is parallel to the unit vector along the x -axis, $\hat{i} = 1\hat{i} + 0\hat{j} + 0\hat{k}$.

$$\vec{n} = \vec{AB} \times \hat{i}$$

The normal vector $\vec{n}$ to the plane is perpendicular to both $\vec{AB}$ and $\hat{i}$:

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -4 & 7 \\ 1 & 0 & 0 \end{vmatrix}$$

Expanding the determinant along Row 1:

$$\vec{n} = \hat{i}((-4)(0) - (7)(0)) - \hat{j}((-1)(0) - (7)(1)) + \hat{k}((-1)(0) - (-4)(1))$$

$$= 0\hat{i} + 7\hat{j} + 4\hat{k}$$

So direction ratios of normal are $\langle 0, 7, 4 \rangle$.

Equation of plane passing through $A(2, 3, -4)$ with normal $\langle 0, 7, 4 \rangle$:

$$0(x - 2) + 7(y - 3) + 4(z - (-4)) = 0$$

$$7y - 21 + 4z + 16 = 0$$

$$7y + 4z - 5 = 0$$

Part (ii):

Let the given points be $A(2, 2, -1)$ and $B(3, 4, 2)$.

Vector $\vec{AB}$ is:

$$\vec{AB} = (3 - 2)\hat{i} + (4 - 2)\hat{j} + (2 - (-1))\hat{k}$$

$$= \hat{i} + 2\hat{j} + 3\hat{k}$$

The direction vector of the parallel line is:

$$\vec{v} = 7\hat{i} + 0\hat{j} + 6\hat{k}$$

$$\vec{n} = \vec{AB} \times \vec{v}$$

The normal vector $\vec{n}$ to the plane is:

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 7 & 0 & 6 \end{vmatrix}$$

Expanding the determinant:

$$\vec{n} = \hat{i}(12 - 0) - \hat{j}(6 - 21) + \hat{k}(0 - 14)$$

$$= 12\hat{i} + 15\hat{j} - 14\hat{k}$$

Equation of the plane passing through $A(2, 2, -1)$ with normal $\langle 12, 15, -14 \rangle$:

$$12(x - 2) + 15(y - 2) - 14(z - (-1)) = 0$$

$$12x - 24 + 15y - 30 - 14z - 14 = 0$$

$$12x + 15y - 14z - 68 = 0$$

Final Answer

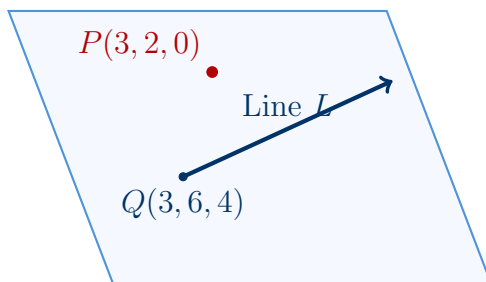
(i) $7y + 4z - 5 = 0$

(ii) $12x + 15y - 14z - 68 = 0$

Question 5

Find the equation of the plane which passes through the point $(3, 2, 0)$ and contains the line $\frac{x-3}{1} = \frac{y-6}{5} = \frac{z-4}{4}$.

TYPE 58 Plane Containing a Given Line, Through an External Point



Plane containing a line L and point P

Given: Given point $P(3, 2, 0)$ and line $\frac{x-3}{1} = \frac{y-6}{5} = \frac{z-4}{4}$.

Concept / Formula Used: $Q \quad \vec{b} \quad P \quad \vec{n} = \vec{PQ} \times \vec{b}$

Solution:

From the given equation of the line:

$$\frac{x-3}{1} = \frac{y-6}{5} = \frac{z-4}{4}$$

A point on the line is $Q(3, 6, 4)$ and the direction vector of the line is:

$$\vec{b} = \hat{i} + 5\hat{j} + 4\hat{k}$$

The given point outside the line is $P(3, 2, 0)$.

Vector connecting P to Q : $\vec{PQ} = (3-3)\hat{i} + (6-2)\hat{j} + (4-0)\hat{k}$
 $= 0\hat{i} + 4\hat{j} + 4\hat{k}$

$$\vec{n} = \vec{PQ} \times \vec{b}$$

The normal vector $\vec{n}$ to the required plane is perpendicular to both $\vec{PQ}$ and $\vec{b}$:

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 4 & 4 \\ 1 & 5 & 4 \end{vmatrix}$$

Expanding the determinant: $\vec{n} = \hat{i}(16-20) - \hat{j}(0-4) + \hat{k}(0-4)$
 $= -4\hat{i} + 4\hat{j} - 4\hat{k}$

Dividing by -4 to simplify the normal vector:

$$\vec{n}' = \hat{i} - \hat{j} + \hat{k}$$

Equation of the plane passing through $P(3, 2, 0)$ with normal direction ratios $\langle 1, -1, 1 \rangle$:

$$1(x - 3) - 1(y - 2) + 1(z - 0) = 0$$

$$x - 3 - y + 2 + z = 0$$

$$x - y + z - 1 = 0$$

Final Answer

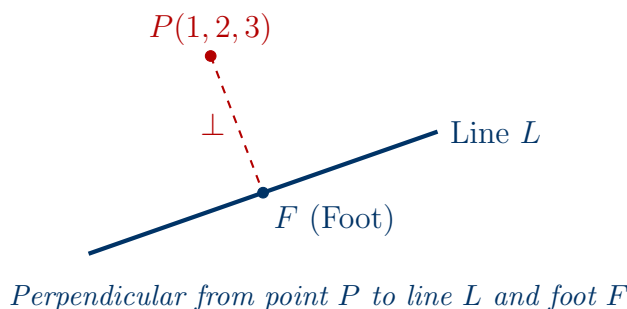
$$x - y + z - 1 = 0$$

Question 6

Find the foot of perpendicular drawn from the point $P(1, 2, 3)$ on the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$. Also obtain the equation of the plane containing the line and the point $(1, 2, 3)$.

TYPE 60 Foot of Perpendicular to a Line, Then Plane Containing Line & Point

★ Likely Exam Question



Given: Point $P(1, 2, 3)$ and line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$.

Concept / Formula Used: $(x_1 + a\lambda, y_1 + b\lambda, z_1 + c\lambda)$ $\vec{PF} \cdot \vec{b} = 0$

Solution:

Let the given line be:

$$\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2} = \lambda$$

Any general point F on this line has coordinates:

$$F(3\lambda + 6, 2\lambda + 7, -2\lambda + 7)$$

Let F be the foot of the perpendicular from $P(1, 2, 3)$ to the line.

The vector $\vec{PF}$ is given by: $\vec{PF} = (3\lambda + 6 - 1)\hat{i} + (2\lambda + 7 - 2)\hat{j} + (-2\lambda + 7 - 3)\hat{k}$
 $= (3\lambda + 5)\hat{i} + (2\lambda + 5)\hat{j} + (-2\lambda + 4)\hat{k}$

The direction vector of the given line is:

$$\vec{b} = 3\hat{i} + 2\hat{j} - 2\hat{k}$$

Since PF is perpendicular to the line, $\vec{PF} \cdot \vec{b} = 0$:

$$3(3\lambda + 5) + 2(2\lambda + 5) - 2(-2\lambda + 4) = 0$$

Expanding the expression:

$$9\lambda + 15 + 4\lambda + 10 + 4\lambda - 8 = 0$$

$$17\lambda + 17 = 0$$

$$17\lambda = -17$$

$$\lambda = -1$$

$$x = 3(-1) + 6 = 3$$

Substituting $\lambda = -1$ into the coordinates of F : $y = 2(-1) + 7 = 5$

$$z = -2(-1) + 7 = 9$$

Thus, the foot of perpendicular is $F(3, 5, 9)$.

Now, to find the equation of the plane containing the line and point $P(1, 2, 3)$:

The line passes through $Q(6, 7, 7)$ and has direction $\vec{b} = 3\hat{i} + 2\hat{j} - 2\hat{k}$.

Vector $\vec{PQ}$ is: $\vec{PQ} = (6 - 1)\hat{i} + (7 - 2)\hat{j} + (7 - 3)\hat{k}$
 $= 5\hat{i} + 5\hat{j} + 4\hat{k}$

$$\vec{n} = \vec{PQ} \times \vec{b}$$

Normal vector $\vec{n}$ to the plane:
$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & 5 & 4 \\ 3 & 2 & -2 \end{vmatrix}$$

$$\vec{n} = \hat{i}(5(-2) - 4(2)) - \hat{j}(5(-2) - 4(3)) + \hat{k}(5(2) - 5(3))$$

Expanding the determinant: $= \hat{i}(-10 - 8) - \hat{j}(-10 - 12) + \hat{k}(10 - 15)$
 $= -18\hat{i} + 22\hat{j} - 5\hat{k}$

Equation of plane passing through $P(1, 2, 3)$ with normal vector $\langle -18, 22, -5 \rangle$:

$$-18(x - 1) + 22(y - 2) - 5(z - 3) = 0$$

$$-18x + 18 + 22y - 44 - 5z + 15 = 0$$

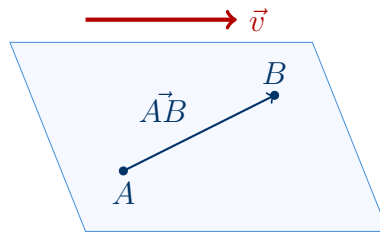
$$-18x + 22y - 5z - 11 = 0$$

Multiplying by -1 :

$$18x - 22y + 5z + 11 = 0$$

Final AnswerFoot of perpendicular: $(3, 5, 9)$ Equation of plane: $18x - 22y + 5z + 11 = 0$ **Question 7**

- (i) Find the equation of the plane passing through the points $(-1, 2, 0)$, $(2, 2, -1)$ and parallel to the line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$.
- (ii) Find the vector equation of the plane passing through the points $(0, -4, 1)$ and $(2, 3, -2)$ and parallel to the line $\vec{r} = 5\hat{i} - 2\hat{j} + \hat{k} + \lambda(2\hat{i} - \hat{k})$.

TYPE 57 Plane Through Two Points, Parallel to a Given Direction*Plane through points A, B parallel to line vector $\vec{v}$*

- Given:** (i) Points $A(-1, 2, 0)$, $B(2, 2, -1)$, line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$.
- (ii) Points $A(0, -4, 1)$, $B(2, 3, -2)$, line $\vec{r} = 5\hat{i} - 2\hat{j} + \hat{k} + \lambda(2\hat{i} - \hat{k})$.

Concept / Formula Used: $\vec{n} = \vec{AB} \times \vec{v}$ $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$ **Solution:****Part (i):**Let points be $A(-1, 2, 0)$ and $B(2, 2, -1)$.

Vector $\vec{AB}$ is:
$$\begin{aligned}\vec{AB} &= (2 - (-1))\hat{i} + (2 - 2)\hat{j} + (-1 - 0)\hat{k} \\ &= 3\hat{i} + 0\hat{j} - \hat{k}\end{aligned}$$

The line equation is given as:

$$\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$$

Rewriting the second ratio to standard form $\frac{y-y_0}{b}$:

$$\frac{2y+1}{2} = \frac{2(y+1/2)}{2} = \frac{y+1/2}{1}$$

Thus, direction ratios of the line are $\langle 1, 1, -1 \rangle$, and its direction vector is:

$$\vec{v} = \hat{i} + \hat{j} - \hat{k}$$

$$\vec{n} = \vec{AB} \times \vec{v}$$

The normal vector $\vec{n}$ to the plane:

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 0 & -1 \\ 1 & 1 & -1 \end{vmatrix}$$

Expanding the determinant:

$$\vec{n} = \hat{i}(0 - (-1)) - \hat{j}(-3 - (-1)) + \hat{k}(3 - 0) \\ = \hat{i} + 2\hat{j} + 3\hat{k}$$

Equation of the plane passing through $A(-1, 2, 0)$ with normal direction ratios $\langle 1, 2, 3 \rangle$:

$$1(x - (-1)) + 2(y - 2) + 3(z - 0) = 0$$

$$x + 1 + 2y - 4 + 3z = 0$$

$$x + 2y + 3z - 3 = 0$$

Part (ii):

Points are $A(0, -4, 1)$ and $B(2, 3, -2)$. Position vector of A is $\vec{a} = 0\hat{i} - 4\hat{j} + \hat{k}$.

Vector $\vec{AB}$:

$$\vec{AB} = (2 - 0)\hat{i} + (3 - (-4))\hat{j} + (-2 - 1)\hat{k} \\ = 2\hat{i} + 7\hat{j} - 3\hat{k}$$

The line direction vector is:

$$\vec{v} = 2\hat{i} + 0\hat{j} - \hat{k}$$

$$\vec{n} = \vec{AB} \times \vec{v}$$

Normal vector $\vec{n}$ to the plane:

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 7 & -3 \\ 2 & 0 & -1 \end{vmatrix}$$

Expanding the determinant:

$$\vec{n} = \hat{i}(-7 - 0) - \hat{j}(-2 - (-6)) + \hat{k}(0 - 14) \\ = -7\hat{i} - 4\hat{j} - 14\hat{k}$$

Taking $\vec{n} = 7\hat{i} + 4\hat{j} + 14\hat{k}$ as the normal vector:

$$\vec{a} \cdot \vec{n} = (0\hat{i} - 4\hat{j} + \hat{k}) \cdot (7\hat{i} + 4\hat{j} + 14\hat{k})$$

Evaluating $\vec{a} \cdot \vec{n}$:

$$= (0)(7) + (-4)(4) + (1)(14) \\ = 0 - 16 + 14 \\ = -2$$

Vector equation of the plane is $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$:

$$\vec{r} \cdot (7\hat{i} + 4\hat{j} + 14\hat{k}) = -2$$

$$\vec{r} \cdot (7\hat{i} + 4\hat{j} + 14\hat{k}) + 2 = 0$$

Final Answer

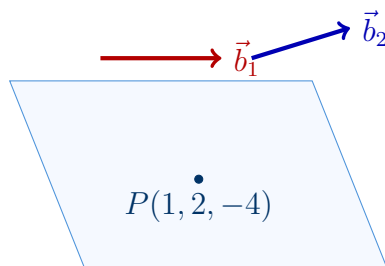
(i) $x + 2y + 3z - 3 = 0$

(ii) $\vec{r} \cdot (7\hat{i} + 4\hat{j} + 14\hat{k}) + 2 = 0$

Question 8

Find the equation of the plane passing through the point $(1, 2, -4)$ and parallel to the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$ and $\frac{x-1}{1} = \frac{y+3}{1} = \frac{z}{-1}$.

TYPE 59 Plane Through a Point, Parallel to Two Given Lines



Plane parallel to two given non-parallel lines

Given: Point $P(1, 2, -4)$, parallel to lines L_1 with d.r. $\langle 2, 3, 6 \rangle$ and L_2 with d.r. $\langle 1, 1, -1 \rangle$.

Concept / Formula Used: $\vec{n} = \vec{b}_1 \times \vec{b}_2$

Solution:

Direction vectors of the two parallel lines are:

$$\vec{b}_1 = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

$$\vec{b}_2 = \hat{i} + \hat{j} - \hat{k}$$

The normal vector $\vec{n}$ to the plane is perpendicular to both $\vec{b}_1$ and $\vec{b}_2$:

$$\vec{n} = \vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 6 \\ 1 & 1 & -1 \end{vmatrix}$$

$$\vec{n} = \hat{i}(3(-1) - 6(1)) - \hat{j}(2(-1) - 6(1)) + \hat{k}(2(1) - 3(1))$$

Expanding the determinant:

$$= \hat{i}(-3 - 6) - \hat{j}(-2 - 6) + \hat{k}(2 - 3)$$

$$= -9\hat{i} + 8\hat{j} - \hat{k}$$

Taking $\vec{n} = 9\hat{i} - 8\hat{j} + \hat{k}$ as the normal vector:

Equation of the plane passing through $P(1, 2, -4)$ with normal $\langle 9, -8, 1 \rangle$:

$$9(x - 1) - 8(y - 2) + 1(z - (-4)) = 0$$

$$9x - 9 - 8y + 16 + z + 4 = 0$$

$$9x - 8y + z + 11 = 0$$

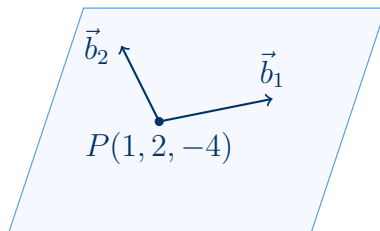
Final Answer

$$9x - 8y + z + 11 = 0$$

Question 9

Find the vector as well as the cartesian equation of the plane passing through the point $(1, 2, -4)$ and parallel to the lines $\vec{r} = \hat{i} + 2\hat{j} - \hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$ and $\vec{r} = \hat{i} - 3\hat{j} + 5\hat{k} + \mu(\hat{i} + \hat{j} - \hat{k})$.

TYPE 59 Plane Through a Point, Parallel to Two Given Lines



Plane defined by point P and direction vectors $\vec{b}_1, \vec{b}_2$

Given: Point $P(1, 2, -4)$ with position vector $\vec{a} = \hat{i} + 2\hat{j} - 4\hat{k}$, parallel to lines with directions $\vec{b}_1 = 2\hat{i} + 3\hat{j} + 6\hat{k}$ and $\vec{b}_2 = \hat{i} + \hat{j} - \hat{k}$.

Concept / Formula Used: $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$ $\vec{n} = \vec{b}_1 \times \vec{b}_2$

Solution:

Position vector of the given point $P(1, 2, -4)$ is:

$$\vec{a} = \hat{i} + 2\hat{j} - 4\hat{k}$$

Direction vectors of the two given lines are:

$$\vec{b}_1 = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

$$\vec{b}_2 = \hat{i} + \hat{j} - \hat{k}$$

$$\vec{n} = \vec{b}_1 \times \vec{b}_2$$

The normal vector $\vec{n}$ to the plane is:

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 6 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= \hat{i}(-3 - 6) - \hat{j}(-2 - 6) + \hat{k}(2 - 3)$$

$$= -9\hat{i} + 8\hat{j} - \hat{k}$$

Multiplying by -1 , we use $\vec{n} = 9\hat{i} - 8\hat{j} + \hat{k}$.

$$\vec{a} \cdot \vec{n} = (\hat{i} + 2\hat{j} - 4\hat{k}) \cdot (9\hat{i} - 8\hat{j} + \hat{k})$$

Calculating $\vec{a} \cdot \vec{n}$:

$$= (1)(9) + (2)(-8) + (-4)(1)$$

$$= 9 - 16 - 4$$

$$= -11$$

Vector equation of the plane:

$$\vec{r} \cdot (9\hat{i} - 8\hat{j} + \hat{k}) = -11$$

$$\vec{r} \cdot (9\hat{i} - 8\hat{j} + \hat{k}) + 11 = 0$$

Cartesian equation: Substituting $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$:

$$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (9\hat{i} - 8\hat{j} + \hat{k}) + 11 = 0$$

$$9x - 8y + z + 11 = 0$$

Final Answer

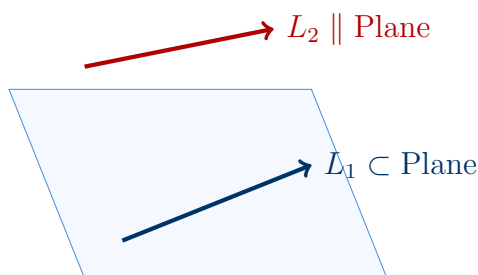
Vector equation: $\vec{r} \cdot (9\hat{i} - 8\hat{j} + \hat{k}) + 11 = 0$

Cartesian equation: $9x - 8y + z + 11 = 0$

Question 10

Find the equation of the plane containing the line $\frac{x-4}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and parallel to the line $\frac{x}{1} = \frac{y-1}{3} = \frac{z-2}{3}$.

TYPE 61 Plane Containing One Line, Parallel to Another Line



Plane containing line L_1 and parallel to line L_2

Given: Plane contains $L_1 : \frac{x-4}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and is parallel to $L_2 : \frac{x}{1} = \frac{y-1}{3} = \frac{z-2}{3}$.

Concept / Formula Used: $\vec{n} = \vec{b}_1 \times \vec{b}_2$ $\vec{b}_1, \vec{b}_2$ L_1, L_2

Solution:

From L_1 , a point on the line is $A(4, 2, 3)$ and its direction vector is:

$$\vec{b}_1 = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

From L_2 , its direction vector is:

$$\vec{b}_2 = \hat{i} + 3\hat{j} + 3\hat{k}$$

$$\vec{n} = \vec{b}_1 \times \vec{b}_2$$

Normal vector $\vec{n}$ to the plane:

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 1 & 3 & 3 \end{vmatrix}$$

$$\vec{n} = \hat{i}(3(3) - 4(3)) - \hat{j}(2(3) - 4(1)) + \hat{k}(2(3) - 3(1))$$

Expanding the determinant:

$$= \hat{i}(9 - 12) - \hat{j}(6 - 4) + \hat{k}(6 - 3)$$

$$= -3\hat{i} - 2\hat{j} + 3\hat{k}$$

Taking $\vec{n} = 3\hat{i} + 2\hat{j} - 3\hat{k}$ as the normal vector:

Equation of the plane passing through $A(4, 2, 3)$ with normal $\langle 3, 2, -3 \rangle$:

$$3(x - 4) + 2(y - 2) - 3(z - 3) = 0$$

$$3x - 12 + 2y - 4 - 3z + 9 = 0$$

$$3x + 2y - 3z - 7 = 0$$

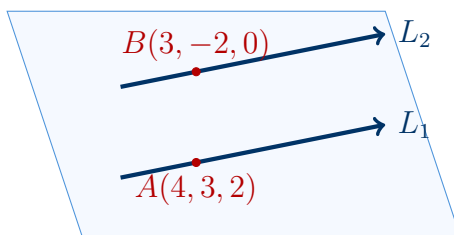
Final Answer

$$3x + 2y - 3z - 7 = 0$$

Question 11

Find the vector equation of the plane which contains the two parallel lines $\frac{x-4}{1} = \frac{y-3}{-4} = \frac{z-2}{5}$ and $\frac{x-3}{1} = \frac{y+2}{-4} = \frac{z}{5}$.

TYPE 62 Plane Containing Two Parallel Lines



Plane containing two parallel lines

Given: Parallel lines $L_1 : \frac{x-4}{1} = \frac{y-3}{-4} = \frac{z-2}{5}$ and $L_2 : \frac{x-3}{1} = \frac{y+2}{-4} = \frac{z}{5}$.

Concept / Formula Used: $\vec{b} \quad A \quad B \quad \vec{n} = \vec{b} \times \vec{AB}$

Solution:

Line L_1 passes through $A(4, 3, 2)$ with position vector $\vec{a} = 4\hat{i} + 3\hat{j} + 2\hat{k}$. Line L_2 passes through $B(3, -2, 0)$.

Both lines are parallel with common direction vector:

$$\vec{b} = \hat{i} - 4\hat{j} + 5\hat{k}$$

Vector $\vec{AB}$ joining point A on L_1 to point B on L_2 :
$$\begin{aligned}\vec{AB} &= (3-4)\hat{i} + (-2-3)\hat{j} + (0-2)\hat{k} \\ &= -\hat{i} - 5\hat{j} - 2\hat{k}\end{aligned}$$

$$\vec{n} = \vec{b} \times \vec{AB}$$

The normal vector $\vec{n}$ to the plane is perpendicular to both $\vec{b}$ and $\vec{AB}$:
$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -4 & 5 \\ -1 & -5 & -2 \end{vmatrix}$$

$$\vec{n} = \hat{i}((-4)(-2) - 5(-5)) - \hat{j}(1(-2) - 5(-1)) + \hat{k}(1(-5) - (-4)(-1))$$

Expanding the determinant:
$$\begin{aligned}&= \hat{i}(8 + 25) - \hat{j}(-2 + 5) + \hat{k}(-5 - 4) \\ &= 33\hat{i} - 3\hat{j} - 9\hat{k}\end{aligned}$$

Dividing by 3 to obtain a simpler normal vector:

$$\vec{n}' = 11\hat{i} - \hat{j} - 3\hat{k}$$

Calculating $\vec{a} \cdot \vec{n}'$:
$$\begin{aligned}\vec{a} \cdot \vec{n}' &= (4\hat{i} + 3\hat{j} + 2\hat{k}) \cdot (11\hat{i} - \hat{j} - 3\hat{k}) \\ &= (4)(11) + (3)(-1) + (2)(-3) \\ &= 44 - 3 - 6 \\ &= 35\end{aligned}$$

Vector equation of the plane $\vec{r} \cdot \vec{n}' = \vec{a} \cdot \vec{n}'$:

$$\vec{r} \cdot (11\hat{i} - \hat{j} - 3\hat{k}) = 35$$

Final Answer

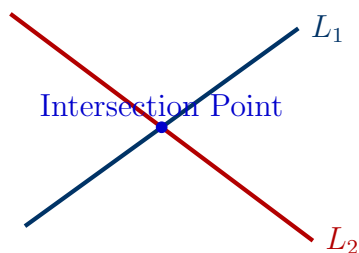
$$\vec{r} \cdot (11\hat{i} - \hat{j} - 3\hat{k}) = 35$$

Question 12

If the lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$ intersect at a point, find the values of k .

TYPE 39 Condition for Two Lines to Intersect – Triple Product Test

★ Likely Exam Question



Intersecting lines at a point in 3D space

Given: Intersecting lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$.

Concept / Formula Used: $\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$ $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$

Solution:

Given line 1 passes through $(x_1, y_1, z_1) = (1, 2, 3)$ with direction ratios $\langle a_1, b_1, c_1 \rangle = \langle k, 2, 3 \rangle$.

Given line 2 passes through $(x_2, y_2, z_2) = (2, 3, 1)$ with direction ratios $\langle a_2, b_2, c_2 \rangle = \langle 3, k, 2 \rangle$.

$$x_2 - x_1 = 2 - 1 = 1$$

Calculating coordinate differences: $y_2 - y_1 = 3 - 2 = 1$

$$z_2 - z_1 = 1 - 3 = -2$$

Using the condition for coplanarity / intersection of two lines:

$$\begin{vmatrix} 1 & 1 & -2 \\ k & 2 & 3 \\ 3 & k & 2 \end{vmatrix} = 0$$

Expanding the determinant along Row 1:

$$1(2(2) - 3(k)) - 1(k(2) - 3(3)) + (-2)(k(k) - 2(3)) = 0$$

$$1(4 - 3k) - 1(2k - 9) - 2(k^2 - 6) = 0$$

$$4 - 3k - 2k + 9 - 2k^2 + 12 = 0$$

Combining like terms:

$$-2k^2 - 5k + 25 = 0$$

Multiplying by -1 :

$$2k^2 + 5k - 25 = 0$$

Factoring the quadratic equation by splitting the middle term:

$$2k^2 + 10k - 5k - 25 = 0$$

$$2k(k + 5) - 5(k + 5) = 0$$

$$(2k - 5)(k + 5) = 0$$

Setting each factor equal to zero: $2k - 5 = 0 \implies k = \frac{5}{2}$

$$k + 5 = 0 \implies k = -5$$

Final Answer

$$k = -5 \quad \text{or} \quad k = \frac{5}{2}$$

Question 13

Show that the lines $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$ and $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$ are coplanar. Also find the equation of the plane containing the lines.

TYPE 63 Coplanarity of Two Lines & Equation of Containing Plane

To Prove: Show that the lines are coplanar and find the equation of the plane containing them.

Concept / Formula Used: $\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0 \quad \vec{n} = \vec{b}_1 \times \vec{b}_2$

Solution:

Line 1 passes through $A(-3, 1, 5)$ with d.r. $\langle a_1, b_1, c_1 \rangle = \langle -3, 1, 5 \rangle$. Line 2 passes through $B(-1, 2, 5)$ with d.r. $\langle a_2, b_2, c_2 \rangle = \langle -1, 2, 5 \rangle$.

$$x_2 - x_1 = -1 - (-3) = 2$$

Calculating coordinate differences: $y_2 - y_1 = 2 - 1 = 1$

$$z_2 - z_1 = 5 - 5 = 0$$

Evaluating the coplanarity determinant:

$$\Delta = \begin{vmatrix} 2 & 1 & 0 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix}$$

$$\begin{aligned} \Delta &= 2(1(5) - 5(2)) - 1((-3)(5) - 5(-1)) + 0 \\ &= 2(5 - 10) - 1(-15 + 5) \\ \text{Expanding along Row 1:} \quad &= 2(-5) - 1(-10) \\ &= -10 + 10 = 0 \end{aligned}$$

Since $\Delta = 0$, the given lines are coplanar.

$$\begin{aligned}\vec{n} &= \vec{b}_1 \times \vec{b}_2 \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix} \\ &= \hat{i}(5 - 10) - \hat{j}(-15 - (-5)) + \hat{k}(-6 - (-1)) \\ &= -5\hat{i} + 10\hat{j} - 5\hat{k}\end{aligned}$$

Dividing by -5 :

$$\vec{n}' = \hat{i} - 2\hat{j} + \hat{k}$$

Equation of the plane passing through $A(-3, 1, 5)$ with normal $\langle 1, -2, 1 \rangle$:

$$1(x - (-3)) - 2(y - 1) + 1(z - 5) = 0$$

$$x + 3 - 2y + 2 + z - 5 = 0$$

$$x - 2y + z = 0$$

Hence Proved

The given lines are coplanar, and the equation of the plane containing them is $x - 2y + z = 0$.

Question 14

Show that the lines $\frac{x-2}{1} = \frac{y-2}{3} = \frac{z-3}{1}$ and $\frac{x-2}{1} = \frac{y-3}{4} = \frac{z-4}{2}$ intersect. Find the point of intersection and the equation of the plane containing them.

TYPE 63 Coplanarity of Two Lines & Equation of Containing Plane

To Prove: Show that the lines intersect, find the point of intersection and plane equation.

Concept / Formula Used: λ, μ

Solution:

Let the general point on Line 1 be:

$$\frac{x-2}{1} = \frac{y-2}{3} = \frac{z-3}{1} = \lambda \implies P(\lambda + 2, 3\lambda + 2, \lambda + 3)$$

Let the general point on Line 2 be:

$$\frac{x-2}{1} = \frac{y-3}{4} = \frac{z-4}{2} = \mu \implies Q(\mu + 2, 4\mu + 3, 2\mu + 4)$$

$$\lambda + 2 = \mu + 2 \implies \lambda = \mu$$

If the lines intersect, P and Q must coincide for some λ, μ : $3\lambda + 2 = 4\mu + 3$

$$\lambda + 3 = 2\mu + 4$$

From (1), substitute $\mu = \lambda$ into (2):

$$3\lambda + 2 = 4\lambda + 3$$

$$4\lambda - 3\lambda = 2 - 3$$

$$\lambda = -1$$

Since $\mu = \lambda$, $\mu = -1$.

Now check if $\lambda = -1, \mu = -1$ satisfies equation (3):

$$\text{LHS} = \lambda + 3 = -1 + 3 = 2$$

$$\text{RHS} = 2\mu + 4 = 2(-1) + 4 = 2$$

Since LHS = RHS, the system of equations is consistent, hence the lines intersect.

$$x = -1 + 2 = 1$$

Substituting $\lambda = -1$ into point P : $y = 3(-1) + 2 = -1$ Point of intersection is $(1, -1, 2)$.

$$z = -1 + 3 = 2$$

Normal vector $\vec{n}$ to the plane containing the lines:

$$\begin{aligned}\vec{n} &= \vec{b}_1 \times \vec{b}_2 \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 1 \\ 1 & 4 & 2 \end{vmatrix} \\ &= \hat{i}(6 - 4) - \hat{j}(2 - 1) + \hat{k}(4 - 3) \\ &= 2\hat{i} - \hat{j} + \hat{k}\end{aligned}$$

Plane equation passing through $(1, -1, 2)$ with normal $\langle 2, -1, 1 \rangle$:

$$2(x - 1) - 1(y - (-1)) + 1(z - 2) = 0$$

$$2x - 2 - y - 1 + z - 2 = 0$$

$$2x - y + z - 5 = 0$$

Hence Proved

The lines intersect at $(1, -1, 2)$ and the containing plane is $2x - y + z - 5 = 0$.

Question 15

If the lines $\frac{x-1}{-3} = \frac{y-2}{-2k} = \frac{z-3}{1}$ and $\frac{x-1}{k} = \frac{y-2}{1} = \frac{z-3}{5}$ are perpendicular, find the value of k and hence find the equation of the plane containing these lines.

TYPE 64 Perpendicular Lines Condition, Combined with Plane Equation

Given: Perpendicular lines L_1 with d.r. $\langle -3, -2k, 1 \rangle$ and L_2 with d.r. $\langle k, 1, 5 \rangle$.

Concept / Formula Used: $a_1a_2 + b_1b_2 + c_1c_2 = 0$ $\vec{n} = \vec{b}_1 \times \vec{b}_2$

Solution:

The direction ratios of the two lines are: $\vec{b}_1 = \langle -3, -2k, 1 \rangle$
 $\vec{b}_2 = \langle k, 1, 5 \rangle$

Since the lines are perpendicular, $\vec{b}_1 \cdot \vec{b}_2 = 0$:

$$(-3)(k) + (-2k)(1) + (1)(5) = 0$$

$$-3k - 2k + 5 = 0$$

$$-5k + 5 = 0$$

$$5k = 5 \implies k = 1$$

Substituting $k = 1$, the direction vectors become: $\vec{b}_1 = -3\hat{i} - 2\hat{j} + \hat{k}$
 $\vec{b}_2 = \hat{i} + \hat{j} + 5\hat{k}$

$$\begin{aligned} \vec{n} &= \vec{b}_1 \times \vec{b}_2 \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & -2 & 1 \\ 1 & 1 & 5 \end{vmatrix} \\ \text{Normal vector } \vec{n} \text{ to the plane: } &= \hat{i}(-10 - 1) - \hat{j}(-15 - 1) + \hat{k}(-3 - (-2)) \\ &= -11\hat{i} + 16\hat{j} - \hat{k} \end{aligned}$$

Multiplying by -1 , we take $\vec{n} = 11\hat{i} - 16\hat{j} + \hat{k}$.

Both lines pass through the common point $A(1, 2, 3)$.

Equation of plane passing through $A(1, 2, 3)$ with normal $\langle 11, -16, 1 \rangle$:

$$11(x - 1) - 16(y - 2) + 1(z - 3) = 0$$

$$11x - 11 - 16y + 32 + z - 3 = 0$$

$$11x - 16y + z + 18 = 0$$

Final Answer

$$k = 1$$

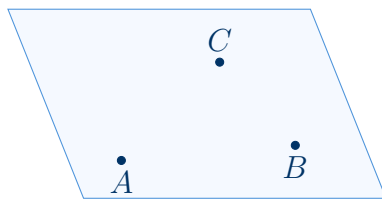
$$\text{Equation of plane: } 11x - 16y + z + 18 = 0$$

Question 16

Find the equations of the planes that pass through the points:

- (i) $(-2, 6, 6)$, $(1, -1, 0)$, and $(1, 2, -1)$
- (ii) $(2, 1, 0)$, $(3, -2, -2)$, and $(3, 1, 7)$

TYPE 65 Plane Through Three Non-Collinear Points – Determinant Form

Plane defined by three points A, B, C

Given: (i) Points $A(-2, 6, 6)$, $B(1, -1, 0)$, $C(1, 2, -1)$.

(ii) Points $A(2, 1, 0)$, $B(3, -2, -2)$, $C(3, 1, 7)$.

Concept / Formula Used: A, B, C $\vec{n} = \vec{AB} \times \vec{AC}$

Solution:

Part (i):

Let $A(-2, 6, 6)$, $B(1, -1, 0)$, $C(1, 2, -1)$.

Vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB} = (1 - (-2))\hat{i} + (-1 - 6)\hat{j} + (0 - 6)\hat{k} = 3\hat{i} - 7\hat{j} - 6\hat{k}$
 $\vec{AC} = (1 - (-2))\hat{i} + (2 - 6)\hat{j} + (-1 - 6)\hat{k} = 3\hat{i} - 4\hat{j} - 7\hat{k}$

$$\begin{aligned}\vec{n} &= \vec{AB} \times \vec{AC} \\ \text{Normal vector } \vec{n}: &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -7 & -6 \\ 3 & -4 & -7 \end{vmatrix} \\ &= \hat{i}(49 - 24) - \hat{j}(-21 - (-18)) + \hat{k}(-12 - (-21)) \\ &= 25\hat{i} + 3\hat{j} + 9\hat{k}\end{aligned}$$

Equation of plane through $B(1, -1, 0)$:

$$25(x - 1) + 3(y - (-1)) + 9(z - 0) = 0$$

$$25x - 25 + 3y + 3 + 9z = 0$$

$$25x + 3y + 9z = 22$$

Part (ii):

Let $A(2, 1, 0)$, $B(3, -2, -2)$, $C(3, 1, 7)$.

Vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB} = (3 - 2)\hat{i} + (-2 - 1)\hat{j} + (-2 - 0)\hat{k} = \hat{i} - 3\hat{j} - 2\hat{k}$
 $\vec{AC} = (3 - 2)\hat{i} + (1 - 1)\hat{j} + (7 - 0)\hat{k} = \hat{i} + 0\hat{j} + 7\hat{k}$

$$\begin{aligned}\vec{n} &= \vec{AB} \times \vec{AC} \\ \text{Normal vector } \vec{n}: &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & -2 \\ 1 & 0 & 7 \end{vmatrix} \\ &= \hat{i}(-21 - 0) - \hat{j}(7 - (-2)) + \hat{k}(0 - (-3)) \\ &= -21\hat{i} - 9\hat{j} + 3\hat{k}\end{aligned}$$

Dividing by -3 :

$$\vec{n}' = 7\hat{i} + 3\hat{j} - \hat{k}$$

Equation of plane through $A(2, 1, 0)$:

$$7(x - 2) + 3(y - 1) - 1(z - 0) = 0$$

$$7x - 14 + 3y - 3 - z = 0$$

$$7x + 3y - z - 17 = 0$$

Final Answer

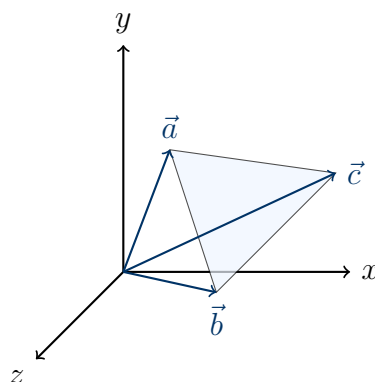
(i) $25x + 3y + 9z = 22$

(ii) $7x + 3y - z - 17 = 0$

Question 17

Find the vector equation of the plane passing through points with position vectors $2\hat{i} + 5\hat{j} - 3\hat{k}$, $-2\hat{i} - 3\hat{j} + 5\hat{k}$, and $5\hat{i} + 3\hat{j} - 3\hat{k}$.

TYPE 65 Plane Through Three Non-Collinear Points – Determinant Form



Plane specified by three position vectors $\vec{a}, \vec{b}, \vec{c}$

Given: Position vectors: $\vec{a} = 2\hat{i} + 5\hat{j} - 3\hat{k}$, $\vec{b} = -2\hat{i} - 3\hat{j} + 5\hat{k}$, $\vec{c} = 5\hat{i} + 3\hat{j} - 3\hat{k}$.

Concept / Formula Used: $(\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})] = 0 \implies \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$

Solution:

Calculating difference vectors $(\vec{b} - \vec{a})$ and $(\vec{c} - \vec{a})$:

$$\vec{b} - \vec{a} = (-2 - 2)\hat{i} + (-3 - 5)\hat{j} + (5 - (-3))\hat{k} = -4\hat{i} - 8\hat{j} + 8\hat{k}$$

$$\vec{c} - \vec{a} = (5 - 2)\hat{i} + (3 - 5)\hat{j} + (-3 - (-3))\hat{k} = 3\hat{i} - 2\hat{j} + 0\hat{k}$$

$$\begin{aligned}\vec{n} &= (\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) \\ \text{Normal vector } \vec{n}: &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} \\ &= \hat{i}(0 - (-16)) - \hat{j}(0 - 24) + \hat{k}(8 - (-24)) \\ &= 16\hat{i} + 24\hat{j} + 32\hat{k}\end{aligned}$$

Dividing by 8:

$$\vec{n}' = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\begin{aligned}\vec{a} \cdot \vec{n}' &= (2\hat{i} + 5\hat{j} - 3\hat{k}) \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) \\ &= (2)(2) + (5)(3) + (-3)(4) \\ \text{Calculating } \vec{a} \cdot \vec{n}': &= 4 + 15 - 12 \\ &= 7\end{aligned}$$

Vector equation of the plane:

$$\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = 7$$

Final Answer

$$\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = 7$$

Question 18

- (i) Show that the points $(1, -1, 1)$, $(2, 3, 1)$, $(1, 2, 3)$, and $(0, -2, 3)$ are coplanar. Also find the equation of the plane in which they lie.
- (ii) Find k such that the four points $A(4, 4, 4)$, $B(5, k, 8)$, $C(5, 4, 1)$, and $D(7, 7, 2)$ are coplanar.

TYPE 66 Coplanarity of Four Points

- To Prove:** (i) Show that four given points are coplanar and find the plane equation.
(ii) Find the value of k for four coplanar points.

Concept / Formula Used: A, B, C, D $[\vec{AB} \quad \vec{AC} \quad \vec{AD}] = 0$

Solution:

Part (i):

Let the points be $P(1, -1, 1)$, $Q(2, 3, 1)$, $R(1, 2, 3)$, $S(0, -2, 3)$.

$$\begin{aligned}\vec{PQ} &= (2 - 1)\hat{i} + (3 - (-1))\hat{j} + (1 - 1)\hat{k} = \hat{i} + 4\hat{j} + 0\hat{k} \\ \text{Vectors } \vec{PQ}, \vec{PR}, \text{ and } \vec{PS}: \vec{PR} &= (1 - 1)\hat{i} + (2 - (-1))\hat{j} + (3 - 1)\hat{k} = 0\hat{i} + 3\hat{j} + 2\hat{k} \\ \vec{PS} &= (0 - 1)\hat{i} + (-2 - (-1))\hat{j} + (3 - 1)\hat{k} = -\hat{i} - \hat{j} + 2\hat{k}\end{aligned}$$

Scalar triple product $[\vec{PQ} \ \vec{PR} \ \vec{PS}]$:

$$\begin{vmatrix} 1 & 4 & 0 \\ 0 & 3 & 2 \\ -1 & -1 & 2 \end{vmatrix}$$

$$= 1(3(2) - 2(-1)) - 4(0(2) - 2(-1)) + 0$$

Expanding along Row 1: $= 1(6 + 2) - 4(0 + 2)$

$$= 8 - 8 = 0$$

Since the scalar triple product is zero, the four points are coplanar.

Normal vector $\vec{n} = \vec{PQ} \times \vec{PR}$:

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 0 \\ 0 & 3 & 2 \end{vmatrix} = 8\hat{i} - 2\hat{j} + 3\hat{k}$$

Equation of plane passing through $P(1, -1, 1)$:

$$8(x - 1) - 2(y - (-1)) + 3(z - 1) = 0$$

$$8x - 8 - 2y - 2 + 3z - 3 = 0$$

$$8x - 2y + 3z - 13 = 0$$

Part (ii):

Points are $A(4, 4, 4)$, $B(5, k, 8)$, $C(5, 4, 1)$, $D(7, 7, 2)$.

$$\vec{AB} = \hat{i} + (k - 4)\hat{j} + 4\hat{k}$$

Vectors $\vec{AB}$, $\vec{AC}$, $\vec{AD}$: $\vec{AC} = \hat{i} + 0\hat{j} - 3\hat{k}$

$$\vec{AD} = 3\hat{i} + 3\hat{j} - 2\hat{k}$$

Coplanarity condition $[\vec{AB} \ \vec{AC} \ \vec{AD}] = 0$:

$$\begin{vmatrix} 1 & k - 4 & 4 \\ 1 & 0 & -3 \\ 3 & 3 & -2 \end{vmatrix} = 0$$

Expanding along Row 2:

$$-1((k - 4)(-2) - 4(3)) + 0 - (-3)(1(3) - (k - 4)(3)) = 0$$

$$-1(-2k + 8 - 12) + 3(3 - 3k + 12) = 0$$

$$-1(-2k - 4) + 3(15 - 3k) = 0$$

$$2k + 4 + 45 - 9k = 0$$

$$-7k + 49 = 0$$

$$7k = 49 \implies k = 7$$

Hence Proved

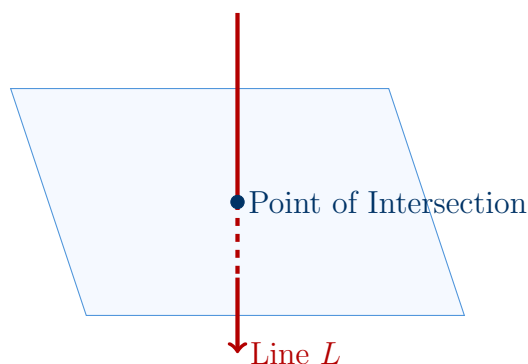
(i) Four points are coplanar; plane equation is $8x - 2y + 3z - 13 = 0$.

(ii) $k = 7$

Question 19

- (i) Find the equations of the line passing through $(2, -1, 2)$ and $(5, 3, 4)$ and of the plane passing through $(2, 0, 3)$, $(1, 1, 5)$, and $(3, 2, 4)$. Also, find their point of intersection.
- (ii) Find the vector and the cartesian equations of the plane passing through the points $(2, 5, -3)$, $(-2, -3, 5)$, and $(5, 3, -3)$. Also find the point of intersection of this plane with the line passing through the points $(3, 1, 5)$ and $(-1, -3, -1)$.

TYPE 50 Point of Intersection of a Line and a Plane



Intersection of line L with plane π

Given: (i) Line through $(2, -1, 2)$ and $(5, 3, 4)$; plane through $(2, 0, 3)$, $(1, 1, 5)$, $(3, 2, 4)$.
(ii) Plane through $(2, 5, -3)$, $(-2, -3, 5)$, $(5, 3, -3)$; line through $(3, 1, 5)$, $(-1, -3, -1)$.

Concept / Formula Used: t

Solution:

Part (i):

Line passing through $L_1(2, -1, 2)$ and $L_2(5, 3, 4)$: Direction ratios $\langle 5 - 2, 3 - (-1), 4 - 2 \rangle = \langle 3, 4, 2 \rangle$. Symmetric equation of line:

$$\frac{x - 2}{3} = \frac{y + 1}{4} = \frac{z - 2}{2} = t$$

Parametric point on line: $P(3t + 2, 4t - 1, 2t + 2)$.

Plane passing through $A(2, 0, 3)$, $B(1, 1, 5)$, $C(3, 2, 4)$: $\vec{AB} = -\hat{i} + \hat{j} + 2\hat{k}$ Normal vector
 $\vec{AC} = \hat{i} + 2\hat{j} + \hat{k}$

$\vec{n} = \vec{AB} \times \vec{AC} = -3\hat{i} + 3\hat{j} - 3\hat{k} \implies \hat{i} - \hat{j} + \hat{k}$. Plane equation through $A(2, 0, 3)$:

$$1(x - 2) - 1(y - 0) + 1(z - 3) = 0 \implies x - y + z - 5 = 0$$

Substituting parametric point P into plane equation:

$$(3t + 2) - (4t - 1) + (2t + 2) - 5 = 0$$

$$3t - 4t + 2t + 2 + 1 + 2 - 5 = 0$$

$$t = 0$$

For $t = 0$: $x = 2$, $y = -1$, $z = 2$. Point of intersection is $(2, -1, 2)$.

Part (ii):

Plane passing through $A(2, 5, -3)$, $B(-2, -3, 5)$, $C(5, 3, -3)$: $\vec{AB} = -4\hat{i} - 8\hat{j} + 8\hat{k}$ Nor-
 $\vec{AC} = 3\hat{i} - 2\hat{j} + 0\hat{k}$
 mal $\vec{n} = \vec{AB} \times \vec{AC} = 16\hat{i} + 24\hat{j} + 32\hat{k} \implies 2\hat{i} + 3\hat{j} + 4\hat{k}$.

Position vector of A : $\vec{a} = 2\hat{i} + 5\hat{j} - 3\hat{k}$. $\vec{a} \cdot \vec{n}' = (2)(2) + (5)(3) + (-3)(4) = 7$. Vector equation: $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = 7$. Cartesian equation: $2x + 3y + 4z - 7 = 0$.

Line through $(3, 1, 5)$ and $(-1, -3, -1)$: Direction ratios $\langle -1 - 3, -3 - 1, -1 - 5 \rangle = \langle -4, -4, -6 \rangle \equiv \langle 2, 2, 3 \rangle$. Line equation: $\frac{x-3}{2} = \frac{y-1}{2} = \frac{z-5}{3} = t \implies (2t+3, 2t+1, 3t+5)$.

Substitute into plane equation:

$$2(2t+3) + 3(2t+1) + 4(3t+5) - 7 = 0$$

$$4t + 6 + 6t + 3 + 12t + 20 - 7 = 0$$

$$22t + 22 = 0 \implies t = -1$$

For $t = -1$: $x = 2(-1) + 3 = 1$, $y = 2(-1) + 1 = -1$, $z = 3(-1) + 5 = 2$. Point of intersection is $(1, -1, 2)$.

Final Answer

(i) Line: $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{2}$; Plane: $x - y + z - 5 = 0$; Intersection: $(2, -1, 2)$

(ii) Vector plane: $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = 7$; Cartesian plane: $2x + 3y + 4z - 7 = 0$; Intersection: $(1, -1, 2)$

Question 20

- Find the equation of the plane passing through the point $(2, 4, 6)$ and making equal intercepts on axes.
- Find the equation of the plane which passes through the point $(1, -2, -1)$ and makes intercepts on the axes such that intercepts on y -axis and z -axis are respectively twice and thrice of the intercept made on x -axis.

TYPE 55 Intercept Form of a Plane – Basic

Given: (i) Passes through $(2, 4, 6)$ with $a = b = c$.

(ii) Passes through $(1, -2, -1)$ with $b = 2a$ and $c = 3a$.

Concept / Formula Used: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Solution:

Part (i):

Let equal intercepts be $a = b = c = a$. Intercept form:

$$\frac{x}{a} + \frac{y}{a} + \frac{z}{a} = 1 \implies x + y + z = a$$

Since plane passes through $(2, 4, 6)$:

$$2 + 4 + 6 = a \implies a = 12$$

Equation of plane:

$$x + y + z = 12$$

Part (ii):

Let intercept on x -axis be a . Intercept on y -axis: $b = 2a$. Intercept on z -axis: $c = 3a$.

Intercept form:

$$\frac{x}{a} + \frac{y}{2a} + \frac{z}{3a} = 1$$

Multiplying by $6a$:

$$6x + 3y + 2z = 6a$$

Since plane passes through $(1, -2, -1)$:

$$6(1) + 3(-2) + 2(-1) = 6a$$

$$6 - 6 - 2 = 6a$$

$$6a = -2$$

Substituting $6a = -2$:

$$6x + 3y + 2z = -2$$

$$6x + 3y + 2z + 2 = 0$$

Final Answer

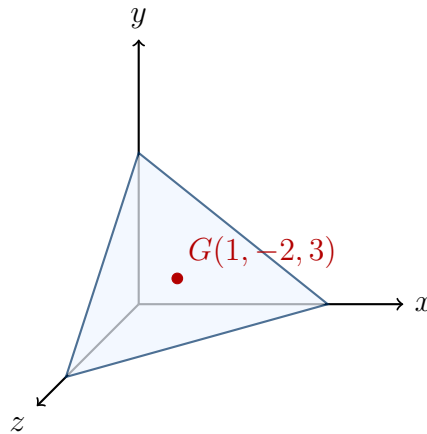
(i) $x + y + z = 12$

(ii) $6x + 3y + 2z + 2 = 0$

Question 21

A plane meets the coordinates axes at A, B , and C respectively such that centroid of triangle ABC is $(1, -2, 3)$. Find the equation of the plane.

TYPE 67 Plane Equation Given Centroid of the Intercept Triangle



Plane meeting axes at A, B, C forming triangle with centroid G

Given: Centroid of $\triangle ABC$ is $(1, -2, 3)$, where $A(a, 0, 0)$, $B(0, b, 0)$, $C(0, 0, c)$.

Concept / Formula Used: (x_1, y_1, z_1) (x_2, y_2, z_2) (x_3, y_3, z_3) $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$

Solution:

Let the plane meet the coordinate axes at $A(a, 0, 0)$, $B(0, b, 0)$, and $C(0, 0, c)$.

The centroid G of triangle ABC is:

$$G \left(\frac{a + 0 + 0}{3}, \frac{0 + b + 0}{3}, \frac{0 + 0 + c}{3} \right) = G \left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3} \right)$$

$$\frac{a}{3} = 1 \implies a = 3$$

Equating to given centroid $(1, -2, 3)$: $\frac{b}{3} = -2 \implies b = -6$

$$\frac{c}{3} = 3 \implies c = 9$$

The equation of the plane in intercept form is:

$$\frac{x}{3} + \frac{y}{-6} + \frac{z}{9} = 1$$

Multiplying the entire equation by the LCM of 3, 6, and 9, which is 18:

$$18 \cdot \frac{x}{3} - 18 \cdot \frac{y}{6} + 18 \cdot \frac{z}{9} = 18 \cdot 1$$

$$6x - 3y + 2z = 18$$

$$6x - 3y + 2z - 18 = 0$$

Final Answer

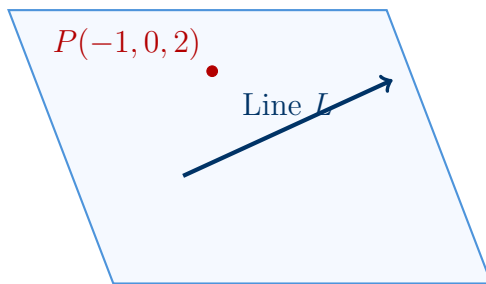
$$6x - 3y + 2z - 18 = 0$$

Question 22

Find the equation of the plane containing the line $\frac{x}{-2} = \frac{y-1}{3} = \frac{1-z}{1}$ and the point $(-1, 0, 2)$.

TYPE 58 Plane Containing a Given Line, Through an External Point

★ Likely Exam Question



Plane containing line L and point $P(-1, 0, 2)$

Given: Point $P(-1, 0, 2)$ and line $\frac{x}{-2} = \frac{y-1}{3} = \frac{1-z}{1}$.

Concept / Formula Used: $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$

Solution:

First, rewrite the given line equation in standard symmetric form:

$$\frac{x-0}{-2} = \frac{y-1}{3} = \frac{z-1}{-1}$$

Point on line is $Q(0, 1, 1)$, and direction vector of line is:

$$\vec{b} = -2\hat{i} + 3\hat{j} - \hat{k}$$

Given point outside the line is $P(-1, 0, 2)$.

Vector $\vec{PQ}$ joining P to Q : $\vec{PQ} = (0 - (-1))\hat{i} + (1 - 0)\hat{j} + (1 - 2)\hat{k}$
 $= \hat{i} + \hat{j} - \hat{k}$

$$\vec{n} = \vec{PQ} \times \vec{b}$$

The normal vector $\vec{n}$ to the plane is perpendicular to both $\vec{PQ}$ and $\vec{b}$:
$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ -2 & 3 & -1 \end{vmatrix}$$

$$\vec{n} = \hat{i}(1(-1) - (-1)(3)) - \hat{j}(1(-1) - (-1)(-2)) + \hat{k}(1(3) - 1(-2))$$

Expanding the determinant: $= \hat{i}(-1 + 3) - \hat{j}(-1 - 2) + \hat{k}(3 + 2)$
 $= 2\hat{i} + 3\hat{j} + 5\hat{k}$

Equation of plane passing through $P(-1, 0, 2)$ with normal $\langle 2, 3, 5 \rangle$:

$$2(x - (-1)) + 3(y - 0) + 5(z - 2) = 0$$

$$2(x + 1) + 3y + 5(z - 2) = 0$$

$$2x + 2 + 3y + 5z - 10 = 0$$

$$2x + 3y + 5z - 8 = 0$$

Final Answer

$$2x + 3y + 5z - 8 = 0$$

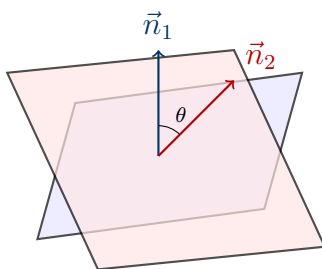
11.7 – Angle Between Two Planes; Angle Between a Line and a Plane

Question 1

Find the acute angle between the planes:

- (i) $\vec{r} \cdot (\hat{i} + \hat{j}) = 1$ and $\vec{r} \cdot (\hat{i} + \hat{k}) = 3$
- (ii) $\vec{r} \cdot (\hat{i} - 2\hat{j} - 2\hat{k}) = 1$ and $\vec{r} \cdot (3\hat{i} - 6\hat{j} + 2\hat{k}) = 0$
- (iii) $\vec{r} \cdot (3\hat{i} - 4\hat{j} + 5\hat{k}) = 0$ and $\vec{r} \cdot (2\hat{i} - \hat{j} - 2\hat{k}) = 7$
- (iv) $2x - y + z = 6$ and $x + y + 2z = 7$

TYPE 68 Angle Between Two Planes – Using Normal Vectors



Acute angle θ between two intersecting planes with normal vectors $\vec{n}_1$ and $\vec{n}_2$

To Find: The acute angle θ between each pair of given planes.

Concept / Formula Used: θ $\vec{n}_1$ $\vec{n}_2$ $\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$

Solution:

- (i) Here, the normal vectors to the given planes are: $\vec{n}_1 = \hat{i} + \hat{j} + 0\hat{k}$ Computing the dot product: $\vec{n}_2 = \hat{i} + 0\hat{j} + \hat{k}$

$$\begin{aligned} \vec{n}_1 \cdot \vec{n}_2 &= (1)(1) + (1)(0) + (0)(1) \\ \text{product:} \quad &= 1 + 0 + 0 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{Computing the magnitudes:} \quad |\vec{n}_1| &= \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2} \\ |\vec{n}_2| &= \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2} \end{aligned}$$

$$\begin{aligned} \cos \theta &= \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|} \\ \text{Using the formula for } \cos \theta: \quad &= \frac{1}{\sqrt{2} \cdot \sqrt{2}} \quad \text{Since } \cos \frac{\pi}{3} = \frac{1}{2}, \text{ we obtain:} \\ &= \frac{1}{2} \\ \theta &= \frac{\pi}{3} = 60^\circ \end{aligned}$$

- (ii) The normal vectors are: $\vec{n}_1 = \hat{i} - 2\hat{j} - 2\hat{k}$ Computing the dot product: $\vec{n}_2 = 3\hat{i} - 6\hat{j} + 2\hat{k}$

$$\begin{aligned} \vec{n}_1 \cdot \vec{n}_2 &= (1)(3) + (-2)(-6) + (-2)(2) \\ &= 3 + 12 - 4 \\ &= 11 \end{aligned}$$

$$\begin{aligned} \text{Computing the magnitudes:} \quad |\vec{n}_1| &= \sqrt{1^2 + (-2)^2 + (-2)^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3 \\ |\vec{n}_2| &= \sqrt{3^2 + (-6)^2 + 2^2} = \sqrt{9 + 36 + 4} = \sqrt{49} = 7 \end{aligned}$$

$$\begin{aligned} \text{Evaluating } \cos \theta: \quad \cos \theta &= \frac{11}{3 \cdot 7} \\ &= \frac{11}{21} \end{aligned}$$

$$\theta = \cos^{-1} \left(\frac{11}{21} \right)$$

- (iii) The normal vectors are: $\vec{n}_1 = 3\hat{i} - 4\hat{j} + 5\hat{k}$ Computing the dot product: $\vec{n}_2 = 2\hat{i} - \hat{j} - 2\hat{k}$

$$\begin{aligned} \vec{n}_1 \cdot \vec{n}_2 &= (3)(2) + (-4)(-1) + (5)(-2) \\ &= 6 + 4 - 10 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \cos \theta &= 0 \\ \text{Since } \vec{n}_1 \cdot \vec{n}_2 = 0, \text{ we have:} \quad \theta &= \frac{\pi}{2} = 90^\circ \end{aligned}$$

- (iv) In Cartesian form, the direction ratios of the normal vectors are: $\vec{n}_1 = 2\hat{i} - \hat{j} + \hat{k}$ $\vec{n}_2 = \hat{i} + \hat{j} + 2\hat{k}$

$$\begin{aligned} \vec{n}_1 \cdot \vec{n}_2 &= (2)(1) + (-1)(1) + (1)(2) \\ \text{Computing the dot product:} \quad &= 2 - 1 + 2 \\ &= 3 \end{aligned}$$

$$\begin{aligned} \text{Computing the magnitudes:} \quad |\vec{n}_1| &= \sqrt{2^2 + (-1)^2 + 1^2} = \sqrt{4 + 1 + 1} = \sqrt{6} \\ |\vec{n}_2| &= \sqrt{1^2 + 1^2 + 2^2} = \sqrt{1 + 1 + 4} = \sqrt{6} \end{aligned}$$

$$\begin{aligned} \cos \theta &= \frac{3}{\sqrt{6} \cdot \sqrt{6}} \\ \text{Evaluating } \cos \theta: \quad &= \frac{3}{6} \\ &= \frac{1}{2} \end{aligned}$$

$$\theta = \frac{\pi}{3} = 60^\circ$$

Final Answer

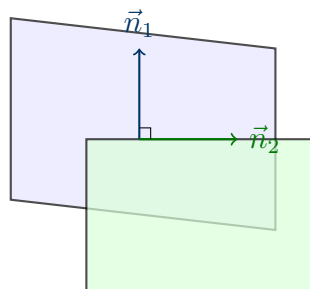
- (i) $\theta = 60^\circ$ or $\frac{\pi}{3}$
 (ii) $\theta = \cos^{-1}\left(\frac{11}{21}\right)$
 (iii) $\theta = 90^\circ$ or $\frac{\pi}{2}$
 (iv) $\theta = 60^\circ$ or $\frac{\pi}{3}$

Common Mistake: Confusing the formula for angle between two planes with the angle between a line and a plane; the latter uses $\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$.

Question 2

Show that the following pairs of planes are at right angles to each other:

- (i) $3x - 5y + 3z - 1 = 0$ and $2x + 3y + 3z = 7$
 (ii) $\vec{r} \cdot (2\hat{i} + 6\hat{j} + 6\hat{k}) = 13$ and $\vec{r} \cdot (3\hat{i} + 4\hat{j} - 5\hat{k}) + 7 = 0$

TYPE 69 Proving Two Planes are Perpendicular

Two perpendicular planes intersecting at right angles (90°)

To Prove: That the given pairs of planes are mutually perpendicular.

Concept / Formula Used: $\vec{n}_1 = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ $\vec{n}_2 = a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$ $\vec{n}_1 \cdot \vec{n}_2 = a_1a_2 + b_1b_2 + c_1c_2 = 0$

Solution:

- (i) For the planes $3x - 5y + 3z - 1 = 0$ and $2x + 3y + 3z = 7$, the normal vectors are:
 $\vec{n}_1 = 3\hat{i} - 5\hat{j} + 3\hat{k}$ $\vec{n}_2 = 2\hat{i} + 3\hat{j} + 3\hat{k}$
 Evaluating the dot product $a_1a_2 + b_1b_2 + c_1c_2$:
 $\vec{n}_1 \cdot \vec{n}_2 = 3(2) + (-5)(3) + 3(3) = 6 - 15 + 9 = 0$
 Since $\vec{n}_1 \cdot \vec{n}_2 = 0$, the normal vectors are orthogonal, which implies the planes are perpendicular to each other.
- (ii) For the vector equations $\vec{r} \cdot (2\hat{i} + 6\hat{j} + 6\hat{k}) = 13$ and $\vec{r} \cdot (3\hat{i} + 4\hat{j} - 5\hat{k}) = -7$,
 the normal vectors are:
 $\vec{n}_1 = 2\hat{i} + 6\hat{j} + 6\hat{k}$ $\vec{n}_2 = 3\hat{i} + 4\hat{j} - 5\hat{k}$
 Evaluating the dot product $\vec{n}_1 \cdot \vec{n}_2$:

$$\begin{aligned}\vec{n}_1 \cdot \vec{n}_2 &= 2(3) + 6(4) + 6(-5) \\ &= 6 + 24 - 30 \\ &= 0\end{aligned}$$

and hence the planes are at right angles.

Hence Proved

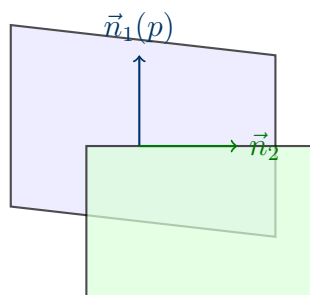
In both cases $\vec{n}_1 \cdot \vec{n}_2 = 0$, hence the given pairs of planes are at right angles to each other.

Question 3

Find the value of p for which the following planes are perpendicular to each other:

- (i) $\vec{r} \cdot (p\hat{i} + 2\hat{j} + 3\hat{k}) = 5$ and $\vec{r} \cdot (\hat{i} + 2\hat{j} - 7\hat{k}) + 11 = 0$
- (ii) $x - 4y + pz + 3 = 0$ and $2x + 2y + 3z = 5$
- (iii) $x - 2y + 4z = 10$ and $18x + 17y + pz = 50$

TYPE 70 Solve for Unknown Constant Using Plane Perpendicularity



Perpendicular planes condition $\vec{n}_1 \cdot \vec{n}_2 = 0$

To Find: The value of the parameter p in each case.

Concept / Formula Used: $\vec{n}_1 \cdot \vec{n}_2 = 0$

Solution:

- (i) Normal vectors: $\vec{n}_1 = p\hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{n}_2 = \hat{i} + 2\hat{j} - 7\hat{k}$ Using the perpendicularity condition $\vec{n}_1 \cdot \vec{n}_2 = 0$:

$$p(1) + 2(2) + 3(-7) = 0$$

$$p + 4 - 21 = 0$$

$$p - 17 = 0$$

$$p = 17$$

- (ii) Direction ratios of the normal vectors are $(1, -4, p)$ and $(2, 2, 3)$. Using $a_1a_2 + b_1b_2 +$

$$1(2) + (-4)(2) + p(3) = 0$$

$$2 - 8 + 3p = 0$$

$$c_1 c_2 = 0:$$

$$-6 + 3p = 0$$

$$3p = 6$$

$$p = 2$$

(iii) Direction ratios of the normal vectors are $(1, -2, 4)$ and $(18, 17, p)$. Using $a_1 a_2 +$
 $1(18) + (-2)(17) + 4(p) = 0$

$$18 - 34 + 4p = 0$$

$$b_1 b_2 + c_1 c_2 = 0:$$

$$-16 + 4p = 0$$

$$4p = 16$$

$$p = 4$$

Final Answer

(i) $p = 17$

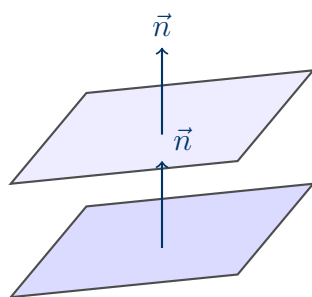
(ii) $p = 2$

(iii) $p = 4$

Question 4

- Write the equation of the plane passing through origin and parallel to the plane $5x - 3y + 7z + 13 = 0$.
- Find the equation of the plane through the point $(1, 4, -2)$ and parallel to the plane $-2x + y - 3z = 7$.
- Find the equation of the plane with intercept 3 on the y-axis and parallel to xz-plane.

TYPE 72 Equation of a Plane Parallel to a Given Plane/Coordinate Plane



Parallel planes having identical normal vector $\vec{n}$

To Find: The Cartesian equations of the required parallel planes.

Concept / Formula Used: $ax + by + cz + d = 0$ $ax + by + cz + \lambda = 0$ λ

Solution:

- (i) The given plane is $5x - 3y + 7z + 13 = 0$. Any plane parallel to it is of the form:

$$5x - 3y + 7z + \lambda = 0$$

Since it passes through the origin $(0, 0, 0)$: $5(0) - 3(0) + 7(0) + \lambda = 0$ Substituting
 $\lambda = 0$

$\lambda = 0$:

$$5x - 3y + 7z = 0$$

- (ii) The given plane is $-2x + y - 3z = 7$, or $-2x + y - 3z - 7 = 0$. Any plane parallel to it has equation:

$$-2x + y - 3z + \lambda = 0$$

$$-2(1) + 4 - 3(-2) + \lambda = 0$$

$$-2 + 4 + 6 + \lambda = 0$$

$$8 + \lambda = 0 \quad \text{Substi-}$$

$$\lambda = -8$$

Since it passes through the point $(1, 4, -2)$:

tuting $\lambda = -8$: $-2x + y - 3z - 8 = 0$
 $2x - y + 3z + 8 = 0$

- (iii) The equation of the xz -plane is $y = 0$. Any plane parallel to the xz -plane is of the form:

$$y = k$$

Since the plane makes an intercept of 3 on the y -axis, it passes through $(0, 3, 0)$:

$$3 = k$$

$k = 3$ Thus, the required equation is:

$$y = 3 \quad \text{or} \quad y - 3 = 0$$

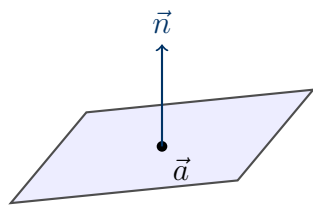
Final Answer

- (i) $5x - 3y + 7z = 0$
 (ii) $2x - y + 3z + 8 = 0$
 (iii) $y = 3$

Question 5

- (i) Find the equation of the plane through the point $3\hat{i} + 4\hat{j} - \hat{k}$ and parallel to the plane $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) + 5 = 0$.
 (ii) Find the equation of the plane passing through the point (a, b, c) and parallel to the plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 2$.

TYPE 72 Equation of a Plane Parallel to a Given Plane/Coordinate Plane



Plane passing through position vector $\vec{a}$ parallel to plane with normal $\vec{n}$

To Find: The vector and Cartesian equations of the required planes.

Concept / Formula Used: $\vec{a}$ $\vec{r} \cdot \vec{n} = d_1$ $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$

Solution:

(i) Here, the given plane is $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) = -5$. Its normal vector is:

$$\vec{n} = 2\hat{i} - 3\hat{j} + 5\hat{k}$$

The position vector of the given point is:

$$\vec{a} = 3\hat{i} + 4\hat{j} - \hat{k}$$

Since the required plane is parallel to the given plane, its equation is:

$$\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$$

$$\begin{aligned} \vec{a} \cdot \vec{n} &= (3\hat{i} + 4\hat{j} - \hat{k}) \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) \\ &= (3)(2) + (4)(-3) + (-1)(5) \quad \text{Thus, the vector equation is:} \\ &= 6 - 12 - 5 \\ &= -11 \end{aligned}$$

$$\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) = -11 \quad \text{or} \quad \vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) + 11 = 0$$

In Cartesian form, substituting $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$:

$$2x - 3y + 5z + 11 = 0$$

(ii) The given plane is $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 2$. Its normal vector is:

$$\vec{n} = \hat{i} + \hat{j} + \hat{k}$$

The position vector of the given point (a, b, c) is:

$$\vec{r}_0 = a\hat{i} + b\hat{j} + c\hat{k}$$

The equation of the parallel plane is:

$$\vec{r} \cdot \vec{n} = \vec{r}_0 \cdot \vec{n}$$

$$\vec{r}_0 \cdot \vec{n} = (a\hat{i} + b\hat{j} + c\hat{k}) \cdot (\hat{i} + \hat{j} + \hat{k})$$

Evaluating $\vec{r}_0 \cdot \vec{n}$: $= a(1) + b(1) + c(1)$ Thus, the required vector equation is:

$$= a + b + c$$

In Cartesian form:

$$x + y + z = a + b + c$$

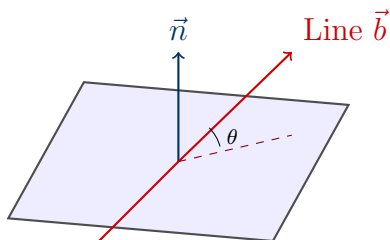
Final Answer

- (i) $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) + 11 = 0$ (Cartesian: $2x - 3y + 5z + 11 = 0$)
 (ii) $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = a + b + c$ (Cartesian: $x + y + z = a + b + c$)

Question 6

- (i) Find the angle between the line $\vec{r} = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda(3\hat{i} - \hat{j} + 2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 3$.
 (ii) Find the angle between the line $\vec{r} = 2\hat{i} - 3\hat{j} + \lambda(\hat{k})$ and the plane $\vec{r} \cdot (\hat{j} - \hat{k}) = 7$.

TYPE 73 Angle Between a Line and a Plane



Angle θ between line with direction vector $\vec{b}$ and plane with normal $\vec{n}$

To Find: The angle θ between the given line and plane.

Concept / Formula Used: θ $\vec{r} = \vec{a} + \lambda\vec{b}$ $\vec{r} \cdot \vec{n} = d$ $\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$

Solution:

- (i) From the line equation $\vec{r} = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda(3\hat{i} - \hat{j} + 2\hat{k})$, the direction vector is:

$$\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$$

From the plane equation $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 3$, the normal vector is:

$$\vec{n} = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{b} \cdot \vec{n} = (3)(1) + (-1)(1) + (2)(1)$$

Evaluating the dot product: $= 3 - 1 + 2$

Computing magni-

$$= 4$$

$$|\vec{b}| = \sqrt{3^2 + (-1)^2 + 2^2} = \sqrt{9 + 1 + 4} = \sqrt{14}$$

$$|\vec{n}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

Using the formula for $\sin \theta$:

$$\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|}$$

$$= \frac{4}{\sqrt{14} \cdot \sqrt{3}}$$

$$= \frac{4}{\sqrt{42}}$$

$$\theta = \sin^{-1} \left(\frac{4}{\sqrt{42}} \right)$$

(ii) From the line equation $\vec{r} = 2\hat{i} - 3\hat{j} + \lambda(\hat{k})$, the direction vector is:

$$\vec{b} = 0\hat{i} + 0\hat{j} + \hat{k}$$

From the plane equation $\vec{r} \cdot (\hat{j} - \hat{k}) = 7$, the normal vector is:

$$\vec{n} = 0\hat{i} + \hat{j} - \hat{k}$$

Evaluating the dot product: $\vec{b} \cdot \vec{n} = (0)(0) + (0)(1) + (1)(-1)$

$$= -1$$

Computing magni-

$$|\vec{b}| = \sqrt{0^2 + 0^2 + 1^2} = 1$$

$$|\vec{n}| = \sqrt{0^2 + 1^2 + (-1)^2} = \sqrt{2}$$

Evaluating $\sin \theta$:

$$\sin \theta = \frac{|-1|}{1 \cdot \sqrt{2}} = \frac{1}{\sqrt{2}}$$

Since $\sin \frac{\pi}{4} =$

$$\frac{1}{\sqrt{2}}:$$

$$\theta = \frac{\pi}{4} = 45^\circ$$

Final Answer

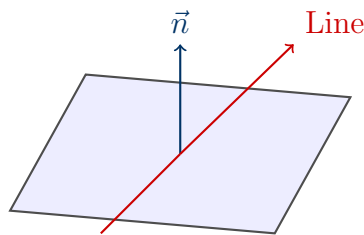
$$(i) \theta = \sin^{-1} \left(\frac{4}{\sqrt{42}} \right)$$

$$(ii) \theta = 45^\circ \text{ or } \frac{\pi}{4}$$

Question 7

- (i) Find the angle between the line $\frac{x-2}{3} = \frac{y+1}{-1} = \frac{z-3}{2}$ and the plane $3x+4y+z+5=0$.
 (ii) Find the angle between the line $\frac{x+1}{2} = \frac{3y+5}{9} = \frac{3-z}{-6}$ and the plane $10x+2y-11z=3$.

TYPE 73 Angle Between a Line and a Plane



Line intersecting plane with direction vector $\vec{b}$ and normal vector $\vec{n}$

To Find: The angle θ between the given Cartesian line and plane.

Concept / Formula Used: $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ $a_2x + b_2y + c_2z + d = 0$
 $\sin \theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$

Solution:

- (i) Direction ratios of the line are $(a_1, b_1, c_1) = (3, -1, 2)$. Direction ratios of the normal to the plane are $(a_2, b_2, c_2) = (3, 4, 1)$. Evaluating $a_1a_2 + b_1b_2 + c_1c_2$:

$$a_1a_2 + b_1b_2 + c_1c_2 = 3(3) + (-1)(4) + 2(1)$$

$$= 9 - 4 + 2$$

$$= 7$$

$$\sin \theta = \frac{7}{\sqrt{14}\sqrt{26}}$$

$$= \frac{7}{\sqrt{364}}$$

Evaluating $\sin \theta$:

$$= \frac{7}{\sqrt{4 \times 91}}$$

$$= \frac{7}{2\sqrt{91}}$$

$$\theta = \sin^{-1} \left(\frac{7}{2\sqrt{91}} \right)$$

$$\begin{aligned} \text{Calculating magnitudes: } \sqrt{a_1^2 + b_1^2 + c_1^2} &= \sqrt{3^2 + (-1)^2 + 2^2} \\ \sqrt{a_2^2 + b_2^2 + c_2^2} &= \sqrt{3^2 + 4^2 + 1^2} \end{aligned}$$

- (ii) First convert the given line equation to standard symmetric form:

$$\frac{x+1}{2} = \frac{3\left(y+\frac{5}{3}\right)}{9} = \frac{-(z-3)}{-6}$$

$$\frac{x+1}{2} = \frac{y+\frac{5}{3}}{3} = \frac{z-3}{6}$$

Thus, direction ratios of the line are $(a_1, b_1, c_1) = (2, 3, 6)$. Direction ratios of normal to the plane are $(a_2, b_2, c_2) = (10, 2, -11)$. Evaluating the dot product term:

$$a_1a_2 + b_1b_2 + c_1c_2 = 2(10) + 3(2) + 6(-11)$$

$$= 20 + 6 - 66$$

$$= -40$$

Calculating magnitudes:

$$\sqrt{a_1^2 + b_1^2 + c_1^2} = \sqrt{2^2 + 3^2 + 6^2}$$

$$\sqrt{a_2^2 + b_2^2 + c_2^2} = \sqrt{10^2 + 2^2 + (-11)^2}$$

$$\begin{aligned} \sin \theta &= \frac{|-40|}{7 \times 15} \\ \text{Evaluating } \sin \theta: &= \frac{40}{105} \\ &= \frac{8}{21} \end{aligned}$$

$$\theta = \sin^{-1} \left(\frac{8}{21} \right)$$

Final Answer

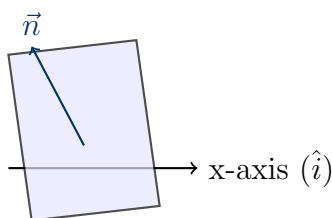
- (i) $\theta = \sin^{-1} \left(\frac{7}{2\sqrt{91}} \right)$
 (ii) $\theta = \sin^{-1} \left(\frac{8}{21} \right)$

Common Mistake: Forgetting to convert $\frac{3y+5}{9}$ and $\frac{3-z}{-6}$ into standard form $\frac{y-y_1}{b}$ and $\frac{z-z_1}{c}$ leads to wrong direction ratios for the line.

Question 8

- (i) If the plane $2x - 3y - 6z = 13$ makes an angle $\sin^{-1}(\lambda)$ with the x-axis, then find the value of λ .
 (ii) If the angle between the line $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and the plane $2x - y + \sqrt{\lambda}z = -4$ is $\sin^{-1} \left(\frac{1}{3} \right)$, find the value of λ .

TYPE 71 Solve for Unknown Parameter Using Line-Plane Angle Formula



Plane making angle θ with x-axis (line along $\hat{i}$)

To Find: The numerical value of λ in both parts.

Concept / Formula Used: θ $\vec{b}$ $\vec{n}$ $\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|}$

Solution:

- (i) The direction of the x-axis is along vector $\vec{b} = \hat{i} = (1, 0, 0)$. The normal vector to the plane $2x - 3y - 6z = 13$ is $\vec{n} = 2\hat{i} - 3\hat{j} - 6\hat{k}$. Given that angle $\theta = \sin^{-1}(\lambda)$, so

$$\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$$

$$\lambda = \frac{|(1)(2) + (0)(-3) + (0)(-6)|}{\sqrt{1^2 + 0^2 + 0^2} \sqrt{2^2 + (-3)^2 + (-6)^2}}$$

$\sin \theta = \lambda$. Using the formula:

$$\begin{aligned} &= \frac{2}{1 \cdot \sqrt{4 + 9 + 36}} \\ &= \frac{2}{\sqrt{49}} \\ &= \frac{2}{7} \end{aligned}$$

(ii) Direction vector of line: $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$. Normal vector to plane: $\vec{n} = 2\hat{i} - \hat{j} + \sqrt{\lambda}\hat{k}$. Given angle $\theta = \sin^{-1}\left(\frac{1}{3}\right) \implies \sin \theta = \frac{1}{3}$. Using the formula:

$$\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$$

$$\frac{1}{3} = \frac{|1(2) + 2(-1) + 2\sqrt{\lambda}|}{\sqrt{1^2 + 2^2 + 2^2} \sqrt{2^2 + (-1)^2 + (\sqrt{\lambda})^2}}$$

Multiplying both sides by 3:

$$\frac{1}{3} = \frac{|2 - 2 + 2\sqrt{\lambda}|}{\sqrt{9}\sqrt{4 + 1 + \lambda}}$$

$$\frac{1}{3} = \frac{2\sqrt{\lambda}}{3\sqrt{5 + \lambda}}$$

$$1 = \frac{2\sqrt{\lambda}}{\sqrt{5 + \lambda}}$$

Cross-multiplying:

$$\sqrt{5 + \lambda} = 2\sqrt{\lambda}$$

$$5 + \lambda = 4\lambda$$

$$4\lambda - \lambda = 5$$

Squaring both sides:

$$3\lambda = 5$$

$$\lambda = \frac{5}{3}$$

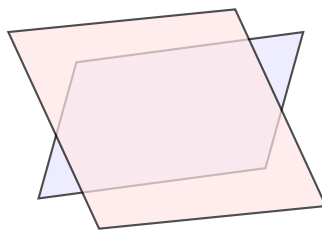
Final Answer

- (i) $\lambda = \frac{2}{7}$
 (ii) $\lambda = \frac{5}{3}$

Question 9

In the following problems, determine whether the given planes are parallel or perpendicular, and in case they are neither, find the (acute) angle between them:

- (i) $2x + y + 3z - 2 = 0$ and $x - 2y + 5 = 0$
 (ii) $4x + 8y + z - 8 = 0$ and $y + z - 4 = 0$
 (iii) $7x + 5y + 6z + 30 = 0$ and $3x - y - 10z + 4 = 0$
 (iv) $2x - 2y + 4z + 5 = 0$ and $3x - 3y + 6z - 1 = 0$

TYPE 68 Angle Between Two Planes – Using Normal Vectors

Testing parallelism / perpendicularity

*Geometric relationship between two planes***To Find:** Classification of each pair of planes and acute angle θ if neither.

Concept / Formula Used: $a_1x + b_1y + c_1z + d_1 = 0$ $a_2x + b_2y + c_2z + d_2 = 0$
 $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ $a_1a_2 + b_1b_2 + c_1c_2 = 0$ θ $\cos \theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$

Solution:

- (i) Here $(a_1, b_1, c_1) = (2, 1, 3)$ and $(a_2, b_2, c_2) = (1, -2, 0)$. Check perpendicularity:
 $a_1a_2 + b_1b_2 + c_1c_2 = 2(1) + 1(-2) + 3(0)$
 $= 2 - 2 + 0$ Since dot product of normals is 0, the planes
 $= 0$
 are **perpendicular**.

- (ii) Here $(a_1, b_1, c_1) = (4, 8, 1)$ and $(a_2, b_2, c_2) = (0, 1, 1)$. Check ratios for parallel: $\frac{4}{0}$ is undefined while $\frac{8}{1} = 8$, so not parallel. Check perpendicularity:

$$a_1a_2 + b_1b_2 + c_1c_2 = 4(0) + 8(1) + 1(1) = 9 \neq 0$$

$$\sqrt{a_1^2 + b_1^2 + c_1^2} = \sqrt{4^2 + 8^2 + 1^2} = \sqrt{16 + 64 + 1} = \sqrt{81} = 9$$

$$\sqrt{a_2^2 + b_2^2 + c_2^2} = \sqrt{0^2 + 1^2 + 1^2} = \sqrt{2}$$

So the planes are **neither**. Now compute acute angle θ :

$$\cos \theta = \frac{9}{9 \cdot \sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ = \frac{\pi}{4}$$

- (iii) Here $(a_1, b_1, c_1) = (7, 5, 6)$ and $(a_2, b_2, c_2) = (3, -1, -10)$. Check ratios for parallel:

$$a_1a_2 + b_1b_2 + c_1c_2 = 7(3) + 5(-1) + 6(-10)$$

 $\frac{7}{3} \neq \frac{5}{-1}$, so not parallel. Check perpendicularity:

$$= 21 - 5 - 60$$

$$= -44 \neq 0$$

$$\sqrt{a_1^2 + b_1^2 + c_1^2} = \sqrt{7^2 + 5^2 + 6^2} = \sqrt{49 + 25 + 36} = \sqrt{110}$$

$$\sqrt{a_2^2 + b_2^2 + c_2^2} = \sqrt{3^2 + (-1)^2 + (-10)^2} = \sqrt{9 + 1 + 100} = \sqrt{110}$$

So the planes are **neither**. Compute angle θ :

$$\cos \theta = \frac{|-44|}{\sqrt{110}\sqrt{110}} = \frac{44}{110} = \frac{2}{5}$$

$$\theta = \cos^{-1} \left(\frac{2}{5} \right)$$

(iv) Here $(a_1, b_1, c_1) = (2, -2, 4)$ and $(a_2, b_2, c_2) = (3, -3, 6)$. Compare normal component ratios:

$$\frac{a_1}{a_2} = \frac{2}{3}, \quad \frac{b_1}{b_2} = \frac{-2}{-3} = \frac{2}{3}, \quad \frac{c_1}{c_2} = \frac{4}{6} = \frac{2}{3}$$

Since $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{2}{3}$, the normal vectors are proportional. Thus, the planes are **parallel**.

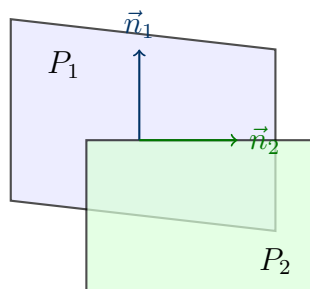
Final Answer

- (i) Perpendicular
- (ii) Neither; acute angle $\theta = 45^\circ$ or $\frac{\pi}{4}$
- (iii) Neither; acute angle $\theta = \cos^{-1} \left(\frac{2}{5} \right)$
- (iv) Parallel

Question 10

Prove that the plane $2x + 3y - 4z = 11$ is perpendicular to each of the planes $x + 2y + 2z - 7 = 0$ and $5x + 6y + 7z = 23$.

TYPE 69 Proving Two Planes are Perpendicular



Plane P_1 perpendicular to both planes P_2 and P_3

To Prove: That the plane $P_1 : 2x + 3y - 4z = 11$ is perpendicular to $P_2 : x + 2y + 2z - 7 = 0$ and $P_3 : 5x + 6y + 7z = 23$.

Concept / Formula Used: P_1 $\vec{n}_1$ P_k $\vec{n}_k$ $\vec{n}_1 \cdot \vec{n}_k = 0$

Solution:

Let the given main plane be $P_1 : 2x + 3y - 4z = 11$. Its normal vector is:

$$\vec{n}_1 = 2\hat{i} + 3\hat{j} - 4\hat{k}$$

For the first target plane $P_2 : x + 2y + 2z - 7 = 0$, the normal vector is:

$$\vec{n}_2 = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$\vec{n}_1 \cdot \vec{n}_2 = 2(1) + 3(2) + (-4)(2)$$

Evaluating the dot product $\vec{n}_1 \cdot \vec{n}_2$:

$$= 2 + 6 - 8$$

$$= 0$$

Since $\vec{n}_1 \cdot \vec{n}_2 = 0$, plane

P_1 is perpendicular to P_2 .

For the second target plane $P_3 : 5x + 6y + 7z = 23$, the normal vector is:

$$\vec{n}_3 = 5\hat{i} + 6\hat{j} + 7\hat{k}$$

$$\vec{n}_1 \cdot \vec{n}_3 = 2(5) + 3(6) + (-4)(7)$$

Evaluating the dot product $\vec{n}_1 \cdot \vec{n}_3$:

$$= 10 + 18 - 28$$

$$= 0$$

Since $\vec{n}_1 \cdot \vec{n}_3 = 0$, plane

P_1 is also perpendicular to P_3 .

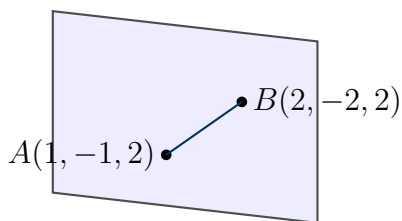
Hence Proved

Since $\vec{n}_1 \cdot \vec{n}_2 = 0$ and $\vec{n}_1 \cdot \vec{n}_3 = 0$, the plane $2x + 3y - 4z = 11$ is perpendicular to each of the given planes.

Question 11

Find the equation of the plane passing through the points $(1, -1, 2)$, $(2, -2, 2)$ and perpendicular to the plane $6x - 2y + 2z = 9$.

TYPE 74 Plane Through Two Points, Perpendicular to Another Plane



Plane passing through points A, B perpendicular to a given plane

To Find: The Cartesian equation of the required plane.

Concept / Formula Used: (x_1, y_1, z_1) $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$
 $aa_p + bb_p + cc_p = 0$

Solution:

The equation of any plane passing through $(1, -1, 2)$ is given by:

$$a(x - 1) + b(y + 1) + c(z - 2) = 0 \quad \text{--- (i)}$$

Since this plane also passes through the second point $(2, -2, 2)$, its coordinates must satisfy

$$a(2 - 1) + b(-2 + 1) + c(2 - 2) = 0$$

$$(i): \quad a(1) + b(-1) + c(0) = 0$$

$$a - b = 0$$

$$a = b \quad \text{--- (ii)}$$

The required plane is perpendicular to

$6x - 2y + 2z = 9$, whose normal has direction ratios $(6, -2, 2)$. Using the perpendicularity condition $a_1a_2 + b_1b_2 + c_1c_2 = 0$:

$$6a - 2b + 2c = 0 \quad \text{--- (iii)}$$

$$6b - 2b + 2c = 0$$

$$4b + 2c = 0$$

$$2c = -4b$$

$$c = -2b$$

Substituting $a = b$ from equation (ii) into equation (iii):

Now substituting

$a = b$ and $c = -2b$ into equation (i):

$$b(x - 1) + b(y + 1) - 2b(z - 2) = 0$$

$$(x - 1) + (y + 1) - 2(z - 2) = 0$$

Dividing by b (since $b \neq 0$):

$$x - 1 + y + 1 - 2z + 4 = 0$$

$$x + y - 2z + 4 = 0$$

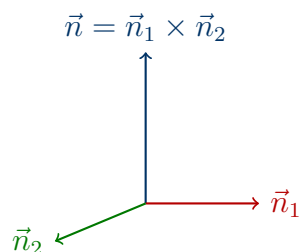
Final Answer

$$x + y - 2z + 4 = 0$$

Question 12

- Find the equation of the plane through the point $(1, 2, 3)$ and perpendicular to each of the planes $x + y + 2z = 3$ and $3x + 2y + z = 4$.
- Find the equation of the plane passing through the points $2\hat{i} - 3\hat{j} + \hat{k}$, $5\hat{i} + 2\hat{j} - \hat{k}$ and perpendicular to the plane $\vec{r} \cdot (\hat{i} - 4\hat{j} + 5\hat{k}) + 3 = 0$.

TYPE 76 Normal of Required Plane via Cross Product of Given Normals



Normal vector $\vec{n} = \vec{n}_1 \times \vec{n}_2$ perpendicular to two planes

To Find: The Cartesian and vector equations of the required planes.

Concept / Formula Used: $\vec{v}_1 \quad \vec{v}_2 \quad \vec{n} = \vec{v}_1 \times \vec{v}_2$

Solution:

- (i) Normals to the given planes $x + y + 2z = 3$ and $3x + 2y + z = 4$ are: $\vec{n}_1 = \hat{i} + \hat{j} + 2\hat{k}$
 $\vec{n}_2 = 3\hat{i} + 2\hat{j} + \hat{k}$

Since the required plane is perpendicular to both, its normal vector $\vec{n}$ is paral-

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 2 \\ 3 & 2 & 1 \end{vmatrix}$$

$$\begin{aligned} \text{lel to } \vec{n}_1 \times \vec{n}_2: &= \hat{i}(1 \cdot 1 - 2 \cdot 2) - \hat{j}(1 \cdot 1 - 2 \cdot 3) + \hat{k}(1 \cdot 2 - 1 \cdot 3) \quad \text{Direction ratios} \\ &= \hat{i}(1 - 4) - \hat{j}(1 - 6) + \hat{k}(2 - 3) \\ &= -3\hat{i} + 5\hat{j} - \hat{k} \end{aligned}$$

of normal are $(a, b, c) = (-3, 5, -1)$. Equation of plane passing through $(1, 2, 3)$:

$$-3(x - 1) + 5(y - 2) - 1(z - 3) = 0$$

$$-3x + 3 + 5y - 10 - z + 3 = 0$$

$$-3x + 5y - z - 4 = 0$$

$$3x - 5y + z + 4 = 0$$

- (ii) Points are $A(2, -3, 1)$ and $B(5, 2, -1)$. Vector joining A and B : $\vec{AB} = (5 - 2)\hat{i} + (2 - (-3))\hat{j} + (-1 - 1)\hat{k}$
 $= 3\hat{i} + 5\hat{j} - 2\hat{k}$

Normal to given plane $\vec{r} \cdot (\hat{i} - 4\hat{j} + 5\hat{k}) + 3 = 0$ is:

$$\vec{n}_1 = \hat{i} - 4\hat{j} + 5\hat{k}$$

$$\vec{n} = \vec{AB} \times \vec{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 5 & -2 \\ 1 & -4 & 5 \end{vmatrix}$$

Normal $\vec{n}$ to required plane is perpendicular to both $\vec{AB}$ and $\vec{n}_1$:

$$\begin{aligned} &= \hat{i}(25 - 8) - \hat{j}(15 - (-2)) + \hat{k}(15 - (-2)) \\ &= 17\hat{i} - 17\hat{j} - 17\hat{k} \end{aligned}$$

Dividing by 17, we take normal vector $\vec{n}' = \hat{i} - \hat{j} - \hat{k}$. Passes through $A(2\hat{i} - 3\hat{j} + \hat{k})$:

$$\vec{r} \cdot (\hat{i} - \hat{j} - \hat{k}) = (2\hat{i} - 3\hat{j} + \hat{k}) \cdot (\hat{i} - \hat{j} - \hat{k})$$

$$= 2(1) + (-3)(-1) + 1(-1)$$

$$= 2 + 3 - 1$$

$$= 4$$

sian equation: $x - y - z = 4$ or $x - y - z - 4 = 0$.

Final Answer

(i) $3x - 5y + z + 4 = 0$

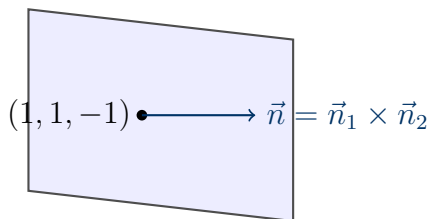
(ii) $\vec{r} \cdot (\hat{i} - \hat{j} - \hat{k}) = 4$ (Cartesian: $x - y - z = 4$)

Question 13

Find the equation of the plane passing through the point $(1, 1, -1)$ and perpendicular to the planes $x + 2y + 3z = 7$ and $2x - 3y + 4z = 0$.

TYPE 75 Plane Through a Point, Perpendicular to Two Given Planes

★ Likely Exam Question



Plane through $(1, 1, -1)$ perpendicular to two intersecting planes

Given: Point $(1, 1, -1)$ and planes $P_1 : x + 2y + 3z = 7$ and $P_2 : 2x - 3y + 4z = 0$.

Concept / Formula Used: $\vec{n} = \vec{n}_1 \times \vec{n}_2$ $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$

Solution:

Normals to the given planes are: $\vec{n}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$ Normal vector $\vec{n}$ to the required
 $\vec{n}_2 = 2\hat{i} - 3\hat{j} + 4\hat{k}$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 2 & -3 & 4 \end{vmatrix}$$

plane is parallel to $\vec{n}_1 \times \vec{n}_2$: $= \hat{i}(2 \cdot 4 - 3(-3)) - \hat{j}(1 \cdot 4 - 3 \cdot 2) + \hat{k}(1(-3) - 2 \cdot 2)$ Thus,
 $= \hat{i}(8 + 9) - \hat{j}(4 - 6) + \hat{k}(-3 - 4)$
 $= 17\hat{i} + 2\hat{j} - 7\hat{k}$

direction ratios of normal are $a = 17, b = 2, c = -7$. Equation of plane passing through
 $17(x - 1) + 2(y - 1) - 7(z - (-1)) = 0$

$$\begin{aligned} (1, 1, -1): \quad & 17(x - 1) + 2(y - 1) - 7(z + 1) = 0 \\ & 17x - 17 + 2y - 2 - 7z - 7 = 0 \\ & 17x + 2y - 7z - 26 = 0 \end{aligned}$$

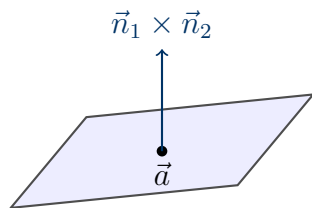
Final Answer

$$17x + 2y - 7z = 26 \quad \text{or} \quad 17x + 2y - 7z - 26 = 0$$

Question 14

Find the equation of the plane through the point $\hat{i} - 3\hat{j} - 2\hat{k}$ and perpendicular to the planes $\vec{r} \cdot (\hat{i} + 2\hat{j} + 2\hat{k}) = 5$ and $\vec{r} \cdot (3\hat{i} + 3\hat{j} + 2\hat{k}) + 7 = 0$.

TYPE 75 Plane Through a Point, Perpendicular to Two Given Planes



Plane through position vector $\vec{a}$ with normal $\vec{n}_1 \times \vec{n}_2$

Given: Position vector $\vec{a} = \hat{i} - 3\hat{j} - 2\hat{k}$ and normals $\vec{n}_1 = \hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{n}_2 = 3\hat{i} + 3\hat{j} + 2\hat{k}$.

Concept / Formula Used: $\vec{a} \quad \vec{n} \quad \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n} \quad \vec{n} = \vec{n}_1 \times \vec{n}_2$

Solution:

$$\begin{aligned} \vec{n} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 3 & 3 & 2 \end{vmatrix} \\ \text{Calculating normal vector } \vec{n} = \vec{n}_1 \times \vec{n}_2: &= \hat{i}(4 - 6) - \hat{j}(2 - 6) + \hat{k}(3 - 6) \quad \text{Computing} \\ &= -2\hat{i} + 4\hat{j} - 3\hat{k} \\ \text{scalar product } \vec{a} \cdot \vec{n} \text{ where } \vec{a} = \hat{i} - 3\hat{j} - 2\hat{k}: &= (1)(-2) + (-3)(4) + (-2)(-3) \\ &= -2 - 12 + 6 \quad \text{Vector} \\ &= -8 \end{aligned}$$

$$\begin{aligned} \text{equation of plane: } \vec{r} \cdot (-2\hat{i} + 4\hat{j} - 3\hat{k}) &= -8 \\ \vec{r} \cdot (2\hat{i} - 4\hat{j} + 3\hat{k}) &= 8 \end{aligned} \quad \text{In Cartesian form: } 2x - 4y + 3z = 8.$$

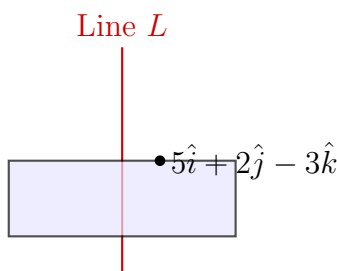
Final Answer

$$\vec{r} \cdot (2\hat{i} - 4\hat{j} + 3\hat{k}) = 8 \quad (\text{Cartesian: } 2x - 4y + 3z = 8)$$

Question 15

Find the vector equation of the plane passing through the point $5\hat{i} + 2\hat{j} - 3\hat{k}$ and perpendicular to the line of intersection of the planes $\vec{r} \cdot (2\hat{i} - \hat{j} + 2\hat{k}) = 0$ and $\vec{r} \cdot (\hat{i} + 3\hat{j} - 5\hat{k}) + 7 = 0$.

TYPE 76 Normal of Required Plane via Cross Product of Given Normals



Plane perpendicular to line of intersection L of two planes

Given: Point $\vec{a} = 5\hat{i} + 2\hat{j} - 3\hat{k}$ and intersecting planes with normals $\vec{n}_1 = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{n}_2 = \hat{i} + 3\hat{j} - 5\hat{k}$.

Concept / Formula Used: $\vec{m} = \vec{n}_1 \times \vec{n}_2$ $\vec{n} = \vec{m}$

Solution:

$$\begin{aligned}\vec{m} = \vec{n}_1 \times \vec{n}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 2 \\ 1 & 3 & -5 \end{vmatrix} \\ &= \hat{i}(5 - 6) - \hat{j}(-10 - 2) + \hat{k}(6 - (-3)) \\ &= -\hat{i} + 12\hat{j} + 7\hat{k}\end{aligned}$$

The direction vector $\vec{m}$ of the line of intersection of given planes is:

Since the required plane is perpendicular to this line of intersection, its normal vector $\vec{n}$ is parallel to $\vec{m}$:

$$\vec{n} = -\hat{i} + 12\hat{j} + 7\hat{k}$$

Using vector equation $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$ with $\vec{a} = 5\hat{i} + 2\hat{j} - 3\hat{k}$:

$$\begin{aligned}\vec{a} \cdot \vec{n} &= (5)(-1) + (2)(12) + (-3)(7) \\ &= -5 + 24 - 21 \\ &= -2\end{aligned}$$

Equation of the plane: $\vec{r} \cdot (-\hat{i} + 12\hat{j} + 7\hat{k}) = -2$

$$\vec{r} \cdot (\hat{i} - 12\hat{j} - 7\hat{k}) = 2$$

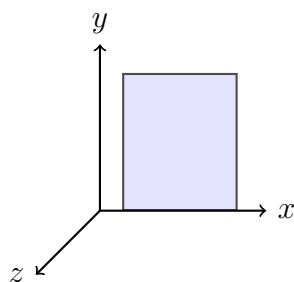
Final Answer

$$\vec{r} \cdot (\hat{i} - 12\hat{j} - 7\hat{k}) = 2$$

Question 16

Find the equation of the plane passing through the points $(2, 3, 1)$ and $(4, -5, 3)$ and parallel to x-axis.

TYPE 77 Plane Parallel to a Coordinate Axis, Through Two Points



Plane parallel to x-axis in 3D coordinate frame

Given: Points $(2, 3, 1)$ and $(4, -5, 3)$, and plane is parallel to x-axis.

Concept / Formula Used: $ax + by + cz + d = 0$ $b(y - y_1) + c(z - z_1) = 0$

Solution:

Since the plane is parallel to the x-axis, its normal vector is perpendicular to $\hat{i} = (1, 0, 0)$. Hence, coefficient of x is 0, so equation of plane through $(2, 3, 1)$ is:

$$b(y - 3) + c(z - 1) = 0 \quad \text{--- (i)}$$

$$b(-5 - 3) + c(3 - 1) = 0$$

Since it also passes through $(4, -5, 3)$:

$$-8b + 2c = 0$$

$$2c = 8b$$

$$c = 4b$$

Substituting $c = 4b$ into

equation (i):

$$b(y - 3) + 4b(z - 1) = 0$$

$$(y - 3) + 4(z - 1) = 0$$

Dividing by $b \neq 0$: $y - 3 + 4z - 4 = 0$

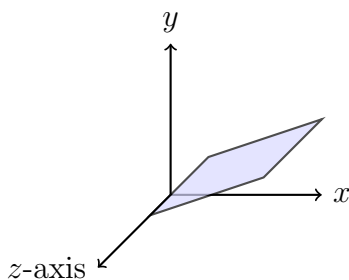
$$y + 4z - 7 = 0$$

Final Answer

$$y + 4z - 7 = 0$$

Question 17

Show that the equation $ax + by + d = 0$ represents a plane parallel to the z-axis. Hence find the equation to a plane through the points $(2, -3, 1)$ and $(-4, 7, 6)$ and parallel to z-axis.

TYPE 77 Plane Parallel to a Coordinate Axis, Through Two Points

Plane $ax + by + d = 0$ parallel to z-axis

To Prove: 1. Show that $ax + by + d = 0$ is parallel to the z-axis.

To Find: 2. Equation of plane through $(2, -3, 1)$ and $(-4, 7, 6)$ parallel to z-axis.

Concept / Formula Used: $\vec{n}$ $\hat{k}$ $\vec{n} \cdot \hat{k} = 0$

Solution:**Part 1: Proof**

The normal vector to the plane $ax + by + d = 0$ is:

$$\vec{n} = a\hat{i} + b\hat{j} + 0\hat{k}$$

The direction vector of the z-axis is $\hat{k} = 0\hat{i} + 0\hat{j} + 1\hat{k}$. Taking dot product $\vec{n} \cdot \hat{k}$:

$$\vec{n} \cdot \hat{k} = a(0) + b(0) + 0(1) = 0$$

Since normal vector $\vec{n}$ is perpendicular to z-axis, the plane itself is parallel to the z-axis.

Part 2: Calculation

The equation of any plane parallel to z-axis is:

$$ax + by + d = 0$$

Since it passes through $(2, -3, 1)$:

$$2a - 3b + d = 0 \quad \text{--- (i)}$$

Since it passes through $(-4, 7, 6)$:

$$-4a + 7b + d = 0 \quad \text{--- (ii)}$$

$$(-4a + 7b + d) - (2a - 3b + d) = 0$$

$$-6a + 10b = 0$$

Subtracting (i) from (ii):

$$6a = 10b \quad \text{From (i), expressing } d \text{ in}$$

$$a = \frac{5}{3}b$$

$$d = 3b - 2a$$

$$= 3b - 2\left(\frac{5}{3}b\right)$$

terms of b :

$$= 3b - \frac{10}{3}b$$

$$= -\frac{1}{3}b$$

Substituting $a = \frac{5}{3}b$ and $d = -\frac{1}{3}b$ into $ax + by + d = 0$:

$$\frac{5}{3}bx + by - \frac{1}{3}b = 0$$

Dividing by $b \neq 0$ and multiplying by 3:

$$5x + 3y - 1 = 0$$

Hence Proved

Since $\vec{n} \cdot \hat{k} = 0$, $ax + by + d = 0$ is parallel to z-axis.

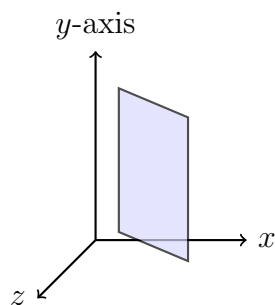
Final Answer

$$5x + 3y - 1 = 0$$

Question 18

Show that the equation $ax + cz + d = 0$ represents a plane parallel to y-axis. Hence find the equation to a plane through the point $(3, -5, -2)$, which makes intercept -4 on x-axis and is parallel to y-axis.

TYPE 77 Plane Parallel to a Coordinate Axis, Through Two Points



Plane $ax + cz + d = 0$ parallel to y-axis

To Prove: 1. Show that $ax + cz + d = 0$ is parallel to y-axis.

To Find: 2. Equation of plane parallel to y-axis passing through $(3, -5, -2)$ with x-intercept -4 .

Concept / Formula Used: $\vec{n} \cdot \hat{j} = 0$

Solution:

Part 1: Proof

The normal vector to the plane $ax + cz + d = 0$ is:

$$\vec{n} = a\hat{i} + 0\hat{j} + c\hat{k}$$

Direction vector of y-axis is $\hat{j} = 0\hat{i} + 1\hat{j} + 0\hat{k}$. Evaluating dot product:

$$\vec{n} \cdot \hat{j} = a(0) + 0(1) + c(0) = 0$$

Since normal vector $\vec{n} \perp \hat{j}$, the plane $ax + cz + d = 0$ is parallel to the y-axis.

Part 2: Calculation

Any plane parallel to y-axis has equation $ax + cz + d = 0$. Given x-intercept is -4 ,
 $a(-4) + c(0) + d = 0$

the plane passes through $(-4, 0, 0)$: $-4a + d = 0$ The plane also passes
 $d = 4a$ — (i)

through $(3, -5, -2)$:

$$a(3) + c(-2) + d = 0 \implies 3a - 2c + d = 0 \quad \text{— (ii)}$$

$$3a - 2c + 4a = 0$$

$$7a - 2c = 0$$

Substituting $d = 4a$ into (ii):

$$2c = 7a$$

Substituting $c = \frac{7}{2}a$ and $d = 4a$

$$c = \frac{7}{2}a \quad \text{--- (iii)}$$

into general equation $ax + cz + d = 0$:

$$ax + \frac{7}{2}az + 4a = 0$$

Dividing by $a \neq 0$ and multiplying by 2:

$$2x + 7z + 8 = 0$$

Hence Proved

Since $\vec{n} \cdot \hat{j} = 0$, $ax + cz + d = 0$ is parallel to y-axis.

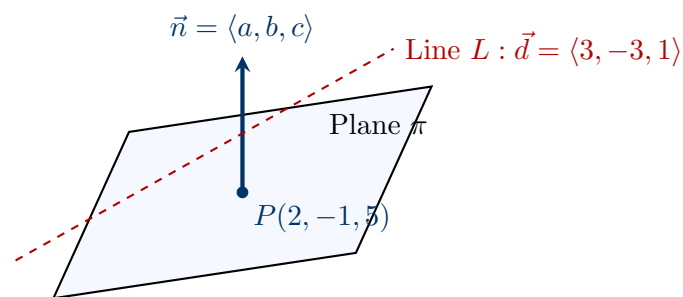
Final Answer

$$2x + 7z + 8 = 0$$

Question 19

Find the equation of a plane passing through the point $(2, -1, 5)$, perpendicular to the plane $x + 2y - 3z = 9$ and parallel to the line $\frac{x-4}{3} = \frac{y+1}{-3} = \frac{z-2}{1}$.

TYPE 78 Equation of a Plane Through a Point – Combined Conditions



Geometry of the required plane π passing through point P , perpendicular to plane π_1 and parallel to line L

Given: Point $P(2, -1, 5)$, perpendicular plane $\pi_1 : x + 2y - 3z = 9$, and parallel line $L : \frac{x-4}{3} = \frac{y+1}{-3} = \frac{z-2}{1}$

Concept / Formula Used: (x_1, y_1, z_1) $\langle a, b, c \rangle$ $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$

Solution:

Let the direction ratios of the normal to the required plane be $\langle a, b, c \rangle$.

Since the required plane passes through the point $(x_1, y_1, z_1) = (2, -1, 5)$, its equation is given by:

$$a(x - 2) + b(y - (-1)) + c(z - 5) = 0$$

$$a(x - 2) + b(y + 1) + c(z - 5) = 0 \quad \text{--- (1)}$$

The normal vector of the given plane $x + 2y - 3z = 9$ has direction ratios $\langle 1, 2, -3 \rangle$. Since the required plane is perpendicular to the given plane, their normal vectors are

orthogonal: $a(1) + b(2) + c(-3) = 0$

$$a + 2b - 3c = 0 \quad \text{--- (2)}$$

The direction vector of the given line $\frac{x-4}{3} = \frac{y+1}{-3} = \frac{z-2}{1}$ has direction ratios $\langle 3, -3, 1 \rangle$. Since the required plane is parallel to the line, the normal to the plane is

perpendicular to the direction vector of the line: $a(3) + b(-3) + c(1) = 0$

$$3a - 3b + c = 0 \quad \text{--- (3)}$$

Solving equations (2) and (3) for a, b, c using the method of cross-multiplication:

$$\frac{a}{(2)(1) - (-3)(-3)} = \frac{b}{(-3)(1) - (3)(3)} = \frac{c}{(3)(-3) - (9)(1)}$$

$$\frac{a}{2 - 9} = \frac{b}{-3 - 9} = \frac{c}{-9 - 9}$$

$$\frac{a}{-7} = \frac{b}{-12} = \frac{c}{-18}$$

$$\frac{a}{7} = \frac{b}{12} = \frac{c}{18}$$

$$a = 7k$$

Thus, we obtain: $b = 10k$

$$c = 9k$$

Substituting the values of a, b , and c back into equation (1): $7k(x - 2) + 10k(y + 1) + 9k(z - 5) = 0$

$$7(x - 2) + 10(y + 1) + 9(z - 5) = 0$$

Dividing the entire equation by k (since $k \neq 0$): $7x - 14 + 10y + 10 + 9z - 45 = 0$

$$7x + 10y + 9z - 49 = 0$$

Final Answer

The required equation of the plane is $7x + 10y + 9z - 49 = 0$.

Common Mistake: Ensure that the condition for a plane to be parallel to a line ($\vec{n} \cdot \vec{d} = 0$) is applied correctly. Students often confuse it with the condition for two lines to be parallel ($\vec{d}_1 \times \vec{d}_2 = \vec{0}$).

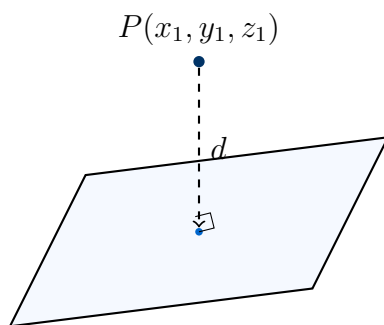
11.8 – Distance of a Point from a Plane

Question 1

In each of the following problems, find the perpendicular distance:

- from the origin to the plane $3x - 4y + 12z = 3$
- from the point $(-6, 0, 0)$ to the plane $2x - 3y + 6z - 2 = 0$
- from the point $(3, -2, 1)$ to the plane $2x - y + 2z + 3 = 0$

TYPE 79 Perpendicular Distance from a Point to a Plane



Perpendicular distance from a point to a plane

Concept / Formula Used: $d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$ d (x_1, y_1, z_1) $Ax + By + Cz + D = 0$

Solution:

- (i) Here, the point is the origin $(x_1, y_1, z_1) = (0, 0, 0)$. The equation of the plane is $3x - 4y + 12z = 3$, which can be written in standard form as:

$$3x - 4y + 12z - 3 = 0$$

Comparing with $Ax + By + Cz + D = 0$, we have:

$$A = 3, \quad B = -4, \quad C = 12, \quad D = -3$$

$$\begin{aligned} d &= \frac{|3(0) - 4(0) + 12(0) - 3|}{\sqrt{3^2 + (-4)^2 + 12^2}} \\ &= \frac{|-3|}{\sqrt{9 + 16 + 144}} \\ &= \frac{3}{\sqrt{169}} \\ &= \frac{3}{13} \text{ units} \end{aligned}$$

Using the perpendicular distance formula:

- (ii) Here, the point is $(x_1, y_1, z_1) = (-6, 0, 0)$. The equation of the plane is:

$$2x - 3y + 6z - 2 = 0$$

Comparing with $Ax + By + Cz + D = 0$, we have:

$$A = 2, \quad B = -3, \quad C = 6, \quad D = -2$$

$$\begin{aligned}
 d &= \frac{|2(-6) - 3(0) + 6(0) - 2|}{\sqrt{2^2 + (-3)^2 + 6^2}} \\
 &= \frac{|-12 - 2|}{\sqrt{4 + 9 + 36}} \\
 &= \frac{|-14|}{\sqrt{49}} \\
 &= \frac{14}{7} \\
 &= 2 \text{ units}
 \end{aligned}$$

Using the perpendicular distance formula:

(iii) Here, the point is $(x_1, y_1, z_1) = (3, -2, 1)$. The equation of the plane is:

$$2x - y + 2z + 3 = 0$$

Comparing with $Ax + By + Cz + D = 0$, we have:

$$A = 2, \quad B = -1, \quad C = 2, \quad D = 3$$

$$\begin{aligned}
 d &= \frac{|2(3) - 1(-2) + 2(1) + 3|}{\sqrt{2^2 + (-1)^2 + 2^2}} \\
 &= \frac{|6 + 2 + 2 + 3|}{\sqrt{4 + 1 + 4}} \\
 &= \frac{|13|}{\sqrt{9}} \\
 &= \frac{13}{3} \text{ units}
 \end{aligned}$$

Using the perpendicular distance formula:

Final Answer

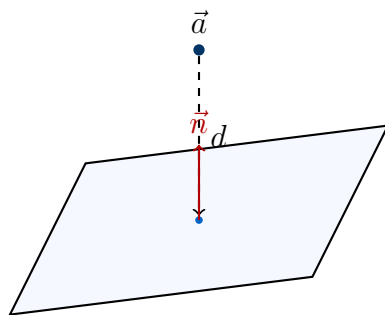
- (i) $\frac{3}{13}$ units
- (ii) 2 units
- (iii) $\frac{13}{3}$ units

Question 2

Solve the following:

- (i) Find the distance of the point whose position vector is $2\hat{i} + \hat{j} - \hat{k}$ from the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + 4\hat{k}) = 9$.
- (ii) Find the length of the perpendicular from origin to the plane $\vec{r} \cdot (3\hat{i} - 4\hat{j} - 12\hat{k}) + 39 = 0$.
- (iii) Find the distance of the point $(3, 4, 5)$ from the plane $\vec{r} \cdot (2\hat{i} - 5\hat{j} + 3\hat{k}) = 13$.
- (iv) Write the distance from the point $(2, 3, -5)$ to the plane XY-plane.

TYPE 79 Perpendicular Distance from a Point to a Plane



Vector distance of a point from a plane

Concept / Formula Used: $d = \frac{|\vec{a} \cdot \vec{n} - q|}{|\vec{n}|}$ d $\vec{a}$ $\vec{r} \cdot \vec{n} = q$

Solution:

- (i) Let the position vector of the point be $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$. The equation of the plane is $\vec{r} \cdot (\hat{i} - 2\hat{j} + 4\hat{k}) = 9$. Here, $\vec{n} = \hat{i} - 2\hat{j} + 4\hat{k}$ and $q = 9$. First, let us calculate

$$\begin{aligned} \vec{a} \cdot \vec{n} &= (2\hat{i} + \hat{j} - \hat{k}) \cdot (\hat{i} - 2\hat{j} + 4\hat{k}) \\ \text{the dot product } \vec{a} \cdot \vec{n}: &= (2)(1) + (1)(-2) + (-1)(4) \quad \text{Next, let us calculate the} \\ &= 2 - 2 - 4 \\ &= -4 \end{aligned}$$

$$\begin{aligned} |\vec{n}| &= \sqrt{1^2 + (-2)^2 + 4^2} \\ \text{magnitude } |\vec{n}|: &= \sqrt{1 + 4 + 16} \quad \text{Using the formula:} \\ &= \sqrt{21} \end{aligned}$$

$$\begin{aligned} d &= \frac{|\vec{a} \cdot \vec{n} - q|}{|\vec{n}|} \\ &= \frac{|-4 - 9|}{\sqrt{21}} \\ &= \frac{|-13|}{\sqrt{21}} \\ &= \frac{13}{\sqrt{21}} \text{ units} \end{aligned}$$

- (ii) Here, the point is the origin, so $\vec{a} = \vec{0} = 0\hat{i} + 0\hat{j} + 0\hat{k}$. The equation of the plane is $\vec{r} \cdot (3\hat{i} - 4\hat{j} - 12\hat{k}) + 39 = 0$, which can be written as:

$$\vec{r} \cdot (3\hat{i} - 4\hat{j} - 12\hat{k}) = -39$$

Here, $\vec{n} = 3\hat{i} - 4\hat{j} - 12\hat{k}$ and $q = -39$. First, let us calculate the dot product $\vec{a} \cdot \vec{n}$:

$$\vec{a} \cdot \vec{n} = 0$$

$$\begin{aligned} |\vec{n}| &= \sqrt{3^2 + (-4)^2 + (-12)^2} \\ &= \sqrt{9 + 16 + 144} \\ \text{Next, let us calculate the magnitude } |\vec{n}|: &= \sqrt{169} \quad \text{Using the} \\ &= 13 \end{aligned}$$

$$\begin{aligned}
 d &= \frac{|\vec{a} \cdot \vec{n} - q|}{|\vec{n}|} \\
 \text{formula: } &= \frac{|0 - (-39)|}{13} \\
 &= \frac{39}{13} \\
 &= 3 \text{ units}
 \end{aligned}$$

(iii) The point is $(3, 4, 5)$, so its position vector is $\vec{a} = 3\hat{i} + 4\hat{j} + 5\hat{k}$. The equation of the plane is $\vec{r} \cdot (2\hat{i} - 5\hat{j} + 3\hat{k}) = 13$. Here, $\vec{n} = 2\hat{i} - 5\hat{j} + 3\hat{k}$ and $q = 13$. First, let us calculate

$$\begin{aligned}
 \vec{a} \cdot \vec{n} &= (3\hat{i} + 4\hat{j} + 5\hat{k}) \cdot (2\hat{i} - 5\hat{j} + 3\hat{k}) \\
 \text{the dot product } \vec{a} \cdot \vec{n}: &= (3)(2) + (4)(-5) + (5)(3) \quad \text{Next, let us calculate} \\
 &= 6 - 20 + 15 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 d &= \frac{|\vec{a} \cdot \vec{n} - q|}{|\vec{n}|} \\
 &= \frac{|1 - 13|}{\sqrt{38}} \\
 &= \frac{|-12|}{\sqrt{38}} \\
 &= \frac{12}{\sqrt{38}} \text{ units}
 \end{aligned}$$

$$\begin{aligned}
 |\vec{n}| &= \sqrt{2^2 + (-5)^2 + 3^2} \\
 \text{the magnitude } |\vec{n}|: &= \sqrt{4 + 25 + 9} \quad \text{Using the formula:} \\
 &= \sqrt{38}
 \end{aligned}$$

(iv) The equation of the XY-plane is $z = 0$. The distance of any point (x_1, y_1, z_1) from the XY-plane is given by $|z_1|$. For the point $(2, 3, -5)$, the distance is:

$$d = |-5| = 5 \text{ units}$$

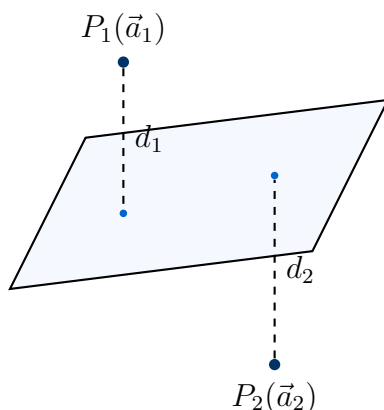
Final Answer

- (i) $\frac{13}{\sqrt{21}}$ units
- (ii) 3 units
- (iii) $\frac{12}{\sqrt{38}}$ units
- (iv) 5 units

Question 3

Show that the points with position vectors $\hat{i} - \hat{j} + 3\hat{k}$ and $3(\hat{i} + \hat{j} + \hat{k})$ are equidistant from the plane $\vec{r} \cdot (5\hat{i} + 2\hat{j} - 7\hat{k}) + 9 = 0$.

TYPE 80 Verify Two Points are Equidistant from a Plane



Equidistant points on opposite sides of a plane

To Prove: The distance of the point $\vec{a}_1 = \hat{i} - \hat{j} + 3\hat{k}$ from the plane is equal to the distance of the point $\vec{a}_2 = 3(\hat{i} + \hat{j} + \hat{k})$ from the plane.

Concept / Formula Used: $d = \frac{|\vec{a} \cdot \vec{n} + D|}{|\vec{n}|}$ d $\vec{a}$ $\vec{r} \cdot \vec{n} + D = 0$

Solution:

Let the given points be represented by position vectors:

$$\vec{a}_1 = \hat{i} - \hat{j} + 3\hat{k}$$

$$\vec{a}_2 = 3(\hat{i} + \hat{j} + \hat{k}) = 3\hat{i} + 3\hat{j} + 3\hat{k}$$

The equation of the plane is:

$$\vec{r} \cdot (5\hat{i} + 2\hat{j} - 7\hat{k}) + 9 = 0$$

Here, the normal vector is $\vec{n} = 5\hat{i} + 2\hat{j} - 7\hat{k}$ and the constant term is $D = 9$.

$$|\vec{n}| = \sqrt{5^2 + 2^2 + (-7)^2}$$

First, let us find the magnitude of the normal vector $|\vec{n}|$:

$$\begin{aligned} &= \sqrt{25 + 4 + 49} \\ &= \sqrt{78} \end{aligned}$$

$$\begin{aligned} \vec{a}_1 \cdot \vec{n} &= (\hat{i} - \hat{j} + 3\hat{k}) \cdot (5\hat{i} + 2\hat{j} - 7\hat{k}) \\ &= (1)(5) + (-1)(2) + (3)(-7) \\ &= 5 - 2 - 21 \\ &= -18 \end{aligned}$$

Now, let us calculate the distance d_1 of the first point $\vec{a}_1$ from the plane:

$$\begin{aligned} d_1 &= \frac{|\vec{a}_1 \cdot \vec{n} + 9|}{|\vec{n}|} \\ &= \frac{|-18 + 9|}{\sqrt{78}} \\ &= \frac{|-9|}{\sqrt{78}} \\ &= \frac{9}{\sqrt{78}} \text{ units} \end{aligned}$$

Using the distance formula:

Next, let us calculate the distance d_2 of the second point $\vec{a}_2$ from the plane:

$$\begin{aligned}\vec{a}_2 \cdot \vec{n} &= (3\hat{i} + 3\hat{j} + 3\hat{k}) \cdot (5\hat{i} + 2\hat{j} - 2\hat{k}) \\ &= (3)(5) + (3)(2) + (3)(-2) \\ &= 15 + 6 - 21 \\ &= 0\end{aligned}$$

$$d_2 = \frac{|\vec{a}_2 \cdot \vec{n} + 9|}{|\vec{n}|}$$

Using the distance formula:
$$\begin{aligned}&= \frac{|0 + 9|}{\sqrt{78}} \\ &= \frac{9}{\sqrt{78}} \text{ units}\end{aligned}$$

Since $d_1 = d_2 = \frac{9}{\sqrt{78}}$ units, the points are equidistant from the given plane.

Hence Proved

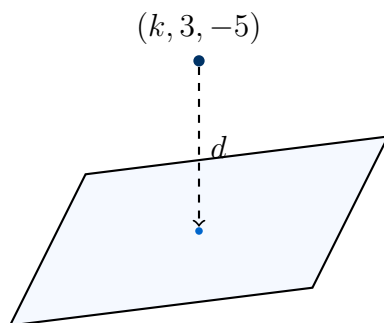
LHS = RHS, hence proved.

Question 4

Solve the following:

- If the distance from the point $(k, 3, -5)$ to the plane $x + 2y - 2z = 9$ is 3 units, then find the value(s) of k .
- If the distance from the point $(-6, 0, 0)$ to the plane $2x - 3y + z - k = 0$ is $\sqrt{14}$ units, find the value(s) of k .

TYPE 81 Solve for Unknown Parameter Using a Given Distance



Distance from a point containing an unknown parameter to a plane

Concept / Formula Used: $d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$ d (x_1, y_1, z_1) $Ax + By + Cz + D = 0$

Solution:

- The given point is $(x_1, y_1, z_1) = (k, 3, -5)$. The equation of the plane is $x + 2y - 2z = 9$, which can be written as:

$$x + 2y - 2z - 9 = 0$$

Here, $A = 1$, $B = 2$, $C = -2$, $D = -9$. The distance is given as $d = 3$ units.

$$3 = \frac{|1(k) + 2(3) - 2(-5) - 9|}{\sqrt{1^2 + 2^2 + (-2)^2}}$$

$$3 = \frac{|k + 6 + 10 - 9|}{\sqrt{1 + 4 + 4}}$$

Substituting these values into the distance formula:

$$3 = \frac{|k + 7|}{\sqrt{9}}$$

$$3 = \frac{|k + 7|}{3}$$

Multiplying both sides by 3:

$$|k + 7| = 9$$

This gives two cases: $k + 7 = 9 \implies k = 2$
 $k + 7 = -9 \implies k = -16$

(ii) The given point is $(x_1, y_1, z_1) = (-6, 0, 0)$. The equation of the plane is:

$$2x - 3y + z - k = 0$$

Here, $A = 2$, $B = -3$, $C = 1$, $D = -k$. The distance is given as $d = \sqrt{14}$ units. Sub-

$$\sqrt{14} = \frac{|2(-6) - 3(0) + 1(0) - k|}{\sqrt{2^2 + (-3)^2 + 1^2}}$$

stituting these values into the distance formula:

$$\sqrt{14} = \frac{|-12 - k|}{\sqrt{4 + 9 + 1}}$$

$$\sqrt{14} = \frac{|-(k + 12)|}{\sqrt{14}}$$

$$\sqrt{14} = \frac{|k + 12|}{\sqrt{14}}$$

Multiplying both sides by $\sqrt{14}$:

$$|k + 12| = 14$$

This gives two cases: $k + 12 = 14 \implies k = 2$
 $k + 12 = -14 \implies k = -26$

Final Answer

(i) $k = 2$ or $k = -16$

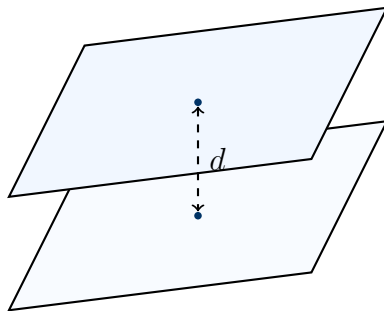
(ii) $k = 2$ or $k = -26$

Question 5

Solve the following:

- Find the distance between the planes $2x + 3y + 4z = 4$ and $4x + 6y + 8z = 12$.
- Find the distance between the planes $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) - 4 = 0$ and $\vec{r} \cdot (6\hat{i} - 9\hat{j} + 18\hat{k}) + 30 = 0$.

TYPE 82 Distance Between Two Parallel Planes



Distance between two parallel planes

Concept / Formula Used: $d = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}$ $Ax + By + Cz + D_1 = 0$ $Ax + By + Cz + D_2 = 0$ d

Solution:

(i) The given planes are:

$$2x + 3y + 4z - 4 = 0 \quad \text{--- (1)}$$

$$4x + 6y + 8z - 12 = 0 \quad \text{--- (2)}$$

Dividing equation (2) by 2 to make the coefficients of x, y, z identical to equation (1):

$$2x + 3y + 4z - 6 = 0 \quad \text{--- (3)}$$

Comparing equations (1) and (3) with the standard forms, we have:

$$A = 2, \quad B = 3, \quad C = 4, \quad D_1 = -4, \quad D_2 = -6$$

$$d = \frac{|-4 - (-6)|}{\sqrt{2^2 + 3^2 + 4^2}}$$

$$\begin{aligned} \text{Using the distance formula:} &= \frac{|-4 + 6|}{\sqrt{4 + 9 + 16}} \\ &= \frac{2}{\sqrt{29}} \text{ units} \end{aligned}$$

(ii) The given planes in vector form are:

$$\vec{r} \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) - 4 = 0$$

$$\vec{r} \cdot (6\hat{i} - 9\hat{j} + 18\hat{k}) + 30 = 0$$

In Cartesian form, these are:

$$2x - 3y + 6z - 4 = 0 \quad \text{--- (1)}$$

$$6x - 9y + 18z + 30 = 0 \quad \text{--- (2)}$$

Dividing equation (2) by 3 to make the coefficients of x, y, z identical to equation (1):

$$2x - 3y + 6z + 10 = 0 \quad \text{--- (3)}$$

Comparing equations (1) and (3) with the standard forms, we have:

$$A = 2, \quad B = -3, \quad C = 6, \quad D_1 = -4, \quad D_2 = 10$$

$$d = \frac{|-4 - 10|}{\sqrt{2^2 + (-3)^2 + 6^2}}$$

$$= \frac{|-14|}{\sqrt{4 + 9 + 36}}$$

Using the distance formula:

$$= \frac{14}{\sqrt{49}}$$

$$= \frac{14}{7}$$

$$= 2 \text{ units}$$

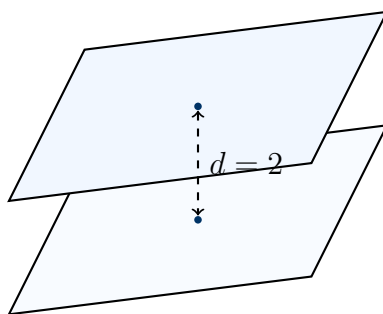
Final Answer

- (i) $\frac{2}{\sqrt{29}}$ units
 (ii) 2 units

Question 6

If the distance between the planes $2x - 2y + z = 3$ and $2x - 2y + z = k$ is 2 units, find the value(s) of k .

TYPE 82 Distance Between Two Parallel Planes



Two parallel planes at distance 2

Concept / Formula Used: $d = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}$ $Ax + By + Cz + D_1 = 0$ $Ax + By + Cz + D_2 = 0$ d

Solution:

The given equations of the planes are:

$$2x - 2y + z - 3 = 0 \quad \text{--- (1)}$$

$$2x - 2y + z - k = 0 \quad \text{--- (2)}$$

Comparing with the standard forms, we have:

$$A = 2, \quad B = -2, \quad C = 1, \quad D_1 = -3, \quad D_2 = -k$$

The distance between the planes is given as $d = 2$ units. Substituting these values into

$$2 = \frac{|-3 - (-k)|}{\sqrt{2^2 + (-2)^2 + 1^2}}$$

the distance formula: $2 = \frac{|k - 3|}{\sqrt{4 + 4 + 1}}$ Multiplying both sides by 3:

$$2 = \frac{|k - 3|}{\sqrt{9}}$$

$$2 = \frac{|k - 3|}{3}$$

$$|k - 3| = 6$$

This gives two cases: $k - 3 = 6 \implies k = 9$
 $k - 3 = -6 \implies k = -3$

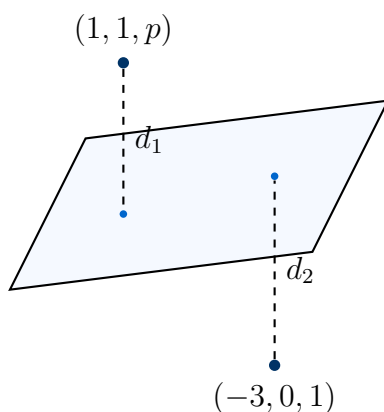
Final Answer

$k = 9$ or $k = -3$

Question 7

If the points $(1, 1, p)$ and $(-3, 0, 1)$ are equidistant from the plane $\vec{r} \cdot (3\hat{i} + 4\hat{j} - 12\hat{k}) + 13 = 0$, find the value(s) of p .

TYPE 80 Verify Two Points are Equidistant from a Plane



Two points equidistant from a plane

Concept / Formula Used: $d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$ d (x_1, y_1, z_1) $Ax + By + Cz + D = 0$

Solution:

The equation of the plane is $\vec{r} \cdot (3\hat{i} + 4\hat{j} - 12\hat{k}) + 13 = 0$. In Cartesian form, this is:

$$3x + 4y - 12z + 13 = 0$$

Here, $A = 3$, $B = 4$, $C = -12$, $D = 13$.

Let d_1 be the perpendicular distance from the point $(1, 1, p)$ to the plane:

$$\begin{aligned} d_1 &= \frac{|3(1) + 4(1) - 12(p) + 13|}{\sqrt{3^2 + 4^2 + (-12)^2}} \\ &= \frac{|3 + 4 - 12p + 13|}{\sqrt{9 + 16 + 144}} \\ &= \frac{|20 - 12p|}{\sqrt{169}} \\ &= \frac{|20 - 12p|}{13} \end{aligned}$$

Let d_2 be the perpendicular distance from the point $(-3, 0, 1)$ to the plane:

$$\begin{aligned} d_2 &= \frac{|3(-3) + 4(0) - 12(1) + 13|}{\sqrt{3^2 + 4^2 + (-12)^2}} \\ &= \frac{|-9 + 0 - 12 + 13|}{13} \\ &= \frac{|-8|}{13} \\ &= \frac{8}{13} \end{aligned}$$

Since the points are equidistant from the plane, we have $d_1 = d_2$: $\frac{|20 - 12p|}{13} = \frac{8}{13}$ This gives $|20 - 12p| = 8$

$$\text{Case 1: } 20 - 12p = 8 \implies 12p = 12 \implies p = 1$$

gives two cases:

$$\text{Case 2: } 20 - 12p = -8 \implies 12p = 28 \implies p = \frac{28}{12} = \frac{7}{3}$$

Final Answer

$$p = 1 \text{ or } p = \frac{7}{3}$$

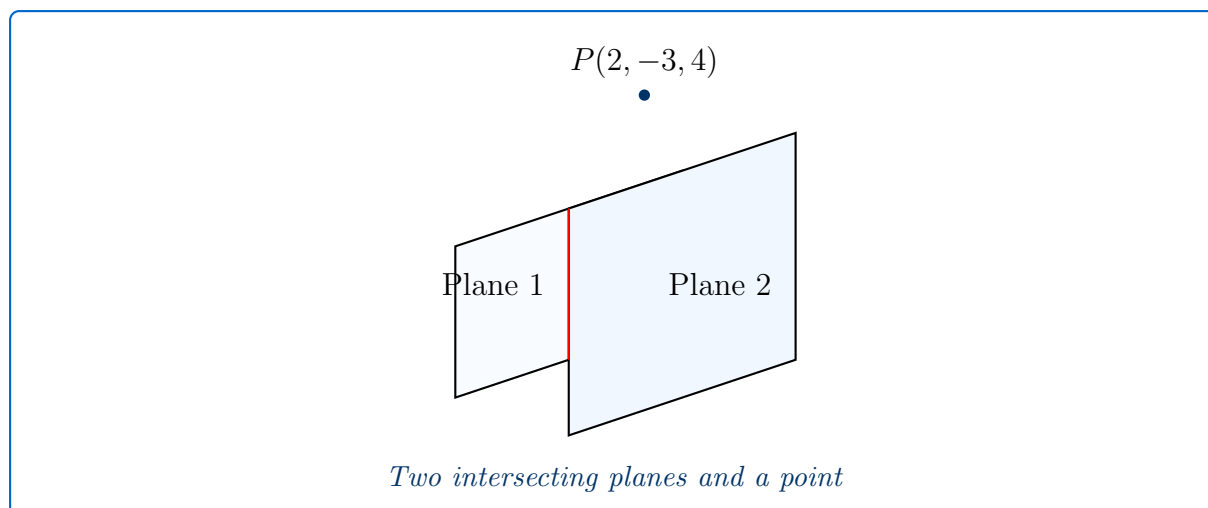
Question 8

Given below are equations of two planes.

$$\text{Plane 1: } \frac{x}{-2} + \frac{3y}{4} - \frac{5z}{4} = 1 \quad \text{and} \quad \text{Plane 2: } \frac{x}{3} - \frac{y}{4} - \frac{z}{1} = 1$$

Is the point $P(2, -3, 4)$ closer to plane 1 or plane 2? Also, find the angle between the planes.

TYPE 83 Compare a Point's Distance to Two Planes & Find Angle Between Them



Concept / Formula Used: $d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$ θ $\vec{n}_1$ $\vec{n}_2$ $\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$

Solution:

First, let us convert the equations of both planes into standard Cartesian form.

For **Plane 1**:

$$\frac{x}{-2} + \frac{3y}{4} - \frac{5z}{4} = 1$$

Multiplying the entire equation by 4:

$$-2x + 3y - 5z = 4 \implies 2x - 3y + 5z + 4 = 0$$

Here, $A_1 = 2$, $B_1 = -3$, $C_1 = 5$, $D_1 = 4$.

For **Plane 2**:

$$\frac{x}{3} - \frac{y}{4} - \frac{z}{1} = 1$$

Multiplying the entire equation by 12:

$$4x - 3y - 12z = 12 \implies 4x - 3y - 12z - 12 = 0$$

Here, $A_2 = 4$, $B_2 = -3$, $C_2 = -12$, $D_2 = -12$.

$$\begin{aligned} d_1 &= \frac{|2(2) - 3(-3) + 5(4) + 4|}{\sqrt{2^2 + (-3)^2 + 5^2}} \\ \text{Let } d_1 \text{ be the distance of the point } P(2, -3, 4) \text{ from Plane 1: } &= \frac{|4 + 9 + 20 + 4|}{\sqrt{4 + 9 + 25}} \\ &= \frac{37}{\sqrt{38}} \text{ units} \end{aligned}$$

Since $\sqrt{38} \approx 6.164$, we have:

$$d_1 \approx \frac{37}{6.164} \approx 6.00 \text{ units}$$

Let d_2 be the distance of the point $P(2, -3, 4)$ from Plane 2:

$$\begin{aligned} d_2 &= \frac{|4(2) - 3(-3) - 12(4) - 12|}{\sqrt{4^2 + (-3)^2 + (-12)^2}} \\ &= \frac{|8 + 9 - 48 - 12|}{\sqrt{16 + 9 + 144}} \\ &= \frac{|-43|}{\sqrt{169}} \\ &= \frac{43}{13} \text{ units} \end{aligned}$$

Since $13 \times 3.307 = 43$, we have:

$$d_2 \approx 3.31 \text{ units}$$

Since $d_2 < d_1$, the point $P(2, -3, 4)$ is closer to Plane 2.

Now, let us find the angle θ between the two planes. The normal vectors are:

$$\vec{n}_1 = 2\hat{i} - 3\hat{j} + 5\hat{k}$$

$$\vec{n}_2 = 4\hat{i} - 3\hat{j} - 12\hat{k}$$

$$\vec{n}_1 \cdot \vec{n}_2 = (2)(4) + (-3)(-3) + (5)(-12)$$

Let us compute $\vec{n}_1 \cdot \vec{n}_2$:

$$\begin{aligned} &= 8 + 9 - 60 \\ &= -43 \end{aligned}$$

The magnitudes are:

$$|\vec{n}_1| = \sqrt{38}$$

$$|\vec{n}_2| = \sqrt{169} = 13$$

$$\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$$

Using the angle formula:

$$\begin{aligned} &= \frac{|-43|}{13\sqrt{38}} \quad \text{Thus, the angle is:} \\ &= \frac{43}{13\sqrt{38}} \end{aligned}$$

$$\theta = \cos^{-1} \left(\frac{43}{13\sqrt{38}} \right)$$

Final Answer

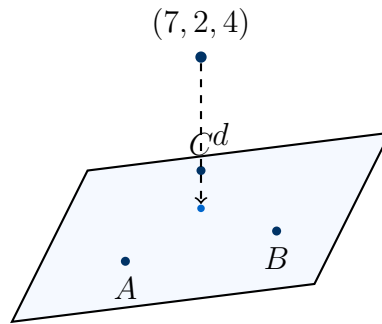
Point P is closer to Plane 2.

$$\text{Angle between planes: } \theta = \cos^{-1} \left(\frac{43}{13\sqrt{38}} \right)$$

Question 9

Find the distance from the point $(7, 2, 4)$ to the plane determined by the points $A(2, 5, -3)$, $B(-2, -3, 5)$ and $C(5, 3, -3)$.

TYPE 79 Perpendicular Distance from a Point to a Plane



Plane through three points and distance from a point

Concept

/

Formula**Used:**

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

0 (x_1, y_1, z_1) (x_2, y_2, z_2) (x_3, y_3, z_3)

Solution:

Let the given points be:

$$(x_1, y_1, z_1) = (2, 5, -3)$$

$$(x_2, y_2, z_2) = (-2, -3, 5)$$

$$(x_3, y_3, z_3) = (5, 3, -3)$$

The equation of the plane is:

$$\begin{vmatrix} x - 2 & y - 5 & z - (-3) \\ -2 - 2 & -3 - 5 & 5 - (-3) \\ 5 - 2 & 3 - 5 & -3 - (-3) \end{vmatrix} = 0$$

$$\begin{vmatrix} x - 2 & y - 5 & z + 3 \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} = 0$$

Expanding the determinant along the first row:

$$(x - 2)[(-8)(0) - (8)(-2)] - (y - 5)[(-4)(0) - (8)(3)] + (z + 3)[(-4)(-2) - (-8)(3)] = 0$$

$$(x - 2)(16) - (y - 5)(-24) + (z + 3)(8 + 24) = 0$$

$$16(x - 2) + 24(y - 5) + 32(z + 3) = 0$$

Dividing the entire equation by 8:

$$2(x - 2) + 3(y - 5) + 4(z + 3) = 0$$

$$2x - 4 + 3y - 15 + 4z + 12 = 0$$

$$2x + 3y + 4z - 7 = 0$$

Now, let us find the perpendicular distance of the point $(7, 2, 4)$ from this plane: Using

$$d = \frac{|2(7) + 3(2) + 4(4) - 7|}{\sqrt{2^2 + 3^2 + 4^2}}$$

$$= \frac{|14 + 6 + 16 - 7|}{\sqrt{4 + 9 + 16}}$$

the distance formula:

$$= \frac{|29|}{\sqrt{29}}$$

$$= \sqrt{29} \text{ units}$$

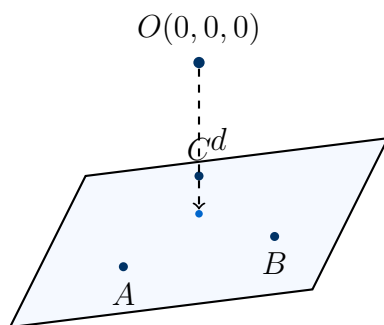
Final Answer

$\sqrt{29}$ units

Question 10

Find the vector equation of the plane determined by the points $A(3, -1, 2)$, $B(5, 2, 4)$ and $C(-1, -1, 6)$. Hence, find the distance of the plane thus obtained from the origin.

TYPE 84 Plane Through Three Points & Its Distance from the Origin



Plane through three points and distance from origin

Concept

/

Formula

Used:

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

$$0 \quad (x_1, y_1, z_1) \quad (x_2, y_2, z_2) \quad (x_3, y_3, z_3)$$

Solution:

Let the given points be:

$$(x_1, y_1, z_1) = (3, -1, 2)$$

$$(x_2, y_2, z_2) = (5, 2, 4)$$

$$(x_3, y_3, z_3) = (-1, -1, 6)$$

The equation of the plane is:

$$\begin{vmatrix} x - 3 & y - (-1) & z - 2 \\ 5 - 3 & 2 - (-1) & 4 - 2 \\ -1 - 3 & -1 - (-1) & 6 - 2 \end{vmatrix} = 0$$

$$\begin{vmatrix} x-3 & y+1 & z-2 \\ 2 & 3 & 2 \\ -4 & 0 & 4 \end{vmatrix} = 0$$

Expanding the determinant along the first row:

$$(x-3)[(3)(4) - (2)(0)] - (y+1)[(2)(4) - (2)(-4)] + (z-2)[(2)(0) - (3)(-4)] = 0$$

$$(x-3)(12) - (y+1)(8+8) + (z-2)(12) = 0$$

$$12(x-3) - 16(y+1) + 12(z-2) = 0$$

Dividing the entire equation by 4:

$$3(x-3) - 4(y+1) + 3(z-2) = 0$$

$$3x - 9 - 4y - 4 + 3z - 6 = 0$$

$$3x - 4y + 3z - 19 = 0$$

The vector equation of the plane is:

$$\vec{r} \cdot (3\hat{i} - 4\hat{j} + 3\hat{k}) = 19$$

Now, let us find the perpendicular distance of this plane from the origin $(0, 0, 0)$: Using

$$d = \frac{|3(0) - 4(0) + 3(0) - 19|}{\sqrt{3^2 + (-4)^2 + 3^2}}$$

$$\begin{aligned} \text{the distance formula:} &= \frac{|-19|}{\sqrt{9+16+9}} \\ &= \frac{19}{\sqrt{34}} \text{ units} \end{aligned}$$

Final Answer

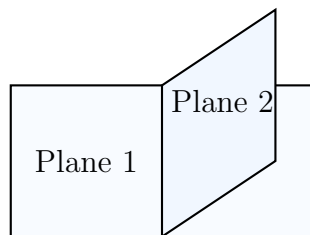
Vector Equation: $\vec{r} \cdot (3\hat{i} - 4\hat{j} + 3\hat{k}) = 19$

Distance from origin: $\frac{19}{\sqrt{34}}$ units

Question 11

Find the equation of the plane passing through the points $(1, 0, -2)$, $(3, -1, 0)$ and perpendicular to the plane $2x - y + z = 8$. Also, find the distance of the plane thus obtained from origin.

TYPE 85 Plane Through Two Points Perpendicular to a Plane, Plus Distance



Two perpendicular planes

Concept / Formula Used: $a(x-x_1)+b(y-y_1)+c(z-z_1) = 0$ $a_1a_2+b_1b_2+c_1c_2 = 0$
 (x_1, y_1, z_1)

Solution:

The equation of any plane passing through the point $(1, 0, -2)$ is:

$$a(x-1) + b(y-0) + c(z-(-2)) = 0$$

$$a(x-1) + by + c(z+2) = 0 \quad \text{--- (1)}$$

Since the plane also passes through the point $(3, -1, 0)$, this point must satisfy equation (1):

$$a(3-1) + b(-1) + c(0+2) = 0$$

$$2a - b + 2c = 0 \quad \text{--- (2)}$$

The plane is perpendicular to the given plane $2x - y + z = 8$. The normal vector of the given plane is $\vec{n} = 2\hat{i} - \hat{j} + \hat{k}$. Since the two planes are perpendicular, the dot product of their normal vectors must be zero:

$$2a - b + c = 0 \quad \text{--- (3)}$$

Now, let us solve equations (2) and (3) using the method of cross-multiplication:

Using cross-multiplication:

$$\frac{a}{(-1)(1) - (2)(-1)} = \frac{b}{(2)(2) - (2)(1)} = \frac{c}{(2)(-1) - (-1)(2)}$$

$$\frac{a}{-1+2} = \frac{b}{4-2} = \frac{c}{-2+2}$$

$$\frac{a}{1} = \frac{b}{2} = \frac{c}{0} = \lambda$$

Thus, we have:

$$a = \lambda, \quad b = 2\lambda, \quad c = 0$$

Substituting these values of a, b, c into equation (1):

$$\lambda(x-1) + 2\lambda y + 0(z+2) = 0$$

Dividing by λ (since $\lambda \neq 0$):

$$x - 1 + 2y = 0 \implies x + 2y - 1 = 0$$

Now, let us find the perpendicular distance of this plane from the origin $(0, 0, 0)$: Using

$$d = \frac{|1(0) + 2(0) - 1|}{\sqrt{1^2 + 2^2 + 0^2}}$$

the distance formula:
$$= \frac{|-1|}{\sqrt{1+4}}$$

$$= \frac{1}{\sqrt{5}} \text{ units}$$

Final Answer

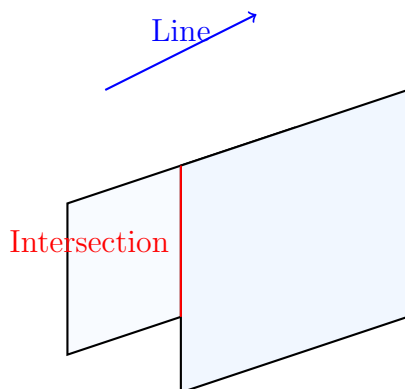
Equation of the plane: $x + 2y - 1 = 0$

Distance from origin: $\frac{1}{\sqrt{5}}$ units

Question 12

Find the equation of the plane passing through the intersection of the planes $4x - y + z = 10$ and $x + y - z = 4$ and parallel to the line with direction ratios $\langle 2, 1, 1 \rangle$. Also find the perpendicular distance of the point $(1, 1, 1)$ from this plane.

TYPE 86 Plane Through Intersection of Two Planes, Plus Distance



Plane through intersection of two planes

Concept / Formula Used: $P_1 + \lambda P_2 = 0$ $P_1 = 0$ $P_2 = 0$

Solution:

The given equations of the planes are:

$$4x - y + z - 10 = 0 \quad \text{--- (1)}$$

$$x + y - z - 4 = 0 \quad \text{--- (2)}$$

The equation of any plane passing through the intersection of these two planes is:

$$(4x - y + z - 10) + \lambda(x + y - z - 4) = 0$$

Rearranging the terms:

$$(4 + \lambda)x + (\lambda - 1)y + (1 - \lambda)z - (10 + 4\lambda) = 0 \quad \text{--- (3)}$$

The direction ratios of the normal to this plane are:

$$\langle 4 + \lambda, \lambda - 1, 1 - \lambda \rangle$$

The plane is parallel to the line with direction ratios $\langle 2, 1, 1 \rangle$. Therefore, the normal to the plane must be perpendicular to this line. Using the condition of perpendicularity:

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

$$2(4 + \lambda) + 1(\lambda - 1) + 1(1 - \lambda) = 0$$

$$8 + 2\lambda + \lambda - 1 + 1 - \lambda = 0$$

$$8 + 2\lambda = 0$$

Expanding and simplifying:

$$2\lambda = -8$$

$$\lambda = -4$$

Substituting $\lambda = -4$ into equation (3):

$$(4 - 4)x + (-4 - 1)y + (1 - (-4))z - (10 + 4(-4)) = 0$$

$$-5y + 5z - (10 - 16) = 0$$

$$-5y + 5z + 6 = 0$$

Multiplying by -1 :

$$5y - 5z - 6 = 0$$

Now, let us find the perpendicular distance of the point $(1, 1, 1)$ from this plane: Using

$$d = \frac{|0(1) + 5(1) - 5(1) - 6|}{\sqrt{0^2 + 5^2 + (-5)^2}}$$

$$= \frac{|-6|}{\sqrt{25 + 25}}$$

the distance formula: $= \frac{6}{\sqrt{50}}$

$$= \frac{6}{5\sqrt{2}}$$

$$= \frac{3\sqrt{2}}{5} \text{ units}$$

Final Answer

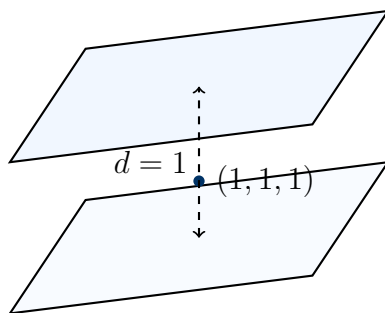
Equation of the plane: $5y - 5z - 6 = 0$

Perpendicular distance: $\frac{3\sqrt{2}}{5}$ units

Question 13

Find the equations of the planes parallel to the plane $x - 2y + 2z = 3$ and at a unit distance from the point $(1, 1, 1)$.

TYPE 87 Planes Parallel to a Given Plane at a Given Distance



Planes parallel to a given plane at unit distance

Concept / Formula Used: $Ax + By + Cz + \lambda = 0$ $Ax + By + Cz + D = 0$

Solution:

The given plane is $x - 2y + 2z - 3 = 0$. Any plane parallel to this plane is of the form:

$$x - 2y + 2z + \lambda = 0 \quad \text{--- (1)}$$

The perpendicular distance from the point $(1, 1, 1)$ to this plane is given as 1 unit. Using

$$1 = \frac{|1(1) - 2(1) + 2(1) + \lambda|}{\sqrt{1^2 + (-2)^2 + 2^2}}$$

$$\text{the distance formula: } 1 = \frac{|1 - 2 + 2 + \lambda|}{\sqrt{1 + 4 + 4}} \quad \text{Multiplying both sides by 3:}$$

$$1 = \frac{|1 + \lambda|}{\sqrt{9}}$$

$$1 = \frac{|1 + \lambda|}{3}$$

$$|1 + \lambda| = 3$$

This gives two cases:

$$1 + \lambda = 3 \implies \lambda = 2$$

$$1 + \lambda = -3 \implies \lambda = -4$$

Substituting these values of λ back into equation (1): For $\lambda = 2$:

$$x - 2y + 2z + 2 = 0$$

For $\lambda = -4$:

$$x - 2y + 2z - 4 = 0$$

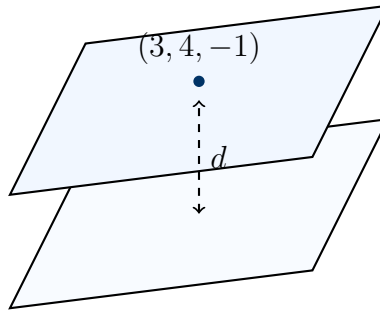
Final Answer

$$x - 2y + 2z + 2 = 0 \text{ and } x - 2y + 2z - 4 = 0$$

Question 14

Find the equation of the plane through the point $(3, 4, -1)$ which is parallel to the plane $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) + 7 = 0$. Also, find the distance between the two planes.

TYPE 87 Planes Parallel to a Given Plane at a Given Distance



Parallel planes and distance

Concept / Formula Used: $\vec{r} \cdot \vec{n} + \lambda = 0$ $\vec{r} \cdot \vec{n} + D = 0$

Solution:

The given plane is:

$$\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) + 7 = 0$$

In Cartesian form, this is:

$$2x - 3y + 5z + 7 = 0 \quad \text{--- (1)}$$

Any plane parallel to this plane is of the form:

$$2x - 3y + 5z + \lambda = 0 \quad \text{--- (2)}$$

Since the parallel plane passes through the point $(3, 4, -1)$, this point must satisfy equation (2):

$$2(3) - 3(4) + 5(-1) + \lambda = 0$$

$$6 - 12 - 5 + \lambda = 0$$

$$-11 + \lambda = 0$$

$$\lambda = 11$$

Substituting $\lambda = 11$ into equation (2), the equation of the parallel plane is:

$$2x - 3y + 5z + 11 = 0$$

In vector form, this is:

$$\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) + 11 = 0$$

Now, let us find the distance d between the two parallel planes:

$$2x - 3y + 5z + 7 = 0 \quad \text{and} \quad 2x - 3y + 5z + 11 = 0$$

$$d = \frac{|11 - 7|}{\sqrt{2^2 + (-3)^2 + 5^2}}$$

Using the distance formula for parallel planes:

$$= \frac{4}{\sqrt{4 + 9 + 25}}$$

$$= \frac{4}{\sqrt{38}} \text{ units}$$

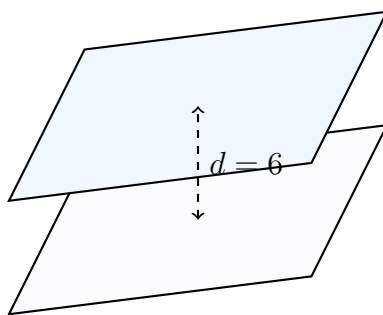
Final Answer

Equation of the plane: $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) + 11 = 0$

Distance between planes: $\frac{4}{\sqrt{38}}$ units

Question 15

Find the equations of the planes parallel to the plane $3x - 6y + 2z = 12$ and 6 units away from it.

TYPE 87 Planes Parallel to a Given Plane at a Given Distance

Parallel planes at distance 6

Concept / Formula Used: $d = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}$ $Ax + By + Cz + D_1 = 0$ $Ax + By + Cz + D_2 = 0$ d

Solution:

The given plane is $3x - 6y + 2z - 12 = 0$. Any plane parallel to this plane is of the form:

$$3x - 6y + 2z + \lambda = 0$$

The distance between these two parallel planes is given as 6 units. Using the distance

$$6 = \frac{|\lambda - (-12)|}{\sqrt{3^2 + (-6)^2 + 2^2}}$$

$$6 = \frac{|\lambda + 12|}{\sqrt{9 + 36 + 4}}$$

formula: $6 = \frac{|\lambda + 12|}{\sqrt{49}}$ Multiplying both sides by 7:

$$6 = \frac{|\lambda + 12|}{7}$$

$$|\lambda + 12| = 42$$

This gives two cases: $\lambda + 12 = 42 \implies \lambda = 30$
 $\lambda + 12 = -42 \implies \lambda = -54$

Substituting these values of λ back into the parallel plane equation: For $\lambda = 30$:

$$3x - 6y + 2z + 30 = 0$$

For $\lambda = -54$:

$$3x - 6y + 2z - 54 = 0$$

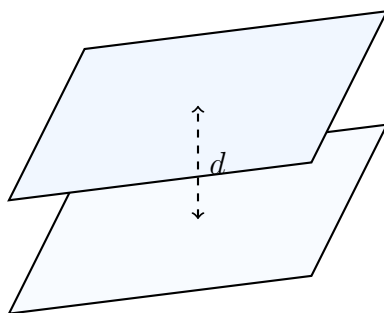
Final Answer

$$3x - 6y + 2z + 30 = 0 \text{ and } 3x - 6y + 2z - 54 = 0$$

Question 16

Find the shortest distance between the planes $6x + 2y - 3z = 12$ and $12x + 4y - 6z = 17$.

TYPE 82 Distance Between Two Parallel Planes



Shortest distance between two parallel planes

Concept / Formula Used: $d = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}$ $Ax + By + Cz + D_1 = 0$ $Ax + By + Cz + D_2 = 0$ d

Solution:

The given equations of the planes are:

$$6x + 2y - 3z - 12 = 0 \quad \text{--- (1)}$$

$$12x + 4y - 6z - 17 = 0 \quad \text{--- (2)}$$

Multiplying equation (1) by 2 to make the coefficients of x, y, z identical to equation (2):

$$12x + 4y - 6z - 24 = 0 \quad \text{--- (3)}$$

Comparing equations (2) and (3) with the standard forms, we have:

$$A = 12, \quad B = 4, \quad C = -6, \quad D_1 = -17, \quad D_2 = -24$$

$$\begin{aligned} d &= \frac{|-17 - (-24)|}{\sqrt{12^2 + 4^2 + (-6)^2}} \\ &= \frac{|-17 + 24|}{\sqrt{144 + 16 + 36}} \end{aligned}$$

Using the distance formula:

$$\begin{aligned} &= \frac{7}{\sqrt{196}} \\ &= \frac{7}{14} \\ &= \frac{1}{2} \text{ units} \end{aligned}$$

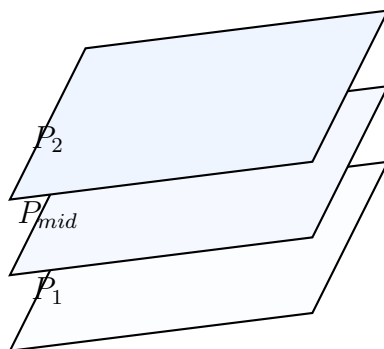
Final Answer

$$\frac{1}{2} \text{ units}$$

Question 17

Find the equation of the planes mid-parallel to the planes $2x - 2y + z + 3 = 0$ and $2x - 2y + z + 9 = 0$.

TYPE 88 Equation of the Mid-Parallel Plane



Mid-parallel plane between two parallel planes

Concept / Formula Used: $Ax + By + Cz + \frac{D_1 + D_2}{2} = 0$ $Ax + By + Cz + D_1 = 0$
 $Ax + By + Cz + D_2 = 0$

Solution:

The given parallel planes are:

$$2x - 2y + z + 3 = 0 \quad \text{--- (1)}$$

$$2x - 2y + z + 9 = 0 \quad \text{--- (2)}$$

Comparing with the standard forms, we have:

$$A = 2, \quad B = -2, \quad C = 1, \quad D_1 = 3, \quad D_2 = 9$$

The mid-parallel plane lies exactly halfway between the two planes. Its constant term D

$$D = \frac{D_1 + D_2}{2}$$

is the average of D_1 and D_2 : $\quad \quad \quad$ Thus, the equation of the mid-parallel plane

$$= \frac{3 + 9}{2}$$

$$= \frac{12}{2}$$

$$= 6$$

is:

$$2x - 2y + z + 6 = 0$$

Final Answer

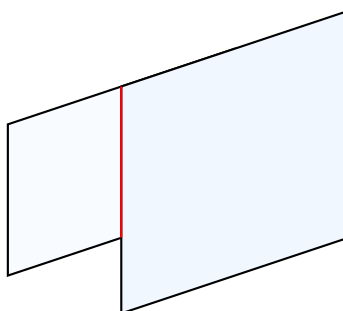
$$2x - 2y + z + 6 = 0$$

Question 18

Find the equation of the planes passing through the intersection of the planes $\vec{r} \cdot (2\hat{i} + 6\hat{j}) + 12 = 0$ and $\vec{r} \cdot (3\hat{i} - \hat{j} + 4\hat{k}) = 0$ and at a unit distance from the origin.

TYPE 86 Plane Through Intersection of Two Planes, Plus Distance

$O(0,0,0)$



Planes through intersection of two planes at unit distance from origin

Concept / Formula Used: $P_1 + \lambda P_2 = 0$ $P_1 = 0$ $P_2 = 0$

Solution:

First, let us write the equations of the given planes in Cartesian form. For the first plane:

$$\vec{r} \cdot (2\hat{i} + 6\hat{j}) + 12 = 0 \implies 2x + 6y + 12 = 0 \implies x + 3y + 6 = 0 \quad \text{--- (1)}$$

For the second plane:

$$\vec{r} \cdot (3\hat{i} - \hat{j} + 4\hat{k}) = 0 \implies 3x - y + 4z = 0 \quad \text{--- (2)}$$

The equation of any plane passing through the intersection of these two planes is:

$$(x + 3y + 6) + \lambda(3x - y + 4z) = 0$$

Rearranging the terms:

$$(1 + 3\lambda)x + (3 - \lambda)y + 4\lambda z + 6 = 0 \quad \text{--- (3)}$$

The perpendicular distance from the origin $(0, 0, 0)$ to this plane is given as 1 unit. Using

$$1 = \frac{|(1 + 3\lambda)(0) + (3 - \lambda)(0) + 4\lambda(0) + 6|}{\sqrt{(1 + 3\lambda)^2 + (3 - \lambda)^2 + (4\lambda)^2}}$$

the distance formula: $1 = \frac{6}{\sqrt{(1 + 6\lambda + 9\lambda^2) + (9 - 6\lambda + \lambda^2) + 16\lambda^2}}$ Squaring both sides:

$$1 = \frac{6}{\sqrt{26\lambda^2 + 10}}$$

$$1 = \frac{36}{26\lambda^2 + 10}$$

$$26\lambda^2 + 10 = 36$$

$$26\lambda^2 = 26$$

$$\lambda^2 = 1$$

$$\lambda = \pm 1$$

Now, let us substitute these values of λ back into equation (3):

Case 1: For $\lambda = 1$

$$(1 + 3(1))x + (3 - 1)y + 4(1)z + 6 = 0$$

$$4x + 2y + 4z + 6 = 0$$

Dividing by 2:

$$2x + y + 2z + 3 = 0$$

In vector form:

$$\vec{r} \cdot (2\hat{i} + \hat{j} + 2\hat{k}) + 3 = 0$$

Case 2: For $\lambda = -1$

$$(1 + 3(-1))x + (3 - (-1))y + 4(-1)z + 6 = 0$$

$$-2x + 4y - 4z + 6 = 0$$

Dividing by -2 :

$$x - 2y + 2z - 3 = 0$$

In vector form:

$$\vec{r} \cdot (\hat{i} - 2\hat{j} + 2\hat{k}) - 3 = 0$$

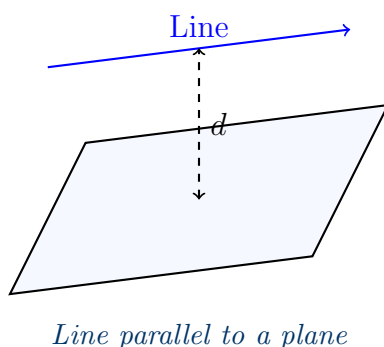
Final Answer

$$\vec{r} \cdot (2\hat{i} + \hat{j} + 2\hat{k}) + 3 = 0 \text{ and } \vec{r} \cdot (\hat{i} - 2\hat{j} + 2\hat{k}) - 3 = 0$$

Question 19

State when the line $\vec{r} = \vec{a} + \lambda\vec{b}$ is parallel to the plane $\vec{r} \cdot \vec{n} = d$. Show that the line $\vec{r} = \hat{i} + \hat{j} + \lambda(3\hat{i} - \hat{j} + 2\hat{k})$ is parallel to the plane $\vec{r} \cdot (2\hat{j} + \hat{k}) = 3$. Also, find the distance between the line and the plane.

TYPE 45 Line Parallel to a Plane – Condition Check



Concept / Formula Used: $\vec{b} \cdot \vec{n} = 0$ and $\vec{a} \cdot \vec{n} \neq d$ $D = \frac{|\vec{a} \cdot \vec{n} - d|}{|\vec{n}|}$ $\vec{r} = \vec{a} + \lambda\vec{b}$ $\vec{r} \cdot \vec{n} = d$ $\vec{a}$

Solution:

Condition for Parallelism: A line $\vec{r} = \vec{a} + \lambda\vec{b}$ is parallel to the plane $\vec{r} \cdot \vec{n} = d$ if the direction vector of the line $\vec{b}$ is perpendicular to the normal vector of the plane $\vec{n}$, which means:

$$\vec{b} \cdot \vec{n} = 0$$

and the point $\vec{a}$ on the line does not lie on the plane, which means:

$$\vec{a} \cdot \vec{n} \neq d$$

Proof of Parallelism: The given line is:

$$\vec{r} = (\hat{i} + \hat{j}) + \lambda(3\hat{i} - \hat{j} + 2\hat{k})$$

Comparing with $\vec{r} = \vec{a} + \lambda\vec{b}$, we have:

$$\vec{a} = \hat{i} + \hat{j}$$

$$\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$$

The given plane is:

$$\vec{r} \cdot (2\hat{j} + \hat{k}) = 3$$

Comparing with $\vec{r} \cdot \vec{n} = d$, we have:

$$\vec{n} = 2\hat{j} + \hat{k}$$

$$d = 3$$

$$\vec{b} \cdot \vec{n} = (3\hat{i} - \hat{j} + 2\hat{k}) \cdot (2\hat{j} + \hat{k})$$

$$\begin{aligned} \text{Let us compute the dot product } \vec{b} \cdot \vec{n}: &= (3)(0) + (-1)(2) + (2)(1) \quad \text{Since } \vec{b} \cdot \vec{n} = 0, \text{ the} \\ &= 0 - 2 + 2 \\ &= 0 \end{aligned}$$

line is parallel to the plane.

$$\vec{a} \cdot \vec{n} = (\hat{i} + \hat{j}) \cdot (2\hat{j} + \hat{k})$$

$$\begin{aligned} \text{Let us also check if the point } \vec{a} \text{ lies on the plane:} &= (1)(0) + (1)(2) + (0)(1) \quad \text{Since} \\ &= 2 \neq 3 \end{aligned}$$

$\vec{a} \cdot \vec{n} \neq 3$, the line does not lie on the plane. Hence, the line is strictly parallel to the plane.

Distance between the line and the plane: The distance between the parallel line and the plane is the perpendicular distance from the point $\vec{a} = \hat{i} + \hat{j}$ to the plane: First, let

$$|\vec{n}| = \sqrt{0^2 + 2^2 + 1^2}$$

$$\begin{aligned} \text{us find the magnitude of the normal vector } |\vec{n}|: &= \sqrt{4 + 1} \quad \text{Using the distance} \\ &= \sqrt{5} \end{aligned}$$

$$\begin{aligned} D &= \frac{|\vec{a} \cdot \vec{n} - d|}{|\vec{n}|} \\ &= \frac{|2 - 3|}{\sqrt{5}} \\ \text{formula:} \quad &= \frac{|-1|}{\sqrt{5}} \\ &= \frac{1}{\sqrt{5}} \text{ units} \end{aligned}$$

Final Answer

Condition: $\vec{b} \cdot \vec{n} = 0$ and $\vec{a} \cdot \vec{n} \neq d$

Distance: $\frac{1}{\sqrt{5}}$ units